

COMPARATIVE STUDY OF PLASMA FOCUS MACHINES

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DECLARATION

I hereby declare that the work has been done by myself and no portion of the work contained in this thesis has been submitted in support of any application for any other degree or qualification of this or any other university or institute of learning.

Arwinder Singh

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ABSTRACT

The Lee Model code is a very powerful and convenient numerical experiment facility. This thesis analysed 44 Mather-type plasma focus (PF) machines operating in Deuterium, Neon and Argon; obtaining reasonably good fits for all machines so that realistic PF properties are computed and presented. Even the Speed-2 PF machine which uses a Marx bank is reasonably modelled and analysed. Also for Filippov-type PF machines, the computed current traces could be fitted to the measured only up to about half of the rising current trace.

For optimised low-inductance Type-1 machines (typically those in the region of or below 50 nH) the Lee Model 5-phase code is sufficient to fit the measured current traces. Those not properly time-matched may require the Lee Model 6-phase code to complete the fitting of the extended parts of the measured current dip. Type-2 high inductance machines (typically those above 100 nH) require the Lee Model 6-phase code even for optimised condition. The Lee Model 5-phase code is used to start the analysis of a machine, and when necessary proceeds to use the Lee Model 6-phase code to complete the fitting of any extended current dip. The code is found useful for testing whether machines are optimised or not. This has been the case of the India-Assam PF machine with electrodes designed to work in a different energy range. The code also assists in guiding the experimental process towards optimised operation as was the case for NX3. In analyzing radiation yields, it is found that the computed optimised neutron yield agrees with the measured maximum yield (difference between computed and measured being a factor about 3); the computed and available measured neon SXR yields agree in both optimised yields as well as yield versus pressure curves. Data from the machines studied was combined with that from existing machines to obtain neutron, Ne lines and Ar all-lines scaling laws.

One final observation is that, the Slow Focus Mode (SFM) is encountered at the highest pressure at which a pinch just occurs. This SFM produces a much larger pinch cross-section which may be a desirable property for deposition of advanced materials and nano-materials.

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CHAPTER 1

Introduction

1.1 Background

Ever since the industrial revolution, the demand for electrical power has increased tremendously. With the ever depleting amount of petroleum, gas and coal sources, the need to find a long-term source of energy is of extreme importance [1]. Controlled nuclear fusion is one of the alternatives as it is environmentally friendly and practically inexhaustible. To understand fusion reaction, dense plasma focus devices have been built in various parts of the world and studies have been carried out on high energy density nuclear fusion in plasmas as well as applications with fusion neutrons and with other emissions from high density plasmas in various gases.

A plasma focus device is a machine that produces by electromagnetic acceleration and compression, short lived plasma that is so hot and dense that it emits radiation including x-rays and undergoes nuclear fusion when operated in deuterium. These outputs can be used in various applications such as microelectronics, lithography, medical applications, material modifications and various other applications [2].

There are presently many research centres and universities around the world which operate various dense plasma focus (DPF) machines. This includes a network of ten identical DPF machines operating in eight countries under the Asian African Association for Plasma Training [3]; International Centre for Dense Magnetised Plasmas (ICDMP) in Warsaw, Poland which operates the largest DPF machine in the world, PF 1000 [4]; National Institute of Education, Nanyang Technological University, Singapore which operates NX1 [5-7], NX2 [7-9], NX3 [10-11] and FMPF-1 [11-12]; INTI International University in Malaysia which operates the INTI PF [13-18]; Tsinghua University in China which operates the DPF 2.2 [19]; Amirkabir University in Iran which operates the Amirkabir PF [20]; and India

Bhabha Atomic Research Centre which operates the BARC PF1 [21] and BARC PF2 [22]. The Inter-institutional Program for Plasma Focus Research coordinated by the National Laboratory of Dense Magnetized Plasmas in Buenos Aires which cooperates with the Chilean Nuclear Energy Commission has various DPF machines such as PF50 [23-25], PF 400 [26-27], Speed 2 [25], Argentina Nano focus machine [28-29], Chile Nano focus machine [23-25] and GN1 [30-32].

1.2 Problem Statement

With the increasing number and types of DPF machines all over the world, the performance of these machines comes into question. Are these machines operating at the optimal yields or performing according to any existing or new scaling laws?

1.3 Objective

The main objective of the thesis is to carry out a comparative study on the DPF machines worldwide with the aid of the Lee Model codes to compare the operations, yields and performance of the machines according to any existing or new scaling laws. It requires obtaining the current waveforms, including results of yields where available, of DPF machines from published articles or directly from research groups.

A comprehensive collection of DPF machines data will provide reference data on the machine parameters, operating parameters, current waveforms, fitting parameters and plasma parameters. These reference data contribute to enhancing the understanding of plasma focus dynamics. It will also confirm the radiation computation of the Lee Model code specifically neutron yield, soft X-ray yield and plasma focus scaling laws.

1.4 Literature Review

Ever since the initial discovery by J W Mather [33] during his research with the Marshall Gun in United States and independently by N.V.Filippov [34] at Kurchatov Institute, Russia in the early 60's there has been great interest in the plasma focus. This has resulted in more than 20 countries being actively involved in plasma research, including Malaysia and Singapore. The plasma focus machine has two distinct electrode geometries named after their founders. The obvious differences are in the electrode dimensions and the diameter to axial length aspect ratio of the inner electrode (see Figure 1.1). A Mather type Plasma focus has a small aspect ratio (anode radius/anode length) of typically 0.25 or below, with an inner electrode (anode) diameter typically of between 2 to 22 cm. A Filippov type plasma focus, on the other hand has a large aspect ratio, typically greater than 5 with inner electrode (anode) diameter in the range of 50 to 200 cm.

For plasma focus machines to operate and scale efficiently, the configurations and parameters of each machine have to be designed and adjusted within a range not too far from an optimum (efficient performance range). This can be done through numerical experiments using a proven code; as has been the case with several existing machines. Many existing machines have also been designed in the following way: starting with available capacitor bank (bank parameters capacitance C_0 , maximum charging voltage, circuit static inductance L_0 and circuit stray resistance r_0), experimenting until tube parameters are decided upon (tube parameters: anode radius 'a'; cathode radius b; anode length z_0 and then experimenting with the operational parameters (charging voltage V_0 , gas type and gas fill pressure P_0). The combination of the above parameters within an efficient performance range for each machine will produce plasma parameters which are suitable for the efficient production of radiations or plasma streams, depending on the objective or applications of each machine.

The anode and cathode are usually made of stainless steel or copper. The anode is either a cylindrical tube (to reduce hard X-ray emission) or a solid cylinder

whereas the cathode mostly consisting of 6 to 24 rods arranged around the anode may be designed in the form of either a squirrel cage or tubular form. The insulator of the plasma focus device is commonly made of ceramic, glass or Pyrex.

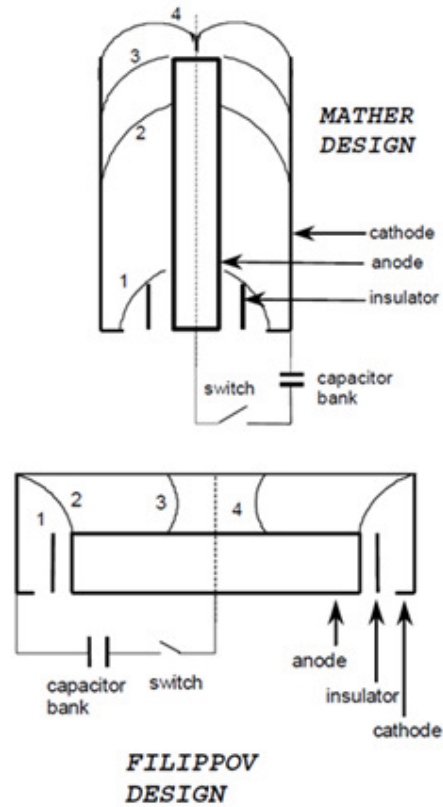


Figure 1.1 Mather and Filippov type Dense Plasma Focus devices schematic drawing [35, 36].

The working concept of the Dense Plasma Focus is as follows: the energy stored electrically in a capacitor bank is quickly discharged to the electrodes by means of a fast switch. In both Mather and Filippov type devices, the current discharge starts along the surface of the insulator in a coaxial electrode region (1 in Figure 1.1). By the Lorentz force action the conducting plasma sheath quickly travels from position 1 through position 2 to position 3. In position 4 the shock wave preceding the sheath reaches the axis of symmetry of the electrodes. To optimize the machine one chooses the dimensions of the electrodes (length and diameters) and the operational pressure in relationship with the characteristics of the energy source

so that the current is maximum when the shock wave reaches the axis. Following this a filament of hot and dense plasma (pinch) is formed in front of the inner electrode (usually the anode). After a very short time due to instabilities the plasma column breaks up and decays [35, 36].

Similarities in these devices are in: (1) the dynamics of the current sheath, (2) the neutron emission scaling laws and (3) the release of ions and electron beams, x- rays etc.

The plasma focus quickly became well known as both a source of X-rays and a source of fusion neutrons. The study on plasma focus has resulted in different machines being designed and made all around the world from the small nanofocus machine [23, 24] to the large Poseidon plasma focus [37].

The research on neutron yield in plasma focus machine for example has resulted in researchers from Los Alamos aggressively scaling up their capacitor energy storage as stated in the paper by Bruce Freeman [38].

DPF 6.5 [39] machine was designed with maximizing neutron yield in mind. This was then followed by Limeul plasma focus [40] machine in 1975 and followed by the 1MJ Frascati machine [41].

In 1988, AAAPT (Asian African Association for Plasma Training) was formed. Their hands-on training programmes which had already started in 1986 and continued intensively through the late 80's had initiated and strengthened plasma research activities in several countries and thus allowed smaller groups of researchers to have an important role in plasma focus research [42]. The International Centre for Dense Magnetized Plasma (ICDMP) was established in Warsaw [35] in 1997 and it now runs several plasma focus machines including PF1000 which is one of the largest plasma focus machines in the world [4].

Chile currently owns the SPEED-2 plasma focus device [25, 43, 44], the largest plasma focus machine of the southern hemisphere.

In recent years, several new machines have been designed and commissioned including the DPF device called "*Focus Fusion-I*" by Eric J. Lerner [45], KSU Plasma Focus at Kansas State University [46-48] and NX3 in Singapore [10, 11].

A recent review by M Krishnan gives a good perspective on the history of plasma focus, results obtained from the past decades of research and ideas for development into the future [49].

1.4.1. Numerical Experiments

In terms of numerical experimentation, David Potter in 1971 wrote in his *Numerical Studies of the Plasma Focus*, a two-dimensional fluid model which estimated the neutron yield in agreement with experimental yields, and concluded that neutrons were the result of thermally reacting deuterons in the hot pinch region [50]. Note that his conclusion about thermally reacting neutrons is generally at variance with recent work [35] including our own work. M Trunk [51, 52] modelled the Stuttgart devices in his numerical studies in 1975.

In 1983, S Lee introduced a 2-phase (axial and radial) model [53, 54] which became a component of a 3 kJ plasma focus experimental package known as the United Nations University/International Centre for Theoretical Physics Plasma Fusion Facility (UNU/ICTP PFF) [55]. This model was written as a 2-phase (non-radiative) model (in GWBASIC) for an experimental program at the 1991 Spring College in Plasma Physics at the International Centre for Theoretical Physics (ICTP) [56]. It was later rewritten in Microsoft EXCEL VISUAL BASIC in order to make it available for wider usage. This 5-phase radiative plasma focus model is now widely known as the Lee Model code [57-60].

This model is very convenient and powerful as it can be configured to any conventional Mather-type plasma focus by inputting the relevant machine parameters which includes the capacitance, inductance, length and electrode radii and the operating parameters such as fill gas pressure and charging voltage.

According to Sharif Al-Hawat [61] the plasma focus has mostly been described by the magnetohydrodynamic MHD model. However the snow plow model is often sufficient to give a good description of the axial acceleration phase. According to this paper, the focus formation is accomplished by two main phases; the axial phase, where an axisymmetric current sheath is accelerated by the electromagnetic force toward the end of the anode; and the radial phase, where a rapid collapse of the current sheath at the end of the axial phase results in the formation of a thin filament of extremely hot and dense plasma. As the dynamics of the current sheath collapse depends on the state of the gas just before the focus action, it is very important to determine the temporal distribution of current and velocity in any plasma focus device.

The details of the plasma focus mechanisms are also described in the websites [45, 57] and by review papers such as [62] in which Lee Model code has been explained.

The physical basis, the structure of the model, the results and extensive scope of the code is given in Figure 1.2 [57, 58, 63].

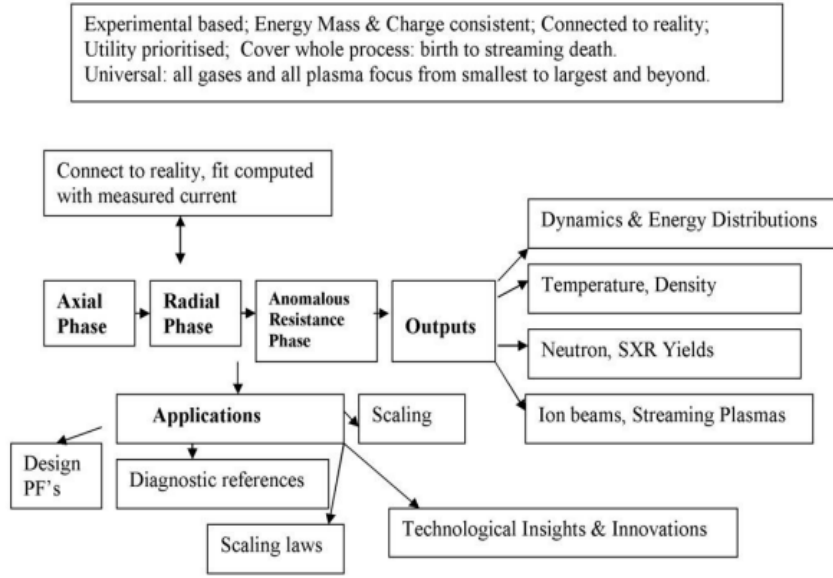


Figure 1.2 The physical basis, structure, results and extensive scope of the Lee Model code [57, 58, 63].

The Lee Model code which has been widely used in numerical experiments has resulted in the following insights into plasma fusion devices including: (1) a previous unsuspected “pinch current limitation” effect [64]; (2) the existence of an optimum L_0 below which pinch current and neutron yield of plasma focus would not increase, but instead decrease [65]; (3) an innovative tool to obtain the pinch current [66]; (4) scaling laws of neutron yield and soft x-ray yield as functions of storage energies and current [67]; (5) a global scaling law for neutron yield as a function of storage energy combining experimental and numerical data showing scaling deterioration has probably been interpreted as neutron ‘saturation’ [35, 37] and (6) a fundamental cause of neutron ‘saturation’[68]

The importance and general acceptance of Lee Model code for numerical experiments has resulted in numerical experiment workshops being held worldwide such as in Malaysia, Singapore, Turkey, Italy, United States and most recently in Nepal [57].

1.4.2 Neutron Yield

The first neutron scaling law was proposed by H Rapp in 1973 in his paper *Measurements referring to plasma scaling laws*, proposing the relationship between neutron yield and energy input into the device [69]. In the same year, H. Krompholz et al in their paper *A scaling law for plasma focus devices* meanwhile proposed the scaling law between neutron yield and sheet current [70]. In year 1988, H. Herold [71] presented in his paper *Physics and technology of large plasma focus devices* the scaling law of neutron yield ($Y_n \sim I_{pinch}^{3.2}$) by compiling the experimental data from the plasma focus machines over a wide range of energies. However, there was uncertainty with this scaling law because the recorded I_{pinch} from those experimental data were inaccurately measured, or estimated from the total peak current. In 1991, S. P. Moo et al in their paper *An investigation of the ion beam of a plasma focus using a metal obstacle and deuterated target* indicated that neutron yield from 3 kJ PF is predominantly beam-target and not thermonuclear [72]. S Lee and S H Saw in their paper *Neutron scaling laws from numerical experiments* [73] assembles the experimental data of neutron yield Y_n against pinch current I_{pinch} to produce a more global scaling law than available. From the data a mid-range point is obtained to calibrate the neutron production mechanism of the Lee Model code. This code is then used for numerical experiments on a range of focus devices to derive neutron scaling laws. The results are the following: $Y_n \sim I_{pinch}^{4.7}$ and $Y_n \sim I_{peak}^{3.9}$. In their paper *Scaling laws for plasma focus machines from numerical experiments* [67] the Lee Model code is applied to the MJ machine PF1000 over a range of capacitance to study the neutrons emitted by PF1000-like bank energies from 10kJ to 25 MJ. The scaling of Y_n with I_{peak} and I_{pinch} over the whole range of energies investigated up to 25 MJ obtained is as follows: $Y_n \sim I_{pinch}^{4.5}$ and $Y_n \sim I_{peak}^{3.8}$.

1.4.3 Radiation Yield

In 1998, L Mahe et al in their paper *Soft X-ray measurement in a small plasma focus operated in neon*, [74] stated that in neon 64% of the total soft X-ray

(SXR) yield is contributed by the Ly- α and He- α lines and 36% by the rest, mainly the radiative recombination. The temperature around 300 eV is optimum for neon characteristic SXR production. S Bing in his Phd thesis [36] in 2000 suggested that 2×10^6 K (below 200 eV) would be better. Hence unlike neutron scaling characteristic SXR scaling has a small range of temperature to operate. S Lee and S H Saw [67] in 2010 proposed the scaling laws for neon SXR from numerical experiments over a range of energy from 0.2 kJ to 1 MJ.

1.5 The Lee Model

In this thesis, in order to study and compare the performances of various DPF machines, the computed current waveforms of the machines are fitted to the experimental current waveforms with the aid of Lee Model 5-phase and 6-phase codes (version: RADPF6.1) [57, 58, 64, 67, 68, 73, 75, 76] and the plasma parameters are obtained.

The Lee Model code combines the electrical circuit with plasma focus dynamics, thermodynamics, and radiations, enabling a consistent and realistic representation of all gross plasma focus properties. These codes were widely used in designing several machines including UNU/ICTP PFF [5, 42, 55, 74, 77-79], NX1 [6, 80], NX2 [6, 36, 81] and NX3 [10, 11] and have been adapted for the Filippov-type plasma focus DENA [60, 82]. Figure 1.3 shows a typical plasma focus device with current sheet position in the axial and radial phase.

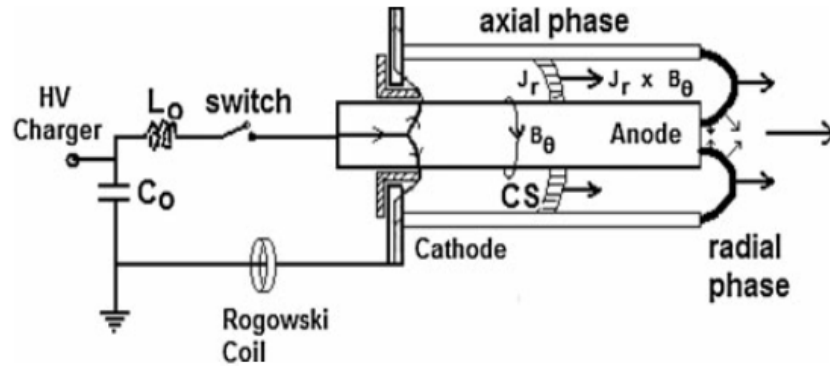


Figure 1.3 Plasma Focus device with current sheet position [83].

A brief description of the Lee Model 5-phase and extended 6-phase model (RADPFV6.1b) is explained below [13, 14, 57-59, 65, 67, 68, 73, 75, 76, 84-89].

a. Axial Phase: The equation of motion in this phase combines the axial phase model parameters: mass and current factors f_m and f_c [90, 91]. The mass swept-up factor f_m accounts for not only the porosity of the current sheet but also for the inclination of the moving current sheet-shock front structure, boundary layer effects, and all other unspecified effects which have effects equivalent to increasing or reducing the amount of mass in the moving structure, during the axial phase. The current factor f_c accounts for the fraction of current effectively flowing in the moving structure (due to all effects such as current shedding at or near the back-wall, and current sheet inclination). This defines the fraction of current effectively driving the structure, during the axial phase. Equations 1 and 2 describe this phase [58].

$$\frac{d^2z}{dt^2} = \left[\frac{f_c^2}{f_m} \frac{\mu(\ln c)}{4\pi^2 \rho_0 (c^2 - 1)} \left(\frac{I}{a} \right)^2 - \left(\frac{dz}{dt} \right)^2 \right] / z \quad (1)$$

$$\frac{dI}{dt} = \left[V_0 - \frac{\int I dt}{C_0} - r_0 I - I f_c \frac{\mu}{2\pi} (\ln c) \frac{dz}{dt} \right] / \left[L_0 + \frac{f_c \mu}{2\pi} (\ln c) z \right] \quad (2)$$

The terms f_m is the fraction of mass swept down the tube in the axial direction; f_c is the fraction of current flowing in piston (or current sheet CS); $c = b/a$ = cathode radius/anode radius, ρ_0 = ambient density, I = time varying circuit current, μ = permeability and r_0 = circuit stray resistance.

The equation of motion is affected by the electric current I . The circuit equation is affected by the current sheath motion dz/dt and position z [58].

b. Radial Inward Shock Phase: The equation of motion in this phase takes into consideration the thermodynamic effects due to ionization and excitation. The model

parameters, radial phase mass swept-up and current factors f_{mr} and f_{cr} are combined in this radial phases. The mass swept-up factor f_{mr} takes into account all mechanisms which increase or reduce the amount of mass in the moving slug, during the radial phase. The current factor f_{cr} is the fraction of current effectively flowing in the moving piston forming the back of the slug (due to all effects). This defines the fraction of current effectively driving the radial slug. Equations 3 to 6 describe this phase [58].

$$\frac{dr_s}{dt} = - \left[\frac{\mu(\gamma+1)}{\rho_0} \right]^{1/2} \frac{f_c}{\sqrt{f_{mr}}} \frac{I}{4\pi r_p} \quad (3)$$

$$\frac{dz_f}{dt} = - \left(\frac{2}{\gamma+1} \right) \frac{dr_s}{dt} \quad (4)$$

$$\frac{dr_p}{dt} = \frac{\frac{2}{\gamma+1} \frac{r_s}{r_p} \frac{dr_s}{dt} - \frac{r_p}{\gamma I} \left(1 - \frac{r_s^2}{r_p^2} \right) \frac{dI}{dt} - \frac{r_p}{z_f} \left(1 - \frac{r_s^2}{r_p^2} \right) \frac{dz_f}{dt}}{\frac{\gamma-1}{\gamma} + \frac{1}{\gamma} \frac{r_s^2}{r_p^2}} \quad (5)$$

$$\frac{dI}{dt} = \frac{V_o - \frac{\int I dt}{C_0} - r_0 I - f_c \frac{\mu}{2\pi} \left(\ln \frac{b}{r_p} \right) I \frac{dz_f}{dt} + f_c \frac{\mu}{2\pi} \frac{z_f}{r_p} I \frac{dr_p}{dt}}{L_0 + f_c \frac{\mu}{2\pi} (\ln c) z_0 + f_c \frac{\mu}{2\pi} \left(\ln \frac{b}{r_p} \right) z_f} \quad (6)$$

where γ is the specific heat ratio of the gas, z_f is the position of the axial current sheet, z_0 is the length of the anode, r_s is the position of the radial shock front and r_p is the position of the radial current sheet

The four generating equations (3, 4, 5, 6) form a closed set of equations which is integrated for r_s , r_p , z_f and I .

c. Radial Reflected Shock (RS) Phase: When the shock front hits the axis, because the focus plasma is collisional, a reflected shock develops which moves radially outwards, whilst the radial current sheath piston continues to move inwards. The model parameters f_{mr} and f_{cr} are also used in this phase. Equations 7 to 10 describe this phase [58].

$$\frac{dr_r}{dt} = -0.3 \left(\frac{dr_s}{dt} \right)_{on-axis} \quad (7)$$

$$\frac{dr_p}{dt} = \frac{-\frac{r_p}{\mathcal{H}} \left(1 - \frac{r_s^2}{r_p^2} \right) \frac{dI}{dt} - \frac{r_p}{z_f} \left(1 - \frac{r_s^2}{r_p^2} \right) \frac{dz_f}{dt}}{\frac{\gamma-1}{\gamma} + \frac{1}{\gamma} \frac{r_s^2}{r_p^2}} \quad (8)$$

$$\frac{dz_f}{dt} = - \left(\frac{2}{\gamma+1} \right) \left(\frac{dr_s}{dt} \right)_{on-axis} \quad (9)$$

$$\frac{dI}{dt} = \frac{V_0 - \frac{\int I dt}{C_0} - r_0 I - f_c \frac{\mu}{2\pi} \left(\ln \frac{b}{r_p} \right) I \frac{dz_f}{dt} + f_c \frac{\mu}{2\pi} \frac{z_f}{r_p} I \frac{dr_p}{dt}}{L_o + f_c \frac{\mu}{2\pi} (\ln c) z_0 + f_c \frac{\mu}{2\pi} \left(\ln \frac{b}{r_p} \right) z_f} \quad (10)$$

The integration of these 4 coupled generating equations 7-10 is carried out step-by-step as in the radial inward shock phase [58].

d. Pinch Phase: When the out-going reflected shock hits the inward moving piston, the compression enters a radiative phase. This phase is described by equation 11.

$$\frac{dr_p}{dt} = \frac{-\frac{r_p}{\gamma I} \frac{dI}{dt} - \frac{1}{\gamma+1} \frac{r_p}{z_f} \frac{dz_f}{dt} + \frac{4\pi(\gamma-1)}{\mu\gamma z_f} \frac{r_p}{f_c^2 I^2} \frac{dQ}{dt}}{\frac{\gamma-1}{\gamma}} \quad (11)$$

It should be noted that an energy loss/gain terms (dQ/dt) has been added into the equation of motion [58]. The plasma gains energy from Joule heating; and loses energy through Bremsstrahlung and line radiation

e. Unstable region: At the end of the pinch phase the system becomes unstable, develops a high “anomalous” resistivity and breaks up. This phase is described by equations 12 and 13.

$$\frac{dz_f}{dt} = \left[\frac{\mu}{4\pi^2(\gamma+1)\rho_0} \right]^{1/2} \frac{If_c}{r_p} \quad (12)$$

$$\frac{dI}{dt} = \frac{V_0 - \frac{\int Idt}{C_0} - \frac{\mu}{2\pi} \left(\ln \frac{b}{r_p} \right) \frac{dz_f}{dt} If_c + \frac{\mu}{2\pi} \frac{z_f}{r_p} \frac{dr_p}{dt} If_c - I(Rf_c + r_0)}{L_0 + \frac{\mu}{2\pi} f_c \left((\ln c)z_0 + \left(\ln \frac{b}{r_p} \right) z_f \right)} \quad (13)$$

In equation 13, the term R is the plasma resistance. It should be noted that step-by-step integration was terminated at the end of a period related to the transit time of small disturbance speed across the plasma pinch column [58].

f. Expanded Column Phase: This phase is not considered important as it occurs after the focus pinch.

Using the four model parameters f_c , f_m , f_{mr} , f_{cr} good fitting can be obtained between the computed total current waveform to the measured total current waveform since Lee Model code combines the energy and mass balances equivalent, at least in the gross sense, to all the processes which are not even specifically modeled. Hence the computed gross features such as speeds, trajectories and

integrated neutron and soft x-ray yields can be obtained; and are found to be comparable with measured values.

The neutron yield in the Lee Model is computed using a phenomenological beam-target neutron generating mechanism described by V A Gribkov *et al* [4] and adapted to yield the following equation

$$Y_{b-t} = C_n n_i I_{\text{pinch}}^2 z_p^2 \ln(b/r_p) \sigma / U^{0.5} \quad (14)$$

where n_i is the ion density, b is the cathode radius, r_p is the radius of the plasma pinch with length z_p , σ the cross-section of the D-D fusion reaction, and U , the beam energy. C_n is treated as a calibration constant combining various constants in the derivation process. [71]

Line radiation Q_L is calculated as

$$\frac{dQ_L}{dt} = -4.6 \times 10^{-31} n_i^2 Z Z_n^4 (\pi r_p^2) z_f / T \quad (15)$$

The SXR energy generated within the plasma pinch depends on the properties: number density n_i , effective ion number density Z , the atomic number Z_n pinch radius r_p , pinch length z_f and temperature T . It also depends on the pinch duration since in the Lee Model code the Q_L is obtained by integrating over the pinch duration.

1.6 Layout of the thesis

This thesis is divided into seven chapters. Chapter 1 explains the aims of our investigation whereas Chapter 2 deals with the methodology used in this thesis as well as the classification of Type 1 and Type 2 machines. Chapters 3, 4 and 5 show how Lee Model codes have been used to fit the existing current waveforms obtained from the dense plasma machines operated in different gas medium at various

pressures. Chapter 6 discusses the results obtained from the various current fittings whereas Chapter 7 presents the conclusion obtained from comparing the outcomes of these machines.

CHAPTER 2

Methodology

2.1 Introduction

According to S Lee and S H Saw [92] the current trace of the plasma focus is one of the best gauges of gross performance of the plasma focus machine. The axial and radial phase dynamics and the critical energy transfer into the focus pinch are among the most vital information that is quickly obvious from these current trace.

The precise time outline of the whole current trace is determined by the focus tube dimensions, by the capacitor bank parameters and the operational constraints. The current trace is also reliant on the portion of sheath current and mass swept-up and the variation of these fractions through both the axial and radial phases. These constraints determine the axial and radial dynamics, explicitly the current sheet speeds which in turn affect the shape and amounts of the discharge current.

The current waveform contains all the data of the dynamic, thermodynamic, electrodynamic and radiation mechanisms that happen in the numerous phases of the plasma focus. Thus, this clarifies the significance involved in matching the computed total current trace to the measured total current trace in the technique embraced by the Lee Model code. Once matched, the fitted model parameters convince that the computation continues with all physical mechanisms accounted for, at least in the gross energy and mass balance sense. One of the most important processes therefore is to link the numerical experiment to the certainty of the actual machine by fitting the computed current trace to a measured current trace. The fitting technique is described below, step-by-step, on how the current waveform computed from the Lee Model code is fitted with a measured current waveform.

2.2 The current fitting technique

The current fitting techniques starts with a measured current (or current derivative) signal. This can be either a published or measured signal (without data file) or a measured signal using digital oscilloscope where data file is available. Figure 2.1 shows the current fitting process. For published (measured) signals without data files, the signal is digitalized using open source digital software called “Engauge” [93].

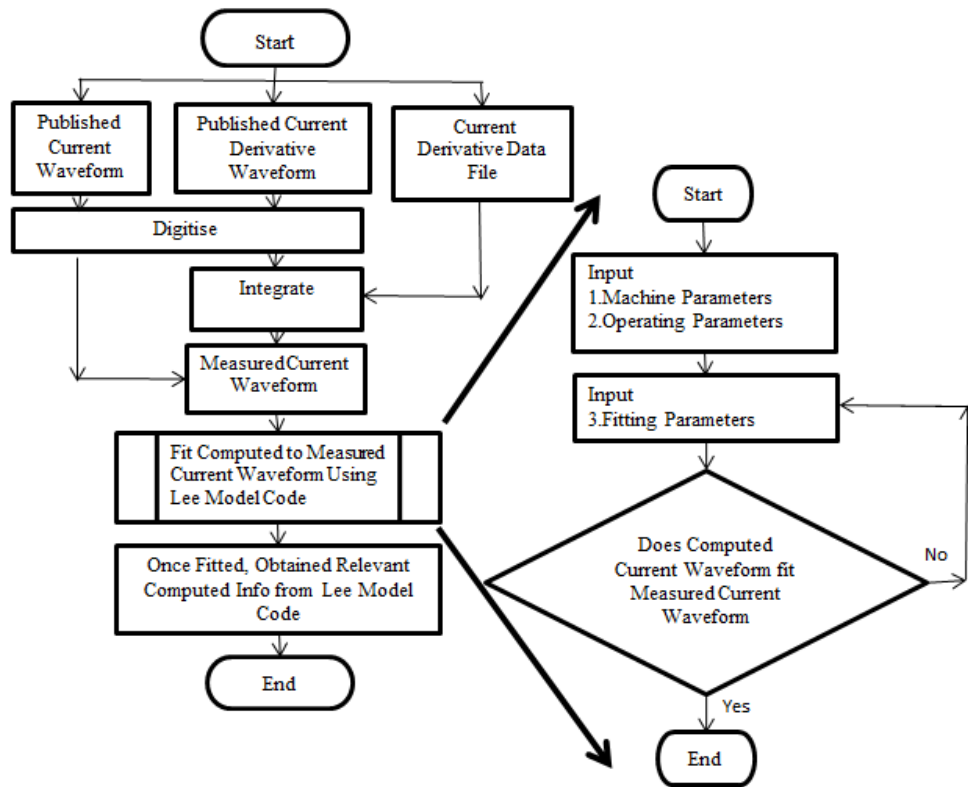


Figure 2.1 Flow chart showing the fitting procedure using the Lee Model codes.

The Lee Model code is designed to work as that plasma focus by entering:
 Bank parameters: inductance L_0 , capacitance C_0 and stray circuit resistance r_0 ,
 Tube parameters: cathode radius b , anode radius ‘ a ’ and anode length z_0 and
 Operational parameters: charge voltage V_0 and pressure P_0 and the fill gas.

The computed total current waveform is fitted to the measured waveform by changing model factors f_m , f_c , f_{mr} and f_{cr} one by one, till the computed waveform agrees with the measured waveform.

First, f_m , f_c are tuned until (see Figure 2.2) the features (1) computed rising slope of the total current trace and (2) the rounding off of the peak current as well as (3) the peak current itself are in reasonable fit with the measured total current trace (see Figures 2.2 and 2.3, measured trace fitted with computed trace).

Then we continue to fit the radial f_{mr} and f_{cr} until features (4) the computed slope and (5) the depth of the dip agree with the measured. For the Lee Model 5-phase code the fitting ends here though the computed current trace goes beyond into the expanded column phase.

In some cases (high static inductance plasma focus designated as Type 2; more of this will be discussed below in the section “Type 1 (T1) and Type 2 (T2) machine”) the computed current dip is fitted to the measured current dip but the extent of the computed dip is not sufficient for the whole measured current dip (see Figure 2.2). We need then to proceed to use the Lee Model 6-phase code to fit part of the instability region (6) by varying the anomalous resistance values [75] (see Figure 2.3).

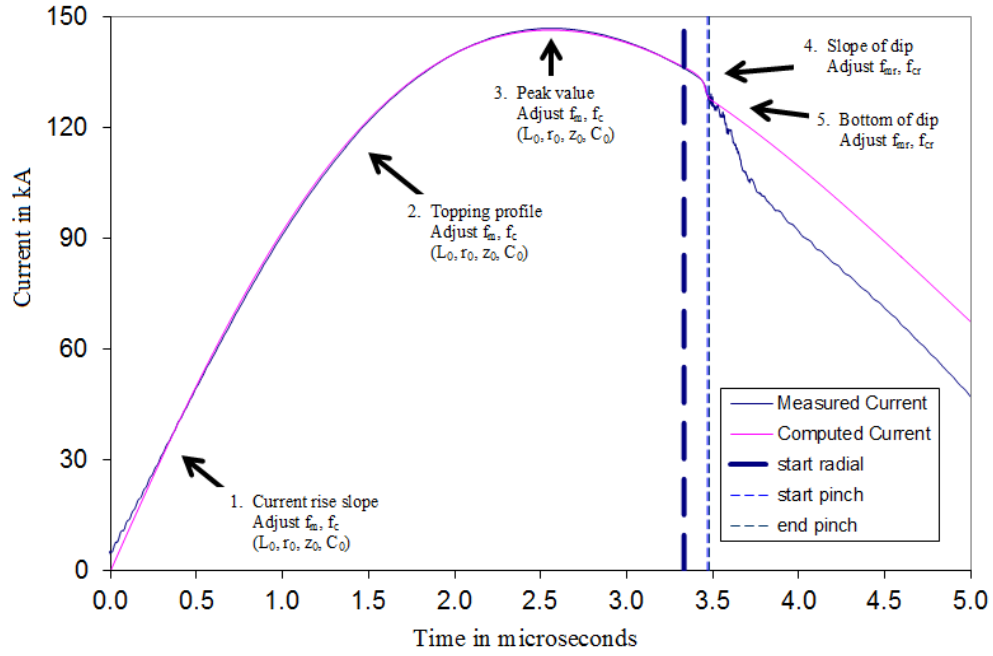


Figure 2.2 The 5-point fitting of current trace to the measured current trace obtained from INTI Plasma Focus (INTI PF) machine (a T2) fired at 2.0 Torr in neon gas.

Figure 2.2 shows the fitting uses Lee Model 5-phase code. It is shown that the Lee Model 5-phase code does fit the computed current dip to the first part of the measured current dip. But there is an extended part of the measured current dip attributed by the anomalous resistivity (due to large amount of remnant inductive energy that is transferred to instabilities unlike a Type 1 machine which transfers energy efficiently through the regular dynamics as evidenced by the large regular current dip and small extended dip [75]) which cannot be fitted by the computed current dip no matter how the model parameters are varied. In such a case, the Lee Model 6-phase code is used. The Lee Model 6-phase code has an additional phase between the end of pinch and the expanded column phase. This additional phase is fitted by adding anomalous resistance terms into the circuit equation (typically 3 sequential anomalous resistance terms). Adjustments are made to the amplitude of each anomalous resistance terms and also the rise and fall time characteristics until a good fit is obtained down to the bottom of the measured current dip.

Fitting of the computed and measured current traces past this point (point 6 in Figure 2.3) is not done. If there is substantial difference of the computed trace with the measured trace beyond the end of the anomalous resistance phase, this difference is not considered vital.

The following are the bank, tube and operational parameters which we input to configure INTI PF to fit the measured current dip of Figure 2.2, first using the Lee Model 5-phase code:

Table 2.1 The actual machine, operational and fitting parameters configured for the INTI Plasma Focus.

Capacitance C_0 (μF)=30	Fill gas(molecular weight)=20
Static inductance L_0 (nH)=114	Fill gas(atomic number)=10
Circuit resistance r_0 ($\text{m}\Omega$)=13	Fill gas(atom)=1
Cathode radius 'b'(cm) =3.2	The fitted model parameters are:
Anode radius 'a'(cm)=0.95	Axial phase mass factor, $f_m=0.0408$
Anode length 'z ₀ '(cm)=16	Axial phase current factor, $f_c=0.75$
Charging voltage V_0 (kV)=12	Radial phase mass factor, $f_{mr}=0.41$
Fill pressure P_0 (Torr)=2	Radial phase current factor, $f_{cr}=0.78$

The model parameters given in Table 2.1 are typically constant for a machine operated with a fixed gas [94]. Configuring the Lee Model 5-phase code according to Table 2.1 gives the computed current waveform and the fitting to the measured current waveform as seen in Figure 2.2. To complete the fit of the current dip, the Lee Model 6-phase code is configured according to Table 2.1 and Table 2.2; which then gives the fit shown in Figure 2.3. Figure 2.4 is the zoom in version of this extended fit.

Table 2.2 An example of the values of the anomalous resistance used to fit the INTI Plasma Focus shot for Figures 2.3 and 2.4.

	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	1	25	5	0.2
Dip 2	0.14	100	20	0.5
Dip 3	0.36	120	25	1

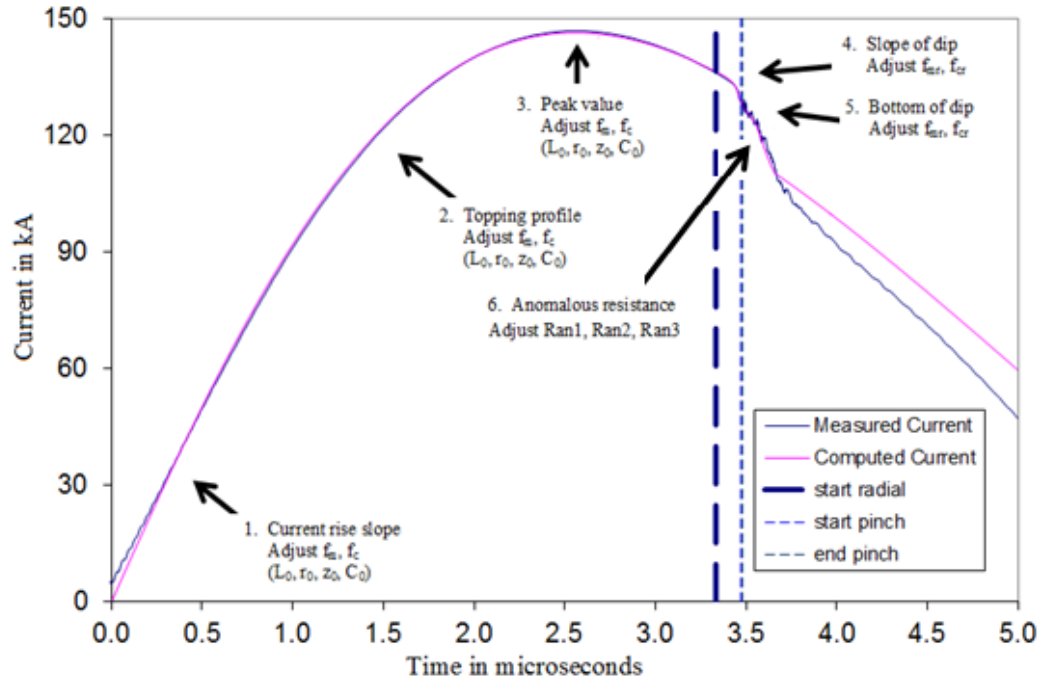


Figure 2.3 The 6-point fitting of computed current trace to the measured current trace obtained from INTI PF machine fired at 2.0 Torr in neon gas.

In Figure 2.3, point 1 shows the current rise slope. Point 2 shows the topping profile. Point 3 shows the peak value of the current. Point 4 shows the slope of the current dip. Point 5 shows the bottom of the current regular dip [75, 92] while point 6 is the instability region. Fitting is done up to point 6 only.

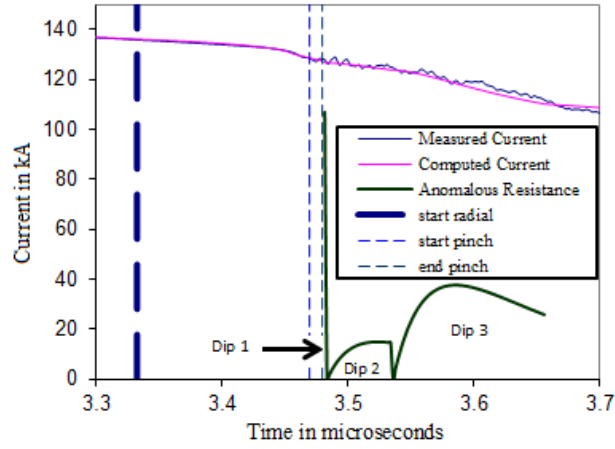


Figure 2.4 Expanded (zoom in) region of Point 6 to show how the anomalous resistance phase was fitted.

Figure 2.4 shows the radial start time, the pinch start time and the pinch end time. The extended dip in current signal is fitted using Lee Model 6-phase code.

We have noted that the parameters of the anomalous resistance (typically fitted as 3 sequential resistances as shown in Table 2.2) vary considerably from shot to shot even for the same machine operated with a given gas. This observation is consistent with the nature of the instabilities causing these anomalous resistances.

2.3 Type 1 (T1) and Type 2 (T2) machines

For modeling reasons plasma focus machines may be appropriately categorised into T1 and T2 [75]; the previous having low static inductance L_0 typically less than 100 nH while the latter normally having L_0 of more than 100 nH. The Lee Model 5-phase code fits well with the measured current waveforms of the T1 plasma focus devices whereas T2 devices have a small computed ‘regular’ current dip (RD) which could only be fitted to the first part of the measured current dip. The measured current dip then continues beyond the computed dip (RD) into a longer and deeper extended dip (ED) which cannot be fitted using the standard Lee Model 5-phase code. To complete the fit for T2 devices the Lee Model 5-phase code

was extended to the Lee Model 6-phase code with the addition of a post-pinch phase of anomalous resistances.

2.4 Type 1 (T1) machine that shows Type 2 (T2) characteristics.

These machines have low static inductance (below 100 nH) but have a measured current dip that continues beyond the regular computed dip (RD) into a longer and deeper extended dip (ED) thus requiring Lee Model 6-phase code to fit the current waveform.

In this project two plasma focus machines are used as examples for a first discussion: namely NX2 (Type 1 machine, low L_0) and INTI PF (Type 2 machine, high L_0) donated to INTI International University by Nanyang Technological University.

2.5 NX2 Plasma Focus machine

The NX2 machine [6] from National Institute of Education, NTU, Singapore, is a medium energy plasma focus device initially intended to operate in neon, for microlithography experiments. It has been adjusted with a new electrode system to make it operate well with deuterium and other working gases. This Mather-type medium energy plasma focus machine has a repetition rate up of 16 Hz using four pseudo-spark switches to switch stored energy in the 28.8 μF capacitor bank. The device has a total static inductance of 12 nH to 20 nH depending on whether rail-gaps or pseudo-spark switches are used.

Two types of anodes are used, the normal anode (length 4, 5 and 7 cm stainless steel electrodes) and the tapered anode (hollow copper) which has an effective length of 4 cm. A Pyrex insulator sleeve of 3.5 cm in diameter and 6 cm in height is used. The coaxial cathode comprises eight copper cathode rods arranged in a squirrel cage configuration. The general overview of the NX2 is shown in Figure

2.5 whereas the schematic of focus electrodes and chamber of NX2 Plasma focus designed for neon soft-x-ray operation is shown in Figure 2.6.

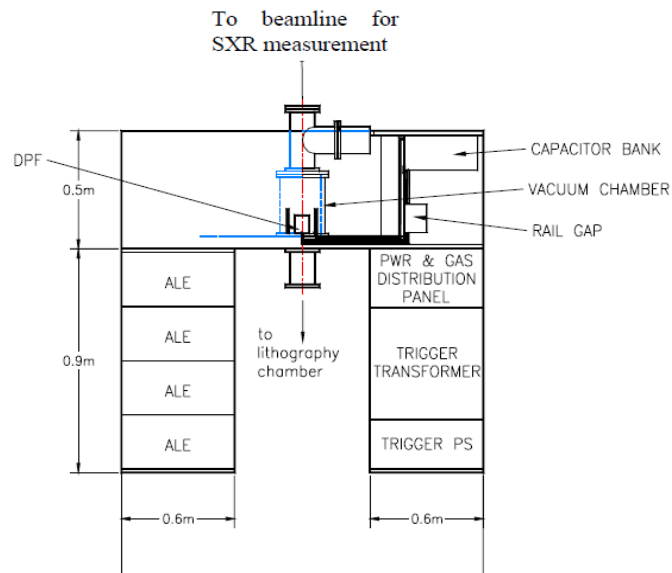


Figure 2.5 General overview of the NX2 apparatus [6].

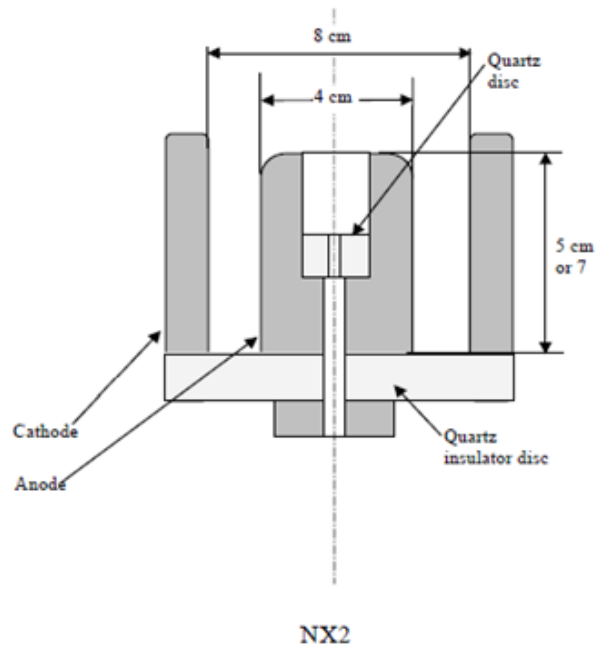


Figure 2.6 NX2 Plasma Focus schematic of electrodes and chamber [6].

2.6 INTI Plasma Focus machine

The schematic diagram of INTI Plasma focus machine [14, 15, 55] is shown in Figure 2.7. This 3 kJ plasma focus machine is energized by a single 30 μF capacitor. The static inductance L_0 of the system is 114 nH and its stray circuit resistance r_0 is 13 m Ω . The electrodes system consists of a central copper tube (anode) of 1.9 cm diameter and 16 cm in length, and a cathode of six copper rods arranged in a circle of 6.4 cm diameter concentric with the anode, the diameter of each cathode is 8 mm. A Pyrex glass tube of 2.4 cm diameter and 5 cm in total length insulates the anode from the cathode. This plasma focus is normally operated with neon. The device is evacuated to base pressure (2×10^{-3} Torr) by a vacuum pump and filled to a particular pressure before operation. To decrease the contamination influence, after several shots, the chamber is pumped down to base pressure and fresh neon is filled. A 3-turn Rogowski coil [15-17] and a voltage probe are used to monitor the derivative of the current in the discharge circuit and the voltage across the INTI PF versus time respectively during the plasma focus process. Magnetic probes are used to determine the current sheath velocities. All diagnostic outputs were connected to a 4-channel 300 MHz Digital Phosphor Oscilloscope (Tektronix-TDS3034C) set at a sampling rate of 1 giga samples per second (1 GS/s).

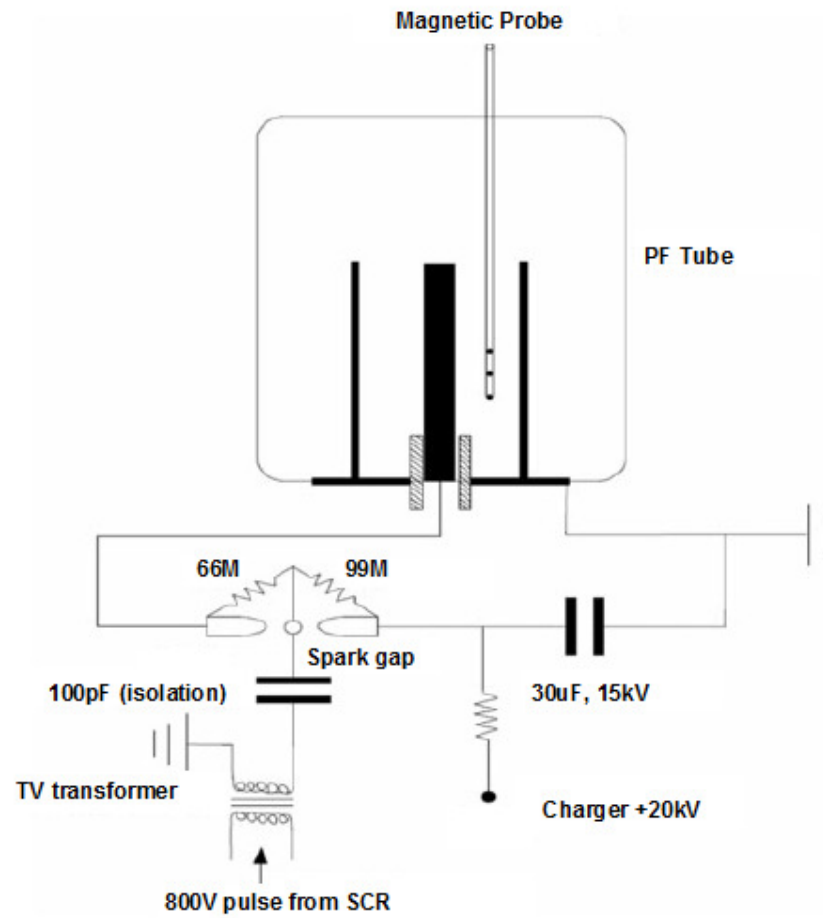


Figure 2.7 INTI Plasma Focus device schematic diagram [84].

CHAPTER 3

Results from dense plasma focus machines below 2.5 kJ for deuterium, argon and neon.

3.1 Introduction

Plasma focus machine are widely available all over the world. In our survey, we have found records of plasma focus machines in all continents except in the Antarctica. The 6 major continents where the machines are found are Asia (Japan, China, Korea, Malaysia, Singapore, India, Pakistan, Iran and Egypt), Europe (England, Poland, Bulgaria, Estonia, Italy, Russia and Germany), Africa (Zimbabwe), North America (United States, Mexico) and South America (Chile and Argentina). These machines work in deuterium (D), neon (Ne), argon (Ar), nitrogen (N), air, helium (He), hydrogen (H), oxygen (O) gas medium. A plasma focus was also operated at Australia Institute of Nuclear Science and Technology in the 70's [95].

The plasma focus machines studied include the Mather-type, Filippov-type and various machines with storage energy ranging from a few J to MJ. These 44 machines are discussed in Chapters 3, 4 and 5. Chapter 3 and 4 discuss Mather types machines operating in D, Ne and Ar while Chapter 5 discuss special configuration machines working in gases such as N, air, He, H and O.

In Chapter 3, sixteen Mather-type machines in the energy range below 2.5 kJ will be presented. For these machines, the bank, tube, relevant operational parameters and the corresponding measured current traces are available. The computed current waveform using the Lee Model 5-phase or 6-phase code is fitted to the measured current waveform. The plasma dynamics and yields obtained from the code are compared to the experimental values where available.

In Chapter 4, sixteen Mather-type machines working in the energy range above 2.5 kJ starting with the PF12 from Estonia to the largest Mather type machine PF 1000 from Poland are presented.

Chapter 5 presents machines operating in other gases (N, air, He, H and O) and machines with other configurations such as Filippov-type, hybrid type, machine using Marx capacitor bank (Speed 2) and the pioneering Plasma Focus machines operated at University Malaya, Malaysia in the Seventies and Eighties. [96-106].

The 16 machines presented in this chapter are listed in increasing energy input into the plasma focus starting from the low energy PF50 from Chile increasing until the DPF 2.2 from China:

- 3.2 PF50 Plasma Focus machine (50 J) [23, 24,25]
- 3.3 BARC Plasma Focus 1 machine (77 J) [21]
- 3.4 AASC Plasma Focus machine (110 J) [107, 108]
- 3.5 Argentina nanofocus machine (140 J) [28, 29]
- 3.6 FMPF-1 Plasma Focus machine (170 J) [12]
- 3.7 India's smallest sealed type Plasma Focus machine (200 J) [109]
- 3.8 PF400 Plasma Focus machine (400 J) [26, 27]
- 3.9 NX1 Plasma Focus machine (1.6 kJ-3.1 kJ) [5, 6, 7]
- 3.10 PPRC Plasma Focus machine (1.8 kJ) [110]
- 3.11 Kansas State University Plasma Focus machine (1.8 kJ) [47, 48]
- 3.12 Imperial college Plasma Focus machine (1.9 kJ) [111, 112]
- 3.13 NX2 Plasma Focus machine (1.7 kJ-2.2 kJ) [7, 8, 9]
- 3.14 Paco Plasma Focus machine (1.9 kJ) [113]
- 3.15 University of Teheran Plasma Focus machine (2.0 kJ) [116]
- 3.16 Montecucolino Plasma Focus machine (2.2 kJ) [117]
- 3.17 DPF 2.2 Plasma Focus machine (2.2 kJ) [19]

3.2 PF50 Plasma Focus machine (50 J -Type 1 machine - D gas)

PF-50 is a 50 J Mather type plasma focus device designed and constructed at the Chilean Nuclear Energy Commission (CCHEN). It has a 160 nF capacitor bank and an internal inductance of 38 nH [13]. The electrodes consist of a copper tube central anode, of 6 mm diameter and 28.8 mm length (The effective anode is 4.8 mm long- (after deducting the 24 mm length of the insulator alumina tube)), and an outer cathode formed by eight copper rods, 5 mm diameter, 29 mm long, uniformly spaced on a coaxial circumference of 27 mm diameter. The anode and the cathode are separated by an insulator alumina tube of 24 mm length, which covers the anode. The size of this table-top device is 25 cm $\times$ 25 cm $\times$ 50 cm. The complete machine parameters are stated in Table 3.1. It should be noted that the Lee Model 5-phase code is configured using a 0.2 μ F capacitor.

The measured current waveform of the PF 50 plasma focus operating at 29 kV with 6.8 Torr D is available. Since its internal inductance is low (38 nH), it is a Type 1 machine. Lee Model 5-phase code is used to fit the measured current waveform. When the computed current waveform was fitted to the measured current waveform, the maximum computed current obtained was 63 kA as compared to the experimental current value of (60 ± 3) kA. Another important information compared was that the computed neutron yield obtained was 2.4×10^4 n/shot as compared to the experimental neutron yield of $(3.6 \pm 1.6) \times 10^4$ n/shot. The other computed information is presented in Table 3.2.

Lee Model 5-phase code was then configured to operate from 2.0 Torr to 9.0 Torr at 29 kV (Figure 3.2) and 2.0 Torr to 7.0 Torr at 25 kV (Figure 3.3) respectively and the computed neutron yield versus pressure were compared to the measured values [23].

Table 3.1 The machine, operational and fitting parameters for PF50 Plasma Focus machine.	Figure 3.1 Measured current waveform of PF50 at 29 kV, 6.8 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform. Note: The measured current waveform information was extracted from Figure 2, page 3 of reference [23].
Capacitance C_0 (μF) = 0.2 Static inductance L_0 (nH) = 38 Circuit resistance r_0 (mΩ) = 20 Cathode radius 'b' (cm) = 1.35 Anode radius 'a' (cm) = 0.3 Anode length 'z₀' (cm) = 0.48 Charging voltage V_0 (kV) = 29 Fill pressure P_0 (Torr) = 6.8 Fill gas (molecular weight) = 4 Fill gas (atomic number) = 1 Fill gas (molecule) = 2 The fitted model parameter are: Axial phase mass factor, $f_m = 0.13$ Axial phase current factor, $f_c = 0.7$ Radial phase mass factor, $f_{mr} = 0.1$ Radial phase current factor, $f_{cr} = 0.75$	

Table 3.2 Information obtained from Lee Model 5-phase code configured for PF50 Plasma Focus machine at 29 kV, 6.8 Torr deuterium gas.

Peak current (kA) = 63
Pinch start current (kA) = 42
Pinch minimum temperature (10^6 K) = 6.58
Pinch maximum temperature (10^6 K) = 6.88
Peak axial speed (cm/ μ s) = 4.8
Peak radial shock speed (cm/ μ s) = 36.3
Peak radial piston speed (cm/ μ s) = 24.5
Final pinch radius $r_{\min}$ (cm) = 0.04
Pinch length $z_{\max}$ (cm) = 0.4
Pinch duration (ns) = 2.5
Peak induced voltage (kV) = 10.3
Neutron yield ($\times 10^4$ n/shot) = 2.38
Energy inflow into plasma (%) = 4.4
Speed Factor ((kA/cm)/Torr $^{0.5}$) = 81
Current per cm anode radius (kA/cm) = 210

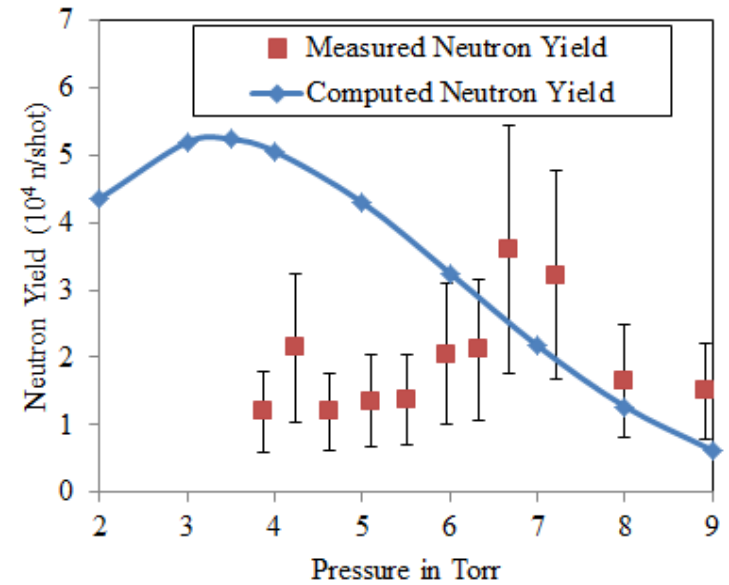


Figure 3.2 Variation of computed and measured neutron yield in PF50 plasma focus machine with respect to pressure when capacitor is charged to 29 kV in deuterium gas.

Note: The measured neutron yield information was extracted from Figure 4, page 4 of reference [23].

Figure 3.2 shows that the magnitudes of average neutron yield are quite similar with the published curve [23]. The main features for comparison include the peak neutron yield (computed value of 5.2×10^4 n/shot against a measured value of $(3.6 \pm 1.6) \times 10^4$ n/shot); optimum pressure (computed value of 3.5 Torr against the measured value of 6.75 Torr (9 mbar) and the drop-off of neutron yield on both sides of the optimum, although the computed drop-offs are more gradual than the measured.

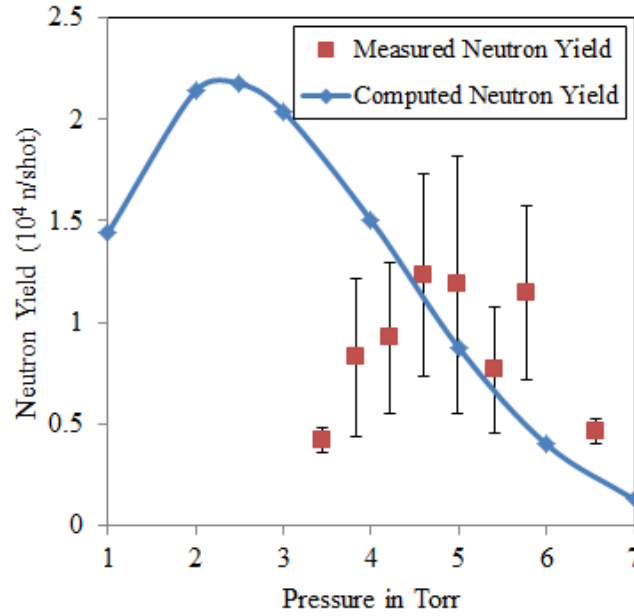


Figure 3.3 Variation of computed and measured neutron yield in PF50 plasma focus machine with respect to pressure when capacitor is charged to 25 kV in deuterium gas.

Note: The measured neutron yield information was extracted from Figure 4, page 4 of reference [23].

Figure 3.3 also shows that the computed average neutron yields are quite similar with the published curve [23]. The main features for comparison include the peak neutron yield (computed value of 2.2×10^4 n/shot against a measured value of $(1.2 \pm 0.5) \times 10^4$ n/shot); optimum pressure (computed value of 2.5 Torr against the measured value of 4.5 Torr (6 mbar)) and the drop-off of neutron yield on both sides of the optimum, although the computed drop-offs are more gradual than the measured.

3.3 BARC Plasma Focus 1 machine (77 J - Type 1 machine - H and D gas)

India Bhabha Atomic Research Center (BARC) has more than one Plasma Focus machine. The machine parameters for one of this plasma focus machine (BARC Plasma Focus 1 machine) where measured current waveform [21] available is compiled in Table 3.3. According to this reference the machine inductance is 60 nH and anode length is 10 mm. Our fitting determined that the value of L_0 is 145 nH and that the length cannot be the short value of 10 mm and is likely to be 10 cm. We have therefore listed these corrected values in Table 3.3.

The maximum computed current was 30 kA (published peak current is 52 kA) and exhibits the radial start time at 2.399 μ s for pinch duration of 0.05 μ s. It has a speed factor of 50 (kA/cm)/Torr^{0.5} hydrogen. The detailed information is presented in Table 3.4. There is no other information to compare from the published paper.

Table 3.3 The machine, operational and fitting parameters for INDIA- BARC Plasma Focus 1 machine.

Capacitance C_0 (μ F) = 8.4	Fill gas (molecular weight) = 2			
Static inductance L_0 (nH) = 145	Fill gas (atomic number) = 1			
Circuit resistance r_0 (m Ω) = 6.5	Fill gas (atom) = 1			
Cathode radius ‘b’ (cm) = 1.25	The fitted model parameters are:			
Anode radius ‘a’ (cm) = 0.6	Axial phase mass factor, f_m = 0.09			
Anode length ‘ z_0 ’ (cm) = 10	Axial phase current factor, f_c = 0.53			
Charging voltage V_0 (kV) = 4.2	Radial phase mass factor, f_{mr} = 0.12			
Fill pressure P_0 (Torr) = 1.2	Radial phase current factor, f_{cr} = 0.9			
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.1	50	10	1
Dip 2	0.01	30	10	1
Dip 3	0.01	20	10	1

Table 3.4 Information obtained from Lee Model 6-phase code configured for INDIA-BARC Plasma Focus 1 machine at 4.2 kV, 1.2 Torr hydrogen gas.	Figure 3.4 The measured current waveform of INDIA-BARC Plasma Focus 1 machine at 4.2 kV, 1.2 Torr hydrogen gas compared with Lee Model 6-phase code computed current waveform. Note: The measured waveform was obtained from the current derivative in Figure 4, page 083501-3 of reference [21].
Peak current (kA) = 30 Pinch start current (kA) = 21 Pinch minimum temperature (10^6 K) = 4.04 Pinch maximum temperature (10^6 K) = 4.11 Peak axial speed (cm/μs) = 5.8 Peak radial shock speed (cm/μs) = 27.3 Peak radial piston speed (cm/μs) = 19 Final pinch radius $r_{\min}$ (cm) = 0.08 Pinch length $z_{\max}$ (cm) = 0.9 Pinch duration (ns) = 6.4 Peak induced voltage (kV) = 3.0 All-line yield (J) ≈ 0 Energy Inflow into Plasma (%) = 3.3 Speed Factor ((kA/cm)/Torr^{0.5}Hy) = 46 Current per cm anode radius (kA/cm) = 50	

A measured waveform is also available [21] for operation in deuterium. For this waveform the operational and fitting parameters are shown in Table 3.5.

Table 3.5 The operational and fitting parameters for INDIA- BARC Plasma Focus 1 machine working in deuterium gas.

Fill pressure P_0 (Torr) = 1.2	The fitted model parameters are:			
Fill gas (molecular weight) = 4	Axial phase mass factor, $f_m = 0.045$			
Fill gas (atomic number) = 1	Axial phase current factor, $f_c = 0.65$			
Fill gas (molecule) = 2	Radial phase mass factor, $f_{mr} = 0.04$			
	Radial phase current factor, $f_{cr} = 0.95$			
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.12	100	10	1
Dip 2	0.01	20	10	1
Dip 3	0.01	50	10	1

The measured current waveform is compared to the computed current waveform using the 6-phase code as shown in Figure 3.5.

The computed maximum current is 30 kA whereas the published maximum current is stated as 53 kA. The detailed information is given in Table 3.6.

The Lee Model 6-phase code is then configured to operate the INDIA-BARC PF1 machine from 0.25 to 1.5 Torr deuterium at 4.3 kV. The variation of the computed neutron yield with respect to pressure is shown in Figure 3.6.

Table 3.6 Information obtained from Lee Model 6-phase code configured for INDIA- BARC Plasma Focus 1 machine at 4.3 kV, 1.2 Torr deuterium gas.

Peak current (kA) = 30
Pinch start current (kA) = 26
Pinch minimum temperature (10^6 K) = 9.74
Pinch maximum temperature (10^6 K) = 9.86
Peak axial speed (cm/ μ s) = 7.3
Peak radial shock speed (cm/ μ s) = 42.1
Peak radial piston speed (cm/ μ s) = 29.1
Final pinch radius $r_{\min}$ (cm) = 0.08
Pinch length $z_{\max}$ (cm) = 0.8
Pinch duration (ns) = 4.1
Peak induced voltage (kV) = 6.2
Neutron yield (n/shot) = 360
Energy Inflow into Plasma (%) = 4.7
Speed Factor ((kA/cm)/Torr ^{0.5}) = 46
Current per cm anode radius (kA/cm) = 51

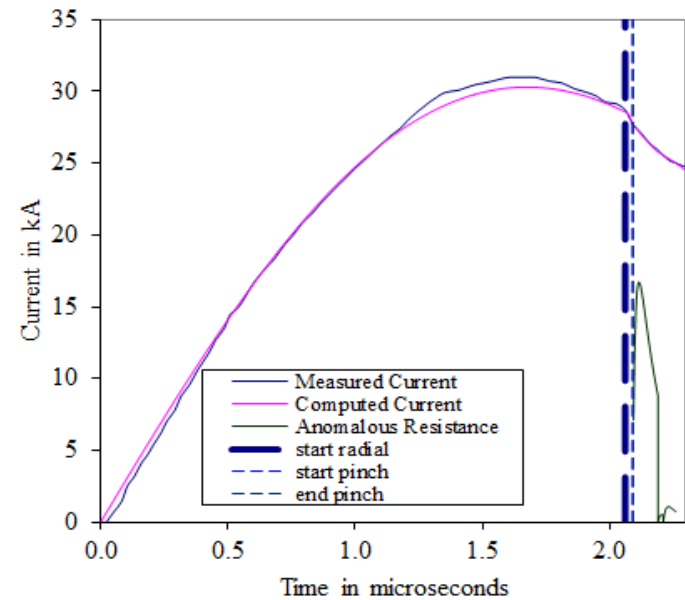


Figure 3.5 The measured current waveform of INDIA-BARC Plasma Focus 1 machine at 4.3 kV, 1.2 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured waveform was obtained from the current derivative in Figure 5, page 083501-3 of reference [21].

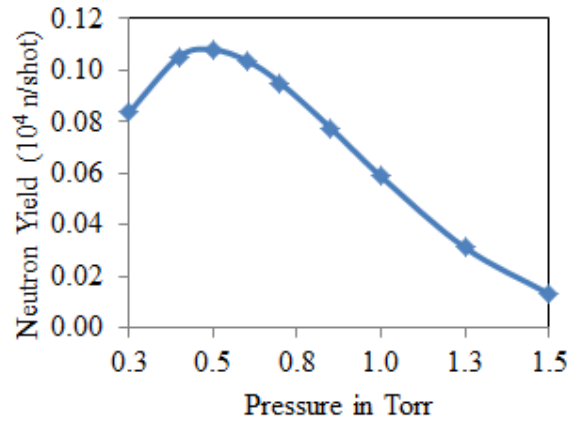


Figure 3.6 Variation of computed neutron yield with respect to pressure operating at 4.3 kV in deuterium for INDIA-BARC PF1.

From Figure 3.6, the maximum neutron yield that can be achieved is 0.108×10^4 n/shot at 0.5 Torr. The energy input into the plasma is 4.5 %.

The author [21] stated that this plasma focus has not been optimally designed for the production of neutrons thus the low computed neutron yield is expected. It also stated that when this machine is fully optimized for production of neutron yield, it is expected to deliver more than 10^4 n/shot as is suggested by various scaling laws that had evolved and had been presented by various plasma focus research scientists, including that as described by Lee and Saw [72] which stated that $Y_n = 2 \times 10^{11} I_{\text{pinch}}^{4.7}$ and $Y_n = 9 \times 10^9 I_{\text{peak}}^{3.9}$, where I_{pinch} and I_{peak} are in MA [74]. We conclude that this machine was a preliminary test on the part of BARC, from the results of which they had then proceeded to build more efficient machines such as the BARC Plasma Focus 2 discussed in Chapter 4.

3.4 AASC Plasma Focus machine (110 J - Type 2 machine - using 6:1 step down transformer - D gas)

This Mather's type machine was charged to 4.5 kV [107] at a pressure of 2.9 Torr deuterium gas with a 6:1 type transformer as shown in Figure 3.7. The transformed secondary values, and the parameters of the equivalent circuit such as capacitor, inductor as well as the machine parameters anode length, anode radius, cathode radius b are obtained from Dr Brian Bures from Alameda Applied Sciences Corporation, San Leandro, CA, USA . Essentially because of the 6:1 transformer, the inductance and stray resistance are each reduced 36 times to get to the equivalent secondary values whilst the capacitance is increased 36 times.



Figure 3.7 The picture of the system used in Alameda Applied Sciences Corporation system which is taken from Figure 1, page 4 of reference [108].

Its equivalent static inductance is 15.8 nH. The Lee Model 5-phase code was used for the current fitting, and where necessary the Lee Model 6-phase code was used to complete the fitting of any extended dip. The machine parameters are shown in Table 3.7. The computed and measured waveforms show a reasonably good fit as presented in Figure 3.8. Except for bulging deviation between computed and

measured waveform near the dip, the 5-phase code could fit to just past half way of the current dip, beyond which the Lee Model 6-phase code was used.

The maximum computed current is 89 kA (published results stated it as 61 kA with a 2.3 % standard deviation [107]) and exhibits the radial start time at 0.575 μ s for pinch duration of 0.054 μ s. It has a speed factor of 70 (kA/cm)/Torr^{0.5} and computed neutron yield of 6.83×10^4 n/shot (compared to the published results of 2.6×10^5 n/shot with a 61 % standard deviation). The detailed information is presented in Table 3.8.

Table 3.7 The machine, operation and fitting parameters for AASC Plasma Focus machine.

Capacitance C_0 (μ F) = 10.88 (ES)	Fill gas (molecular weight) = 4			
Static inductance L_0 (nH) = 15.8 (ES)	Fill gas (atomic number) = 1			
Circuit resistance r_0 (m Ω) = 3.9	Fill gas (molecule) = 2			
Cathode radius ‘b’ (cm) = 2	The fitted model parameters are:			
Anode radius ‘a’ (cm) = 0.75	Axial phase mass factor, f_m = 0.065			
Anode length ‘z ₀ ’ (cm) = 3	Axial phase current factor, f_c = 0.7			
Charging voltage V_0 (kV) = 4.5 (ES)	Radial phase mass factor, f_{mr} = 0.1			
Fill pressure P_0 (Torr) = 2.9	Radial phase current factor, f_{cr} = 0.75			
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.8	24	10	1
Dip 2	0.03	100	10	1
Dip 3	0.01	100	10	1

Note: ES = equivalent secondary

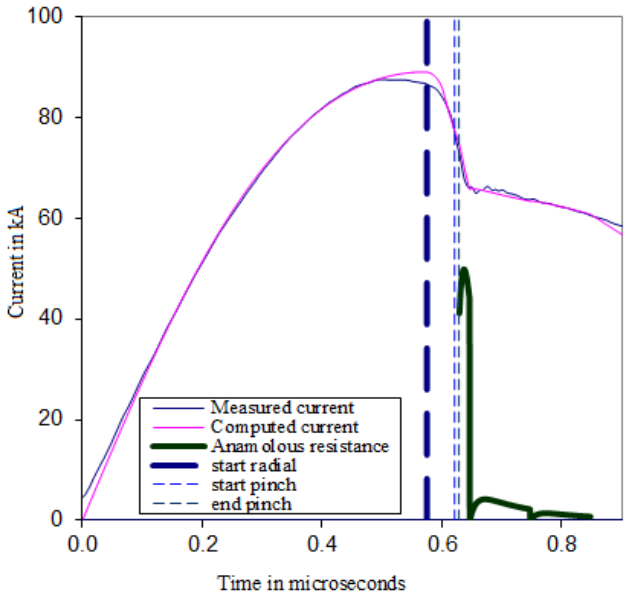
Table 3.8 Information obtained from Lee Model 6-phase code configured for AASC Plasma Focus machine at 4 kV, 2.9 Torr deuterium gas.	
Peak current (kA) = 89 Pinch start current (kA) = 58 Pinch minimum temperature (10^6 K) = 4.73 Pinch maximum temperature (10^6 K) = 5.07 Peak axial speed (cm/μs) = 8.3 Peak radial shock speed (cm/μs) = 32.1 Peak radial piston speed (cm/μs) = 22.6 Final pinch radius $r_{\min}$ (cm) = 0.11 Pinch length $z_{\max}$ (cm) = 1.1 Pinch duration (ns) = 7.4 Peak induced voltage (kV) = 9.3 Neutron yield ($\times 10^4$ n/shot) = 6.83 Energy inflow into plasma (%) = 17.7 Speed Factor ((kA/cm)/Torr$^{0.5}$) = 70 Current per cm anode radius (kA/cm) = 119	

Figure 3.8 The measured current waveform of AASC Plasma Focus machine at 4 kV, 2.9 Torr deuterium gas compared with Lee Model 6-phase code computed current.

Note: The measured current waveform information was provided by *Alameda Applied Sciences Corporation, San Leandro, CA, USA* on 16 May 2012 [107].

We then configured Lee Model 6-phase code using the fitting parameters obtained to work from 0.5 to 3.5 Torr deuterium gas.

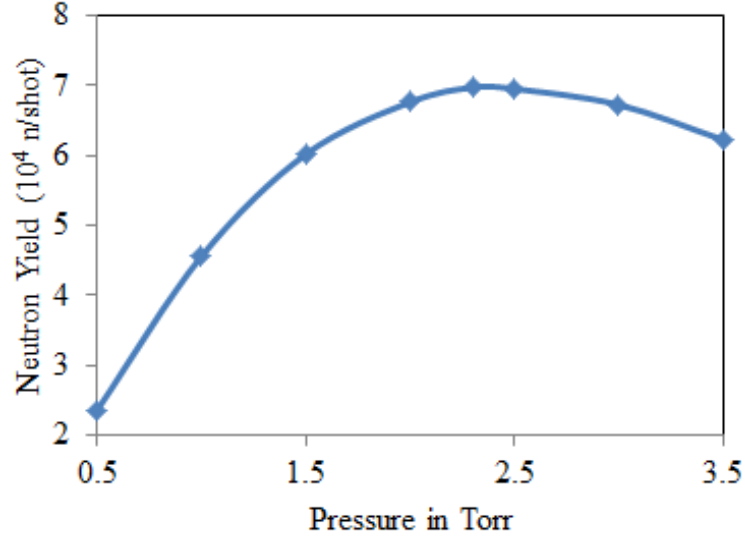


Figure 3.9 Variation of computed neutron yield with respect to pressure operating at 14 kV in deuterium for AASC Plasma Focus machine.

From Figure 3.9, it can be observed that the optimum neutron yield is 6.97×10^4 n/shot at 2.3 Torr while the energy input into plasma is at 16.8 %. This computed neutron yield Y_n differs from that of the published result $Y_n = 2.6 \times 10^5$ n/shot at 2.75 Torr [107] by a factor of 3.7.

3.5 Argentina Nano Focus machine (140 J - Type 1 machine - D gas)

Argentina Nano focus machine shown in Figure 3.10 is a very small transportable dense plasma focus used as a portable intense fast neutron source. The machine parameters are stated in Table 3.9.

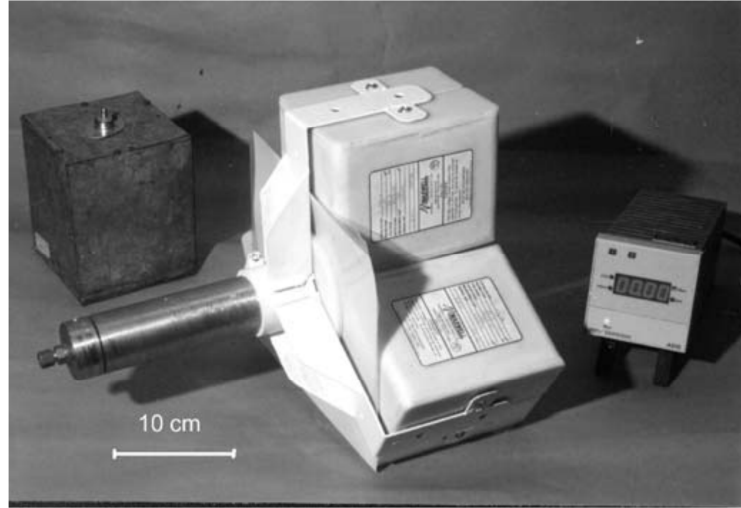


Figure 3.10 Picture of Argentina Nano focus machine taken from Figure 5, page 79 of reference [28].

This machine has a static inductance of 74 nH. The Lee Model 5-phase code was used to fit the measured current waveform and we obtained a good fit as shown in Figure 3.11.

The maximum computed current is 55 kA (compared to 62 kA published [28]) and exhibits the radial start time at 0.408 μ s for pinch duration of 0.049 μ s. It has a speed factor of 60 (kA/cm) / Torr^{0.5} and neutron yield of 5.73×10^4 n/shot. The detailed information is presented in Table 3.10.

Table 3.9 Machines, operational and fitting parameters for Argentina Nano Focus machine.

Capacitance C_0 (μF) = 1.1
Static inductance L_0 (nH) = 74
Circuit resistance r_0 (m Ω) = 25
Cathode radius 'b' (cm) = 2.1
Anode radius 'a' (cm) = 0.75
Anode length 'z ₀ ' (cm) = 1.8
Charging voltage V_0 (kV) = 16
Fill pressure P_0 (Torr) = 1.5
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are :
Axial phase mass factor, $f_m = 0.055$
Axial phase current factor, $f_c = 0.7$
Radial phase mass factor, $f_{mr} = 0.1$
Radial phase current factor, $f_{cr} = 0.95$

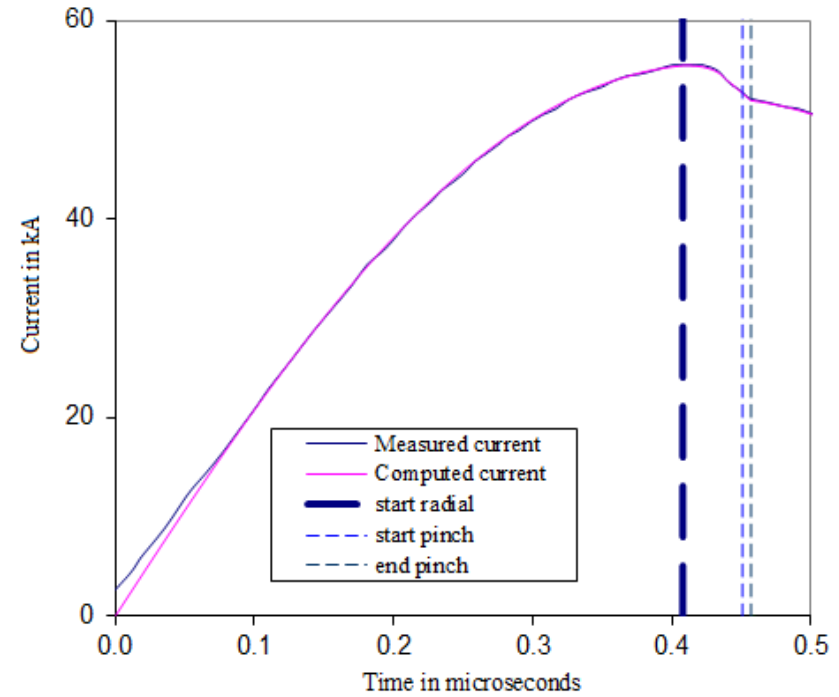


Figure 3.11 The measured current waveform of Argentina Nano Focus machine at 16 kV, 1.5 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: The measured current waveform information was extracted from the current derivative waveform on Figure 7, page 80 of reference [28].

Table 3.10 Information obtained from Lee Model 5-phase code configured for Argentina Nano Focus machine at 16 kV, 1.5 Torr deuterium gas.

Peak current (kA) = 55
Pinch start current (kA) = 50
Pinch minimum temperature (10^6 K) = 6.40
Pinch maximum temperature (10^6 K) = 6.60
Peak axial speed (cm/ μ s) = 7.4
Peak radial shock speed (cm/ μ s) = 34.9
Peak radial piston speed (cm/ μ s) = 23.4
Final pinch radius $r_{\min}$ (cm) = 0.11
Pinch length $z_{\max}$ (cm) = 1.1
Pinch duration (ns) = 6.3
Peak induced voltage (kV) = 10.9
Neutron yield ($\times 10^4$ n/shot) = 5.73
Energy inflow into plasma (%) = 7.2
Speed Factor ((kA/cm)/Torr ^{0.5}) = 60
Current per cm anode radius (kA/cm) = 74

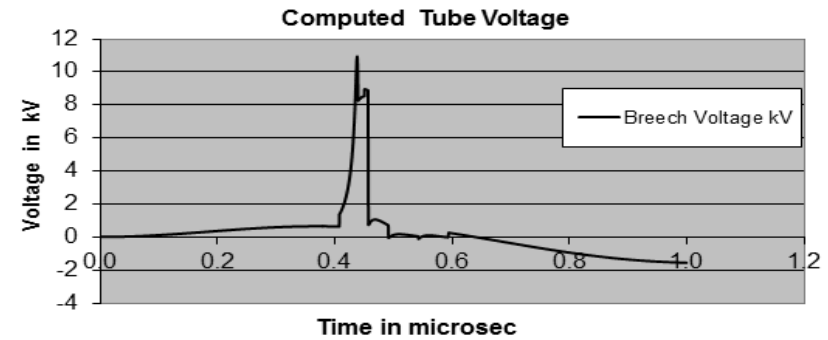


Figure 3.12 Computed tube voltage versus time for Argentina Nano Focus machine at 16 kV, 1.5 Torr deuterium.

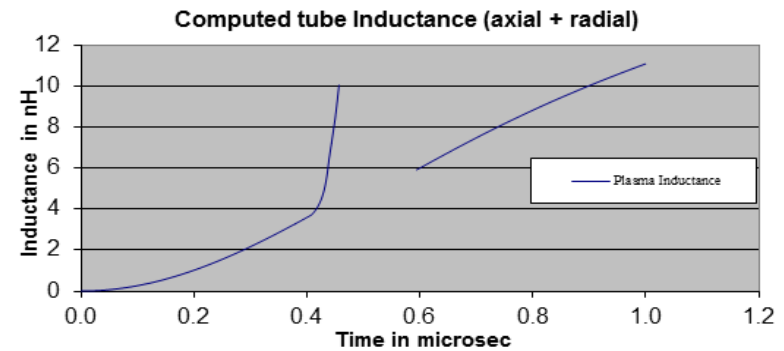


Figure 3.13 Computed inductance variation with time for Argentina Nano Focus machine at 16 kV, 1.5 Torr deuterium.

In the published article [28], it was stated that at 1.5 Torr (2 mbar) pressure, it was observed that when the current derivative curve goes to zero (around 400 ns), a thin and very sharp dip lasting around 50 ns) in the current derivative occurs. The computed results shows that this corresponds to the radial start (occurring at 0.41 μ s) to the end of the pinch (duration of 50 ns) as can be observed from the computed current waveform (which matches exactly the measured current waveform) in Figure 3.11. Correspondingly the computed tube voltage graph in Figure 3.12 shows a rapid growth in voltage at (0.4 μ s) for duration of 50 ns. This also corresponds to a strong inductance variation (computed) because of the fast plasma column compression by pinch effect as shown in Figure 3.13.

Using the information obtained from the current waveform, Lee Model 5-phase code was then configured to operate from 0.5 Torr to 2.0 Torr and the neutron yield versus pressure is plotted in Figure 3.14.

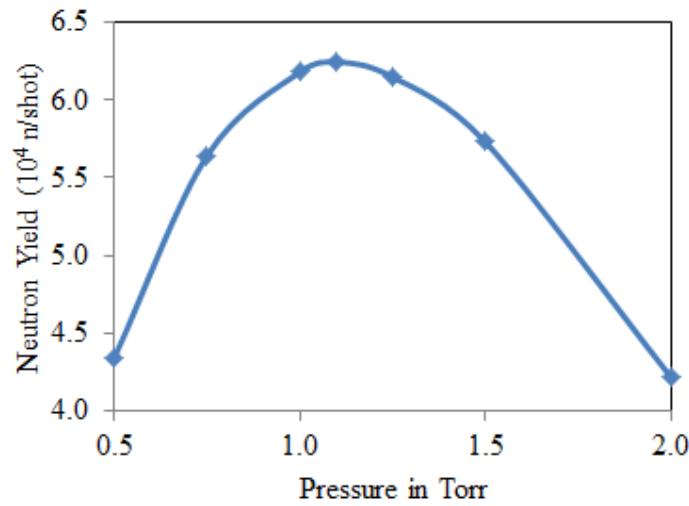


Figure 3.14 Variation of computed neutron yield with respect to pressure operating at 16 kV in deuterium for Argentina Nano Focus machine.

From Figure 3.14, the optimum neutron yield obtained is 6.24×10^4 n/shot at 1.1 Torr with the energy input into plasma at 7.0 %. The published paper does not give a neutron yield.

3.6 FMPF-1 Plasma Focus machine (170 J Taper anode - Type 1 machine - D gas)

The Fast Miniature Plasma Focus device (FMPF-1) from National Institute of Education (NIE) Singapore has its capacitor bank made up of four 0.6 μF , 30 kV low inductance capacitors connected in parallel. The layout is compact using a common transmission plate assembly. Four layers of 125 μm thick Mylar [12] was used as insulation between the transmission plates. The inductance of the connections between capacitor bank, spark gap and plasma focus was minimized by embedding the spark gap within transmission line assembly of the capacitor bank and by directly interfacing plasma focus head to the discharge end of spark gap [12].

The machine parameters and the anode geometry of the FMPF-1 are stated in Table 3.11 and Figure 3.15 respectively. It's $L_0 = 33 \text{ nH}$ is fitted using Lee Model 5-phase code. The Lee Model 5-phase code was used to fit the measured current waveform. A good fit is shown in Figure 3.16.

The detailed information obtained from Lee Model 5-phase code is shown in Table 3.12.

Table 3.11 The machine, operational and fitting parameters for FMPF-1 Plasma Focus machine.

Capacitance C_0 (μF) = 2.4
Static inductance L_0 (nH) = 33
Circuit resistance r_0 (m Ω) = 60
Cathode radius 'b' (cm) = 1.5
Anode radius 'a' (cm) = 0.6
Anode length 'z ₀ ' (cm) = 1.7
Taper starts at (cm) = 1.0
Final tapered radius (cm) = 0.3
Charging voltage V_0 (kV) = 12
Fill pressure P_0 (Torr) = 2.25
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.155$
Axial phase current factor, $f_c = 0.7$
Radial phase mass factor, $f_{mr} = 0.165$
Radial phase current factor, $f_{cr} = 0.75$

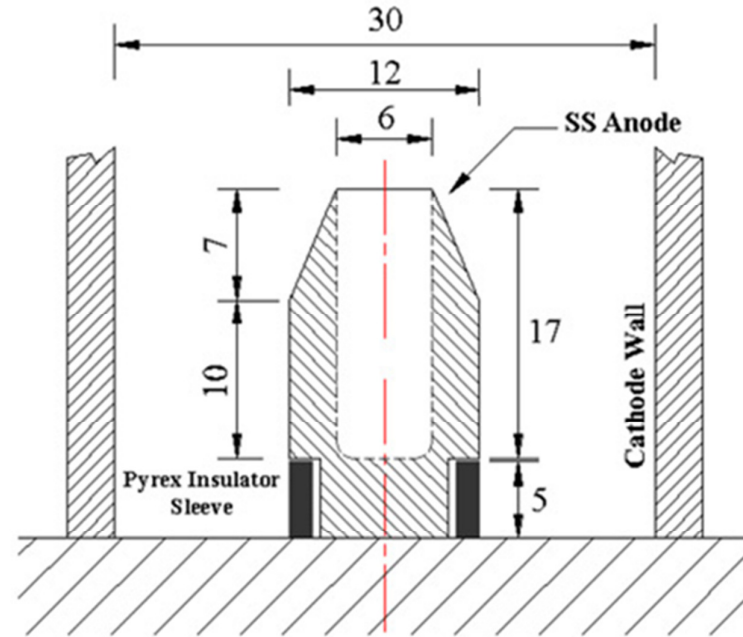


Figure 3.15 Schematic of Composite anode geometry used in FMPF-1

Note: Diagramme was taken from Figure 5(d), page 6 of reference [12]

Table 3.12 Information obtained from Lee Model 5-phase code configured for FMPF-1 Plasma Focus machine at 12 kV, 2.25 Torr deuterium gas.

Peak current (kA) = 69
Pinch start current (kA) = 48
Pinch minimum temperature (10^6 K) = 16.09
Pinch maximum temperature (10^6 K) = 16.61
Peak axial speed (cm/ μ s) = 7.0
Peak radial shock speed (cm/ μ s) = 34.7
Peak radial piston speed (cm/ μ s) = 23.5
Final pinch radius $r_{\min}$ (cm) = 0.06
Pinch length $z_{\max}$ (cm) = 0.6
Pinch duration (ns) = 3.7
Peak induced voltage (kV) = 16.2
Neutron yield ($\times 10^4$ n/shot) = 8.5
Energy Inflow into Plasma (%) = 4
Speed Factor ((kA/cm)/Torr ^{0.5}) = 77

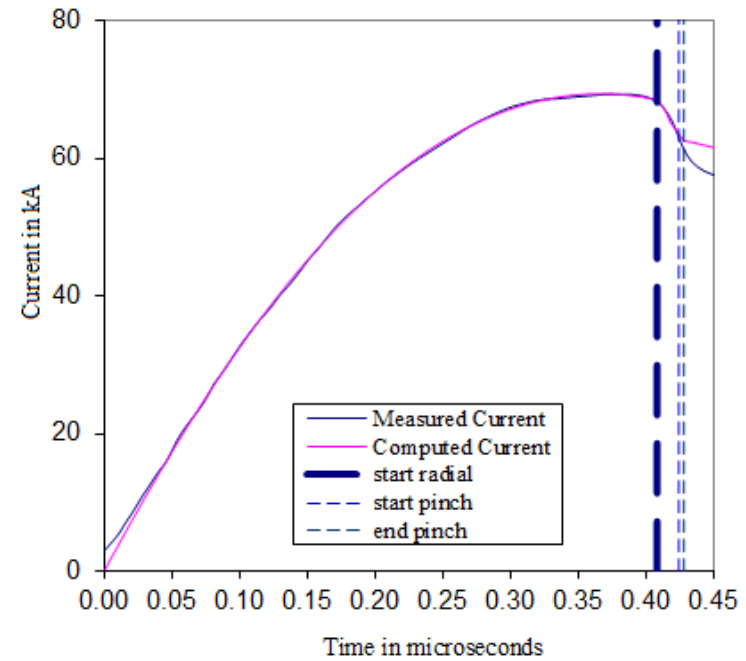


Figure 3.16 The measured current waveform of FMPF-1 Plasma Focus machine at 12 kV, 2.25 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: The measured current waveform information was extracted from the current derivative waveform in Figure 7, page 8 of reference [12].

Using the fitting parameters obtained from fitting the computed current to the measured current waveform, Lee Model 5-phase code was then configured for pressures from 0.5 Torr to 3.0 Torr and its computed neutron yield compared to the measured yield values obtained from the paper “*Compact sub-kilojoule range fast miniature plasma focus as portable neutron source*” [12].

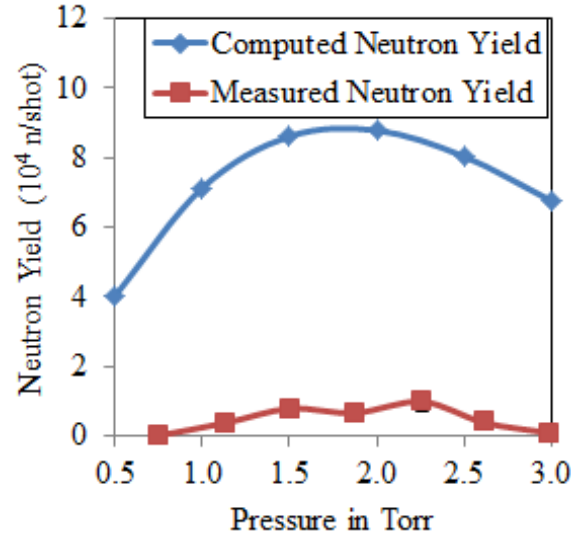


Figure 3.17 Variation of computed and experimental neutron yield with respect to pressure operating at 12 kV in deuterium for FMPF-1 Plasma Focus machine.

Note: The measured values were obtained from Figure 9, page 8 of reference [12].

The computed neutron yield has a maximum value of 8.8×10^4 n/shot which occurs at 2.0 Torr whereas the experimental value is $(1 \pm 0.27) \times 10^4$ n/shot which occurs at 2.25 Torr. The general profiles of neutron yield versus pressure agree well. However the difference between computed peak yield and measured peak yield by factor of 9 is unusually high. The difference for most optimised machines is typically within a factor of 3 or less. This could mean that the particular series of shots which we have analysed are from the FMPF-1 when it was not optimised.

3.7 India's-smallest sealed Plasma Focus machine (200 J - Type 1 machine showing Type 2 characteristics - D gas)

India's smallest Plasma Focus machine [109] has an effective plasma chamber volume of 33 cm^3 including the space occupied by the anode and the insulator. It is an all metal sealed tube and the metal to ceramic sealing is done by silver based active alloy brazing. The photograph of this plasma focus device is shown in Figure 3.18.



Figure 3.18 Photograph of India's smallest sealed- type Plasma Focus machine which was obtained from Figure 2, page 026104-2 of reference [109].

The machine parameters for this Plasma Focus are shown in Table 3.13. Its static inductance is 46 nH. It is a Type 1 machine and the Lee Model 5-phase code was used to fit the computed current waveform to the measured waveform and where necessary the Lee Model 6-phase code was used to complete fitting the whole current dip. The current fitting is shown in Figure 3.19.

Table 3.13 The machine, operation and fitting parameters for India's smallest-seal type Plasma Focus machine

Capacitance C_0 (μF) = 4	Fill gas (molecular weight) = 4																				
Static inductance L_0 (nH) = 46	Fill gas (atomic number) = 1																				
Circuit resistance r_0 (m Ω) = 10	Fill gas (molecule) = 2																				
Cathode radius ‘b’ (cm) = 1.6	The fitted model parameters are:																				
Anode radius ‘a’ (cm) = 0.5	Axial phase mass factor, $f_m = 0.135$																				
Anode length ‘z ₀ ’ (cm) = 2	Axial phase current factor, $f_c = 0.7$																				
Charging voltage V_0 (kV) = 10	Radial phase mass factor, $f_{mr} = 0.1$																				
Fill pressure P_0 (Torr) = 6.0	Radial phase current factor, $f_{cr} = 0.8$																				
The anomalous resistance parameters are:																					
	<table><tr><th></th><th>$R_0(\Omega)$</th><th>Characteristic of fall time τ_2 (ns)</th><th>Characteristic of rise time τ_1 (ns)</th><th>End fraction time</th></tr><tr><td>Dip 1</td><td>0.5</td><td>22</td><td>10</td><td>1</td></tr><tr><td>Dip 2</td><td>0.02</td><td>80</td><td>10</td><td>1</td></tr><tr><td>Dip 3</td><td>0.03</td><td>100</td><td>10</td><td>1</td></tr></table>		$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time	Dip 1	0.5	22	10	1	Dip 2	0.02	80	10	1	Dip 3	0.03	100	10	1
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time																	
Dip 1	0.5	22	10	1																	
Dip 2	0.02	80	10	1																	
Dip 3	0.03	100	10	1																	

The maximum computed current is 82 kA (published experimental value is 83 kA) and exhibits the radial start time at 0.660 μs for pinch duration of 0.033 μs . It has a speed factor of 67 ((kA/cm)/Torr^{0.5}) and neutron yield of 2.43×10^5 n/shot (maximum experimental yield is 7.4×10^4 n/shot). The other detailed information is presented in Table 3.14.

Using the model parameters fitted from the current waveform, Lee Model 6-phase code was then configured to operate from 1.0 Torr to 8.0 Torr and the results are shown in Figure 3.20.

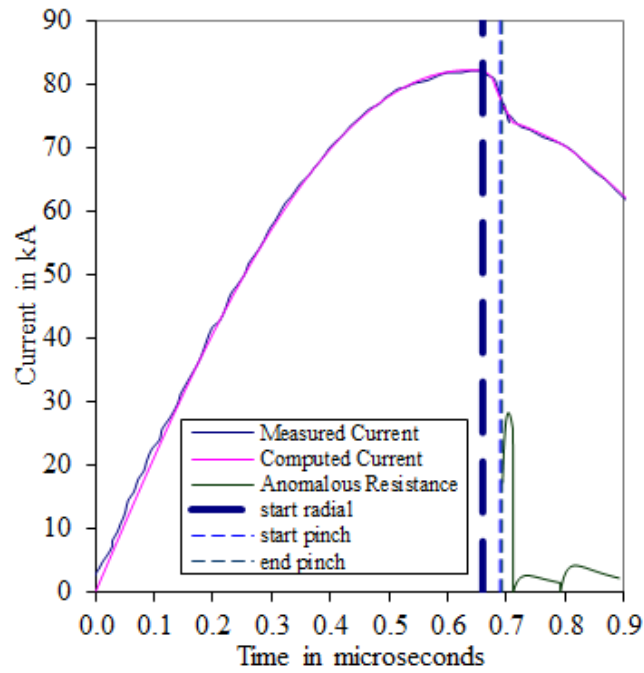


Figure 3.19 The measured current waveform of India's smallest sealed-type Plasma Focus machine at 10 kV, 6.0 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was taken from the current derivative in Figure 3, page 026104-2 of reference [109].

From Figure 3.20, the maximum neutron yield is 2.53×10^5 n/shot at 5.0 Torr. The energy input into plasma is at 7 %. The computed neutron yield is higher than the experimental neutron yield of this device (reported as 7.8×10^4 n/shot) measured at 6.0 Torr (8 mbar) by a factor of 3.

Another thing to note is that in this experiment [109] 200 shots were fired without changing the gas in between the shots and the neutron yield reduced from 7.8×10^4 to 0.1×10^4 n/shot as the series progressed as shown in Figure 3.21.

According to the paper [109] the possible causes of deterioration in neutron yield might be due to the release of impurity into the plasma after the first 50 shots due to evaporation of electrode materials and accumulation of the ^3He gas in the device.

Table 3.14 Information obtained from Lee Model 6-phase code configured for India's smallest sealed-type Plasma Focus machine at 10 kV, 6.0 Torr deuterium gas.

Peak current (kA) = 82
 Pinch start current (kA) = 62
 Pinch minimum temperature (10^6 K) = 6.17
 Pinch maximum temperature (10^6 K) = 6.34
 Peak axial speed (cm/ μ s) = 4.9
 Peak radial shock speed (cm/ μ s) = 34.2
 Peak radial piston speed (cm/ μ s) = 23.5
 Final pinch radius $r_{\min}$ (cm) = 0.07
 Pinch length $z_{\max}$ (cm) = 0.7
 Pinch duration (ns) = 4.3
 Peak induced voltage (kV) = 13.3
 Neutron yield ($\times 10^5$ n/shot) = 2.43
 Energy Inflow into Plasma (%) = 7.1
 Speed Factor ((kA/cm)/Torr $^{0.5}$) = 67

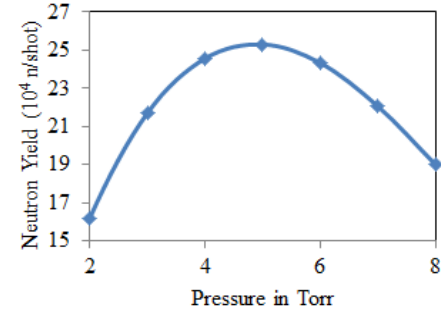


Figure 3.20 Variation of computed neutron yield with respect to pressure operating at 10 kV in deuterium for India's smallest sealed-type Plasma Focus machine.

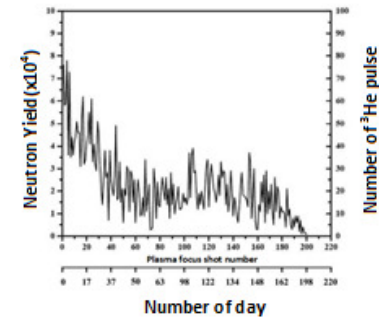


Figure 3.21 Neutron yield variation with shot number and with days was taken from Figure 4, page 026104-3 of reference [109].

3.8 PF400 Plasma Focus machine (400 J - Type 1 machine - D gas)

This Mather's type machine from Chile consists of a 28 mm long, 12 mm diameter copper tube anode and an outer cathode of eight 5 mm diameter copper rods uniformly spaced on a 31 mm diameter circle. The anode and cathode are separated by an alumina tube of 21 mm length. The storage capacitor is 0.95 μ F and is typically charged to 28 kV at an operating pressure of 6.6 Torr deuterium. Its internal inductance is found to be 40 nH so it is a Type 1 machine. Lee Model 5-phase code is used to fit the measured current waveform with the machine parameters as shown in Table 3.15.

When Lee Model 5-phase code was configured with the machine parameters for the PF 400, its effective anode length was found to be only 17 mm whereas the maximum computed current obtained was 126 kA (compared to the measured current of (127 ± 6) kA given in paper [26]). The radial start time was found to be at 0.308 μ s for a radial pinch duration of 0.039 μ s. It has a speed factor of 82 (kA/cm)/Torr^{0.5} and maximum computed neutrons yield of 1.14×10^6 n/shot (compared to the measured neutron yield $(1.06 \pm 0.13) \times 10^6$ n/shot given in paper [26]). The other detailed information is presented in Table 3.16.

Numerical experiments were run from 3.0 Torr to 9.0 Torr. Comparison of our computed neutron yield was made with measured neutron yield published in the paper "*Neutron emission from a fast plasma focus of 400 Joules*" [26].

Figure 3.23 shows that the magnitudes of average neutron yield are quite similar with the published curve [26]. The main features for comparison include the peak neutron yield (computed value of 1.25×10^6 n/shot against a measured value of $(1.06 \pm 0.13) \times 10^6$ n/shot); optimum pressure (computed value of 5.0 Torr against the measured value of 6.75 Torr (9 mbar)) and the drop-off of neutron yield on both sides of the optimum, although the computed drop-offs are more gradual than the measured. These results had already been noted in an earlier paper [89].

Table 3.15 The machine, operation and fitting parameters for PF400 Plasma Focus machine.

Capacitance C_0 (μF) = 0.95
Static inductance L_0 (nH) = 40
Circuit resistance r_0 (m Ω) = 10
Cathode radius 'b' (cm) = 1.55
Anode radius 'a' (cm) = 0.6
Anode length 'z ₀ ' (cm) = 1.7
Charging voltage V_0 (kV) = 28
Fill pressure P_0 (Torr) = 6.6
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.08$
Axial phase current factor, $f_c = 0.7$
Radial phase mass factor, $f_{mr} = 0.11$
Radial phase current factor, $f_{cr} = 0.71$

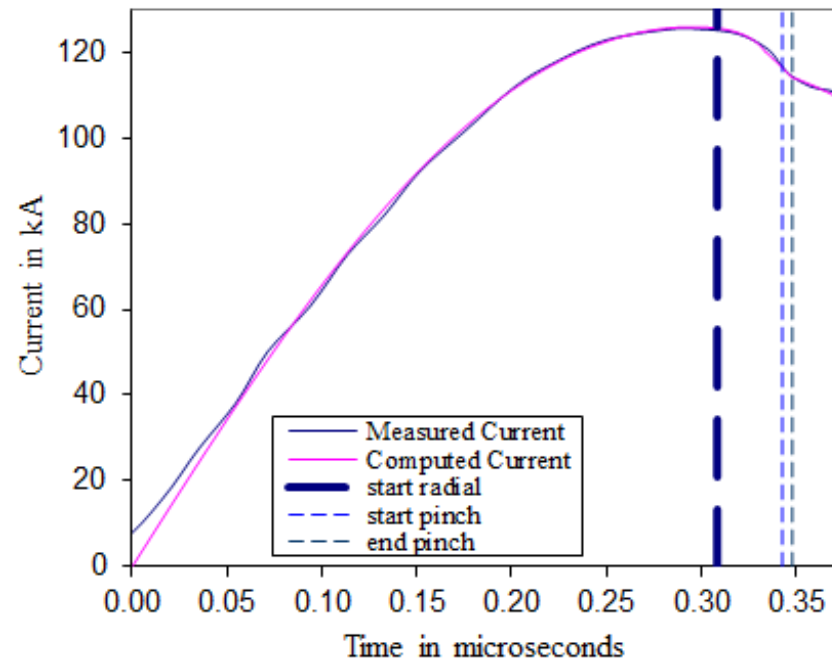


Figure 3.22 Measured current waveform of PF400 Plasma Focus machine at 28 kV, 6.6 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: the digitalized measured current waveform was downloaded from <http://www.plasmafocus.net> [57].

Table 3.16 Information obtained from Lee Model 5-phase code configured for PF400 Plasma Focus machine at 28 kV, 6.6 Torr deuterium gas.

Peak current (kA) = 126
Pinch start current (kA) = 83
Pinch minimum temperature (10^6 K) = 6.14
Pinch maximum temperature (10^6 K) = 6.38
Peak axial speed (cm/ μ s) = 9.0
Peak radial shock speed (cm/ μ s) = 34.8
Peak radial piston speed (cm/ μ s) = 23.2
Final pinch radius $r_{\min}$ (cm) = 0.09
Pinch length $z_{\max}$ (cm) = 0.8
Pinch duration (ns) = 5.2
Peak induced voltage (kV) = 17.2
Neutron yield ($\times 10^6$ n/shot) = 1.14
Energy Inflow into Plasma (%) = 7.1
Speed Factor ((kA/cm)/Torr ^{0.5}) = 82
Current per cm anode radius (kA/cm) = 210

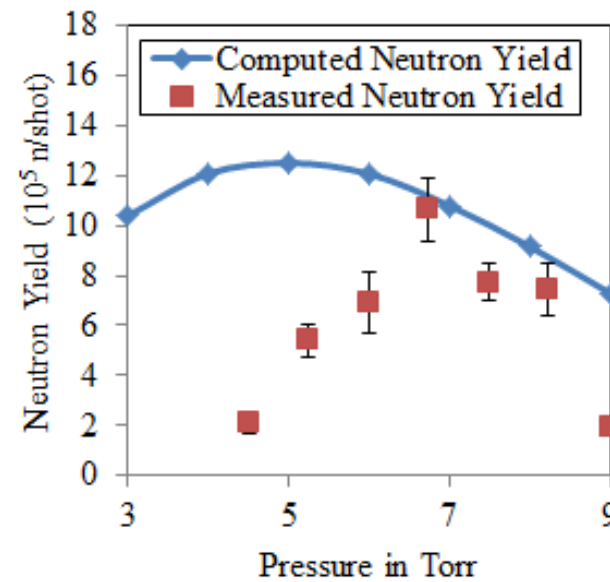


Figure 3.23 Variation of computed and experimental neutron yield with respect to pressure operating at 2.8 kV in deuterium for PF400 Plasma Focus machine.

Note: Measured values were taken from Figure 3, page 3270 of reference [26]

The radial start time and the time to peak current were also computed using Lee Model 5-phase code (Figure 3.24) and compared to the measured values given in Figure 3.25.

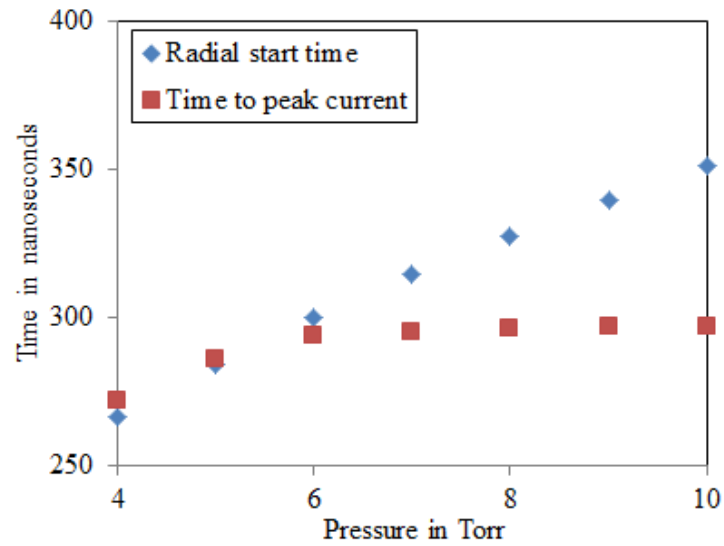


Figure 3.24 Computed PF400 radial start time and time to peak current versus pressure. The time to peak current is almost constant at high pressure as the pinch dimension is almost the same.

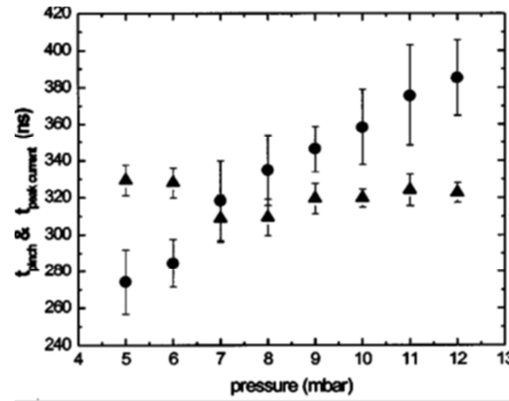


Figure 3.25 Measured Pinch time (●) and time to peak current vs filling pressure (▲) taken from Figure 2, page 3270 of reference [26]

The maximum compression from the computation occurs when the radial start time is close to the current peak time and this occurs at 5.0 Torr (6.7 mbar) compared to 5.25Torr (7 mbar) obtained experimentally.

3.9 NX1 Plasma Focus Machine (1.6 - 3.1 kJ - Type 1 machine - Ne gas)

NX1 is a Mather type plasma machine with a four-module system capacitor bank ($7.8 \mu\text{F} \times 4$). The capacitor banks are connected to the focus via four switches (pseudo spark switches). NX1 electrodes are made of oxygen-free copper and it uses ceramic as its insulator material. This machine operates using neon gas and has $L_0 = 29 \text{ nH}$. The Lee Model 5-phase code was used to fit its measured current waveform as shown in Figure 3.27 and Figure 3.28. In some cases when the Lee Model 5-phase code cannot fit the current dip completely, the Lee Model 6-phase code was used to complete the fit. NX1 machine parameters are shown in Table 3.17.

Table 3.17 Machines bank and focus tube parameters for NX1 Plasma Focus machine operating in Neon

Capacitance C_0 (μF) = 31.2	Anode radius 'a' (cm) = 1.5
Static inductance L_0 (nH) = 29	Fill gas (molecular weight) = 20
Circuit resistance r_0 ($\text{m}\Omega$) = 3	Fill gas (atomic number) = 10
Cathode radius 'b' (cm) = 2.5	Fill gas (atom) = 1

3.9.1 Case of early focus (current waveform showing Type 2 characteristics)

Table 3.18 The machine, operation and fitting parameters for NX1 Plasma Focus machine when focus came early.

Capacitance C_0 (μF) = 31.2	Fill gas (molecular weight) = 20																
Static inductance L_0 (nH) = 29	Fill gas (atomic number) = 10																
Circuit resistance r_0 (m Ω) = 3	Fill gas (atom) = 1																
Cathode radius ‘b’ (cm) = 2.5	The fitted model parameters are:																
Anode radius ‘a’ (cm) = 1.5	Axial phase mass factor, $f_m = 0.112$																
Anode length ‘z ₀ ’ (cm) = 4.5	Axial phase current factor, $f_c = 0.7$																
Charging voltage V_0 (kV) = 10	Radial phase mass factor, $f_{mr} = 0.2$																
Fill pressure P_0 (Torr) = 3.0	Radial phase current factor, $f_{cr} = 0.8$																
The anomalous resistance parameters are:																	
	<table><tr><td>$R_0(\Omega)$</td><td>Characteristic of fall time τ_2 (ns)</td><td>Characteristic of rise time τ_1 (ns)</td><td>End fraction time</td></tr><tr><td>Dip 1</td><td>0.1</td><td>100</td><td>1</td></tr><tr><td>Dip 2</td><td>0.03</td><td>100</td><td>1</td></tr><tr><td>Dip 3</td><td>0.01</td><td>100</td><td>1</td></tr></table>	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time	Dip 1	0.1	100	1	Dip 2	0.03	100	1	Dip 3	0.01	100	1
$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time														
Dip 1	0.1	100	1														
Dip 2	0.03	100	1														
Dip 3	0.01	100	1														

This shot was at 10 kV with a pressure of 3.0 Torr neon gas. The detailed information obtained from this fitting is shown in Table 3.19.

Using the information obtained from the current waveform, Lee Model 6-phase code was then configured to work from 1.0 to 10.0 Torr and the computed soft X-ray yield is compared with the published results [6, 7].

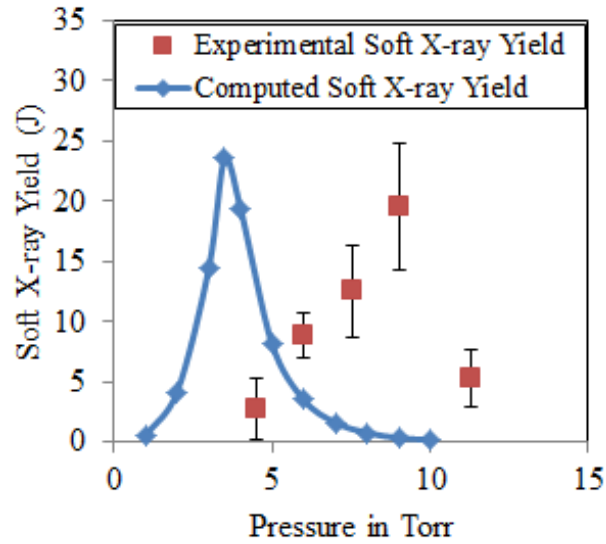


Figure 3.26 Variation of computed and experimental soft X-ray yield with respect to pressure operating at 10 kV in neon for NX1 Plasma Focus machine with anode length of 4.5 cm.

Note: The experimental values data was obtained from Figure 6.65, page 191 of Zhang Guixin, Phd dissertation [7].

Figure 3.26 shows that the experimental and computed results are quite close for optimum yield (agrees to a factor of 1.3): optimum computed soft X-ray yield is 23.6 J and it occurs at 3.5 Torr with the energy input into plasma at 20.8 % whereas the experimental soft X-ray yield is 19 J at 9.0 Torr (12 mbar). The computed soft X-ray yield exhibits a more gradual drop off after the optimum pressure.

Table 3.19 Information obtained from Lee Model 6-phase code configured for NX1 Plasma Focus machine at 10 kV, 3.0 Torr neon gas.

Peak current (kA) = 276
Pinch start current (kA) = 183
Pinch minimum temperature (10^6 K) = 2.41
Pinch maximum temperature (10^6 K) = 2.68
Peak axial speed (cm/ μ s) = 5.5
Peak radial shock speed (cm/ μ s) = 24.6
Peak radial piston speed (cm/ μ s) = 15.6
Final pinch radius $r_{\min}$ (cm) = 0.13
Pinch length $z_{\max}$ (cm) = 2.2
Pinch duration (ns) = 15
Peak induced voltage (kV) = 34.6
Soft X-ray yield (J) = 14.47
Energy Inflow into Plasma (%) = 19.6
Speed Factor ((kA/cm)/Torr ^{0.5} Ne) = 106
Current per cm anode radius (kA/cm) = 184

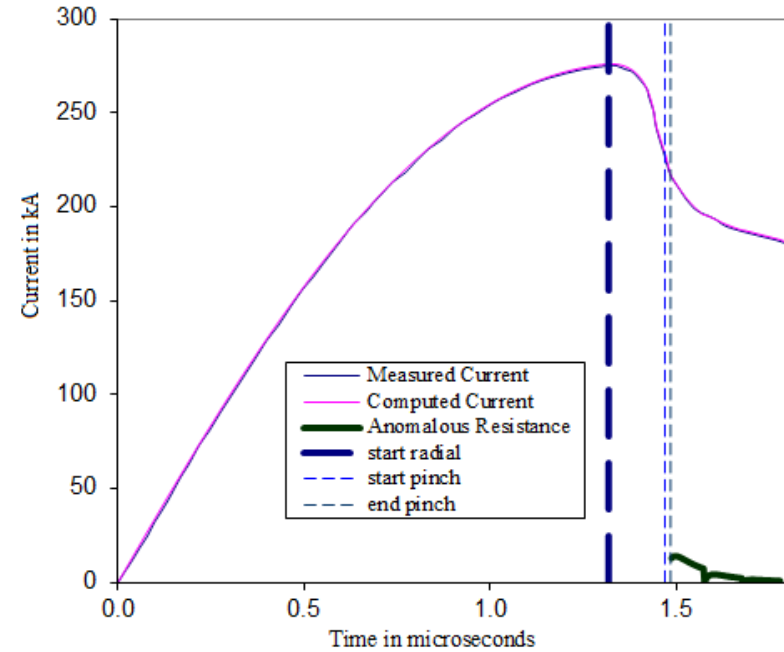


Figure 3.27 The measured current waveform of NX1 Plasma Focus machine at 10 kV, 3.0 Torr neon gas, with anode length of 4.5 cm compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was extracted from Figure 7, page 1123 of reference [6].

3.9.2 Case of late dip (Type 1 characteristics)

Table 3.20 Machine and model parameters for NX1 Plasma Focus machine when dip comes late.

Capacitance C_0 (μF) = 31.2	Fill gas (molecular weight) = 20
Static inductance L_0 (nH) = 29	Fill gas (atomic number) = 10
Circuit resistance r_0 (m Ω) = 3	Fill gas (atom) = 1
Cathode radius 'b' (cm) = 2.5	The fitted model parameters are:
Anode radius 'a' (cm) = 1.5	Axial phase mass factor, $f_m = 0.1$
Anode length 'z ₀ ' (cm) = 5.4	Axial phase current factor, $f_c = 0.8$
Charging voltage V_0 (kV) = 14	Radial phase mass factor, $f_{mr} = 0.1$
Fill pressure P_0 (Torr) = 11.0	Radial phase current factor, $f_{cr} = 0.8$

For this shot NX1 was charged to 14 kV at a pressure of 11.0 Torr neon gas. The detailed information obtained from the Lee Model 5-phase code is shown in Table 3.21.

Using the information obtained from the current waveform, Lee Model 5-phase code is now configured to work from 1.0 to 15.0 Torr and the computed soft X-ray yield, axial speed and focus time are compared with the published results [7] as shown in Figures 3.29, 3.30 and 3.31.

Figure 3.29 shows that the experimental and computed results are quite close (agrees to a factor of 1.5), optimum computed soft X-ray yield is 37 J and it occurs at 11.0 Torr with the energy input into plasma at 20.6 % whereas the experimental soft X-ray yield is 55 J at 11.0 Torr (14 mbar). Both the computed and experimental soft X-ray yield exhibits a gradual drop off after their optimum pressure.

In terms of axial speed and focus time, there are general agreements between both the experimental and computed results even though the experimental results show a slightly longer time to focus and a slightly lower axial speed.

Table 3.21 Information obtained from Lee Model 5-phase code configured for NX1 Plasma Focus machine at 14 kV, 11.0 Torr neon gas.

Peak current (kA) = 387
Pinch start current (kA) = 242
Pinch minimum temperature (10^6 K) = 2.26
Pinch maximum temperature (10^6 K) = 2.56
Peak axial speed (cm/ μ s) = 5.2
Peak radial shock speed (cm/ μ s) = 24.3
Peak radial piston speed (cm/ μ s) = 15.5
Final pinch radius $r_{\min}$ (cm) = 0.12
Pinch length $z_{\max}$ (cm) = 2.1
Pinch duration (ns) = 15
Peak induced voltage (kV) = 44.5
Soft X-ray yield (J) = 37.08
Energy Inflow into Plasma (%) = 20.6
Speed Factor ((kA/cm)/Torr ^{0.5} Ne) = 78
Current per cm anode radius (kA/cm) = 258

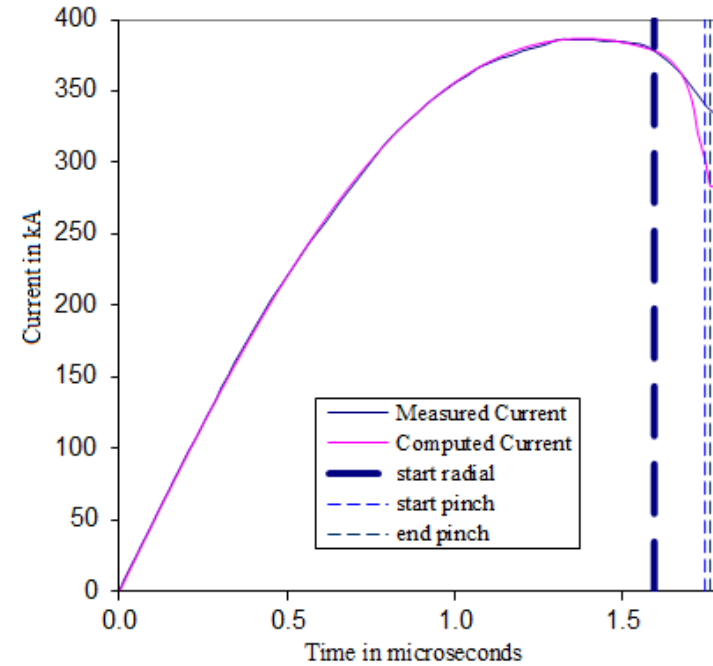


Figure 3.28 The measured current waveform of NX1 Plasma Focus machine at 14 kV, 11.0 Torr neon gas, with anode length of 5.4cm compared with Lee Model 5-phase code computed current.

Note: The measured current waveform information was extracted from Figure 6.36, page 156 of reference [7].

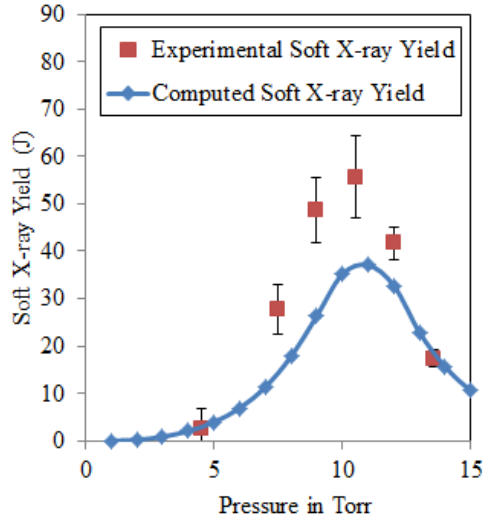


Figure 3.29 Variation of computed and experimental soft X-ray yield with respect to pressure operating at 14 kV in neon for NX1 Plasma Focus machine with anode length of 5.4 cm.

Note: The measured values information was extracted from Figure 6.6(a), page 193 of reference [7].

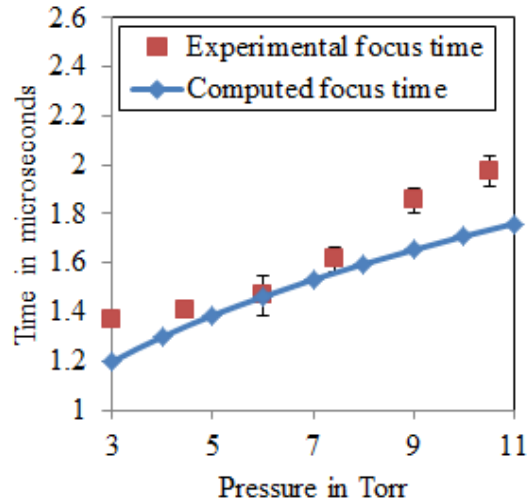


Figure 3.30 Variation of computed and experimental focus time with respect to pressure operating at 14 kV in neon for NX1 Plasma Focus machine with anode length of 5.4 cm.

Note: The measured values information was extracted from Figure 6.6 (b), page 193 of reference [7].

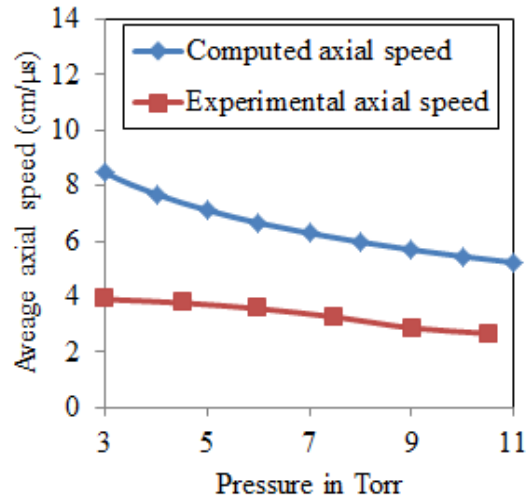


Figure 3.31 Variation of computed and experimental axial speed with respect to pressure operating at 14 kV in neon for NX1 Plasma Focus machine with anode length of 5.4 cm.

Note: The measured values information was extracted from Figure 6.6 (c), page 193 of reference [7].

It should also be noted that the experimental axial speed might have been averaged over the whole axial phase, whilst the computed axial speed is the peak axial speed.

3.10 PPRC Plasma Focus machine (1.8 kJ - Type 2 machine - Ar gas)

PPRC Plasma Focus machine is a plasma focus machine at Azad University, Iran called the Plasma Physics Research Center Mather type PF device. Its static inductance L_0 is 158 nH and we class it as a Type 2 machine. It operates in argon gas.

Lee Model 6-phase code was used to match the various measured current waveforms obtained (Figures 3.32-3.35) at different pressures and voltages. The measured data was made available by Dr. Tania Davari [110]. Table 3.22 shows the machine parameters of this PPRF PF device whereas Table 3.23-3.26 shows the operational parameters and the current and mass factors as well as the anomalous resistance values.

Table 3.22 The machine and operation parameters for PPRC Plasma Focus machine operating in Argon.

Capacitance C_0 (μF) = 10	Anode length ' z_0 ' (cm) = 13.2
Static inductance L_0 (nH) = 158	Fill gas (molecular weight) = 40
Circuit resistance r_0 ($\text{m}\Omega$) = 12.5	Fill gas (atomic number) = 18
Cathode radius ' b ' (cm) = 3	Fill gas (atom) = 1
Anode radius ' a ' (cm) = 1	

Table 3.23	Current and mass factors as well as anomalous resistance values of PPRC Plasma Focus machine working at 14.2 kV, 0.3 Torr argon gas.			
The following are the fitted model parameters:				
Axial phase mass factor, $f_m = 0.007$				
Axial phase current factor, $f_c = 0.75$				
Radial phase mass factor, $f_{mr} = 0.1$				
Radial phase current factor, $f_{cr} = 0.75$				
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time, (ns)	Characteristic of rise time, (ns)	End fraction time
Dip 1	0.37	250	10	1
Dip 2	0.6	200	2	1
Dip 3	0.2	500	10	1

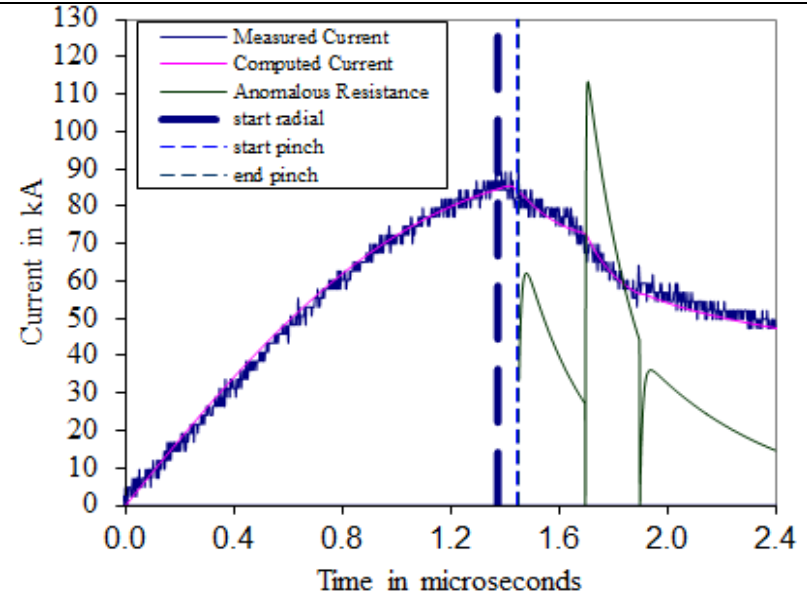


Figure 3.32 Measured current waveform of PPRC Plasma Focus machine at 14.2 kV, 0.3 Torr argon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was provided by Dr Tania Davari from Azad University, Iran on 24th July 2011.

Table 3.24	Current and mass factors as well as anomalous resistance values of PPRC Plasma Focus machine working at 18.5 kV, 0.15 Torr argon gas.			
The following are the fitted model parameters: Axial phase mass factor, $f_m = 0.014$ Axial phase current factor, $f_c = 0.75$ Radial phase mass factor, $f_{mr} = 0.1$ Radial phase current factor, $f_{cr} = 0.75$				
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time, (ns)	Characteristic of rise time, (ns)	End fraction time
Dip 1	0.4	160	10	1
Dip 2	0.45	100	10	1
Dip 3	0.4	500	5	1

Time (μs)	Measured Current (kA)	Computed Current (kA)	Anomalous Resistance (kA)
0.0	10	0	0
0.5	50	50	0
1.0	90	90	0
1.2	100	100	0
1.3	95	95	60
1.4	90	90	30
1.5	85	85	75
1.6	80	80	50
2.0	65	65	30

Figure 3.33 Measured current waveform of PPRC Plasma Focus machine at 18.5 kV, 0.15 Torr argon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was provided by Dr Tania Davari from Azad University, Iran on 24th July 2011.

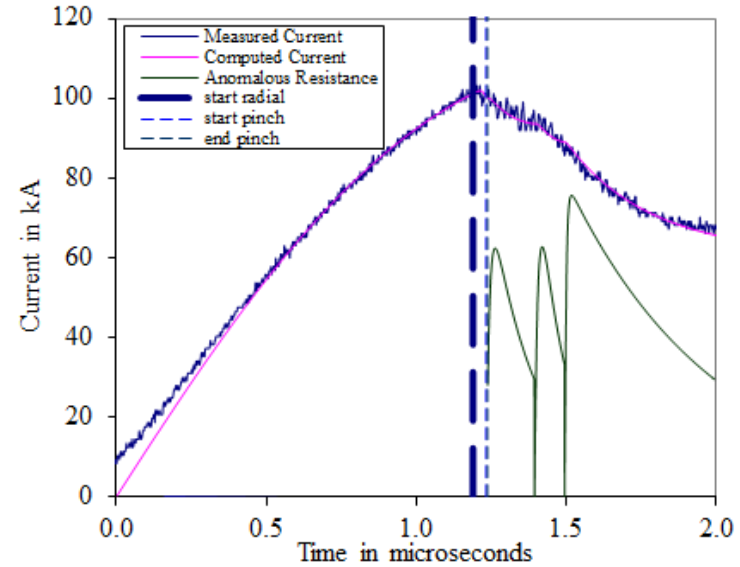


Figure 3.33 Measured current waveform of PPRC Plasma Focus machine at 18.5 kV, 0.15 Torr argon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was provided by Dr Tania Davari from Azad University, Iran on 24th July 2011.

Table 3.25	Current and mass factors as well as anomalous resistance values of PPRC Plasma Focus machine working at 19 kV, 0.7 Torr argon gas.																							
The following are the fitted model parameters: Axial phase mass factor, $f_m = 0.02$ Axial phase current factor, $f_c = 0.75$ Radial phase mass factor, $f_{mr} = 0.1$ Radial phase current factor, $f_{cr} = 0.75$																								
The anomalous resistance parameters are: <table><tr><td></td><td>$R_0(\Omega)$</td><td>Characteristic of fall time, (ns)</td><td>Characteristic of rise time, (ns)</td><td>End fraction time</td></tr><tr><td>Dip 1</td><td>0.17</td><td>500</td><td>3</td><td>1</td></tr><tr><td>Dip 2</td><td>0.23</td><td>480</td><td>10</td><td>1</td></tr><tr><td>Dip 3</td><td>0.4</td><td>300</td><td>5</td><td>1</td></tr></table>						$R_0(\Omega)$	Characteristic of fall time, (ns)	Characteristic of rise time, (ns)	End fraction time	Dip 1	0.17	500	3	1	Dip 2	0.23	480	10	1	Dip 3	0.4	300	5	1
	$R_0(\Omega)$	Characteristic of fall time, (ns)	Characteristic of rise time, (ns)	End fraction time																				
Dip 1	0.17	500	3	1																				
Dip 2	0.23	480	10	1																				
Dip 3	0.4	300	5	1																				

The graph displays the current waveform over time. The measured current (blue line) and computed current (magenta line) are nearly identical, peaking at approximately 130 kA around 1.8 microseconds. The anomalous resistance (green line) shows three distinct dips at approximately 2.0, 2.6, and 3.0 microseconds, corresponding to the start, start, and end of the pinch phase respectively. Vertical dashed lines mark these phases: start radial (solid blue line at ~2.0 μs), start pinch (dashed blue line at ~2.05 μs), and end pinch (dashed blue line at ~2.1 μs).

Figure 3.34 Measured current waveform of PPRC Plasma Focus machine at 19 kV, 0.7 Torr argon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was provided by *Dr Tania Davari* from Azad University, Iran on 24th July 2011.

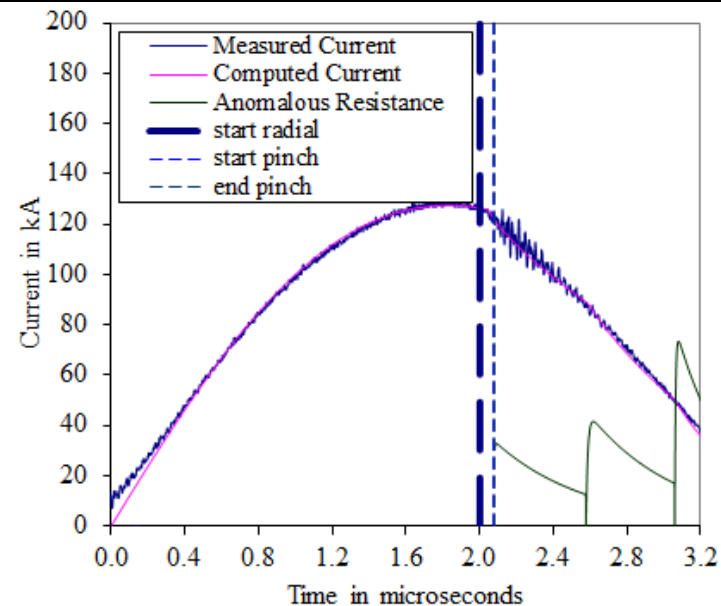


Figure 3.34 Measured current waveform of PPRC Plasma Focus machine at 19 kV, 0.7 Torr argon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was provided by Dr Tania Davari from Azad University, Iran on 24th July 2011.

Table 3.26	Current and mass factors as well as anomalous resistance values of PPRC Plasma Focus machine working at 18 kV, 0.3 Torr argon gas.			
The following are the fitted model parameters: Axial phase mass factor, $f_m = 0.014$ Axial phase current factor, $f_c = 0.75$ Radial phase mass factor, $f_{mr} = 0.1$ Radial phase current factor, $f_{cr} = 0.75$				
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time, (ns)	Characteristic of rise time, (ns)	End fraction time
Dip 1	0.5	100	5	1
Dip 3	0.23	570	5	1

The graph displays the current waveform over time. The measured current (blue line) and computed current (magenta line) are nearly identical, rising from 0 at 0.0 μs to a peak of approximately 110 kA at 1.5 μs, then decaying. The anomalous resistance (green line) shows three distinct peaks: a small one at ~1.5 μs, a larger one at ~1.6 μs, and another at ~1.8 μs. Vertical dashed lines mark the start radial (solid blue at ~1.45 μs), start pinch (dashed blue at ~1.5 μs), and end pinch (dotted blue at ~1.55 μs).

Figure 3.35 Measured current waveform of PPRC Plasma Focus machine at 18 kV, 0.3 Torr argon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was provided by Dr Tania Davari from Azad University, Iran on 24th July 2011.

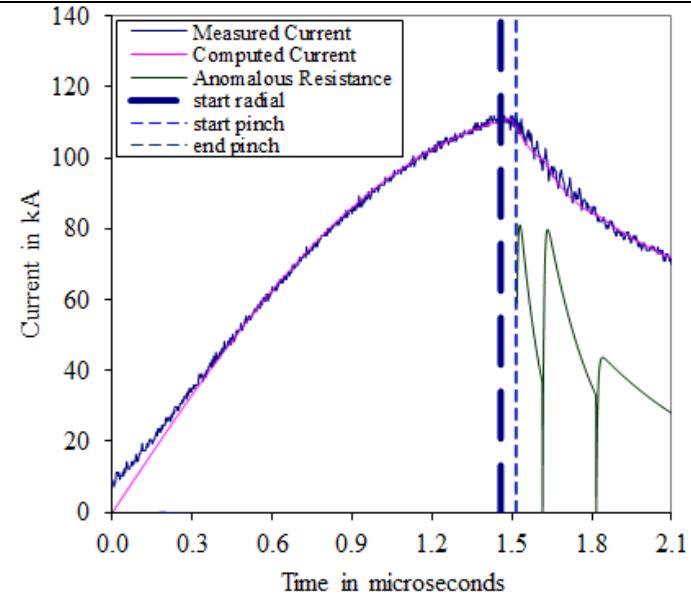


Figure 3.35 Measured current waveform of PPRC Plasma Focus machine at 18 kV, 0.3 Torr argon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was provided by Dr Tania Davari from Azad University, Iran on 24th July 2011.

Table 3.27 Comparison on information about PPRC Plasma Focus machine working at different pressure and charged voltage obtained from Lee Model 6-phase code when computed current waveform is compared to the measured current waveform. The right most data is of 18 kV configured to work at 1.7 Torr.

Argon gas pressure (Torr)	0.3	0.15	0.7	0.3	1.7
Capacitor charged voltage (kV)	14.2	18.5	19	18	18
Peak current (kA)	85	102	127	111	124
Pinch start current (kA)	63	75	92	82	78
Pinch minimum temperature (10^6 K)	7.57	21.42	6.8	12.72	2.13
Pinch maximum temperature (10^6 K)	7.64	21.79	6.9	12.85	3
Peak axial speed (cm/ μ s)	16.7	19.7	10.4	15.6	7.8
Peak radial shock speed (cm/ μ s)	33.3	51.2	31.3	44	17.6
Peak radial piston speed (cm/ μ s)	24.4	33.8	23.5	29.3	13.9
Final pinch radius $r_{\min}$ (cm)	0.08	0.1	0.07	0.09	0.02
Pinch length $z_{\max}$ (cm)	1.4	1.4	1.4	1.4	1.8
Pinch duration (ns)	6.5	4.8	7	5.1	12.2
Peak induced voltage (kV)	23.4	31.7	31.9	32.8	97.7
All-line yield (J)	0.05	0.002	0.34	0.02	34.7
Energy Inflow into Plasma (%)	5.6	4.5	7.8	5.9	9.5
Speed Factor ((kA/cm)/Torr ^{0.5} Ar)	156	263	152	203	95
Current per cm anode radius (kA/cm)	85	102	127	111	124

Using the information from the current curves, Lee Model 6-phase code was configured to work at three different capacitor voltages (14 kV, 18 kV and 19 kV) from 0.1 to 3.0 Torr and the optimum all-line yield obtained when it was configured to work at 1.7 Torr with capacitor charged to 18 kV. The maximum all-line yield obtained is 34.7 J with the energy conversion rate at 9.5 % as shown in Table 3.27 (Right most column).

3.11 KSU Plasma Focus machine (KSU - PF2) (1.8 kJ - Type 2 machine - D and Ne gas)

The Kansas State University Plasma Focus had a static inductance of 123 nH (or 125 nH with different connections). The machine operational and fitting parameters are shown in Table 3.28.

Table 3.28 Machines operational and fitting parameters for KSU-PF2 Plasma Focus machine.

Capacitance C_0 (μF) = 12.5	Fill gas (molecular weight) = 4			
Static inductance L_0 (nH) = 123	Fill gas (atomic number) = 1			
Circuit resistance r_0 (m Ω) = 14.4	Fill gas (molecule) = 2			
Cathode radius ‘b’ (cm) = 2.5	The fitted model parameters are:			
Anode radius ‘a’ (cm) = 0.75	Axial phase mass factor, $f_m = 0.105$			
Anode length ‘z ₀ ’ (cm) = 10	Axial phase current factor, $f_c = 0.7$			
Charging voltage V_0 (kV) = 17	Radial phase mass factor, $f_{mr} = 0.1$			
Fill pressure P_0 (Torr) = 3.0	Radial phase current factor, $f_{cr} = 0.7$			
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	1	45	10	1
Dip 2	0.22	80	40	1
Dip 3	0.52	98	25	1

The Lee Model 6-phase code was used to fit its current waveforms as shown in Figure 3.36. The maximum computed current is 139 kA and exhibits the radial start time at 1.843 μs for pinch duration of 0.035 μs . It has a speed factor of 107 (kA/cm)/Torr^{0.5} and neutron yield of 2.39×10^6 n/shot. The other detailed information is presented in Table 3.29.

Table 3.29 Information obtained from Lee Model 6-phase code configured for KSU-PF2 Plasma Focus machine at 17 kV, 3.0 Torr deuterium gas.

Peak current (kA) = 139
Pinch start current (kA) = 95
Pinch minimum temperature (10^6 K) = 12.88
Pinch maximum temperature (10^6 K) = 13.04
Peak axial speed (cm/ μ s) = 8.7
Peak radial shock speed (cm/ μ s) = 48.4
Peak radial piston speed (cm/ μ s) = 33.3
Final pinch radius $r_{\min}$ (cm) = 0.1
Pinch length $z_{\max}$ (cm) = 1.0
Pinch duration (ns) = 4.5
Peak induced voltage (kV) = 27.9
Neutron yield ($\times 10^6$ n/shot) = 2.39
Energy Inflow into Plasma (%) = 6.2
Speed Factor ((kA/cm)/Torr ^{0.5}) = 107
Current per cm anode radius (kA/cm) = 185

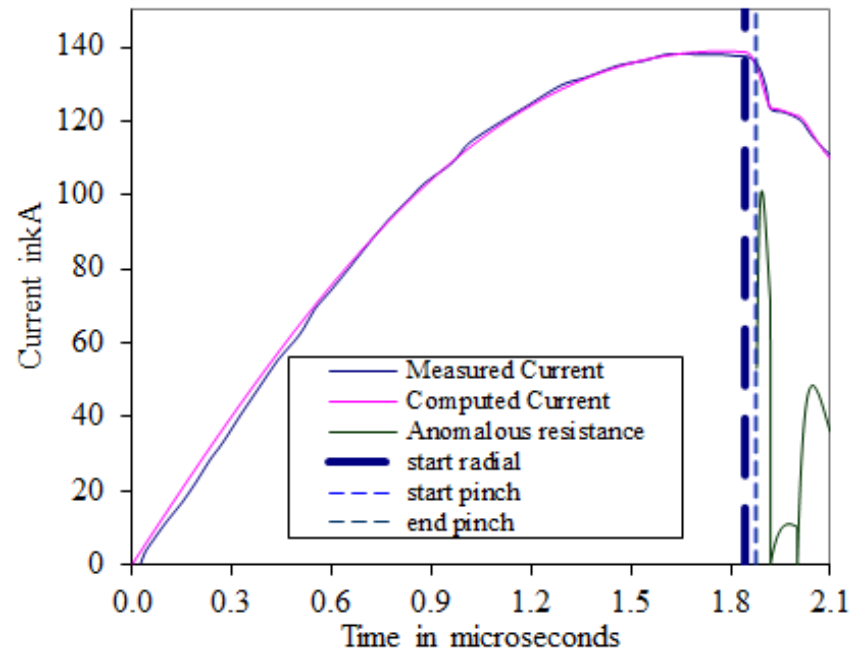


Figure 3.36 The measured current waveform of KSU-PF2 Plasma Focus machine at 17 kV, 3.0 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform was obtained from Figure 5, page 279 of reference [75].

There is no other information to compare from published papers except [47] which shows the variation of neutron yield versus pressure but there was no information about the capacitor charge voltage so comparison cannot be made.

Using the information from our existing current waveform, Lee Model 6-phase code was then configured from 1.0 Torr to 8.0 Torr and the variation of neutron yield with respect to pressure is plotted as shown in Figure 3.37.

From the plot, it is observed that the optimum neutron yield is 2.56×10^6 n/shot and it occurs at 4.0 Torr while the energy input into plasma for this optimum shot was 6.5 %.

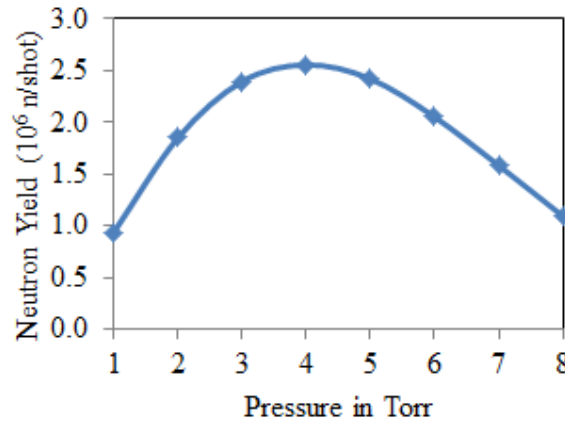


Figure 3.37 Variation of computed neutron yield with respect to pressure operating at 17 kV in deuterium for KSU-PF2 Plasma Focus machine.

Another current waveform for this machine operated in neon at the same voltage of 17 kV was fitted. For this shot the static inductance fitted at 125 nH. The machine parameters are as shown in Table 3.30.

The fitting is shown in Figure 3.38. The maximum computed current was 139 kA and it exhibits the radial start time at 1.943 μ s for pinch duration of 0.043 μ s. It has a speed factor of 175 (kA/cm)/Torr^{0.5} Ne and soft X-ray yield of 4.38 mJ. The other detailed information is presented in Table 3.31. There was no other information to compare from published paper.

Table 3.30 Machines operational and fitting parameters for KSU-PF2 Plasma Focus machine with 125 nH inductance at 1.125 Torr neon gas.

Static inductance L_0 (nH) = 125		The fitted model parameters are: Axial phase mass factor, $f_m = 0.065$ Axial phase current factor, $f_c = 0.7$ Radial phase mass factor, $f_{mr} = 0.1$ Radial phase current factor, $f_{cr} = 0.7$		
Fill pressure P_0 (Torr) = 1.125				
Fill gas (molecular weight) = 20				
Fill gas (atomic number) = 10				
Fill gas (atom) = 1				
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.75	30	10	1
Dip 2	0.08	100	40	1
Dip 3	0.25	100	25	1

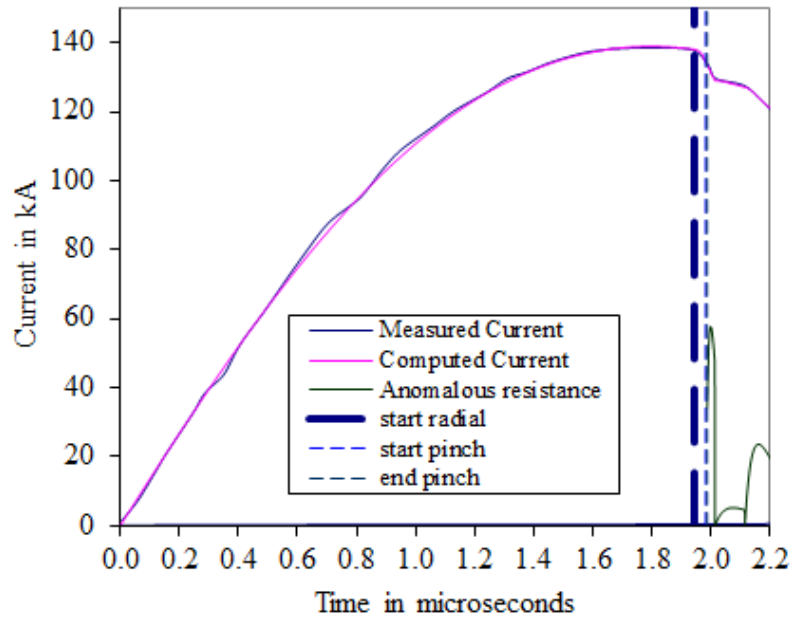


Figure 3.38 The measured current waveform of KSU-PF2 Plasma Focus machine with 125 nH inductance at 1.125 Torr neon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was obtained from Prof Dr Lee Sing research collaboration with KSU.

Table 3.31 Information obtained from Lee Model 6-phase code configured for KSU-PF2 Plasma Focus machine at 17 kV, 1.125 Torr neon gas.

Peak current (kA) = 139	Final pinch radius $r_{\min}$ (cm) = 0.07
Pinch start current (kA) = 94	Pinch length $z_{\max}$ (cm) = 1.0
Pinch minimum temperature (10^6 K) = 12.28	Pinch duration (ns) = 4.6
Pinch maximum temperature (10^6 K) = 13.09	Peak induced voltage (kV) = 32.6
Peak axial speed (cm/ μ s) = 8.1	Soft X-ray yield (mJ) = 4.38
Peak radial shock speed (cm/ μ s) = 40.9	Energy Inflow into Plasma (%) = 6.5
Peak radial piston speed (cm/ μ s) = 29.4	Speed Factor ((kA/cm)/Torr ^{0.5}) = 175
Current per cm anode radius (kA/cm) = 185	

Using the model parameters fitted from the current waveform, Lee Model 6-phase code was then configured from 1.0 Torr to 5.0 Torr and the variation of soft X-ray yield with respect to pressure is plotted as shown in Figure 3.39.

From Figure 3.39, the maximum soft X-ray yield is 0.77 J occurring at 4.0 Torr and the energy input into pinch plasma was 5.9 %.

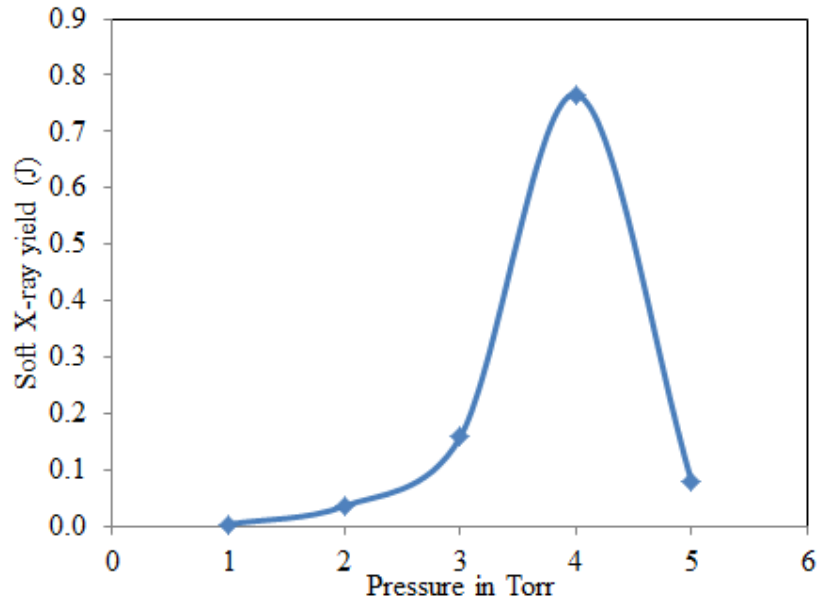


Figure 3.39 Variation of soft X-ray yield with respect to pressure operating at 17 kV in deuterium for KSU-PF2 Plasma Focus machine

3.12 Imperial College Plasma Focus machine (1.9 kJ -Type 1 machine - D gas)

The machine parameters for the Imperial College Plasma Focus machine [111] are shown in Table 3.32.

Table 3.32 The machine, operational and fitting parameters for Imperial College Plasma Focus machine.

Capacitance C_0 (μF) = 2.6	Fill gas (molecular weight) = 4
Static inductance L_0 (nH) = 100	Fill gas (atomic number) = 1
Circuit resistance r_0 (m Ω) = 19.6	Fill gas (molecule) = 2
Cathode radius 'b' (cm) = 2.25	The fitted model parameters are:
Anode radius 'a' (cm) = 1	Axial phase mass factor, $f_m = 0.15$
Anode length 'z ₀ ' (cm) = 5	Axial phase current factor, $f_c = 0.7$
Charging voltage V_0 (kV) = 38	Radial phase mass factor, $f_{mr} = 0.16$
Fill pressure P_0 (Torr) = 2.62	Radial phase current factor, $f_{cr} = 0.7$

Lee Model 5-phase code is used to fit the measured current waveform as shown in Figure 3.40.

The computed and measured waveforms show a good fit when Lee Model 5-phase code has been used.

Table 3.33 Information obtained from Lee Model 5-phase code configured for Imperial College Plasma Focus machine at 38 kV, 2.62 Torr deuterium gas.

Peak current (kA) = 173
Pinch start current (kA) = 113
Pinch minimum temperature (10^6 K) = 7.31
Pinch maximum temperature (10^6 K) = 7.5
Peak axial speed (cm/ μ s) = 9.5
Peak radial shock speed (cm/ μ s) = 37.1
Peak radial piston speed (cm/ μ s) = 25.2
Final pinch radius $r_{\min}$ (cm) = 0.14
Pinch length $z_{\max}$ (cm) = 1.5
Pinch duration (ns) = 7.9
Peak induced voltage (kV) = 24.6
Neutron yield ($\times 10^6$ n/shot) = 6.6
Energy Inflow into Plasma (%) = 5.2
Speed Factor ((kA/cm)/Torr ^{0.5}) = 107
Current per cm anode radius (kA/cm) = 173

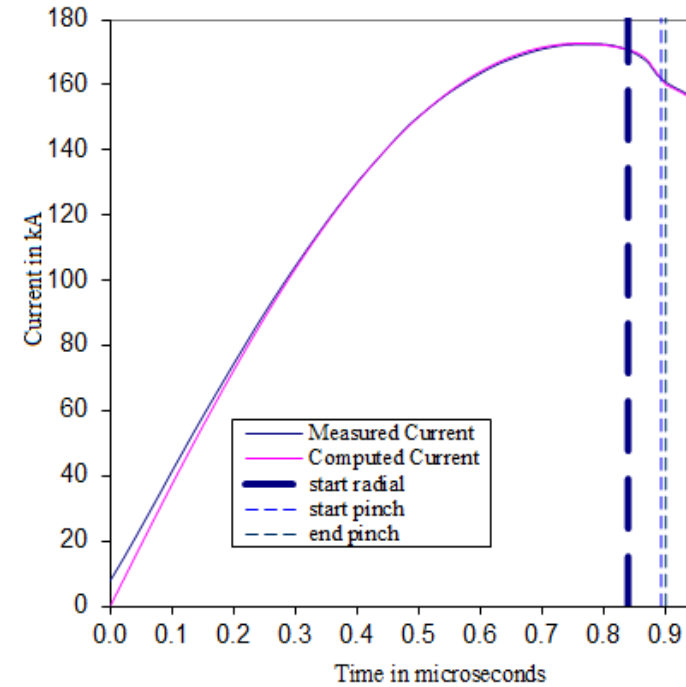


Figure 3.40 The measured current waveform of Imperial College Plasma Focus machine at 38 kV, 2.62 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: The measured current waveform information was extracted from the current derivative waveform in Figure 2, page 3010 of reference [111].

The maximum computed current is 173 kA (measured experimentally at 175 kA) and exhibits the radial start time at 0.839 μ s for pinch duration of 0.061 μ s. It has a speed factor of 107 (kA/cm)/Torr^{0.5} and maximum neutron yield of 6.6×10^6 n/shot (experimental results are $(2.0 \pm 0.5) \times 10^7$ n/shot. The detailed information is presented in Table 3.33.

Lee Model 5-phase code is now configured for 38 kV for 0.5 Torr to 6.5 Torr and the computed neutron yield is compared to the experimental results as shown in Figure 3.41.

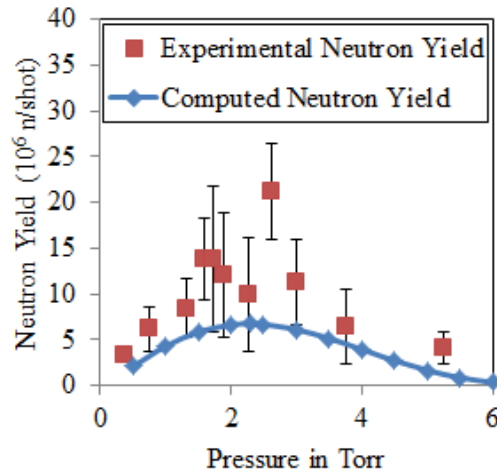


Figure 3.41 Variation of computed and experimental neutron yield with respect to pressure operating at 38 kV in deuterium for Imperial College Plasma Focus machine.

Note: The measured current waveform information was obtained from Figure 3, page 3010 of reference [111].

From Figure 3.41, it can be seen that the experimental and computed result are quite close (factor of 3 in difference). The maximum neutron yield was computed at 2.3 Torr and it gave 6.8×10^6 n/shot whereas the experimental result was $(2 \pm 0.5) \times 10^7$ n/shot at 2.6 Torr. The computed result indicates 5.2 % of energy input into the plasma during the radial phase.

3.13 NX2 Plasma Focus machine (1.7 - 2.2 kJ - Type 1 machine - Ne gas)

The NX2 Plasma Focus machine (NTU/NIE Singapore) is a 3 kJ Mather type high repetition rate machine (up to 16 Hz) whose energy storage is provided by a 28.8 μF capacitor bank (charged by ALE Systems model 802 high voltage capacitor chargers) coupled to the plasma focus electrodes through four rail-gap switches. The machine parameters showing 3 different anode lengths, charging voltages and operating pressures; and model parameters fitted from the measured current waveforms are given in Table 3.34 [7, 8]. These represent 3 different cases (C1, C2 and C3).

Table 3.34 The machine, operational and fitting parameters for NX2 Plasma Focus machine with 3 different anode lengths, charging voltages, operating pressures and model parameters

Case number	C 1	C 2	C 3
Capacitance C_0 (μF)	28.8	28.8	28.8
Static inductance L_0 (nH)	15	15	15
Circuit resistance r_0 (m Ω)	2.2	2.2	2.2
Cathode radius 'b' (cm)	4.1	4.1	4.1
Anode radius 'a' (cm)	1.9	1.9	1.9
Anode length 'z ₀ ' (cm)	5	7	4
Charging voltage V_0 (kV)	11	11.5	11.5
Fill pressure P_0 (Torr)	2.6	1.5	6.0
Fill gas (molecular weight)	20	20	20
Fill gas (atomic number)	10	10	10
Fill gas (atom)	1	1	1
The fitted model parameters are:			
Axial phase mass factor, f_m	0.098	0.17	0.05
Axial phase current factor, f_c	0.7	0.7	0.7
Radial phase mass factor, f_{mr}	0.14	0.1	0.1
Radial phase current factor, f_{cr}	0.69	0.7	0.8

This machine has low inductance and is treated as a Type 1 machine. Lee Model 5-phase code was configured with the machine parameters for the NX2, the information for the 3 shots were obtained and placed in Table 3.35.

Table 3.35 Information obtained from Lee Model 5-phase code configured for NX2 Plasma Focus machine.

Case Number	C1	C2	C3
Peak current (kA)	369	385	390
Pinch start current (kA)	158	137	201
Pinch minimum temperature (10^6 K)	1.61	3.07	1.56
Pinch maximum temperature (10^6 K)	2.05	3.72	2.02
Peak axial speed (cm/ μ s)	6.2	6.2	6.0
Peak radial shock speed (cm/ μ s)	21.6	26.2	21.4
Peak radial piston speed (cm/ μ s)	14.9	17.7	14.9
Final pinch radius $r_{\min}$ (cm)	0.16	0.26	0.16
Pinch length $z_{\max}$ (cm)	2.8	2.7	2.8
Pinch duration (ns)	28.9	23.3	29.4
Peak induced voltage (kV)	22.3	17.1	28.6
Soft X-ray yield (J)	24.5	1.22	42.6
Energy Inflow into Plasma (%)	29.1	26.5	35
Speed Factor ((kA/cm)/Torr ^{0.5} Ne)	121	166	84
Current per cm anode radius (kA/cm)	194	203	205

We now discuss each shot (case) separately.

3.13.1 Case 1 (C1)

For this case the anode length was 5 cm and operation was at 11 kV, 2.6 Torr neon gas, Lee Model 5-phase code was used to fit the measured current waveform.

The computed and the measure current waveforms are shown in Figure 3.43. From this fitting the computed model parameters were obtained as shown in Table 3.35.

Using these fitted model parameters, Lee Model 5-phase code was then configured for pressures from 1.0 Torr to 4.0 Torr and the computed soft X-ray yield versus pressure were compared to the measured values published in [8] as shown in Figure 3.42.

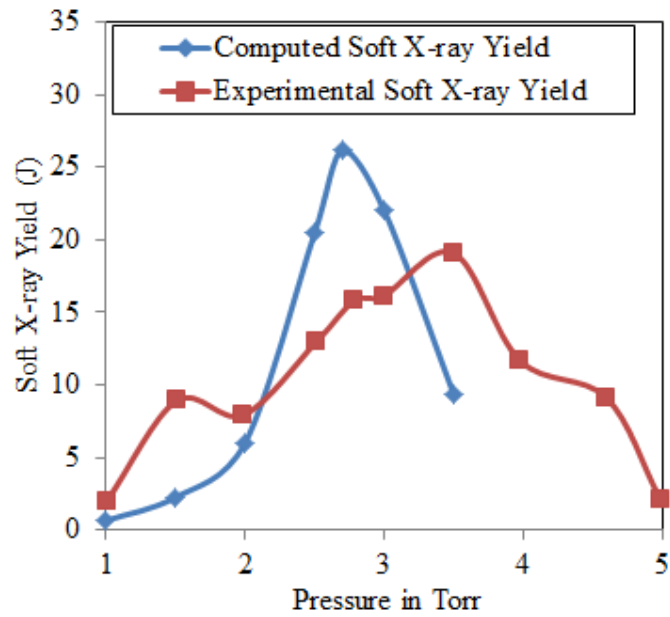


Figure 3.42 Variation of computed and experimental soft X-ray yield with respect to pressure operating with an anode length of 5cm, 11 kV in neon for NX2 Plasma Focus machine (C1).

Note: The experimental soft X-ray yield is obtained from Figure 2, page 023309-5 of reference [8].

Figure 3.42 shows reasonable agreement in the results of the computed and experimentally measured curves. The peak soft X-ray yield computed is 26.2 J at 2.7 Torr compared to 19.1 J obtained experimentally at 3.5 Torr [8]. The computed curve falls off more sharply on both sides of the optimum pressure.

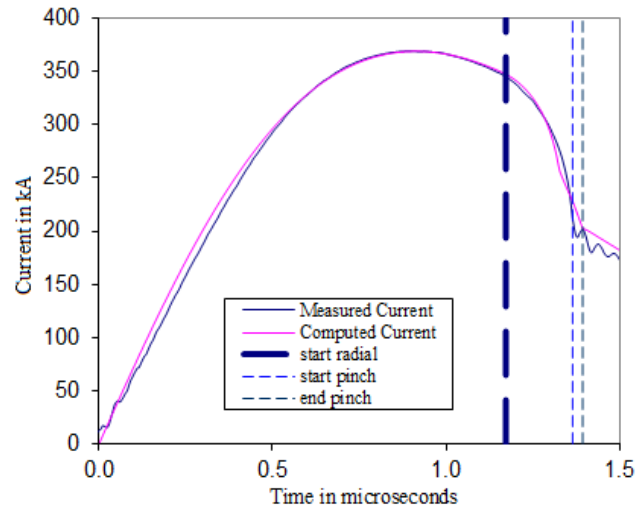


Figure 3.43 Measured current waveform of NX2 at 11 kV, 2.6 Torr neon gas and anode length of 5 cm compared with Lee Model 5-phase code computed current waveform (C1).

Note: The measured current waveform was downloaded from <http://www.plasmafocus.net> [57].

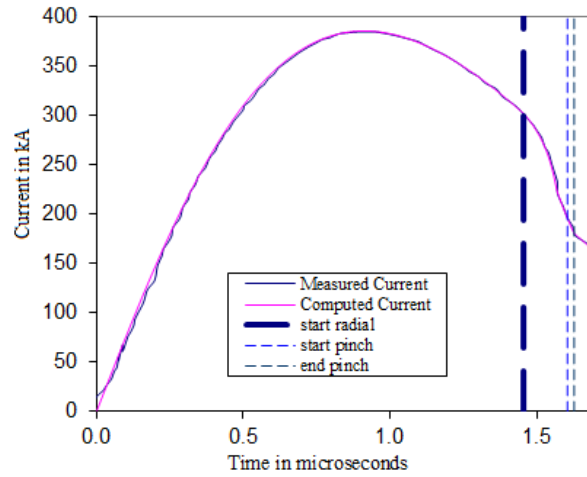


Figure 3.44 Measured current waveform of NX2 at 11.5 kV, 1.5 Torr neon gas and anode length of 7 cm compared with Lee Model 5-phase code computed current waveform (C2).

Note: The measured current waveform was obtained from Figure 6.4(c), page 164, of Zhang Guixin thesis [7].

3.13.2 Case 2 (C2)

In case 2 (C2), the Lee Model 5-phase code was used to fit the measured current waveform of the NX2 machine using anode of 7 cm, 11.5 kV and pressure of 1.5 Torr Ne as shown in Figure 3.44. The model parameters for this configuration of the NX2 are obtained and recorded in Table 3.35, middle column.

Lee Model 5-phase code was then configured for pressures from 0.5 Torr to 2.3 Torr and the computed soft X-ray yield curve was then compared to the measured curve given in Zhang Guixin's PhD thesis [7].

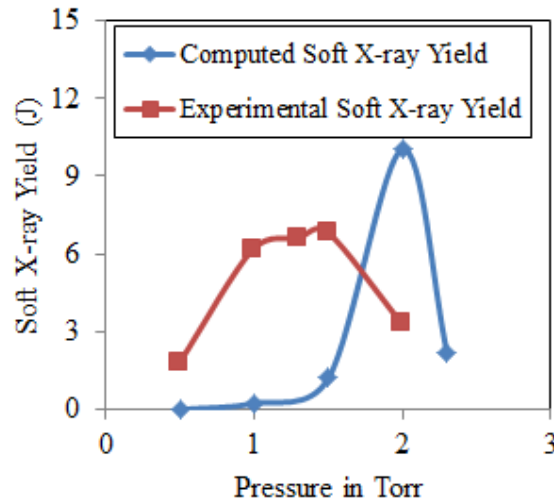


Figure 3.45 Variation of computed and experimental soft X-ray yield with respect to pressure operating with an anode length of 7 cm, 11.5 kV in neon for NX2 Plasma Focus machine (C2).

Note: The experimental result was obtained from Figure 6.67, page 195 of Zhang Guixin PhD thesis [7].

Figure 3.45 shows reasonable agreement in computed and measured peak soft X-ray yields (agrees to better than factor of 1.5) of respectively 7 J at 1.5 Torr compared to 10 J obtained experimentally at 2.0 Torr [7]. The computed curve falls off more sharply on both sides of the optimum pressure. We note that for this series the anode length is 7 cm and the peak soft X-ray yields of both the computed and the

measured curves are about half those of the corresponding values of Case 1 with anode length of 5 cm.

3.13.3 Case 3 (C3)

In case 3 (C3), the anode length was reduced to 4 cm. The Lee Model 5-phase code was used to fit the measured current waveform for this shot at 11.5 kV, 6.0 Torr neon.

Lee Model 5-phase code was then configured to operate from 1.0 Torr to 10.0 Torr and the soft X-ray yield is compared with the published results [7].

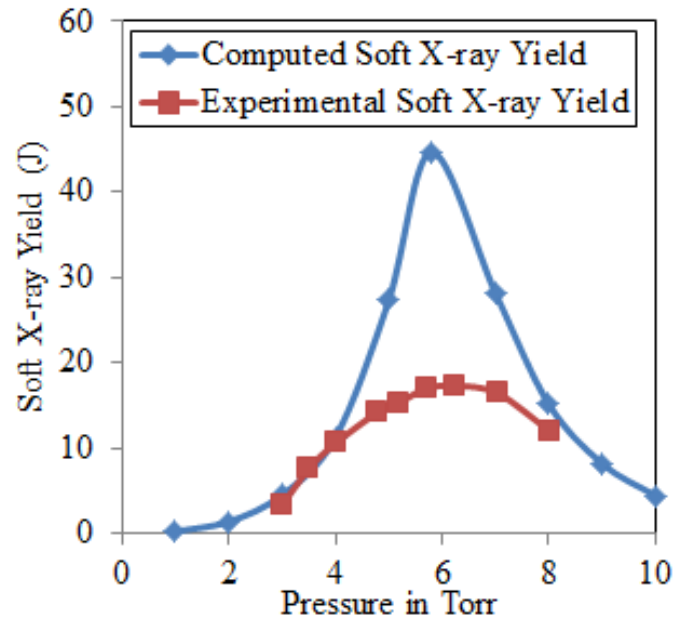


Figure 3.46 Variation of computed and experimental soft X-ray yield with respect to pressure operating with an anode length of 4 cm, 11.5 kV in neon for NX2 Plasma Focus machine (C3).

Note: The experimental results were obtained from Figure 6.69, page 196 Zhang Guixin's thesis [7].

Figure 3.46 shows that the agreement between computed and measured soft X-ray yields for the peak values is less satisfactory in this case than in the other 2 cases. The peak values of yield are: 44.7 J at 5.8 Torr obtained in simulation and 18

J obtained experimentally at 6.2 Torr [7]. The computed curve falls off more sharply on both sides of the optimum pressure. The values of pressure agree better in this case at the wings of the yield versus pressure.

As concluding remarks of the NX2 machine based on the two results obtained from Zhang Guixin thesis (C2 and C3), it can be noted that with the shorter electrode, it is not possible to operate the focus at a higher velocity as the focus would occur too early before the current peaks such that not enough of the capacitor bank energy has been converted to the magnetic field driving the plasma. However, if the anode is made too long and the velocity too high, several factors may then cause the X-ray yield to be reduced such as the plasma focus may not work well due to a possible decoupling of the magnetic piston from the plasma sheath at the end of the axial phase.

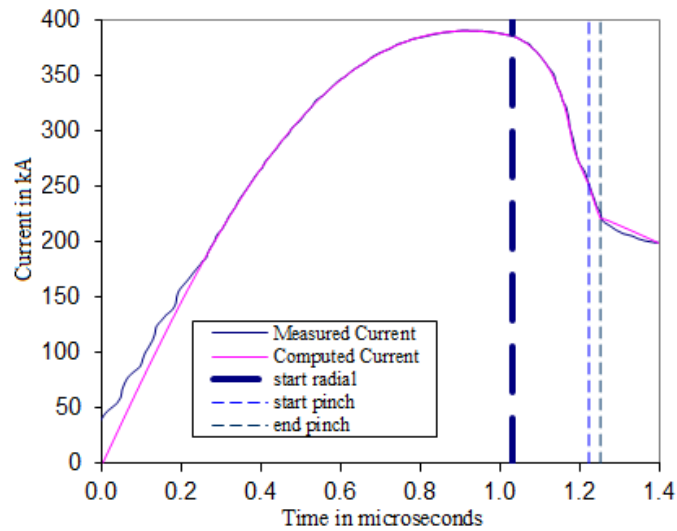


Figure 3.47 Measured current waveform of NX2 at 11.5 kV, 6.0 Torr neon gas compared with Lee Model 5-phase code computed current waveform (C3).

Note: The measured current waveform was obtained from Figure 6.4(a), page 164, of Zhang Guixin thesis [7].

3.13.4 NX2-T Plasma Focus machine (Taper type - Type 1 machine - D gas)

The NX2 plasma focus machine has also been fitted with a tapered anode. This machine is called NX2-T because of its tapered anode and its machine parameters are as stated in Table 3.36.

The computed and measured waveforms are compared and it shows a good fit until the start of the radial phase when Lee Model 5-phase code is used as shown in Figure 3.48.

The maximum computed current obtained is 355 kA and the waveform exhibits the radial start time at 1.067 μ s for pinch duration of 0.117 μ s. The speed factor was 70 (kA/cm)/Torr^{0.5} and the neutron yield obtained was 7.43×10^7 n/shot. The other detailed information is presented in Table 3.37.

Lee Model 5-phase code was then configured for pressures from 1.0 Torr to 19.0 Torr and the computed neutron yield curve was compared to the measured values as published in the reference [9] as shown in Figure 3.49.

From the comparison made, the computed values shows reasonable agreement to the experimentally measured values as the maximum neutron yield computed is 7.7×10^7 n/shot at 13.0 Torr whereas the maximum yield obtained from the experimental result has an average value of $(1 \pm 0.25) \times 10^8$ n/shot at 10.5 Torr. The experimental data [9] is only until 10.5 Torr (14 mbar) [9]. It should also be noted that the experimental values rise at a gentler gradient compared to a steep rise in the experimental results.

Table 3.36 The machine, operation and fitting parameters for NX2-T Plasma Focus machine

Capacitance C_0 (μF) = 28.5
Static inductance L_0 (nH) = 19.8
Circuit resistance r_0 ($\text{m}\Omega$) = 3.3
Cathode radius 'b' (cm) = 3.8
Anode radius 'a' (cm) = 1.55
Anode length 'z ₀ ' (cm) = 4
Taper starts at (cm) = 1.5
Final tapered radius (cm) = 1.15
Charging voltage V_0 (kV) = 12.5
Fill pressure P_0 (Torr) = 10.6
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.137$
Axial phase current factor, $f_c = 0.66$
Radial phase mass factor, $f_{mr} = 0.145$
Radial phase current factor, $f_{cr} = 0.65$

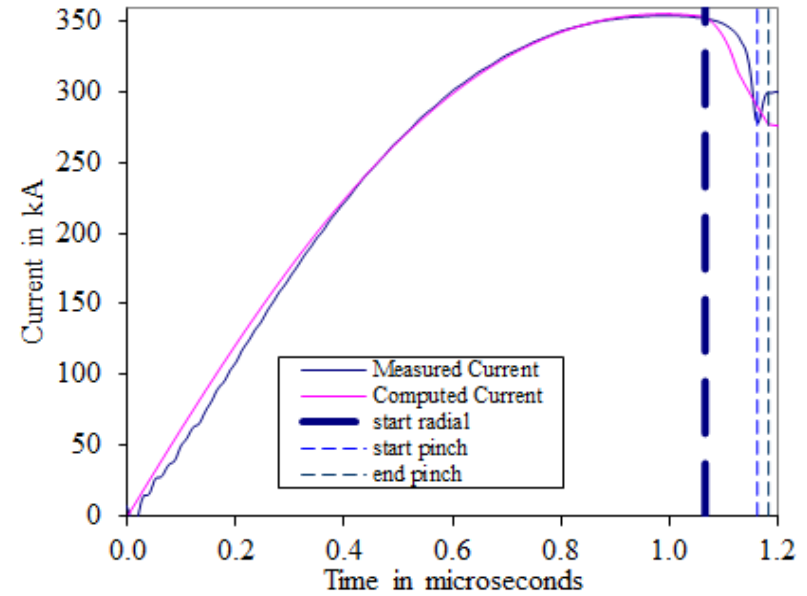


Figure 3.48 Measured current waveform of NX2-T at 12.5 kV, 10.6 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: The digitalized measured current waveform was downloaded from <http://www.intimal.edu.my/school/fas/UFLF/> [59].

Table 3.37 Information obtained from Lee Model 5-phase code configured for NX2-T Plasma Focus machine at 12.5 kV, 10.6 Torr deuterium gas.

Peak current (kA) = 355
Pinch start current (kA) = 189
Pinch minimum temperature (10^6 K) = 3.90
Pinch maximum temperature (10^6 K) = 4.27
Peak axial speed (cm/ μ s) = 6.2
Peak radial shock speed (cm/ μ s) = 22.7
Peak radial piston speed (cm/ μ s) = 16.6
Final pinch radius $r_{\min}$ (cm) = 0.23
Pinch length $z_{\max}$ (cm) = 1.9
Pinch duration (ns) = 20.8
Peak induced voltage (kV) = 23.9
Neutron yield ($\times 10^7$ n/shot) = 7.43
Energy Inflow into Plasma (%) = 16.3
Speed Factor ((kA/cm)/Torr ^{0.5}) = 70

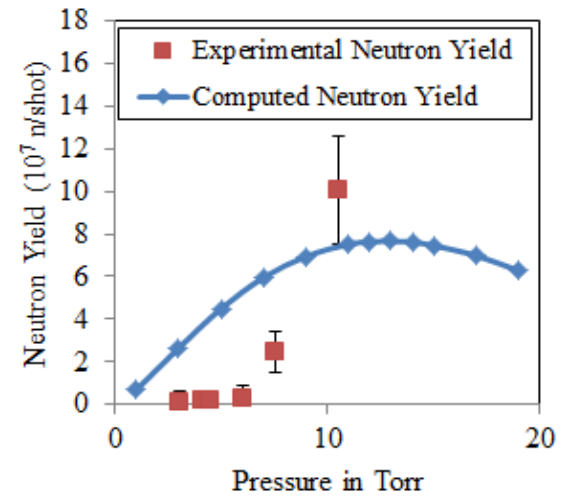


Figure 3.49 Variation of computed and experimental neutron yield with respect to pressure operating at 12.5 kV in deuterium for NX2-T Plasma Focus machine.

Note: The experimental result was obtained from Figure 6, page 053301-3 of reference [9].

3.14 Paco Plasma Focus machine (1.9 kJ - Type 1 machine - D gas)

The machine parameters for the Paco Plasma Focus from Argentina are shown in Table 3.38. Its static inductance L_0 is 55 nH. The Lee Model 5-phase code was used to fit the measured current waveform.

The maximum computed current is 233 kA at pressure of 1.35 Torr (compared to the published maximum current of ≈ 250 kA for filling pressure ranging from 0.8 mbar (0.6 Torr) to 2.3 mbar (1.73 Torr)) and exhibits the radial start time at 0.857 μ s for pinch duration of 0.158 μ s. It has a speed factor of 100 (kA/cm)/Torr^{0.5} and neutron yield of 1.22×10^7 n/shot. The other detailed information is presented in Table 3.39. It should be noted that the published results stated that for the range of 0.6 Torr to 1.73 Torr, the average neutron yield was 2×10^8 n/shot and that the parasitic inductance was 47 nH [113] while our fitted inductance is 55 nH.

Using the information above, Lee Model 5-phase code was then configured for 0.5 Torr to 2.3 Torr and the time when the current dip occurs is compared to the published experimental results.

From Figure 3.51, it can be observed that the experimental time to pinch is slightly longer than the computed results for low pressure but at higher pressure, the current dip time shows good agreement.

Table 3.38 The machine, operation and fitting parameters for Paco Plasma Focus machine.

Capacitance C_0 (μF) = 4
Static inductance L_0 (nH) = 55
Circuit resistance r_0 (m Ω) = 10
Cathode radius 'b' (cm) = 5.5
Anode radius 'a' (cm) = 2
Anode length 'z ₀ ' (cm) = 4
Charging voltage V_0 (kV) = 31
Fill pressure P_0 (Torr) = 1.35
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.19$
Axial phase current factor, $f_c = 0.7$
Radial phase mass factor, $f_{mr} = 0.2$
Radial phase current factor, $f_{cr} = 0.75$

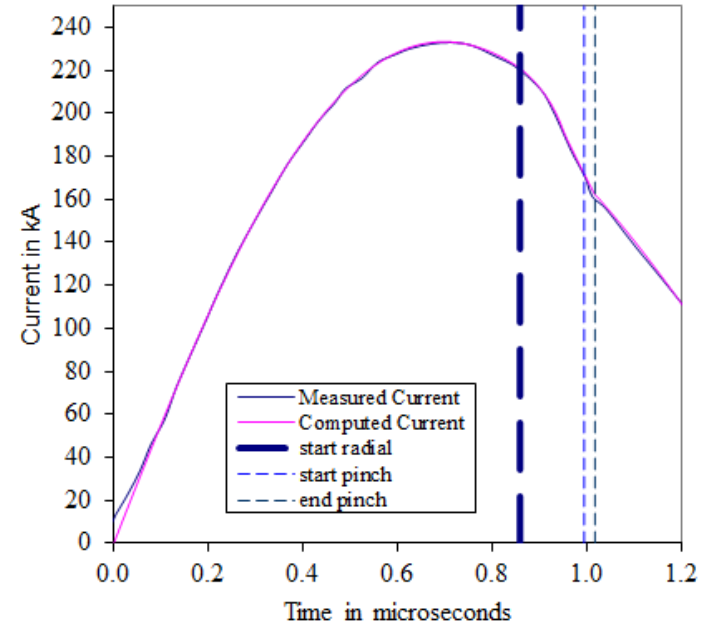


Figure 3.50 Measured current waveform of Paco Plasma Focus machine at 31 kV, 1.35 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: The measured current waveform was extracted from the di/dt waveform of Figure 2, page 18 of reference [113] while the time information was extracted from Figure 3, page 18 of reference [113].

Table 3.39 Information obtained from Lee Model 5-phase code configured for Paco Plasma Focus machine at 31 kV, 1.35 Torr deuterium gas.

Peak current (kA) = 233
Pinch start current (kA) = 129
Pinch minimum temperature (10^6 K) = 3.34
Pinch maximum temperature (10^6 K) = 3.76
Peak axial speed (cm/ μ s) = 7
Peak radial shock speed (cm/ μ s) = 28.9
Peak radial piston speed (cm/ μ s) = 20.3
Final pinch radius $r_{\min}$ (cm) = 0.33
Pinch length $z_{\max}$ (cm) = 3
Pinch duration (ns) = 23
Peak induced voltage (kV) = 20.1
Neutron yield ($\times 10^7$ n/shot) = 1.22
Energy Inflow into Plasma (%) = 13
Speed Factor ((kA/cm)/Torr ^{0.5}) = 100
Current per cm anode radius (kA/cm) = 117

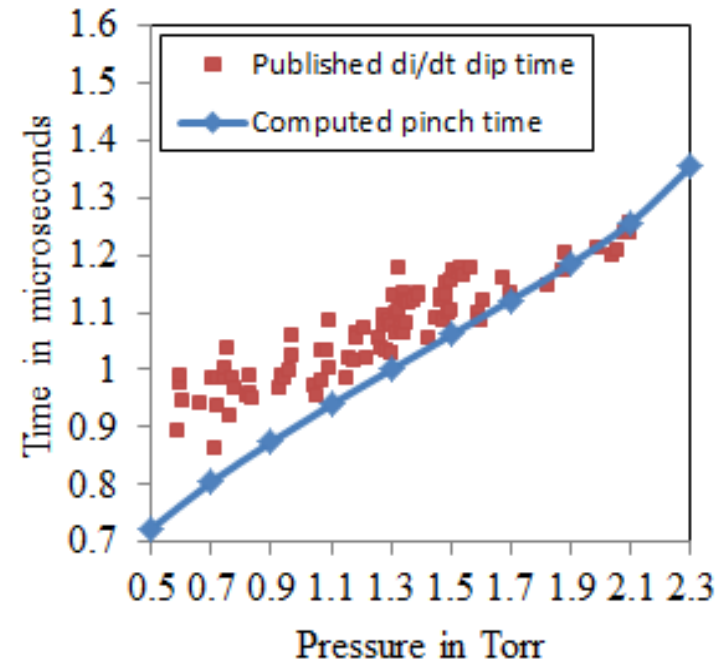


Figure 3.51 The computed time to pinch compared with the experimental results for Paco Plasma focus machine at 31 kV.

Note: The experimental dip time information was obtained from Figure 3, page 18 of reference [113].

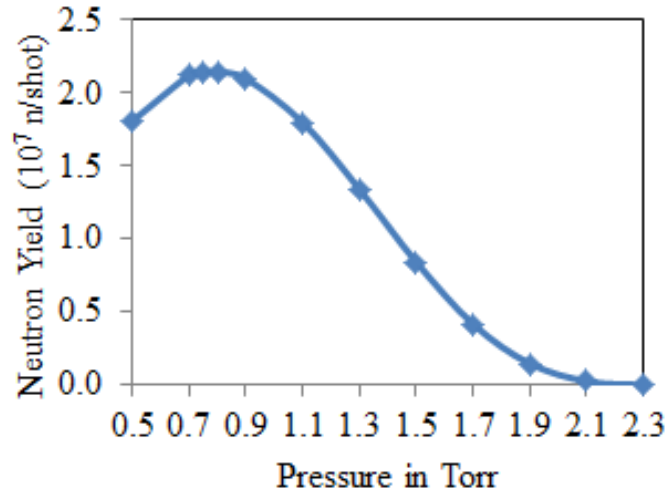


Figure 3.52 Variation of computed neutron yield with respect to pressure operating at 31 kV in deuterium for Paco Plasma Focus machine.

From Figure 3.52, it can be observed that the optimum neutron yield was computed at 2.2×10^7 n/shot at 0.75 Torr while the energy input into plasma is at 14.8 %. The published result states that the optimum yield ($\sim 2 \times 10^8$ n/shot) occurs at about 1.3 Torr (1.7 to 1.8 mbar) [113]. The difference between the optimum computed and experimental neutron yields is a factor of 9. This is a much larger difference than typical.

In the case of PACO we should make a comparison with FNII which has similar static inductance value as PACO. For PACO, the difference between computed and measured neutron yield is larger than typically observed in comparing the computed neutron yield from the Lee code to the measured values in other machines such as FNII. A paper being prepared for publication [114] notes that "...in a detailed study of the neutron yield of PACO, Castillo et al [115] specifically pointed out in their conclusion that the neutron yield of PACO varied widely from shot-to-shot when compared with the FN-II, quoting representative figures for PACO such as (363 ± 214) (in representative units). The same paper measured the total yield of PACO more exhaustively than the typical practice of single-station counting by including both isotropic and anisotropic components. They obtained 3 values of total neutron yield of $(1.26 \pm 1.04) \times 10^8$ n/shot, $(1.62 \pm$

$1.01) \times 10^8$ n/shot and $(0.45 \pm 0.43) \times 10^8$ n/shot. Noting their range of variation it is not surprising that the computed value of neutron yield differs so atypically (1/9) from the claimed experimental value of PACO of 2×10^8 n/shot, which in any case might need to be reduced to a value closer to 10^8 in view of their exhaustive 2003 study [115]; given the 3 neutron yield values they obtained for PACO as shown above.....” The exhaustive studies of Castillo et al has thus isolated the PACO machine as atypically inconsistent in neutron yield; and points to the distinct possibility that the average PACO yield needs to be adjusted to less than 1×10^8 ; which would make our computed yield closer to 1/3 of that adjusted yield.

3.15 University of Tehran Plasma Focus machine (2 kJ - Type 2 machine - Ar gas)

This Mather's type machine from Iran was charged to 19 kV and operated at a fixed pressure of 0.375 Torr argon. We obtained the data of three shots (St1, St2, St3) from R A Behbahani [116] and analyzed these. The machine parameters are as shown in Tables 3.40, 3.41 and 3.42.

Table 3.40 The machine and operational parameters for University of Tehran Plasma Focus machine.

Capacitance C_0 (μF) = 11	Charging voltage V_0 (kV) = 19
Static inductance L_0 (nH) = 126	Fill pressure P_0 (Torr) = 0.375
Circuit resistance r_0 (m Ω) = 10.7	Fill gas (molecular weight) = 40
Cathode radius 'b' (cm) = 3	Fill gas (atomic number) = 18
Anode radius 'a' (cm) = 1	Fill gas (atom) = 1
Anode length 'z ₀ ' (cm) = 15	

Since the static inductance is larger than 100 nH, it is a Type 2 machine. Thus Lee Model 6-phase code is used to fit the measured current waveform.

Table 3.41 Fitted current and mass factors of University of Tehran Plasma Focus machine working at 0.375 Torr argon (st1, st2 and st3).

Shot Number	St1	St2	St3
Axial phase mass factor, f_m	0.031	0.024	0.018
Axial phase current factor, f_c	0.7	0.7	0.7
Radial phase mass factor, f_{mr}	0.08	0.08	0.08
Radial phase current factor, f_{cr}	0.7	0.7	0.7

Table 3.42 Fitted anomalous resistance parameters of University of Tehran Plasma Focus machine working at 0.375 Torr argon (st1, st2 and st3).

	$R_0(\Omega)$			Characteristic of fall time (ns)			Characteristic of rise time (ns)			End fraction time		
	1	0.25	0.9	80	70	120	10	10	10	0.5	1	0.3
Dip 1	1	0.25	0.9	80	70	120	10	10	10	0.5	1	0.3
Dip 2	0.12	0.1	0.09	700	400	800	10	10	10	1	1	1
Dip 3	0.21	0.15	0.1	600	400	700	10	10	10	1	1	1
Shot	St1	St2	St3	St1	St2	St3	St1	St2	St3	St1	St2	St3

There were three shots fired at the same pressure and charging voltage. The results of the current fittings are shown in Figure 3.53, Figure 3.54 and Figure 3.55.

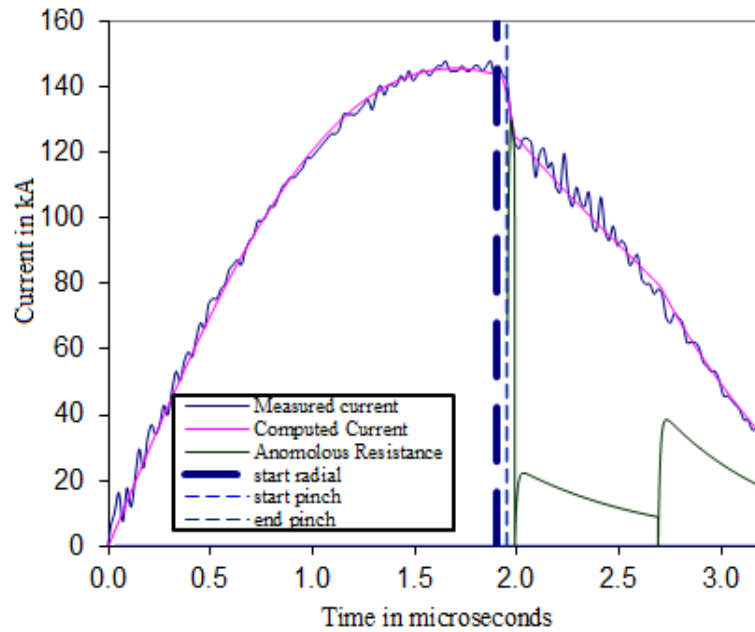


Figure 3.53 Measured current waveform of University of Tehran Plasma Focus machine at 19 kV, 0.375 Torr argon gas compared with Lee Model 6-phase code computed current waveform (St1).

Note: The measured current waveform information was provided by *R A Behbahani* from University of Tehran, Iran on 24th July 2011.

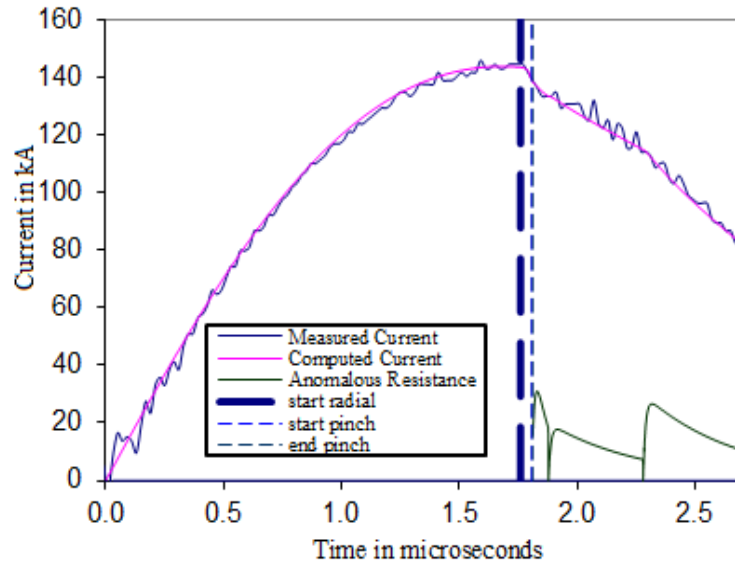


Figure 3.54 Measured current waveform of University of Tehran Plasma Focus machine at 19 kV, 0.375 Torr argon gas compared with Lee Model 6-phase code computed current waveform (St2).

Note: The measured current waveform information was provided by *R A Behbahani* from University of Tehran, Iran on 24th July 2011.

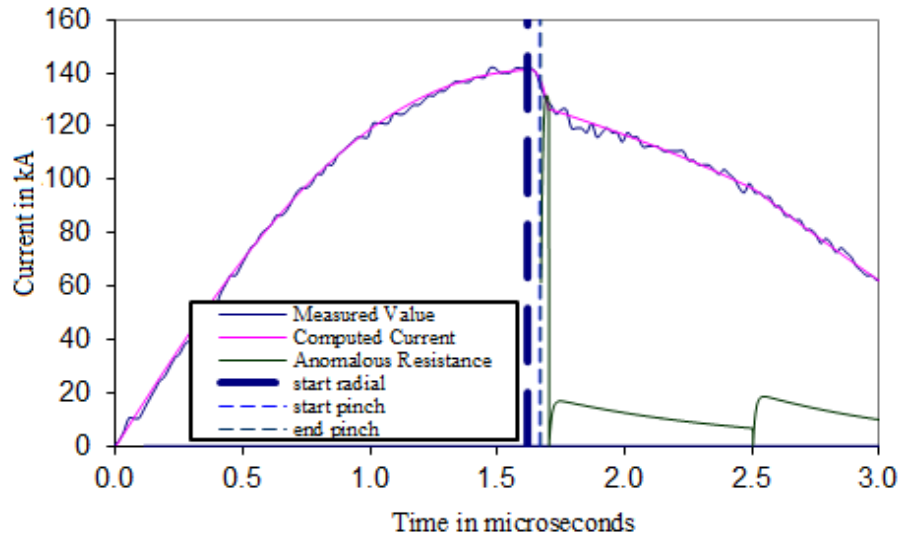


Figure 3.55 Measured current waveform of University of Tehran Plasma Focus machine at 19 kV, 0.375 Torr argon gas compared with Lee Model 6-phase code computed current waveform (St3).

Note: The measured current waveform information was provided by *R A Behbahani* from University of Tehran, Iran on 24th July 2011.

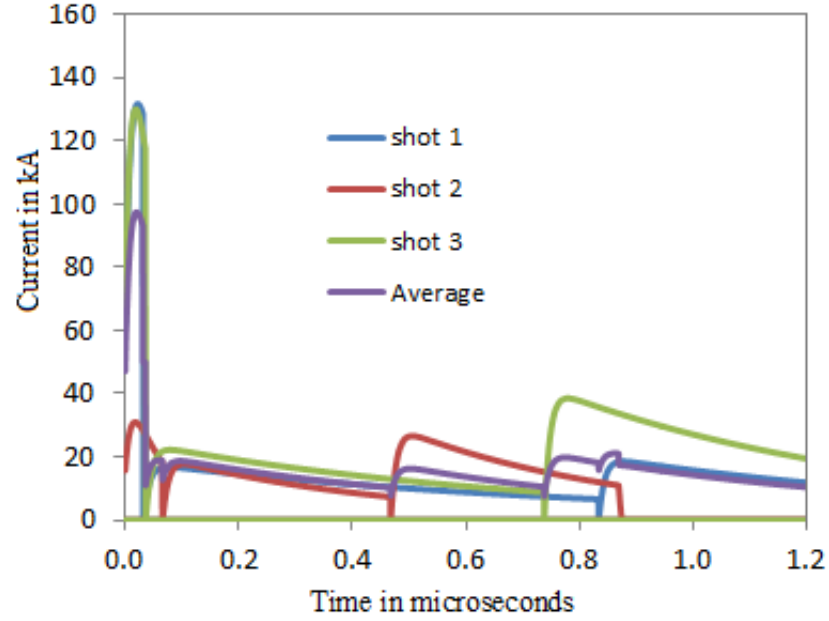


Figure 3.56 Anomalous resistance of all three shots of University of Tehran Plasma Focus machine at 19 kV, 0.375 Torr argon gas and averaged values for the three shots.

When the anomalous resistances for the three shots are drawn on the same graph (starting from the end of the pinch) as shown in Figure 3.56, it was observed that all the first dips have the largest R_0 value, followed by a lower R_0 value for the second dip. The R_0 value increases in the third dip. Thus it can be concluded that at the same pressure and voltage the anomalous resistance exhibits the same time characteristics even though the resistance values are different from shot to shot.

Lee Model 6-phase code obtained various parameters of the plasma as stated in Table 3.43.

Table 3.43 Comparison of computed plasma parameters for 3 different shots fired at 19 kV, 0.375 Torr argon gas in University of Tehran Plasma Focus machine.

Shots at 0.375 Torr argon	St1	St2	St3
Peak current (kA)	146	144	141
Pinch start current (kA)	97	97	96
Pinch minimum temperature (10^6 K)	17.78	17.87	17.49
Pinch maximum temperature (10^6 K)	18.13	18.21	17.79
Peak axial speed (cm/ μ s)	12.3	13.6	15.2
Peak radial shock speed (cm/ μ s)	50.1	50.3	50.0
Peak radial piston speed (cm/ μ s)	32.2	32.2	32
Final pinch radius $r_{\min}$ (cm)	0.10	0.10	0.10
Pinch length $z_{\max}$ (cm)	1.3	1.3	1.3
Pinch duration (ns)	4.8	4.8	4.8
Peak induced voltage (kV)	37.4	38.1	37.7
All-line yield (mJ)	8.07	8.15	8.41
Energy Inflow into Plasma (%)	8.7	8.3	7.7
Speed Factor ((kA/cm)/Torr ^{0.5} Ar)	238	235	231
Current per cm anode radius (kA/cm)	146	144	141

For the 3 waveforms fitted, it was found that the axial phase mass factor decreases(0.031 $\rightarrow$ 0.024 $\rightarrow$ 0.018), while all others factors such as axial phase current factor, f_c , radial phase mass factor f_{mr} and radial phase current factor, f_{cr} are constant. The peak axial speed (V_a) for these shots increase as the fitted axial mass factor decrease (12.3 $\rightarrow$ 13.6 $\rightarrow$ 15.2 cm/ μ s) with an equation of $V_a = 3.18 f_m^{-0.39}$ as shown in Figure 3.57.

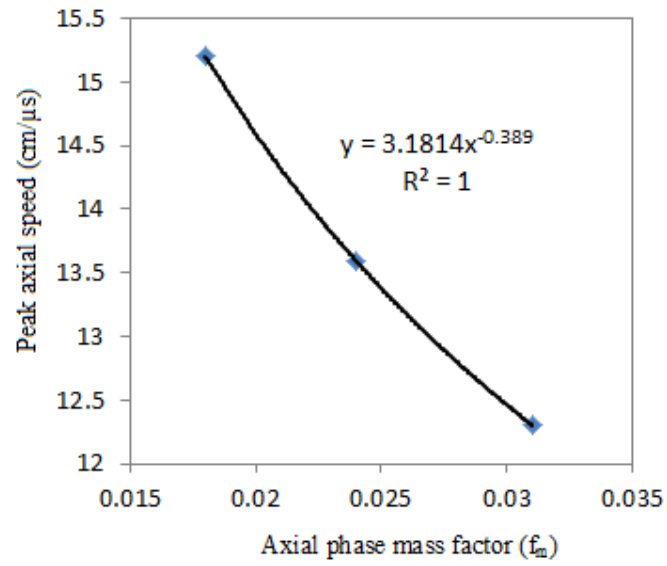


Figure 3.57 The variation of axial phase mass factor (if operational and fitting parameters are constant) and its effect on axial speed on University of Tehran Plasma Focus machine at 19 kV, 0.375 Torr argon gas.

3.16 Montecucolino Plasma Focus machine (2.2 kJ - Type 1 machine - D gas)

The machine parameter for the Montecucolino Plasma Focus machine [117] of Italy is shown in Table 3.44.

Table 3.44 The machine, operational and fitting parameters for Montecucolino Plasma Focus machine.

Capacitance C_0 (μF) = 7.2	Fill gas (molecular weight) = 4
Static inductance L_0 (nH) = 91	Fill gas (atomic number) = 1
Circuit resistance r_0 (m Ω) = 11	Fill gas (molecule) = 2
Cathode radius 'b' (cm) = 2.4	The fitted model parameters are:
Anode radius 'a' (cm) = 1.2	Axial phase mass factor, $f_m = 0.13$
Anode length 'z ₀ ' (cm) = 11.6	Axial phase current factor, $f_c = 0.7$
Charging voltage V_0 (kV) = 25	Radial phase mass factor, $f_{mr} = 0.13$
Fill pressure P_0 (Torr) = 2.0	Radial phase current factor, $f_{cr} = 0.7$

The computed waveform and measured waveform show a good fit when Lee Model 5-phase code was used as shown in Figure 3.58.

The maximum computed current is 192 kA and exhibits the radial start time at 1.478 μs for radial pinch duration of 0.066 μs . It has a speed factor of 113 kA/cm)/Torr^{0.5} and neutron yield of 6.36×10^6 n/shot. The detailed information is presented in Table 3.45.

Table 3.45 Information obtained from Lee Model 5-phase code configured for Montecucolino Plasma Focus machine at 25 kV, 2.0 Torr deuterium gas.

Peak current (kA) = 192
Pinch start current (kA) = 119
Pinch minimum temperature (10^6 K) = 9.03
Pinch maximum temperature (10^6 K) = 9.30
Peak axial speed (cm/ μ s) = 11.8
Peak radial shock speed (cm/ μ s) = 41.6
Peak radial piston speed (cm/ μ s) = 28.8
Final pinch radius $r_{\min}$ (cm) = 0.17
Pinch length $z_{\max}$ (cm) = 1.7
Pinch duration (ns) = 8.5
Peak induced voltage (kV) = 26.1
Neutron yield ($\times 10^6$ n/shot) = 6.36
Energy Inflow into Plasma (%) = 8
Speed Factor ((kA/cm)/Torr ^{0.5}) = 113
Current per cm anode radius (kA/cm) = 160

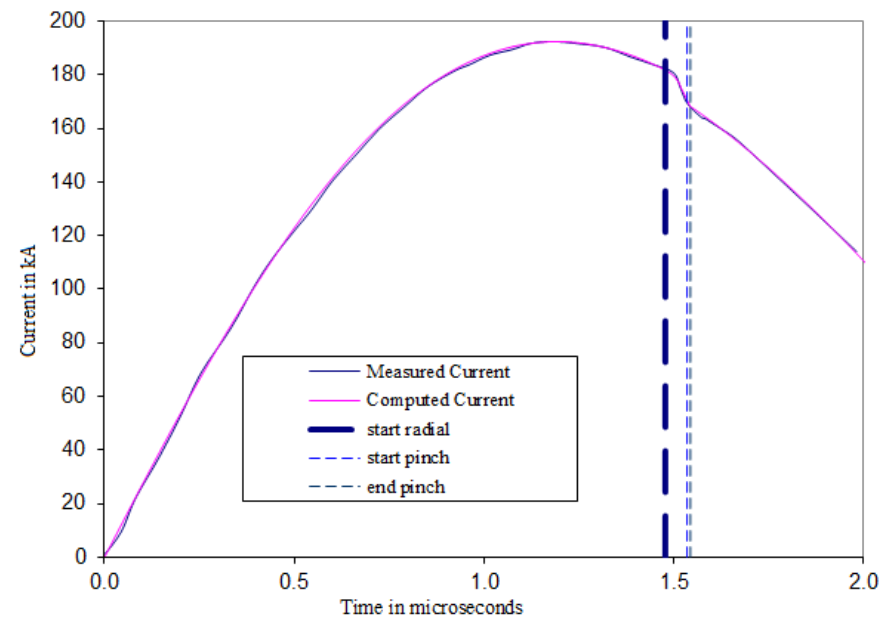


Figure 3.58 The measured current waveform of Montecucolino Plasma Focus machine at 25 kV, 2.0 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: The measured current waveform information was extracted from Figure 4.18, page 68 of reference [117].

We continued with the numerical experiment for varying pressure from 0.5 Torr to 2.5 Torr to obtain the neutron yield versus pressure curve as shown in Figure 3.59.

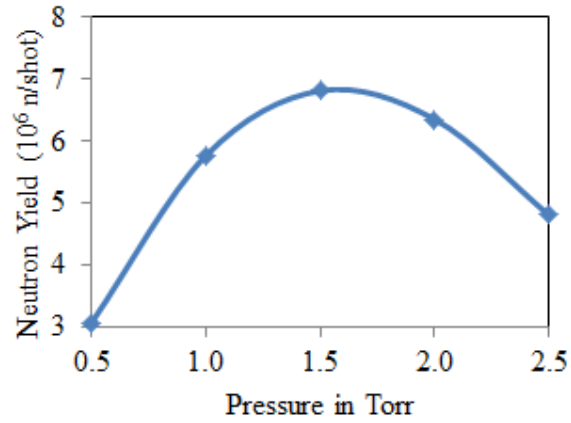


Figure 3.59 Variation of computed neutron yield with respect to pressure operating at 25 kV in deuterium for Montecucolino Plasma Focus machine.

From Figure 3.59, the optimum pressure is found to be 1.5 Torr to produce a neutron yield of 6.8×10^6 n/shot with an energy input into plasma during the radial phase at 8 %.

3.17 DPF 2.2 Plasma Focus machine (2.2 kJ - Type 1 machine showing Type 2 characteristic - D gas)

The DPF 2.2 (Tsinghua University Plasma focus machine from China) is a 2.2 kJ Mather type machine with a cathode which has a squirrel cage cathode configuration (diameter 56 mm) consisting of 8 rods (diameter 14 mm) arranged concentrically around its anode. It has a 30-mm-long alumina insulator. This device is powered by 14 μF capacitor bank that consists of 28, 0.5 μF capacitors. A spark gap is used as a switch [19].

This Mather's type machine was charged to 18 kV at a pressure of 11.0 Torr deuterium gas. Its static inductance is below 76 nH. We first tried to use the 5-phase model to fit; and then found that we had to use the 6-phase to complete the fit. It should be noted that the author has quoted its static inductance of 67 nH but our fitted inductance is 76 nH.

Using the following parameter as shown in Table 3.46, the computed waveform is compared to the measured waveform as shown in Figure 3.60 and the detailed results are shown in Table 3.47.

Lee Model 6-phase code was then configured to work from 5.0 Torr to 19.0 Torr and its result was compared to the published results.

From Figure 3.61, it can be seen that the experimental and computed results for the neutron yield are in agreement. The maximum neutron yield computed at 13.0 Torr is 2.1×10^7 n/shot whereas the experiment measured was $(3.1 \pm 1.5) \times 10^7$ n/shot at 11.25 Torr [19]

Table 3.46 The machine, operational and fitting parameters for DPF2.2 Plasma Focus machine.

Capacitance C_0 (μF) = 14	Charging voltage V_0 (kV) = 18
Static inductance L_0 (nH) = 76	Fill pressure P_0 (Torr) = 11.0
Circuit resistance r_0 (m Ω) = 7	Fill gas (molecular weight) = 4
Cathode radius 'b' (cm) = 2.5	Fill gas (atomic number) = 1
Anode radius 'a' (cm) = 0.8	Fill gas (molecule) = 2
Anode length 'z ₀ ' (cm) = 8.5	

The fitted model parameters are:

Axial phase mass factor, $f_m = 0.06$

Axial phase current factor, $f_c = 0.75$

Radial phase mass factor, $f_{mr} = 0.15$

Radial phase current factor, $f_{cr} = 0.7$

The anomalous resistance parameters are:

	$R_0(\Omega)$	Characteristic of fall time, τ_2 (ns)	Characteristic of rise time, τ_1 (ns)	End fraction time
Dip 1	0.13	100	10	1
Dip 2	0.11	100	10	1
Dip 3	0.09	300	10	1

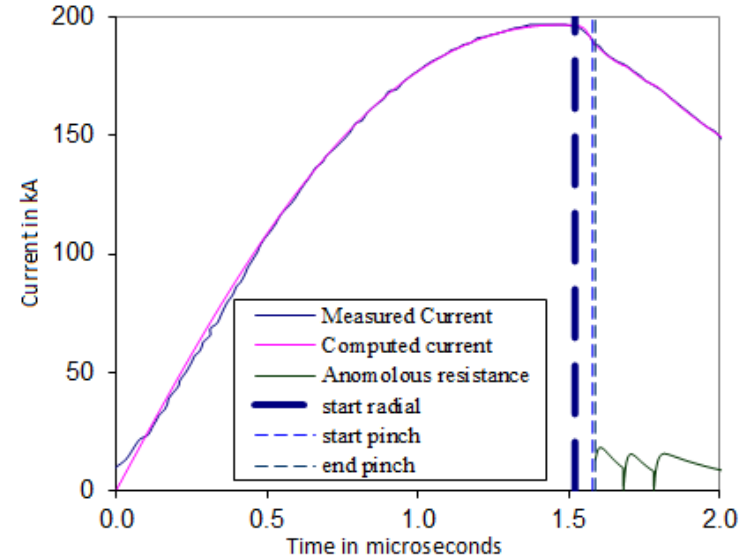


Figure 3.60 Measured current waveform of DPF 2.2 Plasma Focus machine at 18 kV, 11.0 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform was obtained from Figure 2, page 85 of reference [19].

Table 3.47 Information obtained from Lee Model 6-phase code configured for DPF2.2 Plasma Focus machine at 18 kV, 11.0 Torr deuterium gas.

Peak current (kA) = 197
Pinch start current (kA) = 133
Pinch minimum temperature (10^6 K) = 4.02
Pinch maximum temperature (10^6 K) = 4.09
Peak axial speed (cm/ μ s) = 8.9
Peak radial shock speed (cm/ μ s) = 27.3
Peak radial piston speed (cm/ μ s) = 18.6
Final pinch radius $r_{\min}$ (cm) = 0.11
Pinch length $z_{\max}$ (cm) = 1.2
Pinch duration (ns) = 8.5
Peak induced voltage (kV) = 21
Neutron Yield ($\times 10^7$ n/shot) = 2.07
Energy Inflow into Plasma (%) = 9.6
Speed Factor ((kA/cm)/Torr ^{0.5}) = 74
Current per cm anode radius (kA/cm) = 246

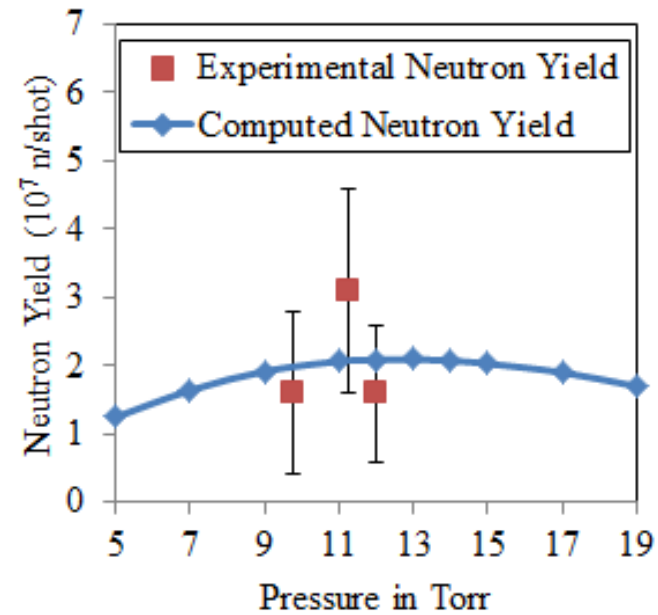


Figure 3.61 Variation of computed and experimental neutron yield with respect to pressure operating at 18 kV in deuterium for DPF2.2 Plasma Focus machine.

Note: The measured neutron yield was obtained from Table 1, page 85 of reference [19].

CHAPTER 4

Results from dense plasma focus machines above 2.5 kJ for deuterium, argon and neon.

4.1 Introduction

In this chapter, we will continue our discussion on Mather-type machines above the energy range 2.5 kJ working in deuterium, argon and neon. We start with the PF12 from Estonia and proceed until we cover the largest plasma machine namely the PF 1000 from Poland. These machines are selected as we have the bank, tube and other relevant operational parameters together with measured current traces. These machines are analysed using the Lee Model 5-phase and 6-phase codes. The plasma dynamics and yields obtained from the code are compared to the experimental values where available.

- 4.2 PF12 Plasma Focus machine (2.6 kJ) [118]
- 4.3 Amirkabir Plasma Focus machine (2.6 kJ) [20]
- 4.4 INTI Plasma Focus machine (2.7 kJ) [13-18]
- 4.5 Syrian Plasma Focus machine (2.8 kJ) [119, 120]
- 4.6 Bora Plasma Focus machine (3.5 kJ) [121]
- 4.7 Hanyang University Plasma Focus machine (4.1 kJ) [122]
- 4.8 GN1 Plasma Focus machine (4.7 kJ) [30-32]
- 4.9 FNII Plasma Focus machine (4.9 kJ) [123]
- 4.10 Assam Plasma Focus machine (5.2 kJ) [124]
- 4.11 NX3 Plasma Focus machine (7.2-9.8 kJ) [10]
- 4.12 BARC Plasma Focus 2 machine (11.5 kJ) [22]
- 4.13 PF143 Plasma Focus machine (20 kJ) [125-127]
- 4.14 DPF78 Plasma Focus machine (31 kJ) [128]
- 4.15 Tamu Plasma Focus machine (126 kJ) [129]
- 4.16 Poseidon Plasma Focus machine (280.8 kJ) [37]
- 4.17 PF1000 Plasma Focus machine (486 kJ) [4]

4.2 PF12 Plasma focus (2.6 kJ - Type 1 machine - D gas)

PF12 is a 20 μF , 80 nH Mather-type plasma focus machine operating at Talinn University, Republic of Estonia.

The machine parameters for this plasma focus machine are compiled in Table 4.1 for a shot in which we were given a measured current waveform by Tõnu Laas [118].

The computed current waveform was fitted to the measured waveform as shown in Figure 4.1. The maximum computed current is 187 kA and exhibits the radial start time at 1.343 μs for pinch duration of 0.066 μs . It has a speed factor of 111 (kA/cm)/Torr^{0.5} and neutron yield of 1.9×10^7 n/shot. The other computed information is presented in Table 4.2.

Using the information obtained from the current waveform, Lee code was configured for PF12 plasma focus machine to work from 2.0 Torr to 16.0 Torr deuterium with the charging voltage of 16 kV to obtain information on neutron yield versus pressure. The resulting neutron yield versus pressure waveform is shown in Figure 4.2.

Figure 4.2 shows that the maximum neutron yield of 3.78×10^7 n/shot and occurs at 11.0 Torr. The energy input into plasma at this instant is 8.2 %.

Table 4.1 The machine, operational and fitting parameters for PF12 Plasma Focus machine.

Capacitance C_0 (μF) = 20
Static inductance L_0 (nH) = 80
Circuit resistance r_0 (m Ω) = 6
Cathode radius 'b' (cm) = 2.7
Anode radius 'a' (cm) = 0.9
Anode length 'z ₀ ' (cm) = 7.2
Charging voltage V_0 (kV) = 16
Fill pressure P_0 (Torr) = 3.5
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.1$
Axial phase current factor, $f_c = 0.7$
Radial phase mass factor, $f_{mr} = 0.26$
Radial phase current factor, $f_{cr} = 0.7$

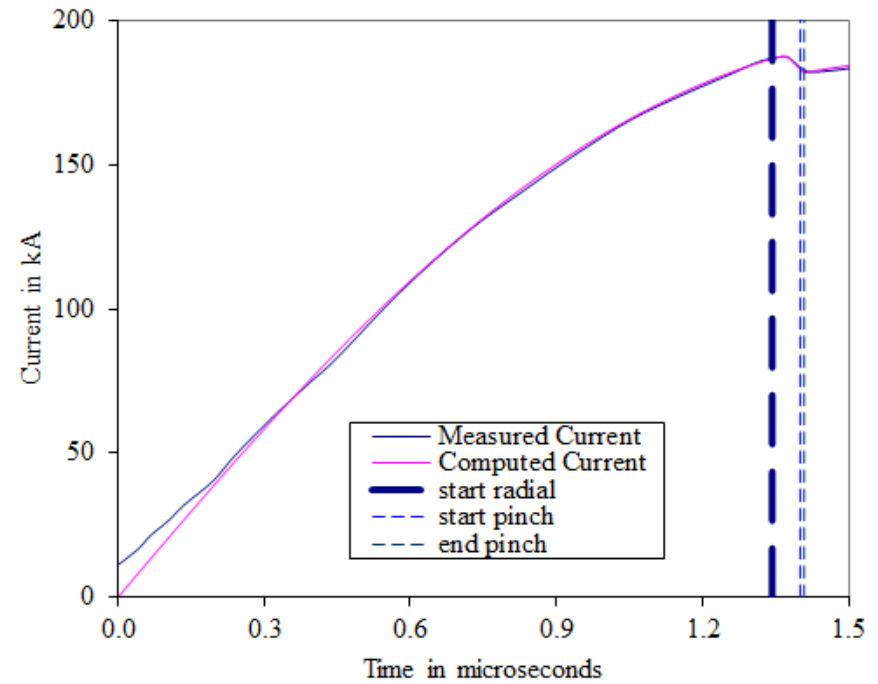


Figure 4.1 The measured current waveform of PF12 at 16 kV, 3.5 Torr deuterium gas compared with Lee Model 5-phase code current waveform.

Note: The measured current waveform was integrated from the information given by Tõnu Laas [118].

Table 4.2 Information obtained from Lee Model 5-phase code configured for PF12 Plasma Focus Machine at 16 kV, 3.5 Torr deuterium gas.

Peak current (kA) = 187
Pinch start current (kA) = 128
Pinch minimum temperature(10^6 K) = 5.37
Pinch maximum temperature(10^6 K) = 5.45
Peak axial speed (cm/ μ s) = 9.4
Peak radial shock speed (cm/ μ s) = 31.3
Peak radial piston speed (cm/ μ s) = 21.3
Final pinch radius $r_{\min}$ (cm) = 0.12
Pinch length $z_{\max}$ (cm) = 1.4
Pinch duration (ns) = 8.3
Peak induced voltage (kV) = 24.4
Neutron yield ($\times 10^7$ n/shot) = 1.9
Energy Inflow into Plasma (%) = 6.1
Speed Factor ((kA/cm)/Torr ^{0.5}) = 111
Current per cm anode radius (kA/cm) = 208

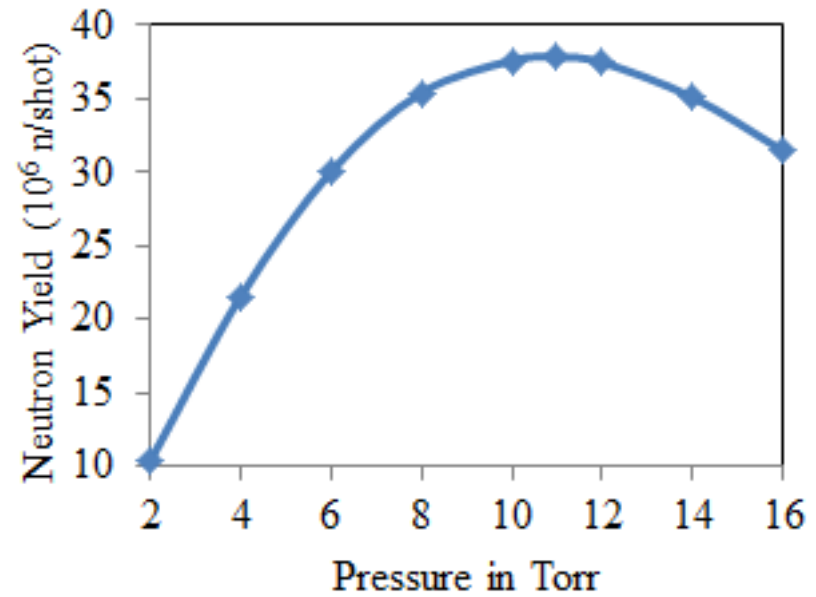


Figure 4.2 Variation of computed neutron yield in PF12 plasma focus machine with respect to pressure when capacitor is charged to 16 kV in deuterium gas.

4.3 Amirkabir Plasma Focus machine (2.6 kJ - Type 2 machine - Ar and Ne gas)

The Amirkabir Plasma Focus machine from Iran parameters are given in Table 4.3. It uses an anode which is a hollow copper cylinder and the outer electrode (cathode) composed of six copper rods which forms the shape of a squirrel cage. The anode and cathode are separated by a Pyrex tube (Pyrex insulator) [20].

This machine is working in argon gas and neon gas.

This machine has a static inductance of 100 nH. The Lee Model 6-phase code was used to fit the measured current waveform for argon gas. A good fit is shown in Figure 4.3.

The maximum computed current is 176 kA with the radial start time at 3.67 μ s for pinch duration of 0.181 μ s. It has a speed factor of 113 (kA/cm)/Torr^{0.5} Ar and total all-line yield of 82.2 J. The detailed information is presented in Table 4.4.

Using this current waveform, the optimum pressures for different voltages are estimated using Lee Model 6-phase code. It should be noted, that the paper [20] did not give its interpretation of optimum pressure but our computed optimum pressure is the pressure at which the highest value of total all-line yield at a particular voltage is obtained assuming all other factors are constant. This was then compared to the measured optimum pressure versus voltage as shown in Figure 4.4.

The variation of total current at pinch time at this optimum pressure is then plotted against the operating voltage as shown in Figure 4.5.

Table 4.3 The machine, operational and fitting parameters for Amirkabir Plasma Focus machine working in argon gas.

Capacitance C_0 (μF) = 36	Fill gas (molecular weight) = 40			
Static inductance L_0 (nH) = 100	Fill gas (atomic number) = 18			
Circuit resistance r_0 (m Ω) = 10	Fill gas (atom) = 1			
Cathode radius ‘b’ (cm) = 4.47	The fitted model parameters are:			
Anode radius ‘a’ (cm) = 1.39	Axial phase mass factor, f_m = 0.03			
Anode length ‘z ₀ ’ (cm) = 14.8	Axial phase current factor, f_c = 0.7			
Charging voltage V_0 (kV) = 12	Radial phase mass factor, f_{mr} = 0.1			
Fill pressure P_0 (Torr) = 1.26	Radial phase current factor, f_{cr} = 0.7			
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.03	140	10	1
Dip 2	0.10	450	10	1
Dip 3	0.04	600	10	1

Table 4.4 Information obtained from Lee Model 6-phase code configured for Amirkabir Plasma Focus machine at 12 kV, 1.26 Torr argon gas.

Peak current (kA) = 176
Pinch start current (kA) = 101
Pinch minimum temperature (10^6 K) = 2.33
Pinch maximum temperature (10^6 K) = 3.38
Peak axial speed (cm/ μ s) = 5.7
Peak radial shock speed (cm/ μ s) = 18.8
Peak radial piston speed (cm/ μ s) = 14.8
Final pinch radius $r_{\min}$ (cm) = 0.02
Pinch length $z_{\max}$ (cm) = 2.7
Pinch duration (ns) = 16.3
Peak induced voltage (kV) = 171.9
All-line yield (J) = 82.18
Energy Inflow into Plasma (%) = 13.6
Speed Factor ((kA/cm)/Torr ^{0.5}) = 113
Current per cm anode radius (kA/cm) = 127

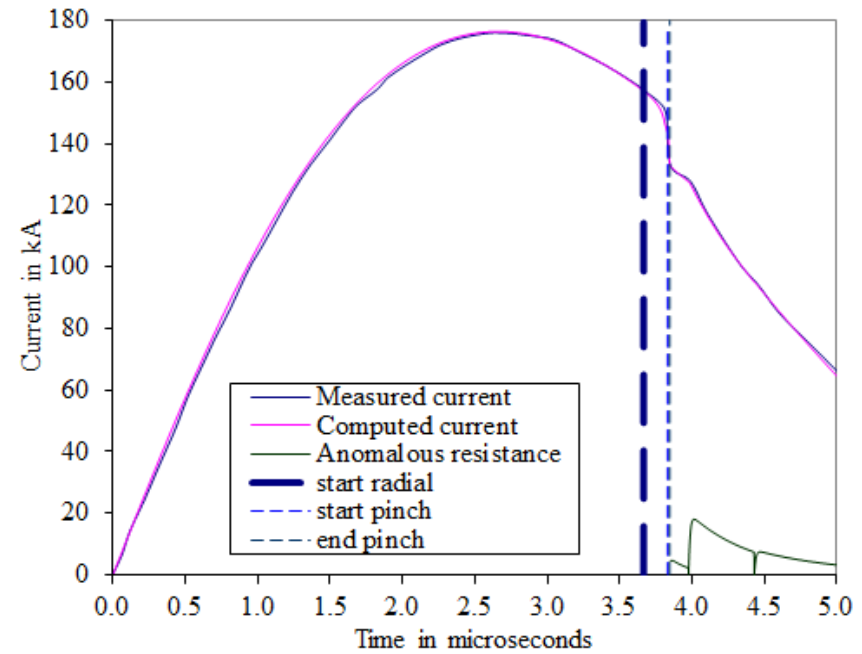


Figure 4.3 The measured current waveform of Amirkabir Plasma Focus machine at 12 kV, 1.26 Torr argon gas compared compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was extracted from Figure 3, page 2 of reference [20].

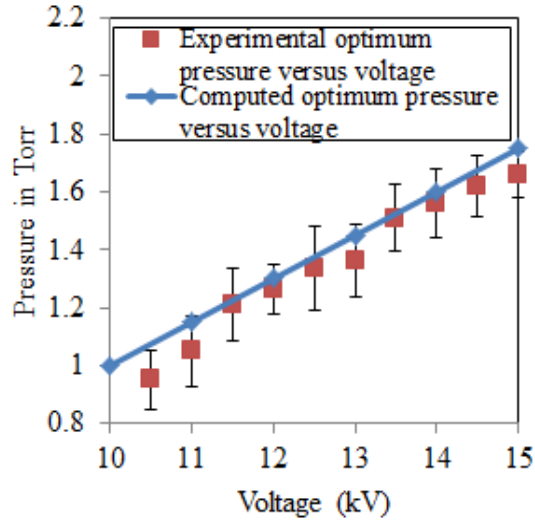


Figure 4.4 Variation of computed and experimental optimum pressure versus capacitor voltage for Amirkabir Plasma Focus machine working in argon gas.

Note: The measured optimum pressure versus capacitor voltage information was extracted from Figure 5, page 3 of reference [20].

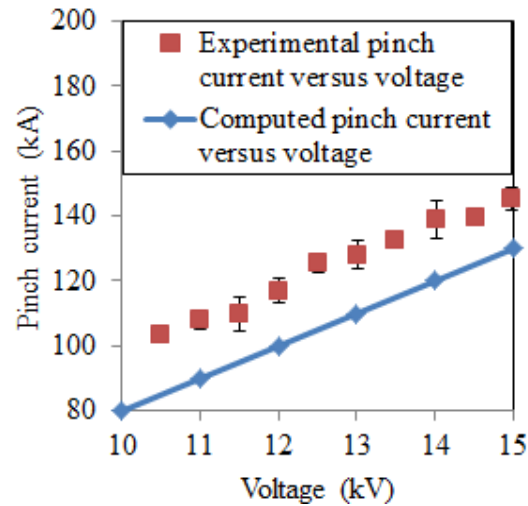


Figure 4.5 Variation of computed and experimental total current at pinch time versus voltage at optimum pressure for Amirkabir Plasma Focus machine working in argon gas.

Note: The measured optimum pressure versus capacitor voltage information was extracted from Figure 7, page 3 of reference [20].

From Figures 4.4 and 4.5, it can be noted that the computed results generally agree with the experimental results.

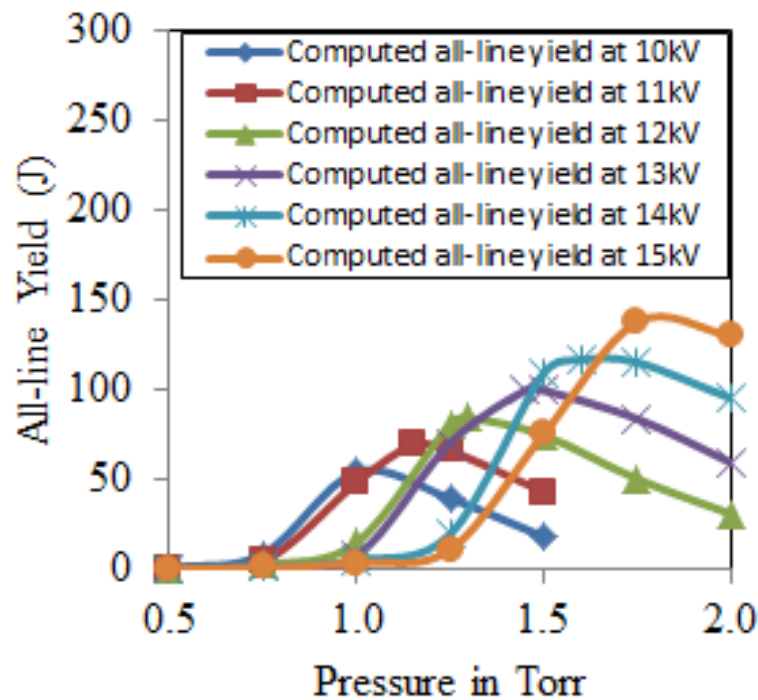


Figure 4.6 Variation of computed total all-line yield with respect to pressure operating at different voltages in argon for Amirkabir Plasma Focus machine.

Table 4.5 Information from Figure 4.6 presented in a table format.

Voltage (kV)	10	11	12	13	14	15
All-line yield (J)	54.37	69.3	81.17	99.8	117	138
Optimum Pressure (Torr)	1.0	1.15	1.3	1.45	1.6	1.75
% Energy input into plasma	12.97	13.28	13.55	13.78	13.96	14.2

A graph of total all-line yield versus pressure at different voltages is then obtained as shown in Figure 4.6. When Figure 4.6 is put into a table form as in Table 4.5, it can be observed that for every 1 kV increase in capacitor voltage, the optimum pressure increases by about 0.15 Torr.

The Lee Model 6-phase code was used to fit the measured current waveform for neon gas and it shows a good fit as in Figure 4.7.

The maximum computed current was 176 kA and exhibits the radial start time at 3.605 μs for pinch duration of 0.277 μs . It has a speed factor of 60 (kA/cm)/Torr^{0.5} Ne and soft X-ray yield of 0.72 J. The other detailed information is presented in Table 4.7.

The optimum pressure for different voltage is computed using Lee Model 6-phase code. This was then compared to the measured optimum pressure versus voltage as shown in Figure 4.8 and their results shows that the computed values are lower than the experimental values.

The variation of total current at pinch time at this optimum pressure is plotted against the operational voltage as shown in Figure 4.9. The computed and measured results show similar characteristic even though the computed value has slightly lower pinch current at each voltage.

Figure 4.10 shows soft X-ray yield versus pressure when capacitors have been charged to different voltages. When Figure 4.10 is put into a table form as in Table 4.8, it can be observed that for every 1 kV increase in capacitor voltage, the optimum pressure increases at least 0.3 Torr.

Table 4.6 The operational and fitting parameters for Amirkabir Plasma Focus machine working in neon gas.

Fill pressure P_0 (Torr) = 4.5	The fitted model parameters are: Axial phase mass factor, $f_m = 0.016$ Axial phase current factor, $f_c = 0.7$ Radial phase mass factor, $f_{mr} = 0.15$ Radial phase current factor, $f_{cr} = 0.7$			
Fill gas(molecular weight) = 20				
Fill gas(atomic number) = 10				
Fill gas(atom) = 1				
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.15	140	10	1
Dip 2	0.06	450	10	1
Dip 3	0.06	600	20	1

Table 4.7 Information obtained from Lee Model 6-phase code configured for Amirkabir Plasma Focus machine at 12 kV, 4.5 Torr neon gas.

Peak current (kA) = 176
Pinch start current (kA) = 100
Pinch minimum temperature (10^6 K) = 0.81
Pinch maximum temperature (10^6 K) = 0.83
Peak axial speed (cm/ μ s) = 5.8
Peak radial shock speed (cm/ μ s) = 13.5
Peak radial piston speed (cm/ μ s) = 10.8
Final pinch radius $r_{\min}$ (cm) = 0.09
Pinch length $z_{\max}$ (cm) = 1.9
Pinch duration (ns) = 20.2
Peak induced voltage (kV) = 18.3
All-line yield (J) = 0.72
Energy Inflow into Plasma (%) = 10.7
Speed Factor ((kA/cm)/Torr ^{0.5}) = 60
Current per cm anode radius (kA/cm) = 127

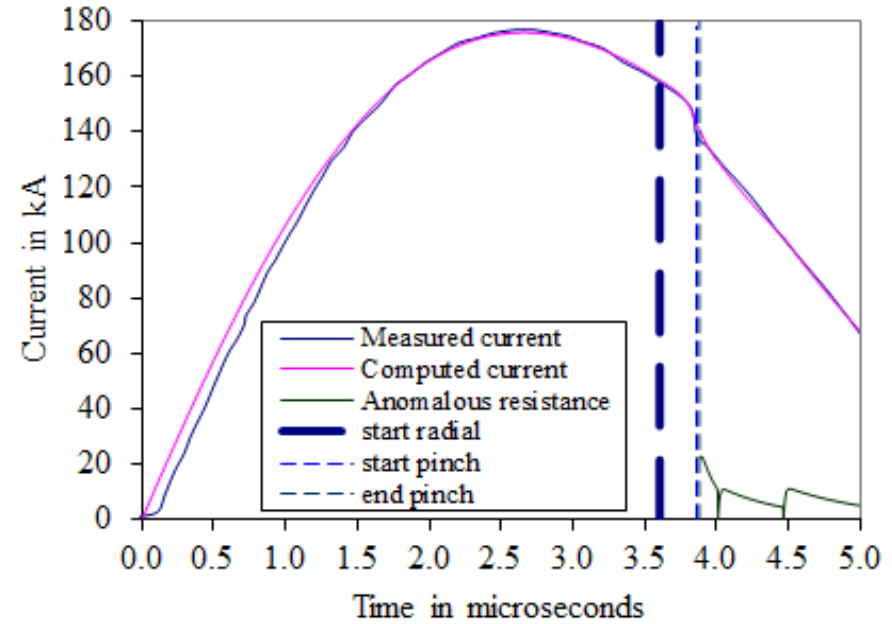


Figure 4.7 The measured current waveform of Amirkabir Plasma Focus machine at 12 kV, 4.5 Torr neon gas compared compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was extracted from Figure 4, page 3 of reference [20].

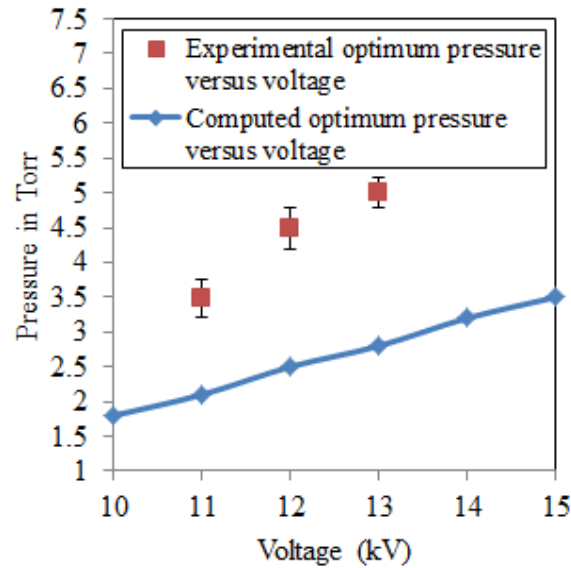


Figure 4.8 Variation of computed and experimental optimum pressure versus capacitor voltage for Amirkabir Plasma Focus machine working in neon gas.

Note: The measured optimum pressure versus capacitor voltage information was extracted from Figure 6, page 3 of reference [20].

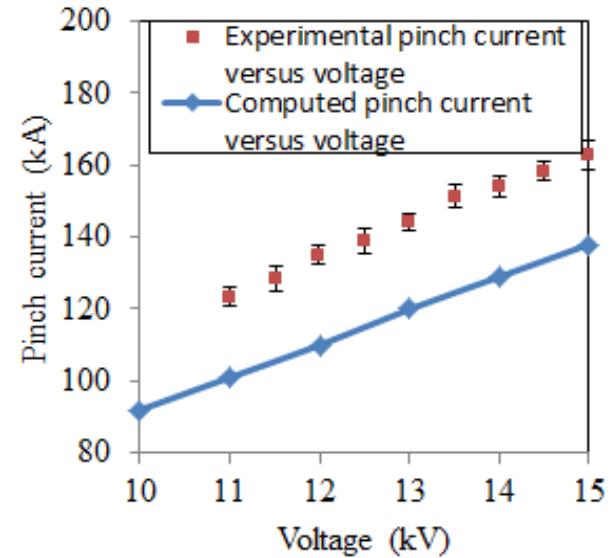


Figure 4.9 Variation of computed and experimental total current at pinch time versus voltage at optimum pressure for Amirkabir Plasma Focus machine working in neon gas.

Note: The measured optimum pressure versus capacitor voltage information was extracted from Figure 8, page 3 of reference [20].

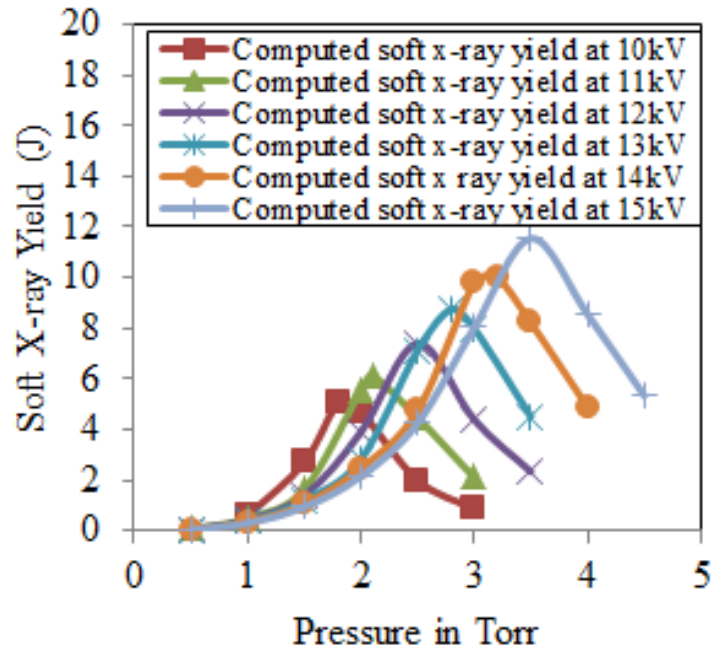


Figure 4.10 Variation of computed soft X-ray yield with respect to pressure operating at different voltages in neon for Amirkabir Plasma Focus machine.

Table 4.8 Information from Figure 4.10 presented in a table format.

Voltage (kV)	10	11	12	13	14	15
Soft X-ray yield (J)	4.77	6.13	7.4	8.72	10.10	11.55
Optimum Pressure (Torr)	1.8	2.1	2.5	2.8	3.2	3.5
% Energy input into plasma	11.1	11	11	10.9	11	10.9

4.4 INTI Plasma Focus machine - UNU/ICTP PFF (2.7 kJ - Type 2 machine - Ar, Ne and D gas).

This machine is part of a network of 3 kJ Mather type plasma machine known as the UNU/ICTP PFF (United Nations University/International Centre for Theoretical Physics Plasma Fusion Facility) [55]. It has a capacitance of 30 μF , static inductance of 110 nH; a circuit resistance of 12 m Ω . This machine has an anode length of 16 cm with a radius of 0.95 cm. The anode is surrounded by cathodes arranged in a circle having a radius of 3.2cm measured from its center.

In operation the UNU/ICTP PFF machine was charged to various voltages, before being discharge in different gas medium at different pressure as stated in Table 4.9. Its measured current waveform was than compared with the computed current waveform using the Lee Model 6-phase code.

Table 4.9 The operational and fitting parameters for UNU/ICTP PFF Plasma Focus machine.

Operational parameters	Type of Gas			
	Argon	Neon	Neon	Deuterium
Charging voltage V_0 (kV)	13.5	12	14	13.5
Fill pressure P_0 (Torr)	1.5	2.0	2.8	3.0
Fill gas (molecular weight)	40	20	20	4
Fill gas (atomic number)	18	10	10	1
Fill gas (atom(1) or molecule(2))	1	1	1	2
The fitted model parameters are:				
Axial phase mass factor, f_m	0.046	0.0385	0.049	0.09
Axial phase current factor, f_c	0.72	0.7	0.7	0.7
Radial phase mass factor, f_{mr}	0.22	0.16	0.2	0.14
Radial phase current factor, f_{cr}	0.7	0.82	0.8	0.7

To enable us to fit the computed waveform to the measured wave current waveform, the anomalous resistance parameters were fitted as shown in Table 4.10.

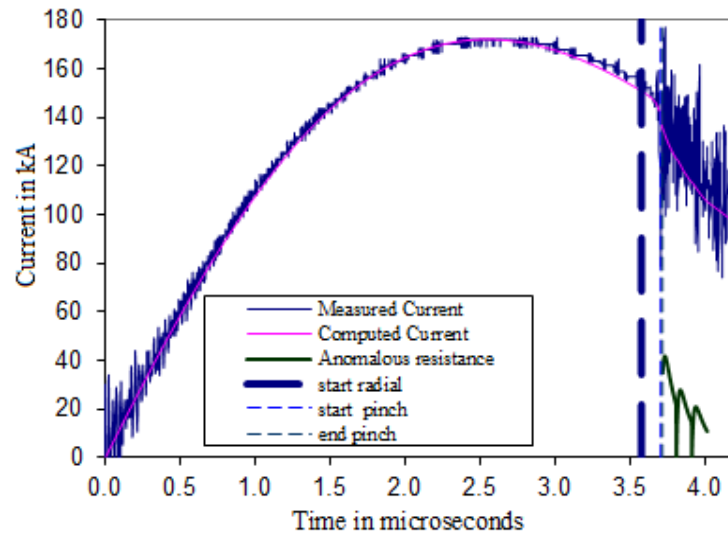


Figure 4.11 Measured current waveform of UNU ICTP PFF machine at 13.5 kV, 1.5 Torr argon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current values were obtained from plasmafocus.net [57].

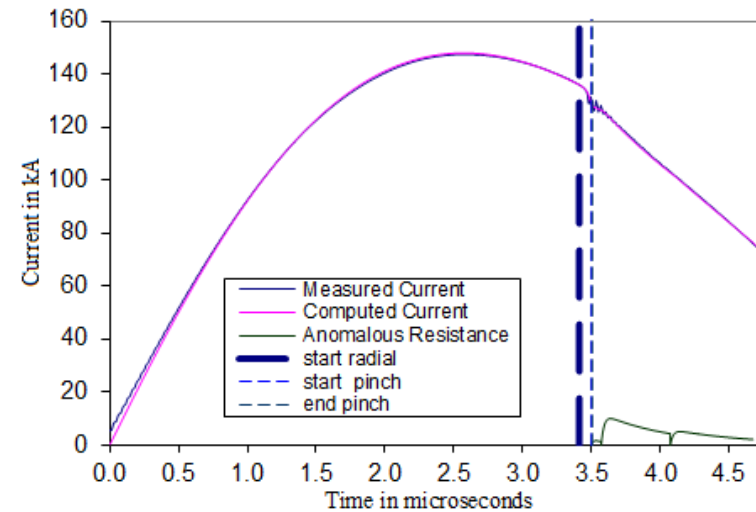


Figure 4.12 Measured current waveform of UNU ICTP PFF machine at 12 kV, 2.0 Torr neon gas compared with Lee Model 6-phase code computed current waveform.

Note: This current waveform from INTI plasma laboratory was numerically integrated from a di/dt waveform obtained using a Rogowski coil.

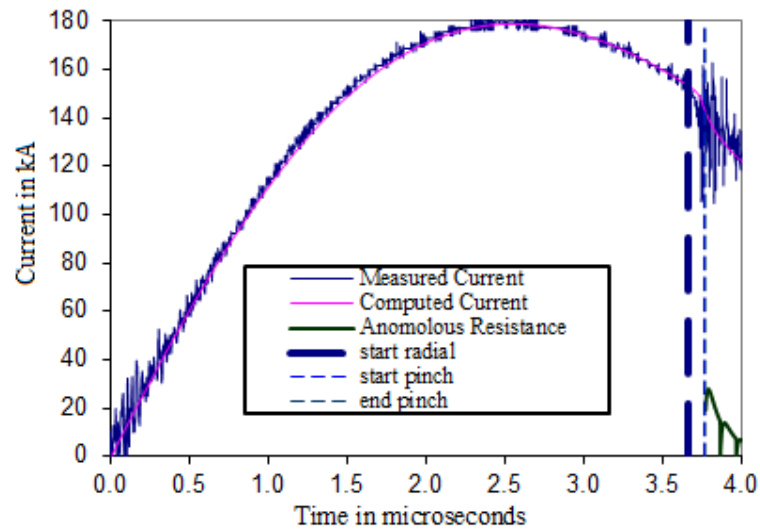


Figure 4.13 Measured current waveform of UNU ICTP PFF machine at 14 kV, 2.8 Torr neon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current values were obtained from plasmafocus.net [57].

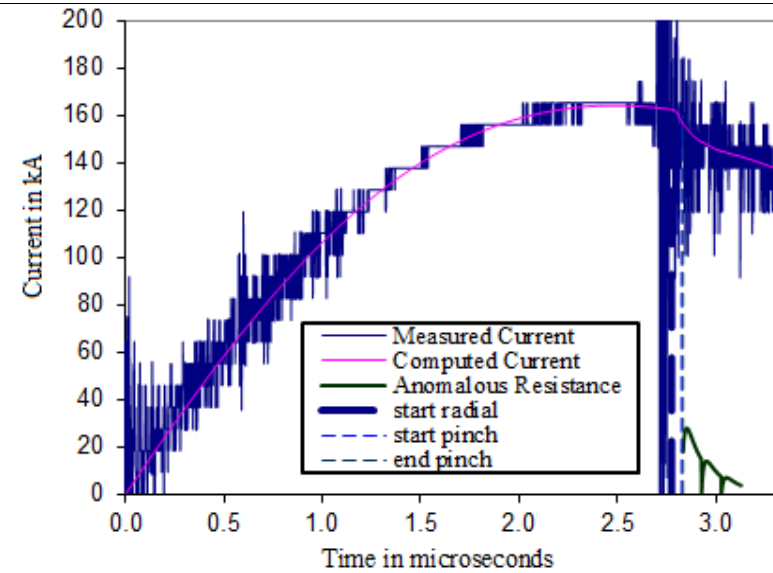


Figure 4.14 Measured current waveform of UNU ICTP PFF machine at 13.5 kV, 3.0 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current values were obtained from plasmafocus.net [57].

Table 4.10 (a) and (b) The anomalous resistances Dip value for UNU/ICTP PFF machine when working in argon gas medium (13.5kV, 1.5 Torr), deuterium gas medium (13.5 kV, 3.0 Torr) and neon gas (12 kV, 2.0 Torr and 14 kV, 2.8 Torr) respectively.

(a)

	$R_0(\Omega)$		Characteristic of fall time τ_2 (ns)		Characteristic of rise time τ_1 (ns)		End fraction time	
Dip 1	0.3	0.2	100	100	10	10	1	1
Dip 2	0.2	0.1	100	100	10	10	1	1
Dip 3	0.15	0.05	100	100	10	10	1	1
Type of gas	Argon	Deuterium	Argon	Deuterium	Argon	Deuterium	Argon	Deuterium
Voltage (kV)	13.5	13.5	13.5	13.5	13.5	13.5	13.5	13.5

(b)

	$R_0(\Omega)$		Characteristic of fall time τ_2 (ns)		Characteristic of rise time τ_1 (ns)		End fraction time	
Dip 1	0.02	0.2	70	100	20	10	1	1
Dip 2	0.06	0.1	500	100	20	10	1	1
Dip 3	0.03	0.05	600	100	20	10	1	1
Type of gas	Neon	Neon	Neon	Neon	Neon	Neon	Neon	Neon
Voltage (kV)	12	14	12	14	12	14	12	14

From analysis of the information obtained in Table 4.10, it is obvious that there is no relationship between the anomalous resistance characteristic and the type of machine as this resistance characteristic differs from shot to shot and also differs when the gas medium changes.

The fitted model parameters were then used for numerical experiments in neon to obtain the neon characteristic soft X-ray yield as a function of pressure. This is shown in Table 4.11 and Figure 4.15.

Table 4.11 Information obtained from Lee Model 6-phase code configured for UNU-ICTP Plasma Focus machine working in neon gas (12 kV, 2.0 Torr and 14 kV, 2.8 Torr) and deuterium gas (13.5 kV, 3.0 Torr).

Operating gas	Neon	Neon	Deuterium
Operating Voltage (kV)	12	14	13.5
Peak current (kA)	148	179	164
Pinch start current (kA)	106	115	110
Pinch minimum temperature (10^6 K)	3.85	2.73	7.73
Pinch maximum temperature (10^6 K)	4.15	2.84	7.85
Peak axial speed (cm/ μ s)	6.7	6	8.9
Peak radial shock speed (cm/ μ s)	26.1	24.8	37.7
Peak radial piston speed (cm/ μ s)	17.1	15.8	26.0
Final pinch radius $r_{\min}$ (cm)	0.09	0.08	0.13
Pinch length $z_{\max}$ (cm)	1.4	1.3	1.4
Pinch duration (ns)	8.5	8.2	7.3
Peak induced voltage (kV)	22	23.7	23.6
Soft X- ray yield (J) or Neutron yield (n/shot)	0.43 J	2 J	6.22×10^6 n
Energy Inflow into Plasma (%)	10.2	10.5	9.1
Speed Factor ((kA/cm)/Torr ^{0.5})	110	113	100
Current per cm anode radius (kA/cm)	156	188	173

Lee Model 6-phase code is configured to work from 1.0 to 6.0 Torr neon at 14 kV (as shown in Figure 4.15) and the results are compared to the experimental

results obtained from a published source [14]. It is observed that the measured curve is broader in pressure range than the computed yield curve. The result also reveals that the experimental optimum point is at 3.0 Torr neon gas with a soft X-ray yield of 5.4 ± 1 J [14]. This compares with the numerical optimum pressure of 3.3 Torr neon gas with a yield of 4.0 J. The drop-off of the soft X-ray on the low-pressure side is very similar, but the computed drop-off on the high pressure side shows a sharper drop-off compared to the experimental values.

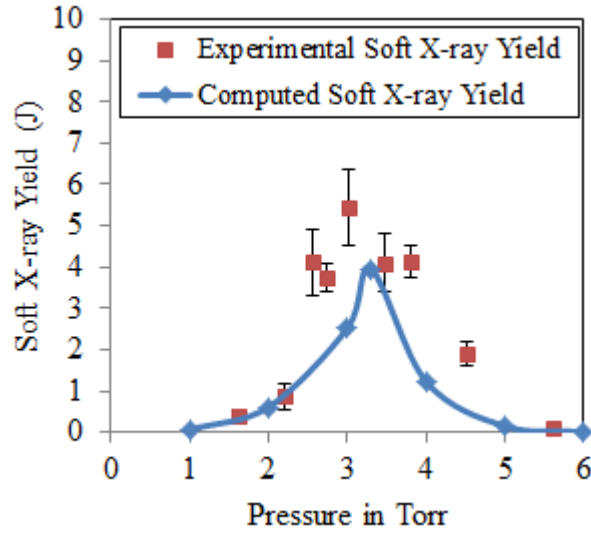


Figure 4.15 Variation of computed and experimental soft X-ray yield with respect to pressure operating at 14 kV in neon for UNU/ICTP Plasma Focus machine.

Note: The measured values were obtained from Figure 1, page 1278 of reference [14].

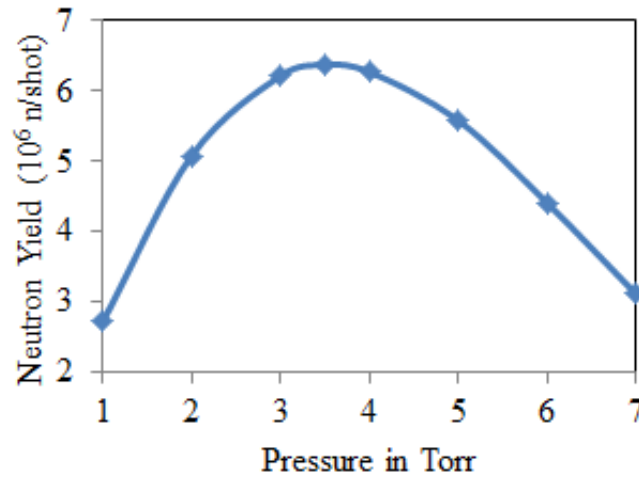


Figure 4.16 Variation of computed neutron yield with respect to pressure operating at 13.5 kV in neon for UNU/ICTP Plasma Focus machine.

When Lee Model 6-phase code was configured to operate the UNU/ICTP plasma focus machine from 1.0 to 7.0 Torr deuterium gas as shown in Figure 4.16 the optimum numerical neutron yield obtained was 6.4×10^6 n/shot which occurs at 3.5 Torr compared to the experimental values of $0.5-1 \times 10^8$ n/shot at 3.0 Torr (using 15 kV) as stated in reference [55]. The energy input into plasma radial phase for this shot is 9.3 %.

4.5 Syrian Plasma Focus machine (2.8 kJ - Type 2 machine - Ar gas)

Syria has 2 types of Plasma focus machine, namely the ACES-PF1 and ACES-PF2. The ACPS-PF1 is similar to ACES-PF2 except that its static inductance is 1430 nH [119]. We now discuss ACES-PF2 in detailed. ACES-PF2 is a Mather-type Plasma Focus whose machine parameters are given in Table 4.12. Its static inductance is 270 nH [120]. The Lee Model 6-phase code was used to fit the computed current waveform to the measured current waveform as shown in Figure 4.17.

Table 4.12 The machine, operational and fitting parameters for AECS-PF2.

Capacitance C_0 (μF) = 25	Fill gas (molecular weight) = 40			
Static inductance L_0 (nH) = 270	Fill gas (atomic number) = 18			
Circuit resistance r_0 (m Ω) = 35	Fill gas (atom) = 1			
Cathode radius ‘b’ (cm) = 3.2	The fitted model parameters are:			
Anode radius ‘a’ (cm) = 0.95	Axial phase mass factor, $f_m = 0.05$			
Anode length ‘z ₀ ’ (cm) = 16	Axial phase current factor, $f_c = 0.7$			
Charging voltage V_0 (kV) = 15	Radial phase mass factor, $f_{mr} = 0.15$			
Fill pressure P_0 (Torr) = 0.41	Radial phase current factor, $f_{cr} = 0.7$			
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.1	500	10	1
Dip 2	0.1	500	10	1
Dip 3	0.1	500	10	1

It should be noted that the measured current waveform is very noisy making fitting difficult and uncertain as shown in this Figure 4.17.

The maximum computed current is 106 kA and exhibits the radial start time at 3.345 μs for pinch duration of 0.086 μs . It has a speed factor of 175 and total all-line yield of 0.32 J. The detailed information is presented in Table 4.13.

Table 4.13 Information obtained from Lee Model 6-phase code configured for AECS-PF2 at 15 kV, 0.41 Torr argon gas.

Peak current (kA) = 106
Pinch start current (kA) = 73
Pinch minimum temperature (10^6 K) = 5.44
Pinch maximum temperature (10^6 K) = 5.94
Peak axial speed (cm/ μ s) = 6.5
Peak radial shock speed (cm/ μ s) = 28.0
Peak radial piston speed (cm/ μ s) = 21.4
Final pinch radius $r_{\min}$ (cm) = 0.07
Pinch length $z_{\max}$ (cm) = 1.4
Pinch duration (ns) = 7.2
Peak induced voltage (kV) = 24.8
All-line yield (J) = 0.324
Energy Inflow into Plasma (%) = 3.8
Speed Factor ((kA/cm)/Torr ^{0.5}) = 175
Current per cm anode radius (kA/cm) = 112

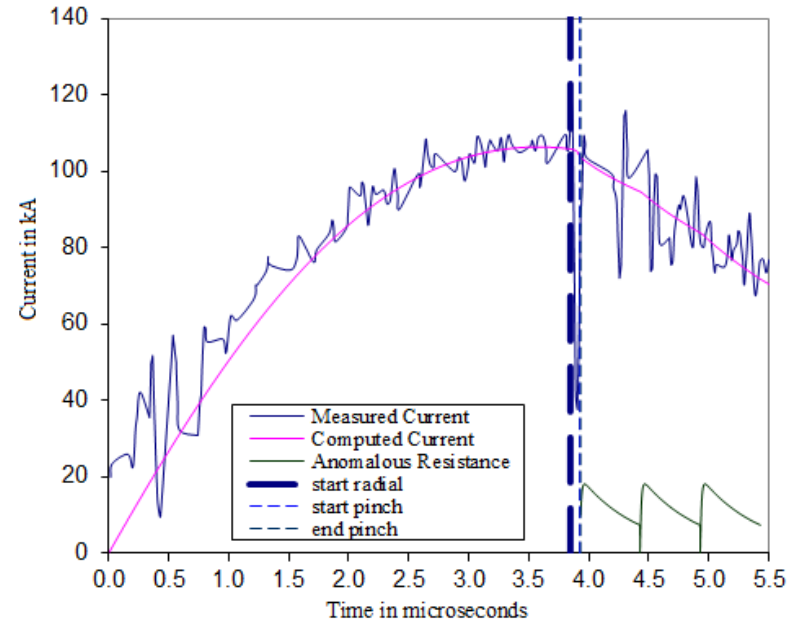


Figure 4.17 The measured current waveform of AECS-PF2 machine at 15 kV, 0.41 Torr argon gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was extracted from Figure 1 of reference [120].

There are no experimental values given in the report for further comparison to be made.

4.6 Bora Plasma Focus machine (3.5 kJ - Type 1 machine - D gas)

A dense Plasma Focus called “Bora” operates at The Abdus Salam International Centre for Theoretical Physics in Trieste, Italy.

The machine parameters for this Plasma Focus machine as stated in Table 4.14 were extracted from the ICTP Training Notes entitled “Plasma Focus Numerical Experiments and BORA” available on plasmafocus.net [121]. The Bora Plasma Focus has a curved shaped anode and cathode which the ICTP Training Notes approximates as straight anode and cathode; with the equivalent dimensions compiled in Table 4.14.

The measured current waveform at 7.6 Torr (extracted from the current derivative waveform) was compared to the computed waveform as shown in Figure 4.18. It can be seen that despite the approximated electrode system the fit is still reasonable; though not as good as we normally get in such current fittings.

The maximum computed current is 302 kA and exhibits the radial start time at 1.562 μs for pinch duration of 0.212 μs . It has a speed factor of 73 (kA/cm)/Torr^{0.5} and neutron yield of 1.8×10^8 n/shot. The detailed information is presented in Table 4.15.

Using the information obtained from the current waveform, Lee code was configured to operate Bora Plasma Focus machine from 5.0 Torr to 10.0 Torr deuterium to get the neutron yield versus pressure curve. The waveform is presented in Figure 4.19.

Figure 4.19 shows that the maximum neutron yield of 1.83×10^8 n/shot occurs at 7.8 Torr when the capacitors are charged to 17 kV and the energy input into plasma is 9.5 %.

Table 4.14 The machine, operational and fitting parameters for Bora Plasma Focus machine.

Capacitance C_0 (μF) = 24.4
Static inductance L_0 (nH) = 55
Circuit resistance r_0 ($\text{m}\Omega$) = 6
Cathode radius 'b' (cm) = 2.5
Anode radius 'a' (cm) = 1.5
Anode length ' z_0 ' (cm) = 6
Charging voltage V_0 (kV) = 17
Fill pressure P_0 (Torr) = 7.6
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.2$
Axial phase current factor, $f_c = 0.7$
Radial phase mass factor, $f_{mr} = 0.4$
Radial phase current factor, $f_{cr} = 0.7$

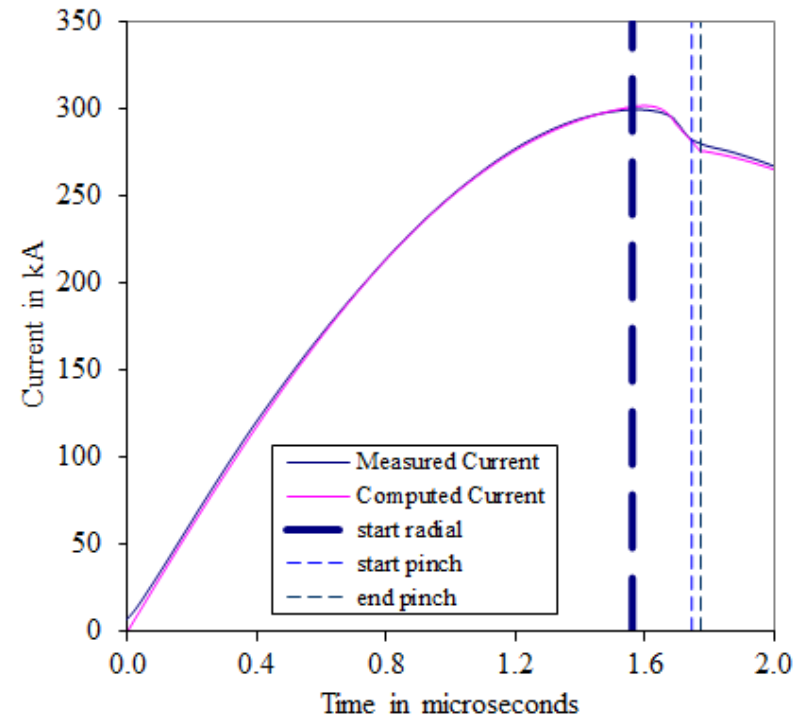


Figure 4.18 The measured current waveform of Bora Plasma focus machine at 17 kV, 7.6 Torr deuterium gas compared with Lee Model 5-phase code computed waveform.

Note: The measured current waveform information was taken from plasmafocus.net [57].

Table 4.15 Information obtained from Lee Model 5-phase code configured for Bora Plasma Focus machine at 17 kV, 7.6 Torr deuterium gas.

Peak current (kA) = 302
 Pinch start current (kA) = 197
 Pinch minimum temperature (10^6 K) = 1.33
 Pinch maximum temperature (10^6 K) = 1.39
 Peak axial speed (cm/ μ s) = 6.5
 Peak radial shock speed (cm/ μ s) = 16.3
 Peak radial piston speed (cm/ μ s) = 10.8
 Final pinch radius $r_{\min}$ (cm) = 0.22
 Pinch length $z_{\max}$ (cm) = 2.5
 Pinch duration (ns) = 27.8
 Peak induced voltage (kV) = 17.1
 Neutron yield ($\times 10^8$ n/shot) = 1.8
 Energy Inflow into Plasma (%) = 9.5
 Speed Factor ((kA/cm)/Torr $^{0.5}$) = 73
 Current per cm anode radius (kA/cm) = 201

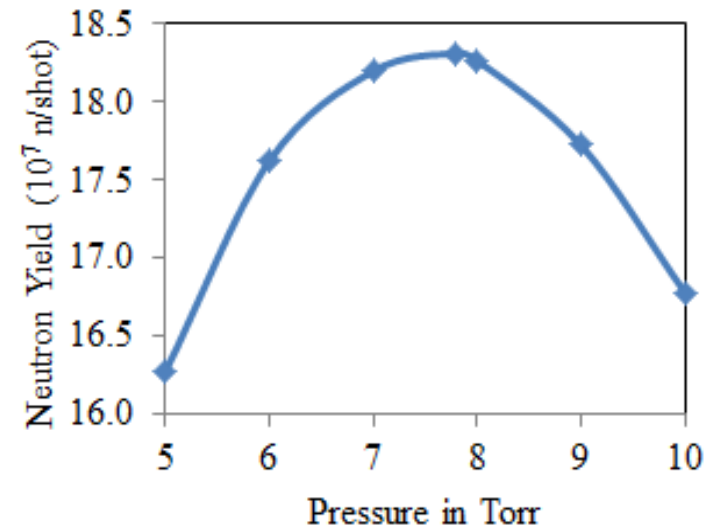


Figure 4.19 Variation of computed neutron yield with respect to pressure operating at 17 kV in deuterium for Bora Plasma Focus machine.

4.7 Hanyang University Plasma Focus machine (4.1 kJ - Type 2 machine - D gas)

The Hanyang University Plasma Focus (HU PF) device is a Mather type Plasma Focus device (32 μ F, 4 kJ) from Korea. It has a copper inner electrode, with a diameter of 20 mm. The outer electrode is composed of sixteen copper rods with diameter of 10 mm, which forms the shape of a squirrel cage with an inner diameter of 70 mm. The 1.5mm thick quartz tube, with a breakdown length of 13 mm, is located between the inner and the outer electrode as an insulator. The machine parameters for anode lengths of 18cm and 22 cm working in different pressure are shown in Table 4.16.

The Lee Model 6-phase code was used to fit the computed current waveform to the measured current waveform as shown in Figure 4.20 and Figure 4.21 respectively.

For the anode length of 18cm, the maximum computed current is 154 kA and exhibits the radial start time at 3.690 μ s for pinch duration of 0.088 μ s. Its neutron yield is 8.15×10^6 n/shot. Compared to the experimental maximum neutron yield of 1.24×10^8 n/shot [122], this results shows a difference of 15 times which is larger than typical value expected. The other detailed information is presented in Table 4.17.

For the anode length of 22cm, the maximum computed current is 152 kA and exhibits the radial start time at 3.684 μ s for pinch duration of 0.055 μ s. Its neutron yield is 5.42×10^6 n/shot. Compared with the experimental yield of 6.67×10^7 n/shot, the difference in the neutron yield is a factor of 12, which is much larger difference than typical compared with most of the machines we have compared. The detailed information is presented in Table 4.18.

It should also be noted that the reference report [122] only took the best of the 5 shots (readings) for each anode length.

Table 4.16 The machine, operational and fitting parameters for HU PF with anode length of 18cm (2.9 Torr) and 22 cm (1.4 Torr) respectively.

Capacitance C_0 (μF)					32		32	
Static inductance L_0 (nH)					230		230	
Circuit resistance r_0 ($\text{m}\Omega$)					9		9	
Cathode radius ‘b’ (cm)					3.5		3.5	
Anode radius ‘a’ (cm)					1		1	
Anode length ‘z_0’ (cm)					18		22	
Charging voltage V_0 (kV)					16		16	
Fill pressure P_0 (Torr)					2.9		1.4	
Fill gas (molecular weight)					4		4	
Fill gas (atomic number)					1		1	
Fill gas (molecule)					2		2	
The fitted model parameters are:								
Axial phase mass factor, f_m					0.078		0.106	
Axial phase current factor, f_c					0.7		0.7	
Radial phase mass factor, f_{mr}					0.25		0.2	
Radial phase current factor, f_{cr}					0.7		0.7	
The anomalous resistance parameters are:								
	$R_0(\Omega)$		Characteristic of fall time, τ_2 (ns)		Characteristic of rise time, τ_1 (ns)		End fraction time	
Dip 1	0.13	0.3	100	90	5	10	1	1
Dip 2	0.15	0.1	100	100	5	15	1	1
Dip 3	0.05	0.01	100	100	5	10	1	1
Anode length	18	22	18	22	18	22	18	22

Table 4.17 Information obtained from Lee Model 6-phase code for HU PF which was configured to work at 16 kV, 2.9 Torr deuterium gas (using 18 cm anode).

Peak current (kA) = 154
 Pinch start current (kA) = 107
 Pinch minimum temperature (10^6 K) = 3.8
 Pinch maximum temperature (10^6 K) = 3.83
 Peak axial speed (cm/ μ s) = 8.1
 Peak radial shock speed (cm/ μ s) = 26.1
 Peak radial piston speed (cm/ μ s) = 17.8
 Final pinch radius $r_{\min}$ (cm) = 0.14
 Pinch length $z_{\max}$ (cm) = 1.6
 Pinch duration (ns) = 11
 Peak induced voltage (kV) = 17.4
 Neutron yield ($\times 10^6$ n/shot) = 8.15
 Energy Inflow into Plasma (%) = 5.7
 Speed Factor ((kA/cm)/Torr^{0.5}) = 91
 Current per cm anode radius (kA/cm) = 154

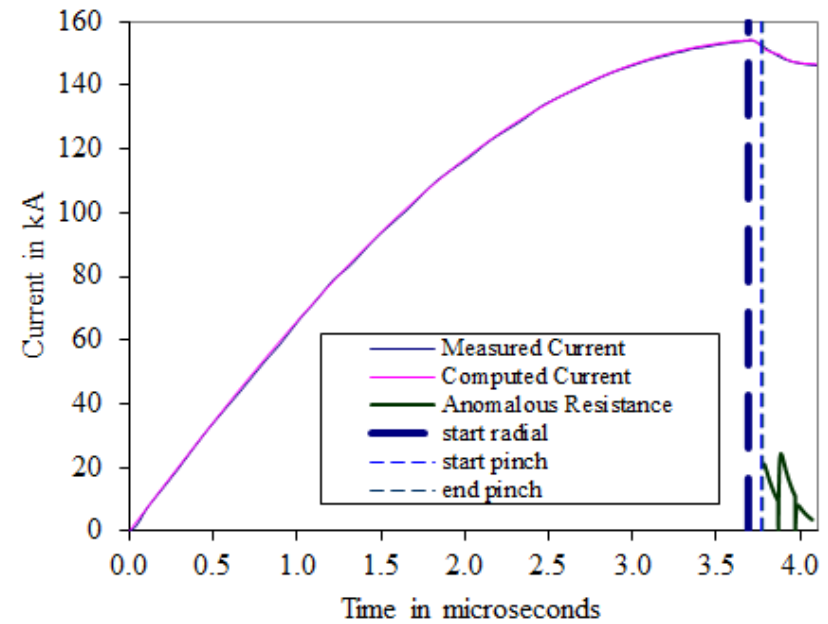


Figure 4.20 The measured current waveform of HU PF at 16 kV, 2.9 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was taken from Figure 3(b), page 1100 of reference [122].

Table 4.18 Information obtained from Lee Model 6-phase code for HU PF which was configured to work at 16 kV, 1.4 Torr deuterium gas (using 22 cm anode).

Peak current (kA) = 152
Pinch start current (kA) = 105
Pinch minimum temperature (10^6 K) = 9.51
Pinch maximum temperature (10^6 K) = 9.59
Peak axial speed (cm/ μ s) = 9.8
Peak radial shock speed (cm/ μ s) = 41.3
Peak radial piston speed (cm/ μ s) = 28.4
Final pinch radius $r_{\min}$ (cm) = 0.14
Pinch length $z_{\max}$ (cm) = 1.5
Pinch duration (ns) = 6.9
Peak induced voltage (kV) = 26.3
Neutron yield ($\times 10^6$ n/shot) = 5.42
Energy Inflow into Plasma (%) = 6.5
Speed Factor ((kA/cm)/Torr ^{0.5}) = 128
Current per cm anode radius (kA/cm) = 152

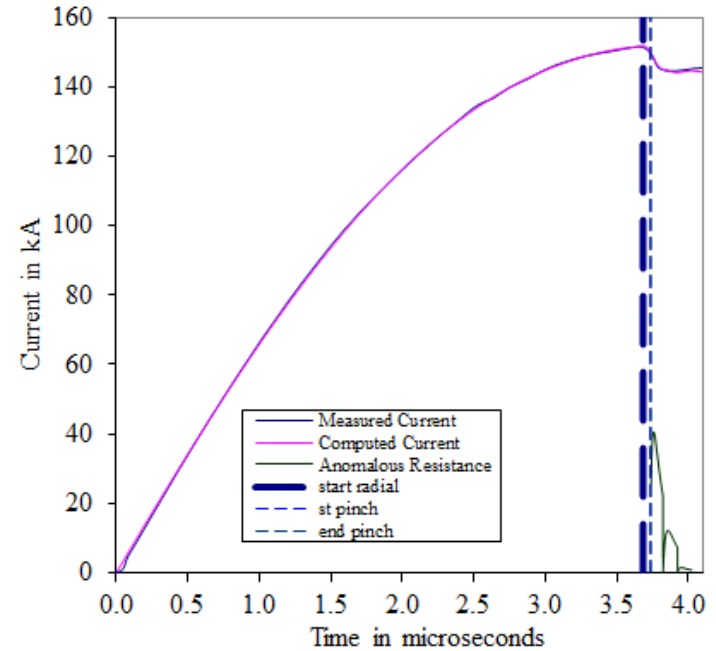


Figure 4.21 The measured current waveform of HU PF at 16 kV, 1.4 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was taken from Figure 3(a), page 1100 of reference [122].

Using the fitting parameters from the current waveform in Figure 4.20, Lee Model 6-phase code is now configured for 14.5, 16, 17, 18, 20.5 and 22 cm anode. For each length of anode, the pressure is varied and the results of the neutron yield versus pressure are shown in Figure 4.22.

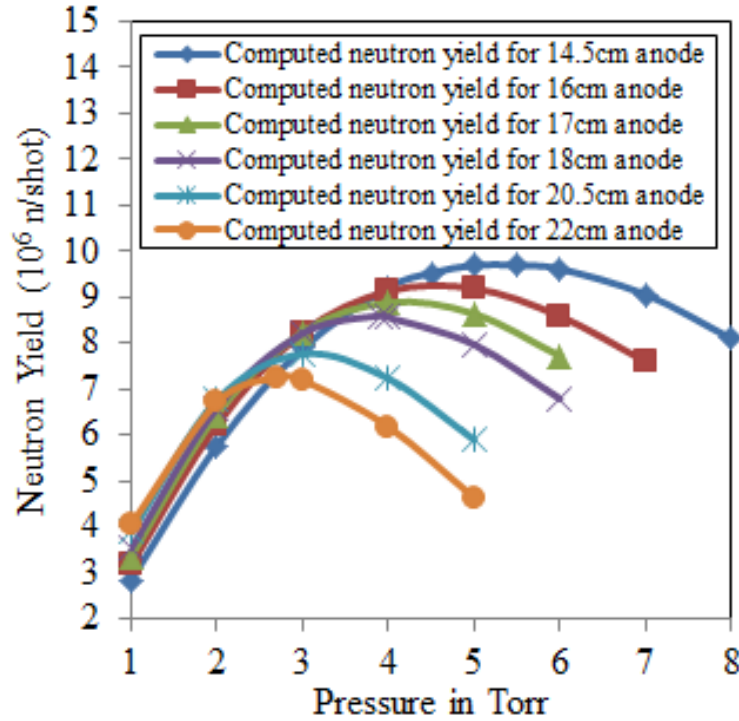


Figure 4.22 Variation of computed neutron yield with pressure for different anode lengths in deuterium for HU PF. The maximum neutron yield is 9.7×10^6 n/shot for 14.5 cm anode with peak $v_a = 6.3$ cm/ μ s and peak $v_s = 20$ cm/ μ s .

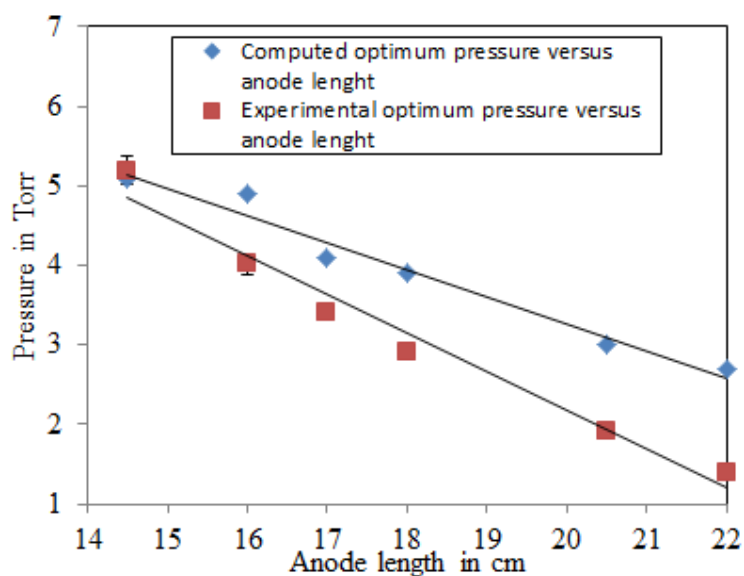


Figure 4.23 Variation of computed and experimental optimum pressure with respect to different anode lengths of HU PF.

Note: The experimental optimum pressure with respect to different anode lengths information was taken from Figure 2, page 1099 of reference [122].

Using the information from Figure 4.22, we obtained for 18cm anode the maximum neutron yield of 8.6×10^6 n/shot (computed) compared to 1.24×10^8 n/shot (experimental) while for 22 cm anode the maximum neutron yield obtained was 6.67×10^6 n/shot (computed) compared to 0.68×10^8 n/shot (experimental) at a pressure of 3.9 Torr and 2.5 Torr respectively. The experimental values are larger than what is typically expected. The optimum pressure versus different anode length was plotted as shown in Figure 4.23 and compared to the experimental data obtained from the published results [122]. The slope of the computed values decays more gently compared to the experimental results.

4.8 GN1 Plasma Focus machine (4.7 kJ - Type 1 machine - D gas)

The machine parameters for the GN1 Plasma Focus from Argentina are compiled in Table 4.19. Its static inductance is 52 nH. Lee Model 5-phase code was used to fit the measured current waveform; and where necessary, the Lee Model 6-phase code was used to complete the fit. The fitting is shown in Figure 4.24. The author stated [30] that the static inductance is 42.6 nH; but our fitted value is 52 nH.

The computed and measured waveforms show a good fit down to 3/4 of the current dip and we needed the 6-phase model to complete the fit of the current dip. The maximum computed current is 317 kA and exhibits the radial start time at 0.910 μ s for pinch duration of 0.065 μ s. It has a speed factor of 176 (kA/cm)/Torr^{0.5} and neutron yield of 4.88×10^7 n/shot. The detailed information is presented in Table 4.20.

Using the model parameters obtained from this current fitting, the code is run for 0.5 to 5.0 Torr and the computed pinch time versus pressure curve is compared to the experimental pinch time curve in Figure 4.25. The neutron yield versus pressure curve is shown in Figure 4.26.

Figure 4.25 shows that the computed arrival time of focus is slightly less than the experimental arrival time of the focus. Quite reasonably, the current sheath arrives later for higher pressures due to the larger mass accumulation in the current sheath.

Figure 4.26 shows that the optimum computed neutron yield for the 18 mm anode is 1.04×10^8 n/shot at 2.25 Torr. The energy input into plasma for this shot was 14.8 %. No experimental neutron yield was published for this series.

Table 4.19 The machine, operational and fitting parameters for GN1 Plasma Focus machine with anode radius 1.8 cm and length 12 cm.

Capacitance C_0 (μF) = 10.5	Fill gas (molecular weight) = 4			
Static inductance L_0 (nH) = 52	Fill gas (atomic number) = 1			
Circuit resistance r_0 (m Ω) = 7	Fill gas (molecule) = 2			
Cathode radius ‘b’ (cm) = 3.6	The fitted model parameters are:			
Anode radius ‘a’ (cm) = 1.8	Axial phase mass factor, $f_m = 0.11$			
Anode length ‘z ₀ ’ (cm) = 12	Axial phase current factor, $f_c = 0.7$			
Charging voltage V_0 (kV) = 30	Radial phase mass factor, $f_{mr} = 0.2$			
Fill pressure P_0 (Torr) = 0.75	Radial phase current factor, $f_{cr} = 0.7$			
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.5	30	10	1
Dip 2	0.03	100	1	1
Dip 3	0.01	150	1	1

Table 4.20 Information obtained from Lee Model 6-phase code configured for GN1 Plasma Focus machine at 30 kV, 0.75 Torr deuterium gas.

Peak current (kA) = 317	Final pinch radius r_{min} (cm) = 0.26
Pinch start current (kA) = 205	Pinch length z_{max} (cm) = 2.7
Pinch minimum temperature (10^6 K) = 20.23	Pinch duration (ns) = 8.5
Pinch maximum temperature (10^6 K) = 21.2	Peak induced voltage (kV) = 65.2
Peak axial speed (cm/ μs) = 21.9	Neutron yield ($\times 10^7$ n/shot) = 4.88
Peak radial shock speed (cm/ μs) = 63.7	Energy Inflow into Plasma (%) = 12.5
Peak radial piston speed (cm/ μs) = 44	Speed Factor ((kA/cm)/Torr ^{0.5}) = 204
Current per cm anode radius (kA/cm) = 176	

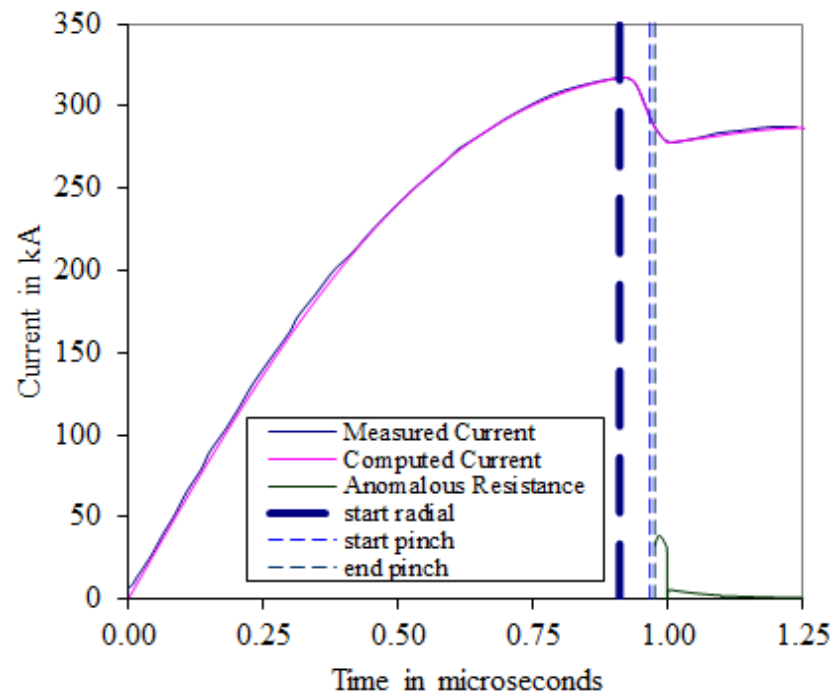


Figure 4.24 The measured current waveform of GN1 Plasma Focus machine at 30 kV, 0.75 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform (using anode radius 1.8 cm and length 12 cm) was obtained from the current derivative in Figure 2, page 1992 of reference [30].

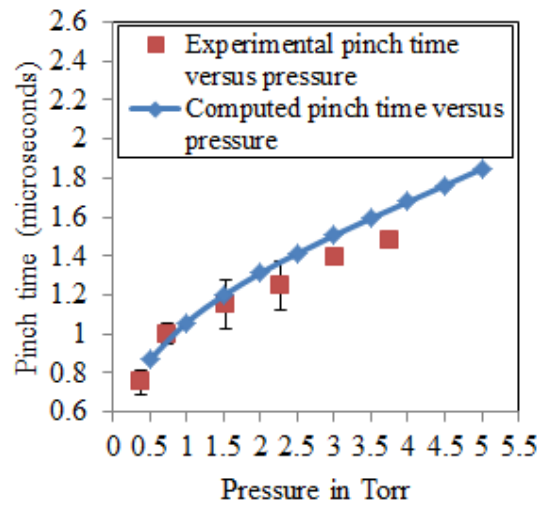


Figure 4.25 Variation of computed and experimental arrival time with respect to pressure operating at 30 kV in deuterium for GN1 Plasma Focus machine.

Note: All the experimental values has been reduce by 0.2835microseconds from Figure 3 to coincide with the Figure 2 which gives the focus time of 0.75 Torr (1 mbar) at 1 microsecond as shown on page 1992 in reference [30].

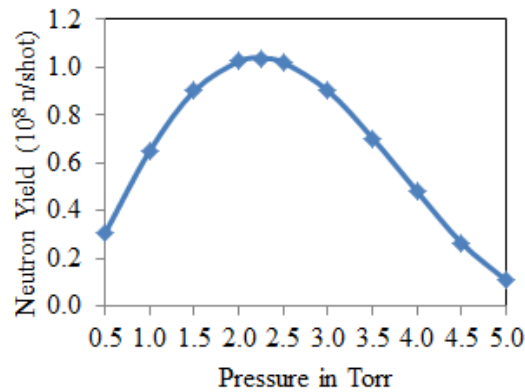


Figure 4.26 Variation of computed neutron yield with respect to pressure operating at 30 kV (using anode radius 1.8 cm and length 12 cm) in deuterium for GN1 Plasma Focus machine.

The results of another series of experiments from this machine using an anode of radius 1.9 cm, length 8.7 cm was available from a publication [31]. A current waveform in 3.0 Torr deuterium was published. This series also measured

neutron yields over a range of pressures. The machine, operational and fitting parameters for this series is compiled in Table 4.21.

Table 4.21 The machine, operational and fitting parameters for GN1 Plasma Focus machine with anode radius 1.9 cm and length 8.7 cm.

Capacitance C_0 (μF) = 10.5	Fill gas (molecular weight) = 4																			
Static inductance L_0 (nH) = 52	Fill gas (atomic number) = 1																			
Circuit resistance r_0 (m Ω) = 7	Fill gas (molecule) = 2																			
Cathode radius ‘b’ (cm) = 3.6	The fitted model parameters are:																			
Anode radius ‘a’ (cm) = 1.9	Axial phase mass factor, $f_m = 0.13$																			
Anode length ‘z ₀ ’ (cm) = 8.7	Axial phase current factor, $f_c = 0.7$																			
Charging voltage V_0 (kV) = 30	Radial phase mass factor, $f_{mr} = 0.15$																			
Fill pressure P_0 (Torr) = 3	Radial phase current factor, $f_{cr} = 0.85$																			
The anomalous resistance parameters are:																				
	<table><tr><td>$R_0(\Omega)$</td><td>Characteristic of fall time τ_2 (ns)</td><td>Characteristic of rise time τ_1 (ns)</td><td>End fraction time</td></tr><tr><td>Dip 1</td><td>0.13</td><td>100</td><td>10</td><td>1</td></tr><tr><td>Dip 2</td><td>0.02</td><td>50</td><td>10</td><td>1</td></tr><tr><td>Dip 3</td><td>0.01</td><td>50</td><td>10</td><td>1</td></tr></table>	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time	Dip 1	0.13	100	10	1	Dip 2	0.02	50	10	1	Dip 3	0.01	50	10	1
$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time																	
Dip 1	0.13	100	10	1																
Dip 2	0.02	50	10	1																
Dip 3	0.01	50	10	1																

The computed and measured waveforms at 3.0 Torr deuterium show a good fit down to half the current dip using the 5-phase code. The 6-phase code is required to complete the fit for the whole current dip. This is shown in Figure 4.27.

The maximum computed current is 357 kA (the experimental peak value is 350 kA) and exhibits the radial start time at 1.184 μs for pinch duration of 0.096 μs . It has a speed factor of 108 (kA/cm)/Torr^{0.5} and neutron yield of 2.09×10^8 n/shot. The maximum experimental neutron yield is 3×10^8 n/shot. The detailed information is presented in Table 4.22.

Using the model parameters obtained from the fitting of the current waveform, we carried out numerical experiments from 0.5 to 6.0 Torr to compare

the focus pinch times and neutron yields with the measured values. The comparisons are shown in Figures 4.28 and 4.29 respectively.

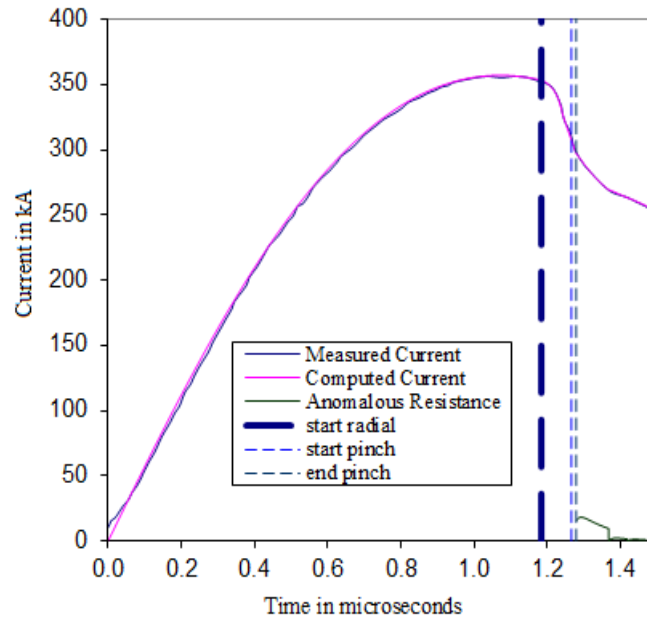


Figure 4.27 The measured current waveform of GN1 Plasma Focus machine at 30 kV, 3.0 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform (using anode radius 1.9 cm and length 8.7 cm) was obtained from the current derivative in Figure 2, page 22 of reference [31].

Table 4.20 Information obtained from Lee Model 6-phase code configured for GN1 Plasma Focus machine at 30 kV, 0.75 Torr deuterium gas.

Peak current (kA) = 357	Final pinch radius $r_{\min}$ (cm) = 0.29
Pinch start current (kA) = 261	Pinch length $z_{\max}$ (cm) = 2.8
Pinch minimum temperature (10^6 K) = 9.58	Pinch duration (ns) = 13.1
Pinch maximum temperature (10^6 K) = 10.3	Peak induced voltage (kV) = 58.1
Peak axial speed (cm/ μ s) = 11.6	Neutron yield ($\times 10^8$ n/shot) = 2.09
Peak radial shock speed (cm/ μ s) = 45.6	Energy Inflow into Plasma (%) = 16.6
Peak radial piston speed (cm/ μ s) = 32.1	Speed Factor ((kA/cm)/Torr ^{0.5}) = 108
Current per cm anode radius (kA/cm) = 188	

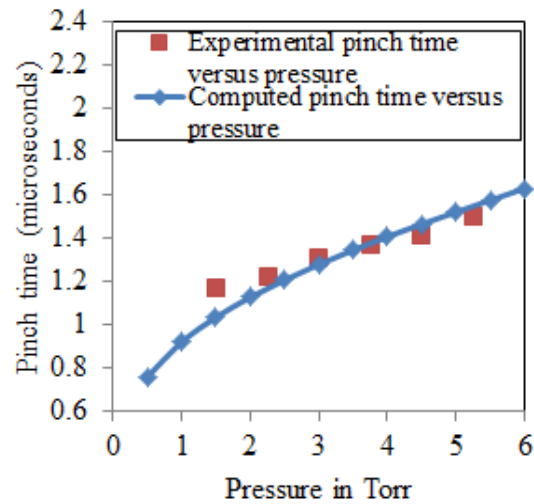


Figure 4.28 Variation of computed and experimental pinch time with respect to pressure operating at 30 kV (using anode radius 1.9 cm and length 8.7 cm) in deuterium for GN1 Plasma Focus machine.

Note: The measured experimental pinch time with respect to pressure was obtained from Figure 3, page 22 of reference [31].

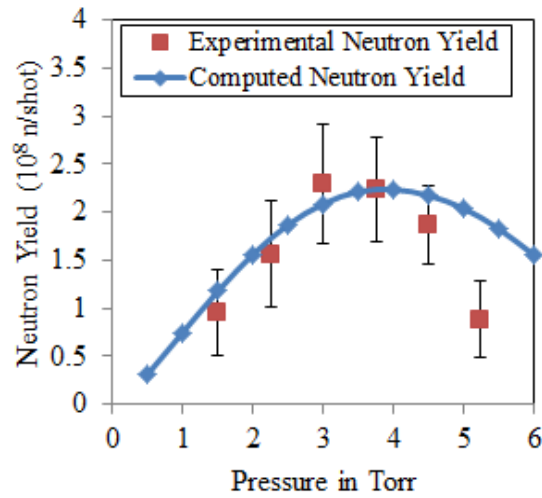


Figure 4.29 Variation of computed and experimental neutron yield with respect to pressure operating at 30 kV (using anode radius 1.9 cm and length 8.7 cm) in deuterium for GN1 Plasma Focus machine.

Note: The measured experimental neutron yield with respect to pressure was obtained from Figure 4, page 22 of reference [31].

From Figure 4.28, the computed time for start of pinch was less than the experimental values from 1.0 to 3.0 Torr whereas the computed time for start of pinch was about the same values for higher pressure.

From Figure 4.29, the computed values show that the optimum yield occurs at 4.0 Torr and produces a neutron yield of 2.24×10^8 n/shot, the energy input into plasma at this instant is 16.2 % whereas the experimental optimum neutron yield occurs at 3.0 Torr (4 mbar) with an average neutron yield of 2.31×10^8 n/shot (maximum is 3×10^8 n/shot as stated in the published results [31]). The agreement of computed and measured neutron yield versus pressure is very good as can be seen from Figure 4.29.

We now fit the GN1 Plasma Focus machine that has an anode length of 8.5cm, but using a capacitor of 12.6 μ F at a pressure of 3.0 Torr [32]. The machine parameters are shown in Table 4.23.

Table 4.23 The machine, operational and fitting parameters for GN1 Plasma Focus machine with 8.5 cm anode, 12.6 μ F capacitor.

Capacitance C_0 (μ F) = 12.6	Fill gas (molecular weight) = 4			
Static inductance L_0 (nH) = 52	Fill gas (atomic number) = 1			
Circuit resistance r_0 (m Ω) = 7	Fill gas (molecule) = 2			
Cathode radius ‘b’ (cm) = 3.6	The fitted model parameters are: Axial phase mass factor, f_m = 0.143 Axial phase current factor, f_c = 0.7 Radial phase mass factor, f_{mr} = 0.2 Radial phase current factor, f_{cr} = 0.7			
Anode radius ‘a’ (cm) = 1.9				
Anode length ‘z ₀ ’ (cm) = 8.5				
Charging voltage V_0 (kV) = 30				
Fill pressure P_0 (Torr) = 3.0				
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.01	0.05	50	1
Dip 2	0.05	0.05	50	1
Dip 3	0.02	0.02	100	10

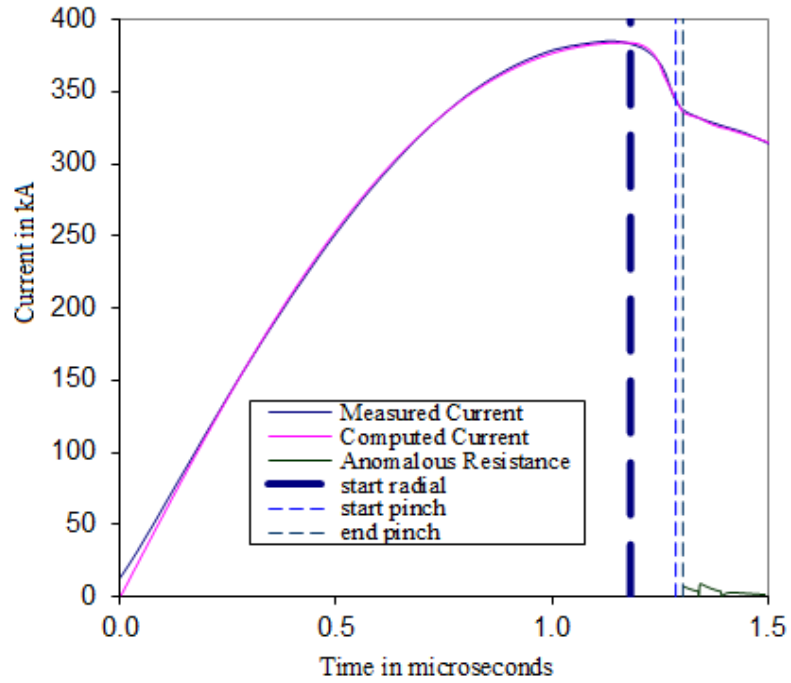


Figure 4.30 The measured current waveform of GN1 Plasma Focus machine at 30 kV, 3.0 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform (using anode radius 1.9 cm and length 8.5 cm, 12.6 μF capacitor) was obtained from the current derivative in Figure 3, page 123303-2 of reference [32].

Table 4.24 Information obtained from Lee Model 6-phase code configured for GN1 Plasma Focus machine at 30 kV, 3 Torr deuterium gas (using anode radius 1.9 cm and length 8.5 cm, 12.6 μ F capacitor).

Peak current (kA) = 384
 Pinch start current (kA) = 241
 Pinch minimum temperature (10^6 K) = 6.23
 Pinch maximum temperature (10^6 K) = 6.58
 Peak axial speed (cm/ μ s) = 11.6
 Peak radial shock speed (cm/ μ s) = 35.9
 Peak radial piston speed (cm/ μ s) = 24.7
 Final pinch radius $r_{\min}$ (cm) = 0.28
 Pinch length $z_{\max}$ (cm) = 2.9
 Pinch duration (ns) = 16.3
 Peak induced voltage (kV) = 44.5
 Neutron yield ($\times 10^8$ n/shot) = 2.33
 Energy Inflow into Plasma (%) = 13
 Speed Factor ((kA/cm)/Torr^{0.5}) = 117
 Current per cm anode radius (kA/cm) = 202

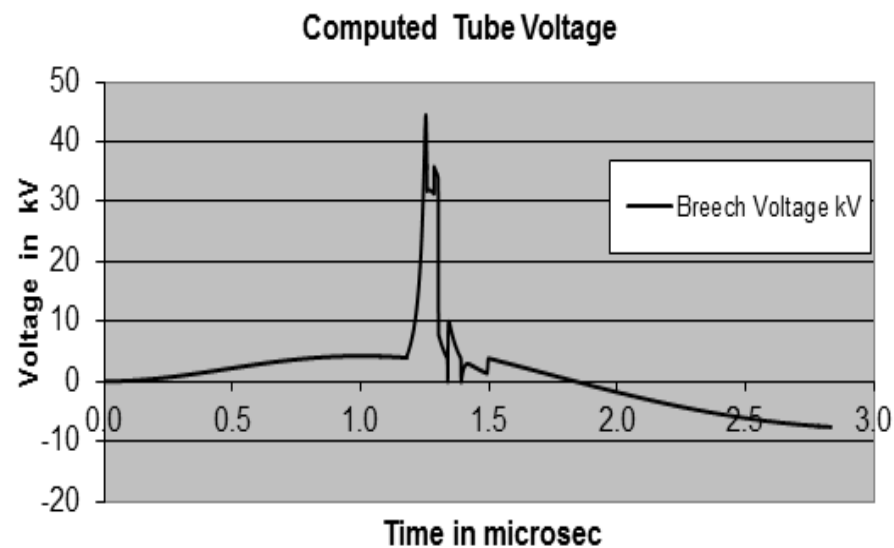


Figure 4.31 Computed tube voltage versus time as obtained from Lee Model 6-phase code configured for GN1 Plasma Focus machine at 30 kV, 3 Torr deuterium gas (using anode radius 1.9 cm and length 8.5 cm, 12.6 μ F capacitor).

Note: This waveform was obtained from Figure on sheet 2 on the Lee code worksheet.

The maximum computed current is 384 kA (compared with the published result of 360 kA) and exhibits the radial start time at 1.182 μs for pinch duration of 0.121 μs . It has a speed factor of 117 (kA/cm)/Torr^{0.5} and neutron yield of 2.33×10^8 n/shot. The detailed information is presented in Table 4.24.

From Lee Model 6-phase code configured for GN1 Plasma Focus machine with the above parameters, it can be noted that the computed plasma focus occurs at 1.3 μs as compared to the experimental dip that occurred at 1.34 μs .

The published result also calculates a tube voltage spike of 180 kV based on the formula $V_p = d(L_p(t)I(t))/dt$; where $I(t)$ is the total current circulating through the pinch and $L_p(t)$ represent the pinch inductance. This is different from the tube voltage estimated by Lee Model 6-phase code, which computed the tube voltage to be 45 kV as shown in Figure 4.31. Their large voltage spike is due to an overestimation of the radial speed calculation in the earlier version on Lee code.

The Lee Model 6-phase code is now configured 0.5 to 8.0 Torr for the GN1 Plasma Focus machine and the tube voltage was compared to the maximum pinch voltage calculation obtained from the paper [32].

When the two curves were compared, it was observed that the tube voltages computed from Lee Model 6-phase code is a factor of 4 smaller than the calculated published results for the range of 2.3 to 3.8 Torr as shown in Figure 4.32.

Figure 4.33 shows that the computed maximum neutron yield obtained was 2.57×10^8 n/shot which occurs at 4.25 Torr.

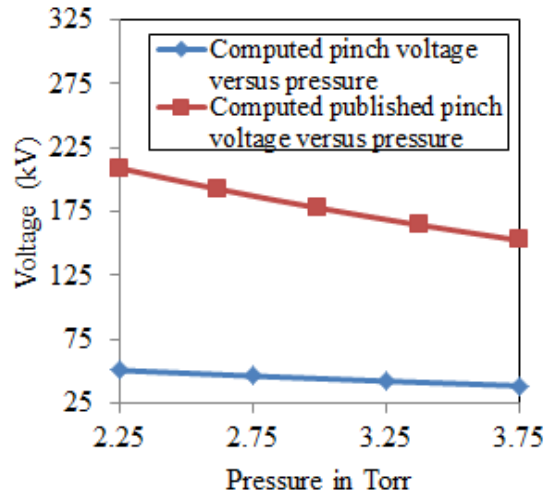


Figure 4.32 The maximum pinch voltage on the deuterium filling pressure (estimated in published paper) compared to maximum tube voltage versus pressure from Lee Model 6-phase code.

Note: The published results in Figure 9, page 123303-5, reference [32] are calculated based on the formula $V_p = d(L_p(t)I(t))/dt$ and are too high due to the use of a pre-1996 version of the Lee Model code.

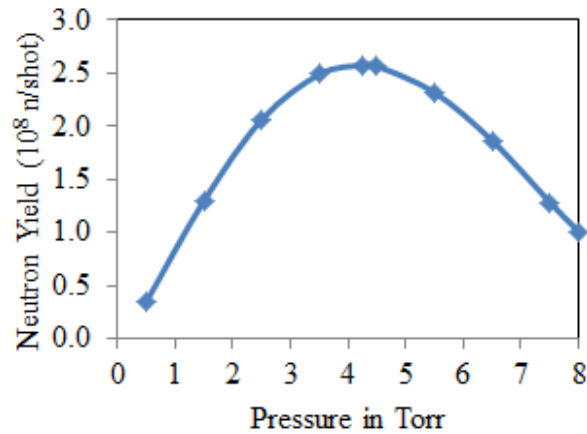


Figure 4.33 Variation of computed neutron yield with respect to pressure operating at 30 kV (using anode radius 1.9 cm and length 8.5 cm, 12.6 μ F capacitor) in deuterium for GN1 Plasma Focus machine.

4.9 FNII Plasma Focus machine (4.9 kJ - Type 1 machine - D gas)

The Fuego Nuevo II (FNII) is a Mather Type Dense Plasma Focus Device from Mexico with an anode of length 40 mm and diameter 50mm. It has a squirrel cage outer electrode (cathode) consisting of 12 copper bars, of 8 mm diameter and 36 mm long each, arrayed on a 100 mm diameter circle. FNII uses a 10 mm long Pyrex sleeve insulator, with an outer diameter of 50 mm [123].

The capacitor bank of the FNII consists of four 1.863 μF capacitors in parallel, with an internal inductance of 24 nH each. The inductance of the system measured by short-circuiting the gun was stated as 54 nH [123] but we found the static inductance value to be 69.5 nH from fitting using the Lee Model 5-phase code. We have published information of this Mather machine (which we took as a Type 1) operated at 36 kV at pressure of 2.75 Torr deuterium. From its measured dI/dt waveform we integrated for the current waveform which was then fitted to the Lee Model 5-phase code as shown in Figure 4.34. The machine, operational and fitting parameters is presented in Table 4.25. The information obtained from the fitting using the Lee code is presented in Table 4.26.

The Lee Model 5-phase code was then configured for pressures from 1.0 Torr to 5.0 Torr at 36 kV and the computed neutron yield versus pressure curve was compared to the measured curve as shown in Figure 4.35. It should be noted that the published values gave the neutron yield as neutron yield per steradian, but it also stated that the maximum yield of $(1.66 \pm 0.14) \times 10^8$ n/shot occurs at 2.75 Torr. Using this information, appropriate scaling factor was used resulting in Figure 4.35. The comparison reveals that the computed peak neutron yield is 1.89×10^8 n/shot (which occurs at 3.5 Torr) which is a factor of 1.14 compared to the measured peak neutron yield (which occurred at 2.75 Torr). The drop-off on both sides of maximum is more gradual for the computed curve than for the measured curve.

Table 4.25 The machine, operational and fitting parameters for FNII Plasma Focus machine.	Figure 4.34 Measured current waveform of FNII Plasma Focus machine at 36 kV, 2.75 Torr deuterium compared with Lee Model 5-phase code computed current waveform. Note: the measured current waveform was integrated from the di/dt waveform obtained from Figure 2 of reference [123].
Capacitance C_0 (μF) = 7.5 Static inductance L_0 (nH) = 69.5 Circuit resistance r_0 (mΩ) = 10 Cathode radius 'b' (cm) = 5 Anode radius 'a' (cm) = 2.5 Anode length 'z₀' (cm) = 4 Charging voltage V_0 (kV) = 36 Fill pressure P_0 (Torr) = 2.75 Fill gas (molecular weight) = 4 Fill gas (atomic number) = 1 Fill gas (molecule) = 2 The fitted model parameters are: Axial phase mass factor, $f_m = 0.145$ Axial phase current factor, $f_c = 0.7$ Radial phase mass factor, $f_{mr} = 0.145$ Radial phase current factor, $f_{cr} = 0.75$	

Table 4.26 Information obtained from Lee Model 5-phase code configured for FNII Plasma Focus machine at 36 kV, 2.75 Torr deuterium gas.

Peak current (kA) = 324
Pinch start current (kA) = 218
Pinch minimum temperature (10^6 K) = 4.38
Pinch maximum temperature (10^6 K) = 4.66
Peak axial speed (cm/ μ s) = 7.2
Peak radial shock speed (cm/ μ s) = 30.4
Peak radial piston speed (cm/ μ s) = 20.3
Final pinch radius $r_{\min}$ (cm) = 0.38
Pinch length $z_{\max}$ (cm) = 3.6
Pinch duration (ns) = 25.5
Peak induced voltage (kV) = 37.1
Neutron yield ($\times 10^8$ n/shot) = 1.77
Energy Inflow into Plasma (%) = 11.9
Speed Factor ((kA/cm)/Torr ^{0.5}) = 78
Current per cm anode radius (kA/cm) = 130

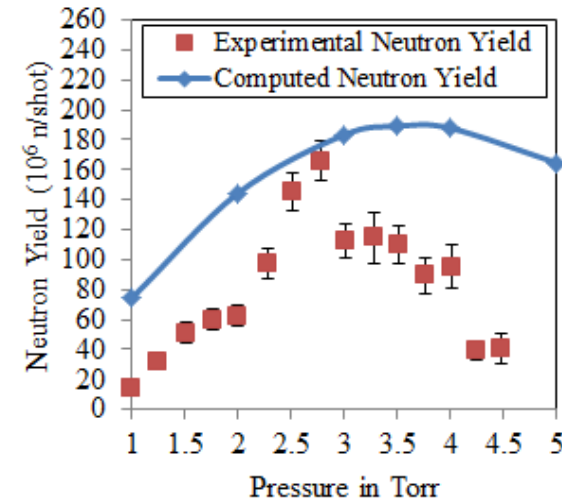


Figure 4.35 Variation of computed and experimental neutron yield with respect to pressure operating at 36 kV in deuterium for FNII Plasma Focus machine.

Note: The measured values were obtained from Figure 3 and multiplied by a suitable scaling factor to obtain the maximum yield of $(1.66 \pm 0.14) \times 10^8$ n/shot as stated in reference [123].

4.10 INDIA-ASSAM Plasma Focus machine (5.2 kJ - Type 2 machine - D gas)

The India-Assam Plasma Focus machine was originally designed as a 22 kJ device but is working at 5 kJ due to its spark-gap current-carrying capacity. This machine uses a hollow cylindrical anode. Its cathode rods are arranged in a squirrel cage fashion using 12 rods each having a diameter of 1 cm. A 4 cm long glass insulator having thickness 2 mm covers the anode at the closed end to separate the 2 electrodes. The machine parameters for this device are presented in Table 4.27.

Table 4.27 The machine, operational and fitting parameters for INDIA-ASSAM Plasma Focus machine.

Capacitance C_0 (μF) = 28.4	Fill gas (molecular weight) = 4																			
Static inductance L_0 (nH) = 113	Fill gas (atomic number) = 1																			
Circuit resistance r_0 (m Ω) = 4	Fill gas (molecule) = 2																			
Cathode radius ‘b’ (cm) = 5	The fitted model parameters are:																			
Anode radius ‘a’ (cm) = 2.5	Axial phase mass factor, $f_m = 0.5$																			
Anode length ‘z ₀ ’ (cm) = 20	Axial phase current factor, $f_c = 0.73$																			
Charging voltage V_0 (kV) = 19	Radial phase mass factor, $f_{mr} = 0.6$																			
Fill pressure P_0 (Torr) = 0.4	Radial phase current factor, $f_{cr} = 0.9$																			
The anomalous resistance parameters are:																				
	<table><tr><th>$R_0(\Omega)$</th><th>Characteristic of fall time τ_2 (ns)</th><th>Characteristic of rise time τ_1 (ns)</th><th>End fraction time</th></tr><tr><td>Dip 1</td><td>0.35</td><td>120</td><td>10</td><td>1</td></tr><tr><td>Dip 2</td><td>0.01</td><td>100</td><td>10</td><td>1</td></tr><tr><td>Dip 3</td><td>0.02</td><td>200</td><td>10</td><td>1</td></tr></table>	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time	Dip 1	0.35	120	10	1	Dip 2	0.01	100	10	1	Dip 3	0.02	200	10	1
$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time																	
Dip 1	0.35	120	10	1																
Dip 2	0.01	100	10	1																
Dip 3	0.02	200	10	1																

Its static inductance is 113 nH as stated in the reference report [127]. The mass factor f_m and the radial current factor f_{cr} are unusually high for this plasma machine. We speculate this may be because the dimension of the electrodes used in this machine was not designed for 5 kJ storage energy but for 22 kJ as indicated in

the published report [127]. The detailed information obtained from the Lee Model 6-phase code is presented in Table 4.28.

Table 4.28 Information obtained from Lee Model 6-phase code configured for INDIA-ASSAM Plasma Focus machine at 19 kV, 0.4 Torr deuterium gas.

Peak current (kA) = 259	Pinch radius $r_{\min}$ (cm) = 0.37
Pinch start current (kA) = 198	Pinch length $z_{\max}$ (cm) = 4.5
Pinch minimum temperature (10^6 K) = 6.07	Pinch duration (ns) = 21.7
Pinch maximum temperature (10^6 K) = 6.38	Peak induced voltage (kV) = 33.3
Peak axial speed (cm/ μ s) = 9	Neutron yield ($\times 10^8$ n/shot) = 1.22
Peak radial shock speed (cm/ μ s) = 35.2	Energy Inflow into Plasma (%) = 16.8
Peak radial piston speed (cm/ μ s) = 24.8	Speed Factor ((kA/cm)/Torr ^{0.5}) = 164
Current per cm anode radius (kA/cm) = 104	

Using the model parameters from the fitted current waveform, Lee Model 6-phase code was configured from 0.2 to 0.6 Torr and the results are compared with the published values which reveals that the maximum computed neutron yield was 1.25×10^8 n/shot at 0.35 Torr deuterium as shown in Figure 4.37. The energy input into the plasma for this maximum yield shot was 17 %. This compares with measured maximum yield of 3.3×10^8 n/shot at 0.4 Torr; a difference by a factor of less than 3. It should be noted that the published results only shows the best shot at each pressure and not the average value with standard deviations or error bars to show neutron emission fluctuations from shot to shot.

Another observation is that the measured neutron yield has a steeper gradient on both side of the optimum yield whereas the computed neutron yield curve shows much gentler gradients.

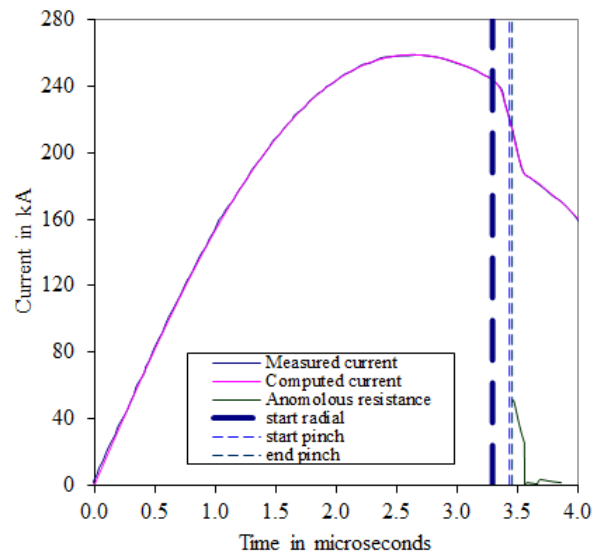


Figure 4.36 The measured current waveform of India Assam Plasma Focus machine working at 19 kV deuterium compared with Lee Model 6-phase code computed current waveform.

Note: The measured waveform was obtained from the current derivative in Figure 2, page 101 of reference [127].

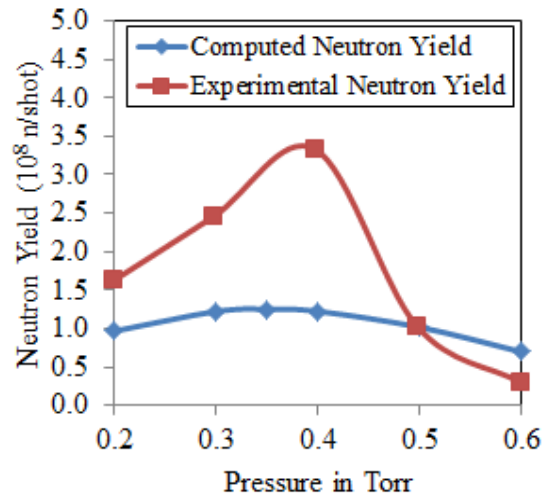


Figure 4.37 Variation of computed and experimental neutron yield with respect to pressure operating at 19 kV in deuterium for India Assam Plasma Focus machine.

Note: The experimental values were taken from Figure 4, page 102 of reference [127].

4.11 NX3 Plasma Focus machine (9.8 kJ – Type 1 machine showing Type 2 characteristics - D gas)

NX3 is a newly developed dense plasma focus device at the Plasma Radiation Source Laboratory, NIE, Nanyang Technological University, Singapore. This Mather's type machine was operated at different voltages and fired in different pressures in deuterium gas during its testing phase.

Its static inductance L_0 is 49 nH. We have several current waveforms and Lee Model 5-phase code was used to fit the measured current waveforms. We found that in several shots which are not time-matched (the start of radial phase is significantly different from the quarter period time) the measured current dip extends significantly beyond the computed current dip despite our stretching the model parameters. In each of these cases the 6-phase code was used which then gives us a complete fit. Table 4.29 below shows the machine parameters for the NX3 plasma focus machine.

Table 4.29 The machine, operational and fitting parameters for NX3 Plasma Focus machine operated at different voltages and different pressures in deuterium gas.

Capacitance C_0 (μF) = 100	Anode length ' z_0 ' (cm) = 16
Static inductance L_0 (nH) = 49	Fill gas (molecular weight) = 4
Circuit resistance r_0 ($\text{m}\Omega$) = 2.2	Fill gas (atomic number) = 1
Cathode radius ' b ' (cm) = 5.1	Fill gas (molecule) = 2
Anode radius ' a ' (cm) = 2.6	

4.11.1 The current waveform fit at 12 kV, 2.28 Torr D gas

For this shot the computed and measured current waveforms show a good fit when Lee Model 5-phase code was used with the mass and current factors as shown in Table 4.30.

Table 4.30 Current and mass factor for the fitting of the computed waveform to the measured current waveform at 12 kV, 2.28 Torr deuterium gas.

Axial phase mass factor, $f_m = 0.18$	Radial phase mass factor, $f_{mr} = 0.1$
Axial phase current factor, $f_c = 0.7$	Radial phase current factor, $f_{cr} = 0.7$

The maximum computed current was 392 kA and exhibits the radial start time at 2.94 μs for pinch duration of 0.14 μs . It has a speed factor of 100 (kA/cm)/Torr^{0.5} and a neutron yield of 1.43×10^8 n/shot. The detailed information is shown in Table 4.34.

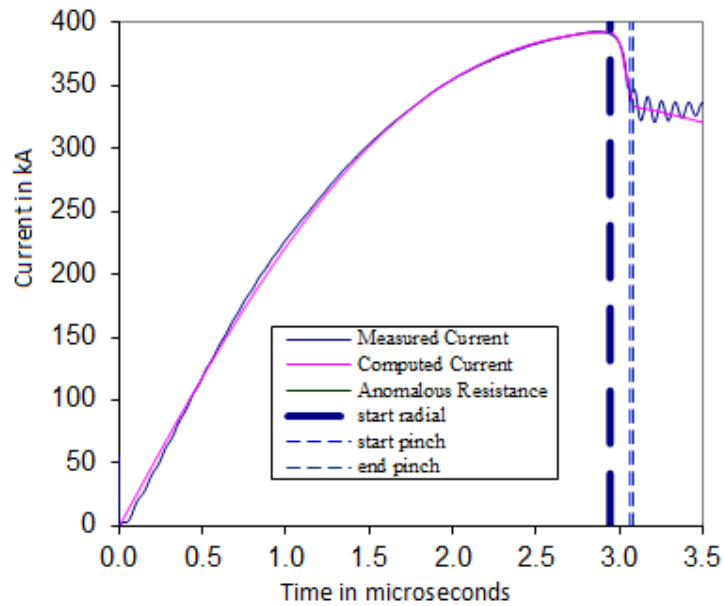


Figure 4.38 Measured current waveform of NX3 at 12 kV, 2.28 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform ($f_m = 0.18$, $f_c = 0.7$, $f_{mr} = 0.1$ and $f_{cr} = 0.7$).

4.11.2 The current waveform fit at 12 kV, 3.75 Torr D gas

This shot cannot be completely fitted with the 5-phase code because of its dip as shown in Figure 4.39. The computed and measured waveforms show a good fit with Lee Model 6-phase code using the mass and current factors as shown in Table 4.31.

Table 4.31 Current and mass factors for the fitting of the computed waveform at 12 kV, 3.75 Torr deuterium gas and anomalous resistance values.

Axial phase mass factor, $f_m = 0.13$		Radial phase mass factor, $f_{mr} = 0.1$		
Axial phase current factor, $f_c = 0.7$		Radial phase current factor, $f_{cr} = 0.7$		
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.28	100	10	1
Dip 2	0.04	100	10	1
Dip 3	0.04	100	10	1

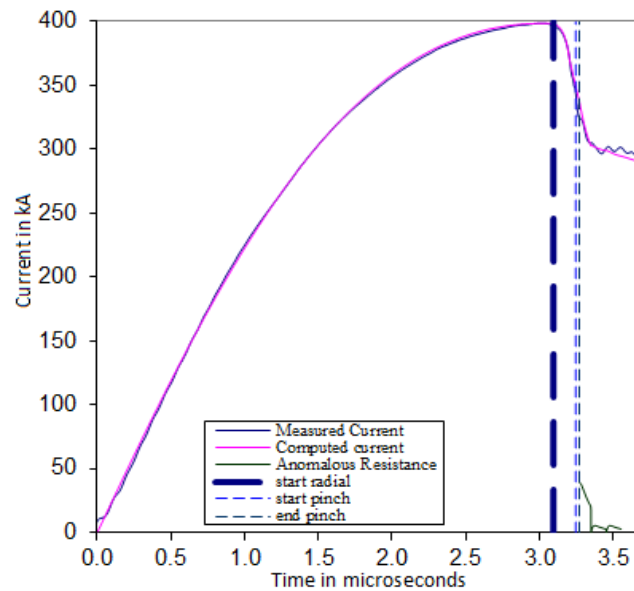


Figure 4.39 Measured current waveform of NX3 at 12 kV, 3.75 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform ($f_m = 0.13$, $f_c = 0.7$, $f_{mr} = 0.1$ and $f_{cr} = 0.7$).

The maximum computed current is 399 kA and exhibits the radial start time at 3.095 μs for pinch duration of 0.175 μs . It has a speed factor of 79 (kA/cm)/Torr^{0.5} and computed neutron yield of 2.08×10^8 n/shot. The detailed information is shown in Table 4.34.

4.11.3 The current waveform fit at 13 kV, 3.0 Torr D gas

This shot also requires the 6-phase code for a complete fit because of its dip as shown in Figure 4.40. The computed and measured waveforms show a good fit using the mass and current factors as shown in Table 4.32.

Table 4.32 Current and mass factors for the fitting of the computed waveform at 13 kV, 3.0 Torr deuterium gas; and anomalous resistance values.

Axial phase mass factor, $f_m = 0.195$		Radial phase mass factor, $f_{mr} = 0.1$		
Axial phase current factor, $f_c = 0.7$		Radial phase current factor, $f_{cr} = 0.7$		
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.05	100	10	1
Dip 2	0.03	100	10	1
Dip 3	0.02	100	10	1

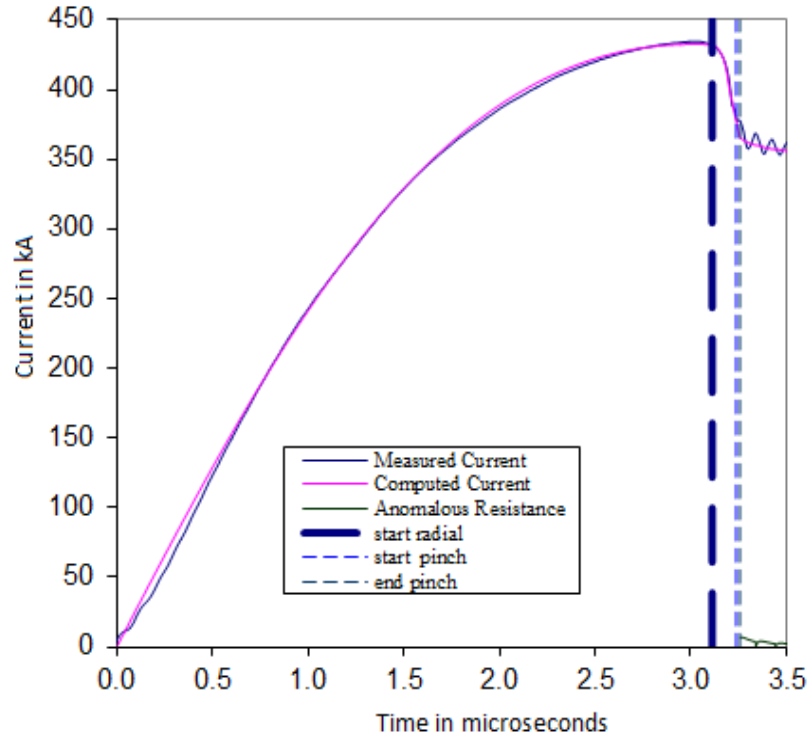


Figure 4.40 Measured current waveform of NX3 at 13 kV, 3.0 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform ($f_m = 0.195$, $f_c = 0.7$, $f_{mr} = 0.1$ and $f_{cr} = 0.7$).

The maximum computed current is 433 kA and exhibits the radial start time at 3.112 μs for pinch duration of 0.145 μs . It has a speed factor of 96 (kA/cm)/Torr^{0.5} and computed neutron yield of 2.33×10^8 n/shot. The detailed information is presented in Table 4.34.

4.11.4 The current waveform fit at 14 kV, 3.75 Torr D gas

This shot also requires the 6-phase code for a complete fit because of its dip as shown in Figure 4.41.

Using the mass and current factors as shown in Table 4.33, the computed neutron yield obtained is 3.43×10^8 n/shot. The detailed information is presented in Table 4.34.

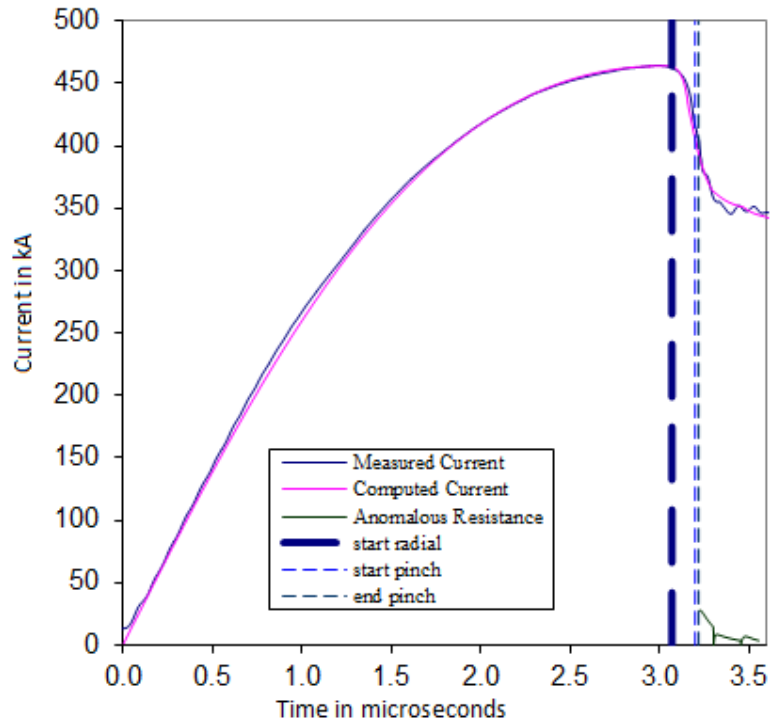


Figure 4.41 Measured current waveform of NX3 at 13 kV, 3.0 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform ($f_m = 0.172$, $f_c = 0.7$, $f_{mr} = 0.1$ and $f_{cr} = 0.7$).

Table 4.33 Current and mass factors for the fitting of the computed waveform at 14 kV, 3.75 Torr deuterium gas; and anomalous resistance values.

Axial phase mass factor, $f_m = 0.172$		Radial phase mass factor, $f_{mr} = 0.1$		
Axial phase current factor, $f_c = 0.7$		Radial phase current factor, $f_{cr} = 0.7$		
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.2	100	10	1
Dip 2	0.05	150	15	1
Dip 3	0.05	100	10	1

Table 4.34 Computed pinch parameters for NX3 Plasma Focus machine working at different pressures and charge voltages.

Deuterium gas pressure (Torr)	2.3	3.8	3.0	3.8
Capacitor charged voltage (kV)	12	12	13	14
Peak current (kA)	392	399	433	464
Pinch start current (kA)	241	244	264	284
Pinch minimum temperature (10^6 K)	8.61	5.39	7.89	7.28
Pinch maximum temperature (10^6 K)	9.2	5.76	8.43	7.78
Peak axial speed (cm/ μ s)	8.7	8.2	8.1	8.3
Peak radial shock speed (cm/ μ s)	43	34.1	41.2	39.6
Peak radial piston speed (cm/ μ s)	30.8	24.3	29.6	28.4
Final pinch radius $r_{\min}$ (cm)	0.39	0.39	0.4	0.39
Pinch length $z_{\max}$ (cm)	3.6	3.6	3.6	3.6
Pinch duration (ns)	19	24	19.8	20.6
Peak induced voltage (kV)	47	37.7	49.3	50.9
Neutron yield ($\times 10^8$ n/shot)	1.43	2.08	2.33	3.43
Energy Inflow into Plasma (%)	16.3	17	17	16.8
Speed Factor ((kA/cm)/Torr ^{0.5})	100	79	96	92
Current per cm anode radius (kA/cm)	151	153	166	178

Lee Model 6-phase code was then configured for the NX3 to work from 0.5 to 10.0 Torr at 12 kV. Figure 4.42 and 4.43 show the respective computed neutron yield variation with pressure and the percentage of energy that is pumped into the plasma (during the radial phase) for this range of pressures. This variation of percentage of energy could be due to the variation of speed of the reflected shock wave that hits the inward moving piston. This causes the variation in the pinch (compression) and thus the amount of energy that is converted into plasma.

From these 2 graphs, it is observed that the maximum neutron yield occurs at 5.5 Torr at 2.37×10^8 n/shot. At this pressure the input energy into the radial phase plasma compression is 18.7 % just below the optimum radial input energy which occurs at 6.0 Torr.

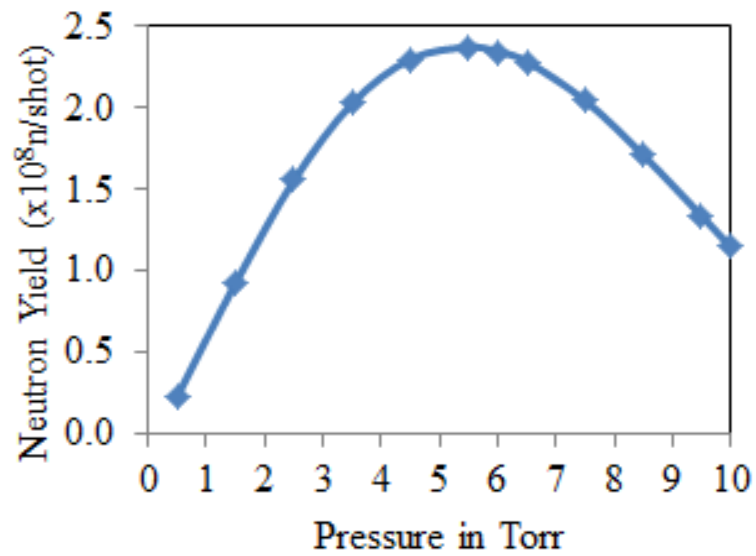


Figure 4.42 Variation of computed neutron yield with respect to pressure operating at 12 kV in deuterium for NX3 Plasma Focus machine.

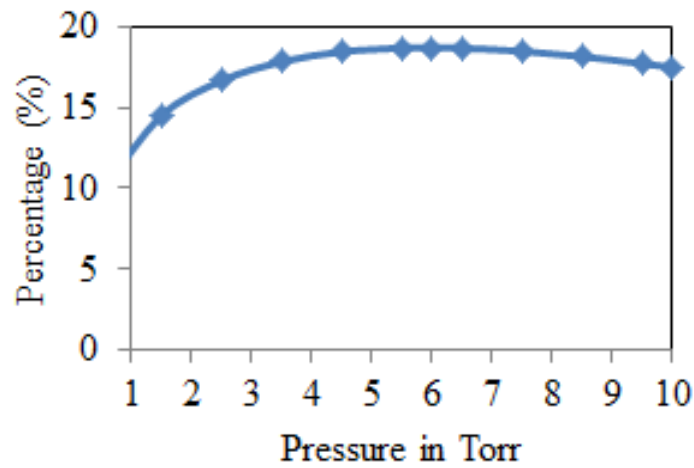


Figure 4.43 Variation of computed energy input into plasma with respect to pressure operating at 12 kV in deuterium for NX3 Plasma Focus machine.

The Lee Model 6-phase code is now configured for the NX3 to work at 14 kV (1.0-15.0 Torr) and the neutron yield and energy input into plasma are then plotted as shown in Figure 4.44 and 4.45.

From these graphs, it is observed that the maximum neutron yield occurs at 8.0 Torr and the computed yield is 5.22×10^8 n/shot. The energy input into the radial phase for this shot is computed as 18.7 %.

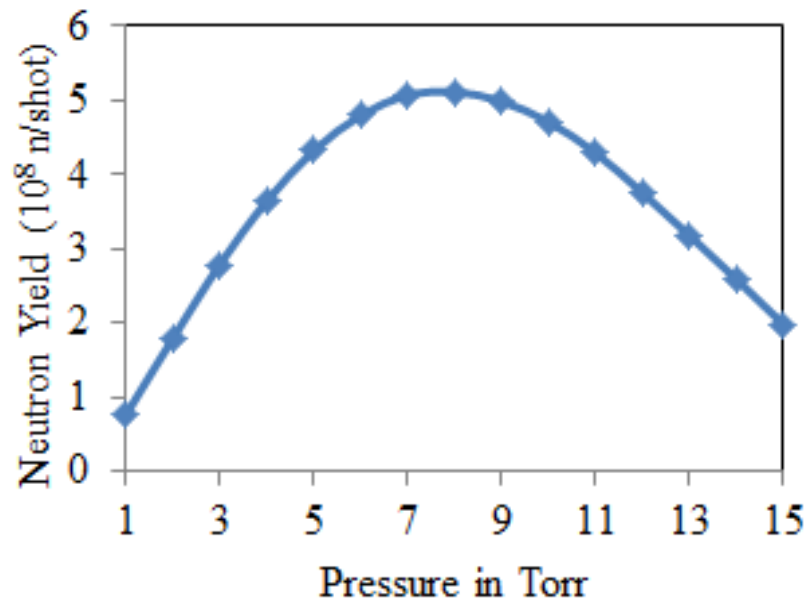


Figure 4.44 Variation of computed neutron yield with respect to pressure operating at 14 kV in deuterium for NX3 Plasma Focus machine.

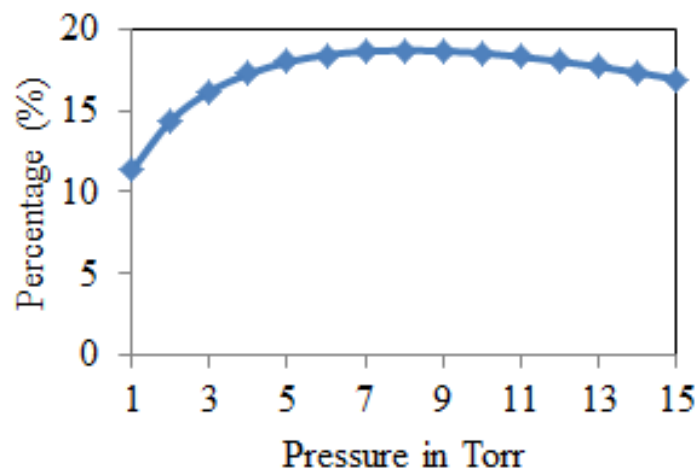


Figure 4.45 Variation of computed energy input into plasma with respect to pressure operating at 14 kV in deuterium for NX3 Plasma Focus machine.

It should also be noted that the dimension of the tube parameters (anode and cathode radii and length) will be changed as NX3 undergo its process of optimisation

4.12 BARC Plasma Focus 2 machine (11.5 kJ - Type 1 machine - D gas)

The parameters of the second plasma focus machine from India Bhabha Atomic Research Centre (BARC Plasma Focus 2 machine) were obtained from Dr Rabindra Kumar Rout [22] from BARC. These parameters are shown in Table 4.35. Its static inductance is 77 nH. The 5-phase model was used to fit the current waveform of this machine.

The detailed information obtained from Lee Model 5-phase code is shown in Table 4.36 and the fitting is shown in Figure 4.46. Using the model parameters obtained from fitting the computed current waveform to the measured current waveform, the Lee Model 5-phase code was then configured to work from 1.0 Torr to 14.0 Torr and the computed neutron yield versus pressure curve was compared to the experimental neutron yield versus pressure obtained from Dr Rabindra Kumar Rout [22]. The comparison is shown in Figure 4.47.

From Figure 4.47, it is observed that the maximum neutron yield computed was 1.63×10^9 n/shot at 7.0 Torr whereas the maximum neutron yield measured (obtained from the best 10 shots) was 1.58×10^9 n/shot at 3.0 Torr. Both the experimental and computed yields show a gradual increase to its peak value before a gradual decrease after this value.

Table 4.35 The machine, operational and fitting parameters for the BARC Plasma Focus 2 machine. The information was obtained from Dr Rabindra Kumar Rout.

Capacitance C_0 (μF) = 40
Static inductance L_0 (nH) = 77
Circuit resistance r_0 ($\text{m}\Omega$) = 4
Cathode radius 'b' (cm) = 6.1
Anode radius 'a' (cm) = 3
Anode length 'z ₀ ' (cm) = 7.7
Charging voltage V_0 (kV) = 24
Fill pressure P_0 (Torr) = 3.0
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.22$
Axial phase current factor, $f_c = 0.7$
Radial phase mass factor, $f_{mr} = 0.22$
Radial phase current factor, $f_{cr} = 0.72$

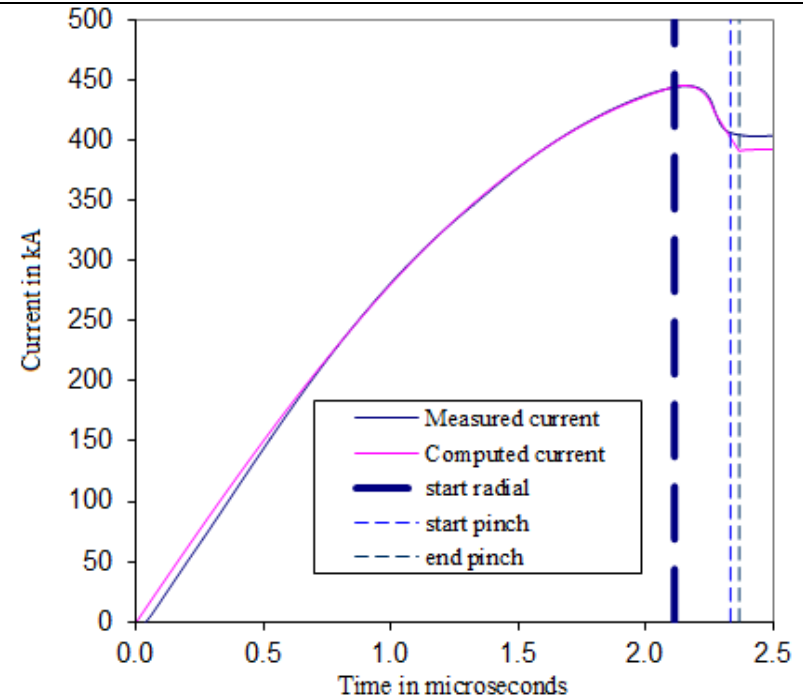


Figure 4.46 The measured current waveform of BARC Plasma Focus 2 machine at 24 kV, deuterium gas compared with Lee Model 5-phase code computed current for deuterium gas.

Note: The measured current waveform information was obtained from Dr Rabindra Kumar Rout from BARC [22].

Table 4.36 Information obtained from Lee Model 5-phase code configured for the BARC Plasma Focus 2 machine at 24 kV, 3.0 Torr deuterium gas.

Peak current (kA) = 446
Pinch start current (kA) = 302
Pinch minimum temperature (10^6 K) = 3.54
Pinch maximum temperature (10^6 K) = 3.75
Peak axial speed (cm/ μ s) = 6.3
Peak radial shock speed (cm/ μ s) = 27.2
Peak radial piston speed (cm/ μ s) = 18.5
Final pinch radius $r_{\min}$ (cm) = 0.45
Pinch length $z_{\max}$ (cm) = 4.6
Pinch duration (ns) = 34.1
Peak induced voltage (kV) = 44.4
Neutron yield ($\times 10^9$ n/shot) = 1.04
Energy Inflow into Plasma (%) = 13.1
Speed Factor ((kA/cm)/Torr $^{0.5}$) = 86
Current per cm anode radius (kA/cm) = 149

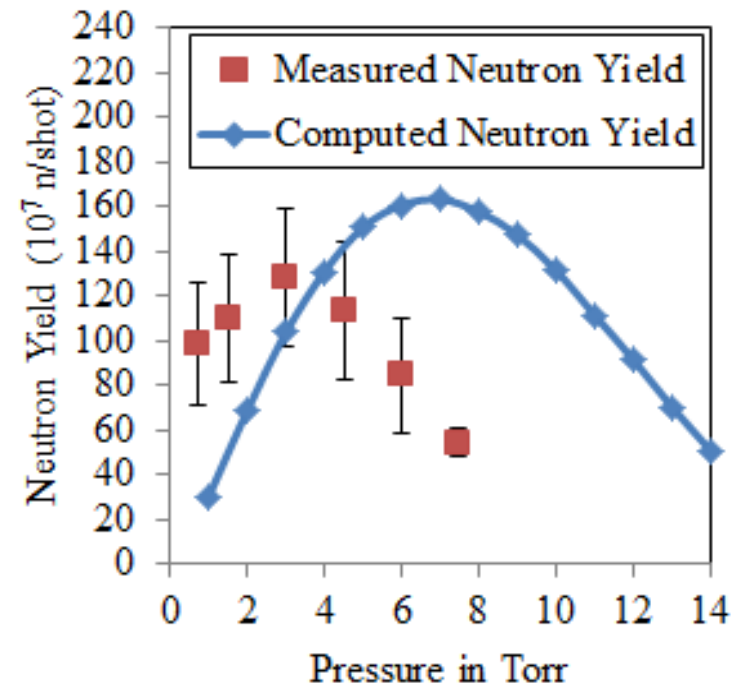


Figure 4.47 Variation of computed and experimental neutron yield with respect to pressure operating at 24 kV in deuterium for BARC Plasma Focus 2.

4.13 PF143 Plasma Focus machine (20 kJ - Type 1 machine - D gas)

The machine parameter for the PF143 Plasma Focus machine from Japan is given in Table 4.37 [125].

Its static inductance found from Lee Model 5-phase code is 45 nH, and we fit the measured current waveform at 1.5 Torr. The Lee Model 6-phase code was used to complete the fit as shown in Figure 4.48.

It should be noted that the anode length stated was 242 mm [106] but its effective length found from Lee code is 202 mm. The maximum computed current is 619 kA and exhibits the radial start time at 1.763 μ s for pinch duration of 0.157 μ s. It has a speed factor of 202 (kA/cm)/Torr^{0.5} and neutron yield of 3.5×10^8 n/shot. The detailed information is presented in Table 4.38.

The published data includes another shot at 9.75 Torr. We collect the machine parameters for this shot in Table 4.37. The current fitting of the measured to the computed waveform is presented in Figure 4.49.

The maximum computed current is 709 kA and exhibits the radial start time at 2.641 μ s for pinch duration of 0.225 μ s. It has a speed factor of 91(kA/cm)/Torr^{0.5} and neutron yield of 5.84×10^8 n/shot. The low neutron yield is because it is not operating at optimum condition. The detailed information is presented in Table 4.39.

We use the information obtained from the current waveform and take the compression time of plasma from the peak value to the end of the radial phase as defined and given in the published paper [125] shown in Figure 4.50. Lee Model 6-phase code was configured to work for 0.5 to 10.5 Torr and the computed compression times (following their definition as shown in Figure 4.50) are compared to the experimental compression times in Figure 4.51.

Table 4.37 The machine, operational and fitting parameters for PF143 Plasma Focus machine working at 1.5 Torr and 9.75 Torr respectively.

Capacitance C_0 (μF)	44.8	44.8
Static inductance L_0 (nH)	45	45
Circuit resistance r_0 ($\text{m}\Omega$)	3	3
Cathode radius 'b' (cm)	5	5
Anode radius 'a' (cm)	2.5	2.5
Anode length ' z_0 ' (cm)	20.2	20.2
Charging voltage V_0 (kV)	30	30
Fill pressure P_0 (Torr)	1.5	9.75
Fill gas (molecular weight)	4	4
Fill gas (atomic number)	1	1
Fill gas (molecule)	2	2
The fitted model parameters are:		
Axial phase mass factor, f_m	0.165	0.1
Axial phase current factor, f_c	0.7	0.7
Radial phase mass factor, f_{mr}	0.25	0.1
Radial phase current factor, f_{cr}	0.45	0.5
The anomalous resistance parameters are:		
	$R_0(\Omega)$	
	Characteristic of fall time, τ_2 (ns)	
	Characteristic of rise time, τ_1 (ns)	
	End fraction time	
Dip 1	0.65	0.2
Dip 2	0.75	0.28
Dip 3	0.5	0.3
Pressure	1.5	9.75

Table 4.38 Information obtained from Lee Model 6-phase code configured for PF143 Plasma Focus machine at 30 kV, 1.5 Torr deuterium gas.

Peak current (kA) = 619
Pinch start current (kA) = 260
Pinch minimum temperature (10^6 K) = 6.8
Pinch maximum temperature (10^6 K) = 7.07
Peak axial speed (cm/ μ s) = 18.1
Peak radial shock speed (cm/ μ s) = 36.6
Peak radial piston speed (cm/ μ s) = 24.8
Final pinch radius $r_{\min}$ (cm) = 0.36
Pinch length $z_{\max}$ (cm) = 3.9
Pinch duration (ns) = 20.6
Peak induced voltage (kV) = 46.7
Neutron yield ($\times 10^8$ n/shot) = 3.5
Energy Inflow in Plasma (in %) = 12.8
Speed Factor ((kA/cm)/Torr ^{0.5}) = 202
Current per cm anode radius (kA/cm) = 248

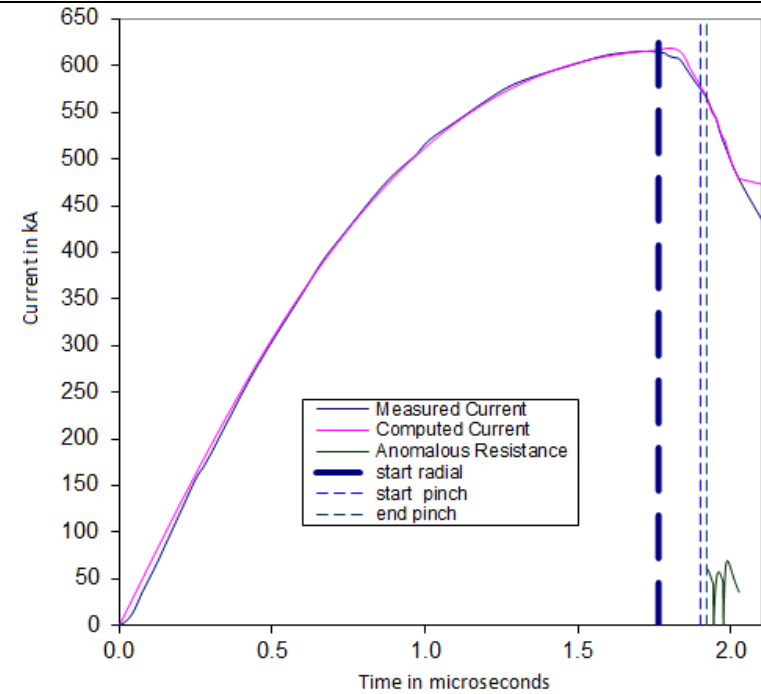


Figure 4.48 The measured current waveform of PF143 Plasma Focus machine at 30 kV, 1.5 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured waveform was obtained from the current derivative in Figure 1, page 250 of reference [125].

Table 4.39 Information obtained from Lee Model 6-phase code configured for PF143 Plasma Focus machine at 30 kV, 9.75 Torr deuterium gas.

Peak current (kA) = 709
Pinch start current (kA) = 284
Pinch minimum temperature (10^6 K) = 3.03
Pinch maximum temperature (10^6 K) = 3.23
Peak axial speed (cm/ μ s) = 10.8
Peak radial shock speed (cm/ μ s) = 25.4
Peak radial piston speed (cm/ μ s) = 17.8
Final pinch radius $r_{\min}$ (cm) = 0.38
Pinch length $z_{\max}$ (cm) = 3.5
Pinch duration (ns) = 30.6
Peak induced voltage (kV) = 31.5
Neutron yield ($\times 10^8$ n/shot) = 5.84
Energy Inflow into Plasma (%) = 17.6
Speed Factor ((kA/cm)/Torr ^{0.5}) = 91
Current per cm anode radius (kA/cm) = 283

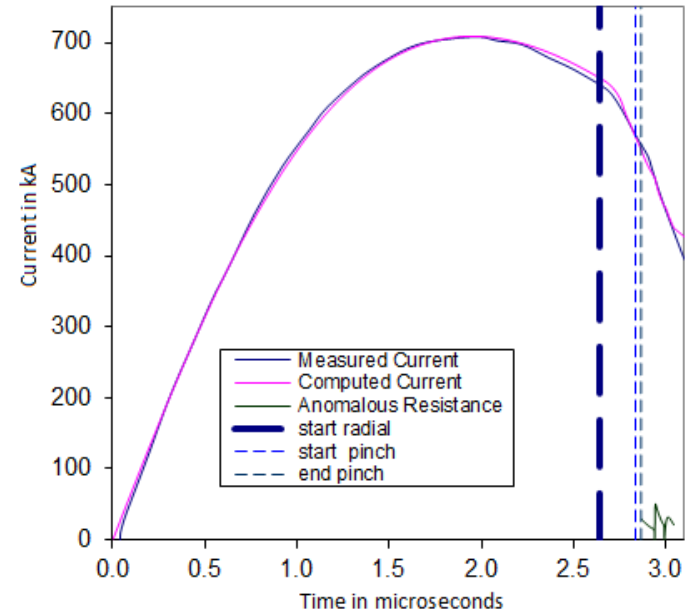


Figure 4.49 The measured current waveform of PF143 Plasma Focus machine at 30 kV, 9.75 Torr deuterium gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured waveform was obtained from the current derivative in Figure 1, page 250 of reference [125].

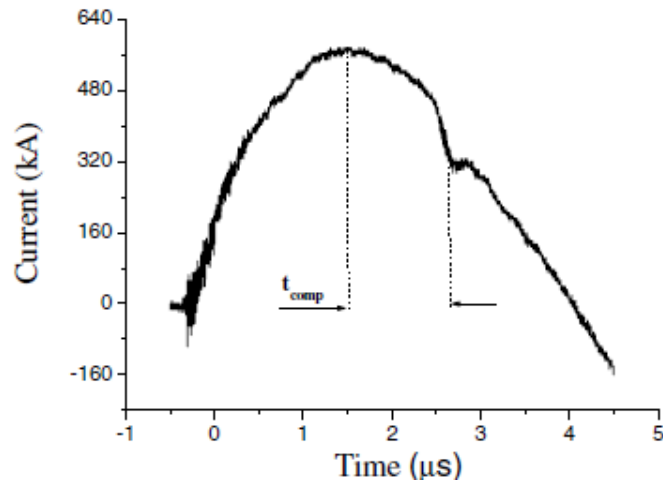


Figure 4.50 The compression time of plasma from the peak value to the end of the radial phase as defined in Figure 1, page 250 of reference [125].

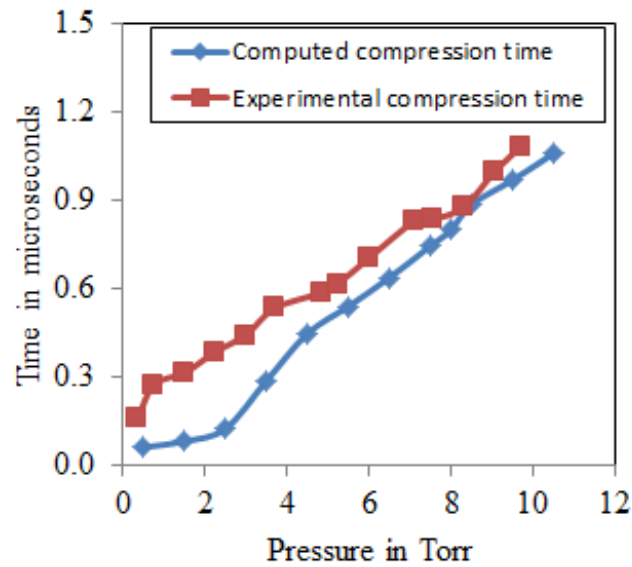


Figure 4.51 Variation of computed and experimental compression time with respect to pressure operating at 30 kV in deuterium for PF143.

Note: The measured waveform was obtained from the current derivative in Figure 2, page 25 of reference [125].

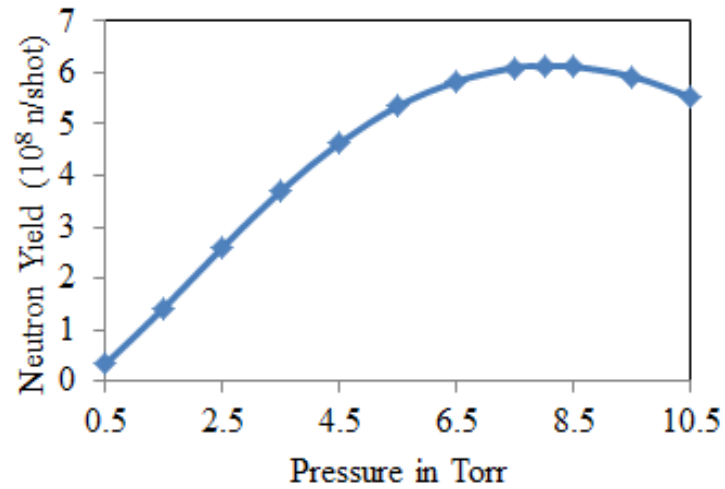


Figure 4.52 Variation of computed neutron yield with respect to pressure operating at 30 kV in deuterium for PF143.

From Figure 4.51, it can be observed that the computed results and the experimental results are very close to each other as both of them decays almost linearly as pressure is reduced even though the experimental curve is comparatively much higher at lower pressure end. The slope of the computed curve also follows closely that of the experimental curve. Using the current waveform, the neutron yield was computed for the pressure range stated above and the results are shown in Figure 4.52. From Figure 4.52, the maximum neutron yield occurs at 8.0 Torr with 6.12×10^8 n/shot. The energy pumped into the plasma by the compression is 17.3 %.

4.14 DPF78 Plasma Focus machine (31 kJ - Type 1 machine - D gas)

DPF78 is a Mather type plasma focus device (60 kV, 28 kJ) which was operated at Stuttgart University Stuttgart, Germany. It had the following geometrical parameters: the anode diameter = 50 mm, the cathode diameter = 100 mm, the length of the electrodes = 150 mm. The initial breakdown forms the current sheath on the surface of the cylindrical glass insulator (50 mm long).

This Mather's type machine was charged to 60 kV at a pressure of 7.6 Torr deuterium gas. Its static inductance L_0 is estimated as 55 nH, and we treat it as a Type 1 machine; using Lee Model 5-phase code to fit this current waveform as shown in Figure 4.53.

From Lee Model 5-phase code, the computed maximum current obtained was 859 kA. The radial phase starts at 1.401 μ s and the pinch duration is 0.155 μ s. It has a speed factor of 125 (kA/cm)/Torr^{0.5} and a computed neutron yield of 3.8×10^9 n/shot. The detailed information is shown in Table 4.40.

We proceed to compare the computed and measured peak currents following the work of S Lee et al [128].

According to S Lee et al [128], the ratio I_p / I_{total} was digitized from Figure 4.54 and plotted as a function of time as shown in Figure 4.55. It should be noted that I_p is the plasma sheath current at the time of pinch. At time = 1.551 μ s, the ratio found was 0.49 (I_{total} measured was 778 kA, whereas $I_{pinch} = 381.2$ kA). The computed I_{pinch} was 4 % larger than the measured I_{pinch} . This difference was expected since f_{cr} used in Lee Model 5-phase code has an average value of 0.51; while the experimental measurement shown in Figure 4.55 has the radial phase I_p / I_{total} varying from 0.63 to 0.4.

Table 4.40 The machine, operational and fitting parameters for DPF78 Plasma Focus machine.

Capacitance C_0 (μF) = 17.2
Static inductance L_0 (nH) = 55
Circuit resistance r_0 (m Ω) = 3.5
Cathode radius 'b' (cm) = 5
Anode radius 'a' (cm) = 2.5
Anode length 'z ₀ ' (cm) = 13.7
Charging voltage V_0 (kV) = 60
Fill pressure P_0 (Torr) = 7.6
Fill gas(molecular weight) = 4
Fill gas(atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.061$
Axial phase current factor, $f_c = 0.6$
Radial phase mass factor, $f_{mr} = 0.16$
Radial phase current factor, $f_{cr} = 0.6$

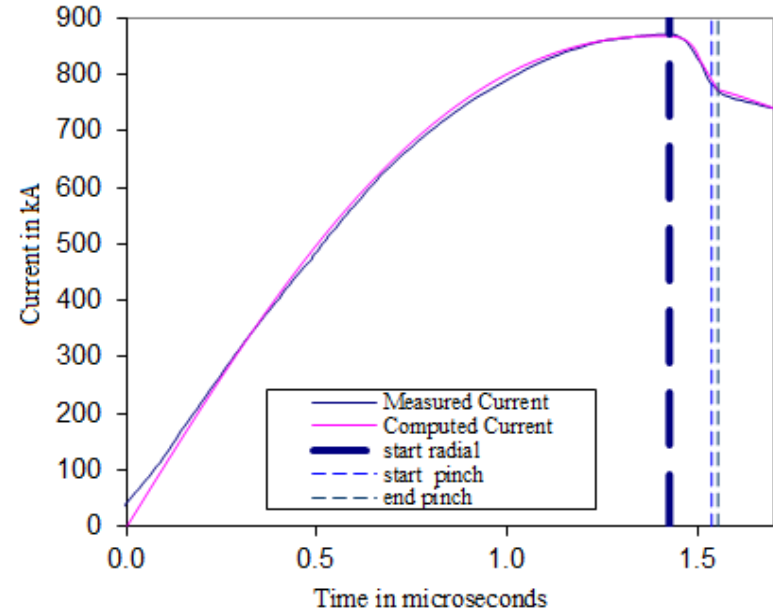


Figure 4.53 Measured current waveform of DPF78 at 60 kV, 7.6 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: the digitalized measured current waveform was downloaded from <http://www.plasmafocus.net> [57] but f_m , f_c , f_{mr} and f_{cr} has been changed for better axial phase fit.

Table 4.41 Information obtained from Lee Model 5-phase code configured for DPF78 Plasma focus machine at 60 kV, 7.6 Torr deuterium gas.

Peak current (kA) = 859
 Pinch start current (kA) = 465
 Pinch minimum temperature (10^6 K) = 6.59
 Pinch maximum temperature (10^6 K) = 6.96
 Peak axial speed (cm/ μ s) = 15.8
 Peak radial shock speed (cm/ μ s) = 36.9
 Peak radial piston speed (cm/ μ s) = 25.2
 Final pinch radius $r_{\min}$ (cm) = 0.37
 Pinch length $z_{\max}$ (cm) = 3.7
 Pinch duration (ns) = 20.8
 Peak induced voltage (kV) = 85.4
 Neutron yield ($\times 10^9$ n/shot) = 3.8
 Energy Inflow into Plasma (%) = 13
 Speed Factor ((kA/cm)/Torr $^{0.5}$) = 125
 Current per cm anode radius (kA/cm) = 344

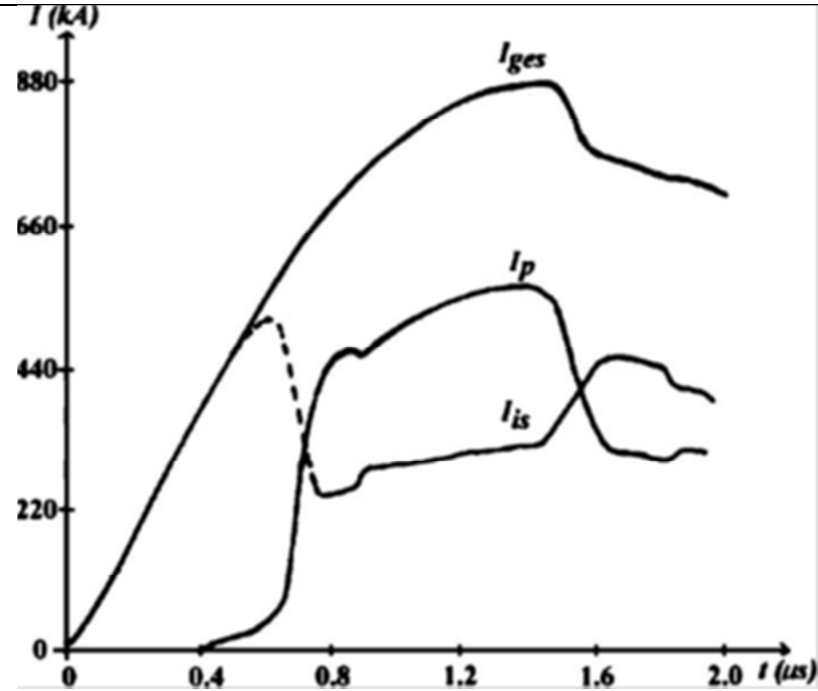


Figure 4.54 DPF78 measured I_{total} (labeled as I_{ges}) and measured I_p waveforms. The third trace I_{is} is the difference of I_{total} and I_p .

Note: This diagramme was downloaded from Figure 1, page 111501-2 of reference [128].

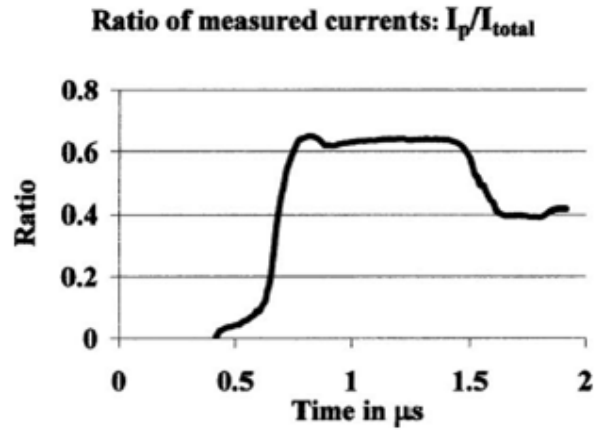


Figure 4.55 Ratio of measured I_p to I_{total} as a function of time [128].

Note: This graph is extracted from Figure 3, page 111501-3 of reference [128].

The Lee Model 5-phase code was then configured to work from 10.0 Torr to 16.0 Torr deuterium gas and the neutron yield versus pressure plotted is as shown in Figure 4.56.

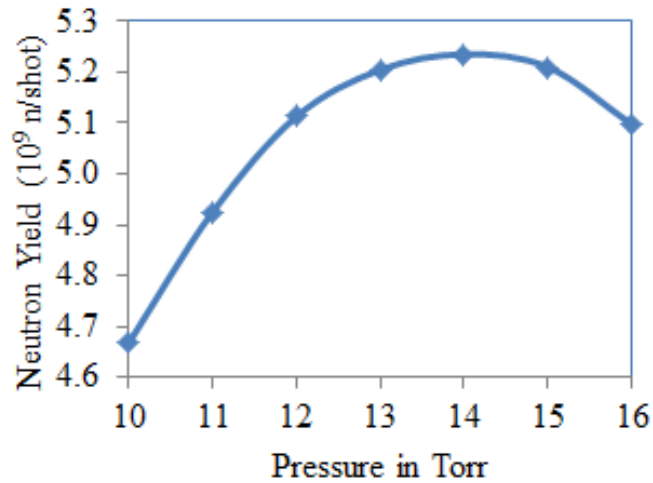


Figure 4.56 Variation of computed neutron yield with respect to pressure operating at 36 kV in deuterium for DPF78.

From Figure 4.56, the maximum neutron yield computed is 5.24×10^9 n/shot at 14.0 Torr.

4.15 Tamu Plasma Focus Machine (126 kJ - Type 1 machine - D gas)

The Tamu Plasma Focus machine at Texas A&M University uses a capacitor bank made up of 6 modules, each consisting of 24, 1.85 μF , 60 kV capacitor. Each module is discharged through a single rail gap switch [129]. The details of the machine parameters are shown in Table 4.42. The fitting of the computed to measured current waveform using Lee Model 5-phase code is shown in Figure 4.58.

The maximum computed current is 1495 kA and it exhibits a radial start time at 4.1 μs for pinch duration of 0.54 μs . It has a speed factor of 85 (kA/cm)/Torr^{0.5} and neutron yield of 5.39×10^{10} n/shot at 12.0 Torr (Compared to the experimental neutron yield of 2×10^{11} n/shot at 12.0 Torr [129]). We note that the computed maximum yield is 1.1×10^{11} n/shot at 32.0 Torr (See Figure 4.57). The detailed information is presented in Table 4.43.

Using the information of the current waveform, Lee Model 5-phase code was configured from 10.0 Torr to 40.0 Torr and the neutron yield versus pressure curve is shown in Figure 4.57.

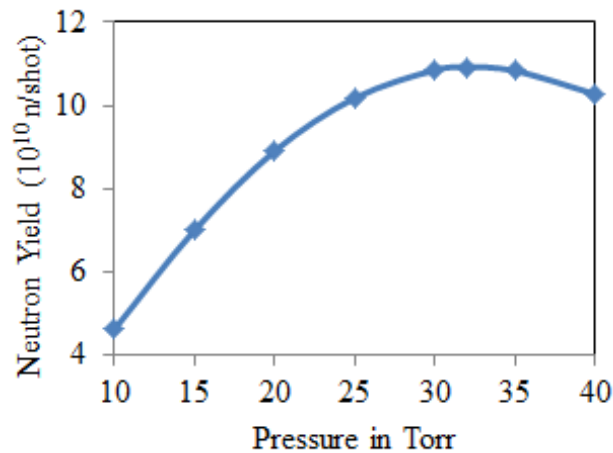


Figure 4.57 Variation of computed neutron yield with respect to pressure operating at 30 kV in deuterium for Tamu Plasma Focus machine.

Table 4.42 The machine, operational and fitting parameters for Tamu Plasma Focus machine.
Capacitance C_0 (μF) = 266.4
Static inductance L_0 (nH) = 40
Circuit resistance r_0 (m Ω) = 2.5
Cathode radius 'b'(cm) = 8.6
Anode radius 'a'(cm) = 5.1
Anode length 'z ₀ '(cm) = 24
Charging voltage V_0 (kV) = 30
Fill pressure P_0 (Torr) = 12.0
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.1625$
Axial phase current factor, $f_c = 0.74$
Radial phase mass factor, $f_{mr} = 0.1944$
Radial phase current factor, $f_{cr} = 0.6$

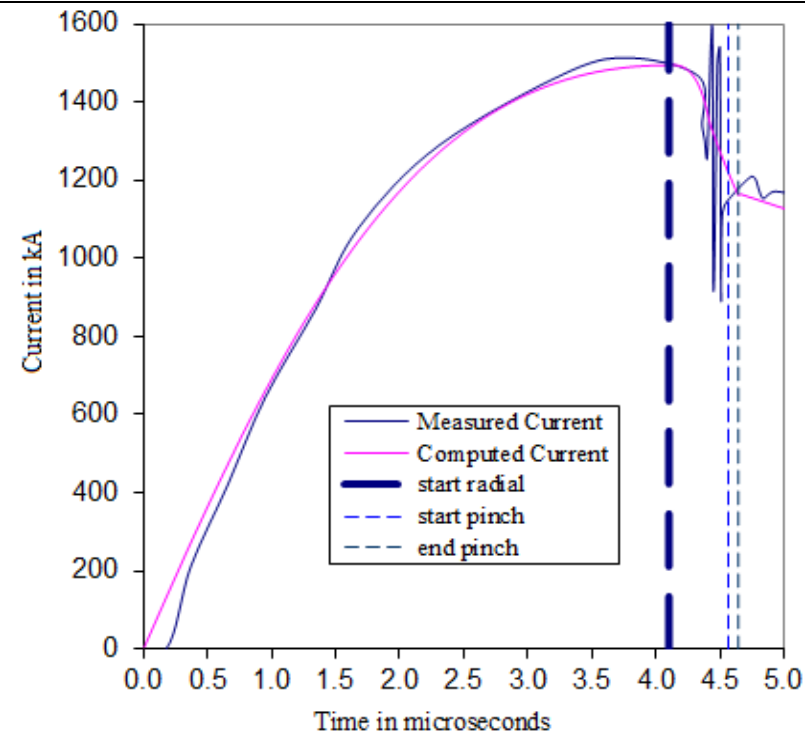


Figure 4.58 The measured current waveform of Tamu Plasma Focus machine at 30 kV, 12.0 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: The measured current waveform information was extracted from Figure 1, page 31, reference [129].

Table 4.43 Information obtained from Lee Model 5-phase code configured for
Tamu Plasma Focus machine at 30 kV, 12.0 Torr deuterium gas.

Peak current (kA) = 1495	Final pinch radius $r_{\min}$ (cm) = 0.84
Pinch start current (kA) = 733	Pinch length $z_{\max}$ (cm) = 7.7
Pinch minimum temperature (10^6 K) = 1.96	Pinch duration (ns) = 78.3
Pinch maximum temperature (10^6 K) = 2.16	Peak induced voltage (kV) = 65.3
Peak axial speed (cm/ μ s) = 9.1	Neutron yield ($\times 10^{10}$ n/shot) = 5.39
Peak radial shock speed (cm/ μ s) = 21.8	Energy Inflow into Plasma (%) = 18.7
Peak radial piston speed (cm/ μ s) = 15.5	Speed Factor ((kA/cm)/Torr ^{0.5}) = 85
Current per cm anode radius (kA/cm) = 293	

From Figure 4.57, the optimum neutron yield obtained is 1.1×10^{11} n/shot at 32.0 Torr with energy input into plasma at 21.6 %.

4.16 POSIDEON Plasma Focus machine (280.8 kJ - Type 1 machine - D gas)

The machine parameters for the POSIDEON Plasma Focus machine of Stuttgart University, Germany is listed in Table 4.44.

Table 4.44 The machine, operational and fitting parameters for Posideon Plasma Focus machine.

Capacitance C_0 (μF) = 156	Fill gas (molecular weight) = 4
Static inductance L_0 (nH) = 17.7	Fill gas (atomic number) = 1
Circuit resistance r_0 (m Ω) = 1.7	Fill gas (atom) = 2
Cathode radius 'b' (cm) = 9.5	The fitted model parameters are:
Anode radius 'a' (cm) = 6.55	Axial phase mass factor, $f_m = 0.275$
Anode length 'z ₀ ' (cm) = 30	Axial phase current factor, $f_c = 0.595$
Charging voltage V_0 (kV) = 60	Radial phase mass factor, $f_{mr} = 0.45$
Fill pressure P_0 (Torr) = 3.8	Radial phase current factor, $f_{cr} = 0.44$

This is a high voltage low static inductance machine. The Lee Model 5-phase code was used to fit the computed current waveform to the measured current waveform as shown in Figure 4.59. The detailed information obtained from Lee Model 5-phase code is shown in Table 4.45.

Using the information obtained from the current waveform, the Lee Model 5-phase code was configured for different pressures and voltages and the computed neutron yield was compared to the measured yield obtained from "Comparative analysis of large plasma focus experiments performed at IPF, Stuttgart, and at IPJ, Swierk"[37].

Table 4.45 Information obtained from Lee Model 5-phase code configured for Posideon Plasma Focus machine at 60 kV, 3.8 Torr deuterium gas.

Peak current (kA) = 3205
Pinch start current (kA) = 1000
Pinch minimum temperature (10^6 K) = 2.88
Pinch maximum temperature (10^6 K) = 3.33
Peak axial speed (cm/ μ s) = 18.6
Peak radial shock speed (cm/ μ s) = 29.1
Peak radial piston speed (cm/ μ s) = 20.9
Final pinch radius $r_{\min}$ (cm) = 1.22
Pinch length $z_{\max}$ (cm) = 11
Pinch duration (ns) = 83.1
Peak induced voltage (kV) = 97.4
Neutron yield ($\times 10^{11}$ n/shot) = 1.12
Energy Inflow into Plasma (%) = 21.6
Speed Factor ((kA/cm)/Torr $^{0.5}$) = 251
Current per cm anode radius (kA/cm) = 486

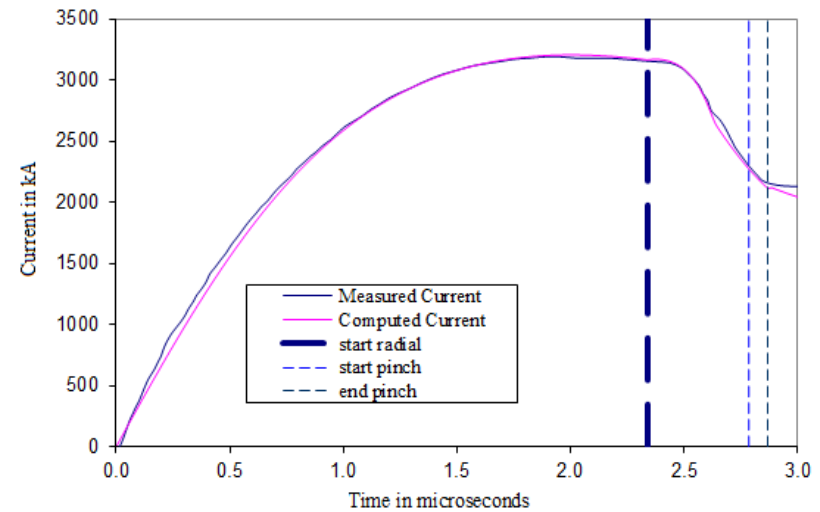


Figure 4.59 The measured current waveform of Posideon Plasma Focus machine at 60 kV, 3.8 Torr deuterium gas compared with Lee Model 5-phase code computed current.

Note: The measured current waveform information was extracted from <http://www.intimal.edu.my/school/fas/UFLF/machines/machines.htm> [59] but f_m , f_c , f_{mr} and f_{cr} has been changed for a closer fit.

Table 4.46 Computed and experimental neutron yield as compared to different pressure and voltages for Posideon Plasma Focus machine [37].

Energy (kJ)	280	280	380	380	380	500
Voltage (kV)	60	60	70	70	70	80
Pressure (Torr)	1.5	3	2.25	3	3.75	4.5
Computed neutron yield ($\times 10^{11}$ n/shot)	0.40	0.89	0.83	1.2	1.5	2.3
Experimental neutron yield ($\times 10^{11}$ n/shot)	0.51	0.94	0.31	1.2	1.4	1.8
Note: The experimental values were extracted from Figure 10, page 1264 of reference [37].						

From Table 4.46, it is observed that the experimental and computed neutron yields are close to each other.

Lee Model 5-phase code was then configured for 60 kV for 1.0 Torr to 10.0 Torr and the computed neutron yield curve compared to the experimental results is as shown in Figure 4.60.

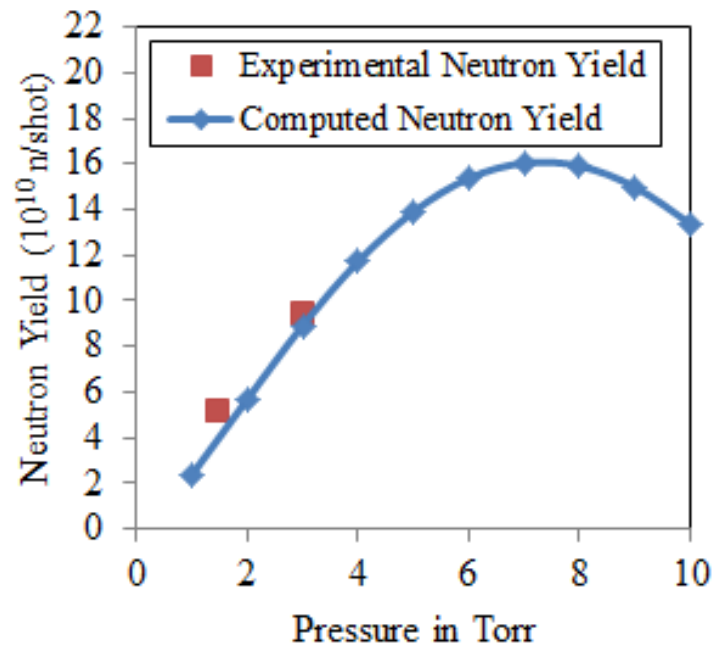


Figure 4.60 Variation of computed and experimental neutron yield with respect to pressure operating at 60 kV in deuterium for POSIDEON Plasma Focus.

Note: The experimental values were extracted from Figure 10, page 1264 of reference [37].

From Figure 4.60 it is seen that the computed optimum pressure when the machine is charged to 60 kV occurs at 7.0 Torr with a neutron yield of 1.6×10^{11} n/shot. The energy inflow for the radial phase for this shot is computed as 23.6 %.

4.17 PF1000 Plasma Focus machine (486 kJ - Type 1 machine - D gas)

PF1000 is seated at the Institute of Plasma Physics and Laser Micro fusion, Warsaw, Poland [4]. We discuss one of its configurations in which it has an outer electrode which consists of 24 stainless steel rods. The rod diameter is 32 mm. The outer electrode and copper center electrode diameters are 320 and 230 mm, respectively, with center electrode length of 600 mm. The outer electrode is attached to a circular grounded cable header. The center electrode attaches to a central header that provides an electrical connection and vacuum seal. The cylindrical alumina insulator sits on the center electrode. The main part of the insulator extends 113 mm along the center electrode into the vacuum chamber. The vacuum vessel of stainless steel surrounds the electrode structure. The condenser bank (1200 kJ) system consists of twelve condenser modules each comprising twenty four 50 kV, 4.625 μ F low inductance condensers connected in parallel. The machine parameters are given in Table 4.47.

Table 4.47 The machine, operational and fitting parameters for PF1000 Plasma Focus machine.

Capacitance C_0 (μ F) = 1332	Fill gas (molecular weight) = 4
Static inductance L_0 (nH) = 33.5	Fill gas (atomic number) = 1
Circuit resistance r_0 (m Ω) = 6.3	Fill gas (molecule) = 2
Cathode radius 'b' (cm) = 16	The fitted model parameters are:
Anode radius 'a' (cm) = 11.5	Axial phase mass factor, $f_m = 0.13$
Anode length 'z ₀ ' (cm) = 60	Axial phase current factor, $f_c = 0.7$
Charging voltage V_0 (kV) = 27	Radial phase mass factor, $f_{mr} = 0.35$
Fill pressure P_0 (Torr) = 3.5	Radial phase current factor, $f_{cr} = 0.65$

A current waveform of this Mather Type 1 machine at 27 kV, 3.5 Torr is published [57]. The Lee Model 5-phase code is used to fit the computed current waveform to the measured current waveform as shown in Figure 4.61.

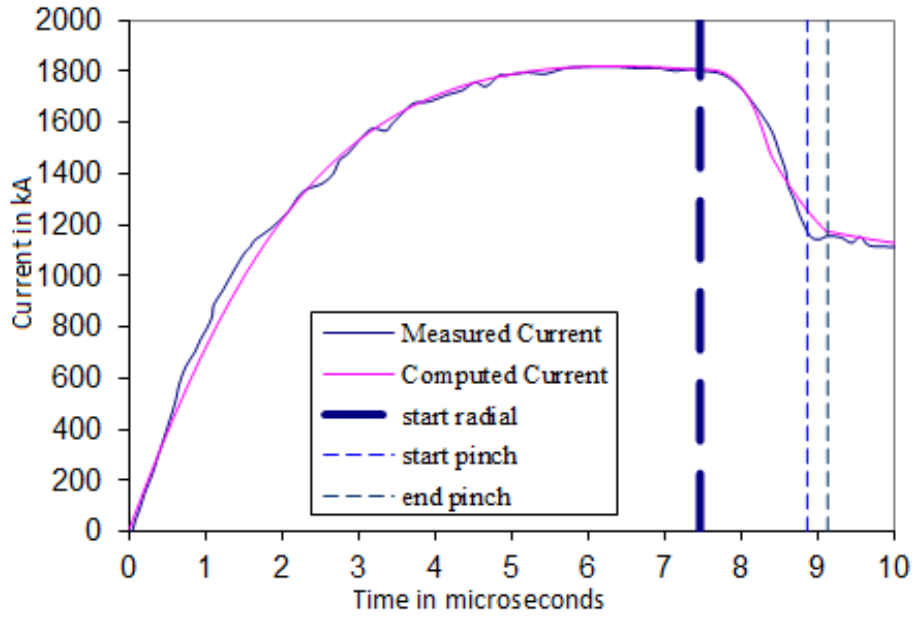


Figure 4.61 Measured current waveform of PF1000 at 27 kV, 3.5 Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: the digitalized measured current waveform was downloaded from <http://www.plasmafocus.net> [57].

The computed and measured waveforms show a good fit. In the process of fitting, the static inductance is fitted as 33 nH, significantly different from their estimated value of ~ 20 nH [4]. The maximum computed current is 1821 kA and exhibits the radial start time at 7.46 μs for pinch duration of 1.67 μs . The Lee Model 5-phase code computed the pinch to start at 8.85 μs and end at 9.13 μs , which is agreeable with the experimental pinch which occurs at 8.61 μs [4]. The speed factor is $85 \text{ (kA/cm)/Torr}^{0.5}$ and the computed neutron yield of $1.03 \times 10^{11} \text{ n/shot}$ may be compared to $2 \times 10^{11} \text{ n/shot}$ published in “Plasma dynamics in the PF-1000 device under full-scale energy storage: II. Fast electron and ion characteristics versus neutron emission parameters and gun optimization perspectives” [4]. Detailed information is presented in Table 4.48.

Table 4.48 Information obtained from Lee Model 5-phase code configured for PF1000 Plasma Focus machine at 27 kV, 3.5 Torr deuterium gas.

Peak current (kA) = 1821	Final pinch radius rmin (cm) = 2.18
Pinch start current (kA) = 817	Pinch length zmax (cm) = 18.7
Pinch minimum temperature (10^6 K) = 0.87	Pinch duration (ns) = 266.5
Pinch maximum temperature (10^6 K) = 1.01	Peak induced voltage (kV) = 39.2
Peak axial speed (cm/ μ s) = 11	Neutron yield ($\times 10^{11}$ n/shot) = 1.03
Peak radial shock speed (cm/ μ s) = 16.2	Energy Inflow into Plasma (%) = 12.1
Peak radial piston speed (cm/ μ s) = 12.1	Speed Factor ((kA/cm)/Torr ^{0.5}) = 85
Current per cm anode radius (kA/cm) = 158	

Using this information Lee Model 5-phase code is now configured to operate from 1 Torr to 15.0 Torr and the variation of neutron yield with respect to the pressure is shown in Figure 4.62.

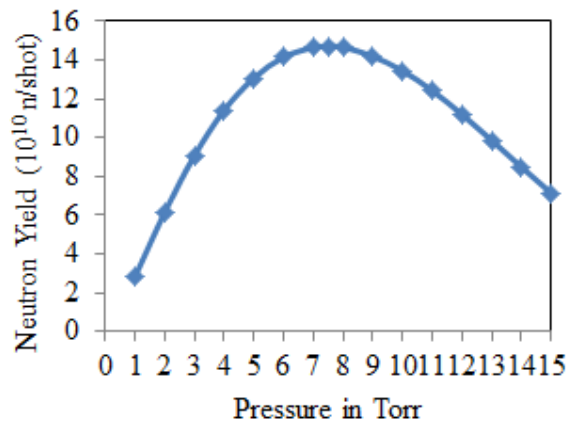


Figure 4.62 Variation of computed neutron yield with respect to pressure operating at 27 kV in deuterium for PF1000 plasma focus machine.

From the computed results shown in Figure 4.62, the peak neutron yield is 1.47×10^{11} n/shot and occurs at 7.5 Torr with the energy input into plasma at 13.5 %.

CHAPTER 5

Results from dense plasma focus machines using other gases and other configurations

5.1 Introduction

Chapter 5 presents machines operating in other gases (N, air, He, H and O) and machines with other configurations such as Filippov-type, hybrid type, machine using Marx capacitor bank (Speed 2) and the pioneering Plasma Focus machines operated at University Malaya, Malaysia in the Seventies and Eighties [96-106].

The machines discussed in this chapter are:

- 5.2 India Mohanty Plasma Focus machine (2.2 kJ) [130, 131]
- 5.3 Sofia Plasma Focus machine (2.6 kJ) [132]
- 5.4 Zimbabwe Plasma Focus machine (2.7 kJ) [133, 134]
- 5.5 Nano Focus machine (a hybrid configuration) (0.1 J) [23, 24, 25]
- 5.6 QUAID I AZAM University Plasma Focus machine (1.8 kJ) [135]
- 5.7 Egypt Plasma Focus machine (112.5 J) [136]
- 5.8 PFMA-1 Plasma Focus machine (71 kJ) [137]
- 5.9 Rico Plasma Focus machine (1 kJ) [138]
- 5.10 PF 3 Plasma Focus machine (a Filippov configuration) (373 kJ) [139]
- 5.11 Dena Plasma Focus machine (a Filippov configuration) (37 kJ) [82, 141]
- 5.12 Speed 2 Plasma Focus machine (67.5 kJ) [25]
- 5.13 University Malaya Plasma Focus machines (12 kJ) [96-106]

5.2 INDIA MOHANTY Plasma Focus machine (2.2 kJ - Type 2 machine - N gas)

We have current waveforms from this machine operated at 25 kV at a pressure of 0.3 Torr nitrogen in different lengths of anode. The lengths are: 110, 115, 120, 125 and 130 mm. The static inductance of this machine is 330 nH (which we fitted from the current waveform) and the 6-phase code was used for the current fitting. Each anode has a hemispherical tip. The machine parameters are shown in Table 5.1 and the fitted anomalous resistances are recorded in Table 5.2.

We fixed the value of L_0 in the following way:

From the given measure current waveforms we estimate the periodic time T (assuming resonance frequency) of the discharge and used the given capacitance C

$$\text{to estimate } L_0 \text{ as } L_0 = \frac{1000 \times (T/2)^2}{\pi^2 \times C} = \frac{25000}{70.074} \approx 360 \text{ nH}$$

Using this value as initial value and then fitting with the Lee Model 6-phase code so that the rise time of the discharge has a good match we found the value of L_0 to be 330 nH.

The computed and measured waveforms for all five anodes show good fit when the Lee Model 6-phase code is used; as shown in Figures 5.1 and 5.2.

We note for the shortest anode, 11 cm length, the radial compression comes too early compared to the time to peak current, so much so that the current continues to climb after the radial phase. The effect of an early current dip is that the energy input into the plasma during the radial phase is low (for the 110 mm anode, it is only 2.4 % compared to the case of the 130 mm anode at 3.8 % when the dip is late).

Table 5.1 The machine, operational and fitting parameters of INDIA-Mohanty's Plasma Focus machine for different anode lengths in nitrogen gas medium.

Capacitance C_0 (μF)	7.1	7.1	7.1	7.1	7.1
Static inductance L_0 (nH)	330	330	330	330	330
Circuit resistance r_0 ($\text{m}\Omega$)	20	20	20	20	20
Cathode radius 'b' (cm)	3	3	3	3	3
Anode radius 'a' (cm)	1	1	1	1	1
Anode length 'z₀' (cm)	11	11.5	12	12.5	13
Charging voltage V_0 (kV)	25	25	25	25	25
Fill pressure P_0 (Torr)	0.3	0.3	0.3	0.3	0.3
Fill gas (molecular weight)	28	28	28	28	28
Fill gas (atomic number)	7	7	7	7	7
Fill gas (molecule)	2	2	2	2	2
The fitted model parameters are:					
Axial phase mass factor, f_m	0.0235	0.0246	0.028	0.03	0.039
Axial phase current factor, f_c	0.7	0.7	0.73	0.73	0.73
Radial phase mass factor, f_{mr}	0.1	0.1	0.15	0.15	0.15
Radial phase current factor, f_{cr}	0.7	0.7	0.8	0.8	0.8

Table 5.2 The parameters of the anomalous resistances for the for INDIA-Mohanty's Plasma Focus machine that uses different anode lengths in nitrogen gas medium.

Anode length		$R_0(\Omega)$	Characteristic of fall time, τ_2 (ns)	Characteristic of rise time, τ_1 (ns)	End fraction time
110 mm	Dip 1	0.45	28	10	1
	Dip 2	0.12	50	10	1
	Dip 3	0.1	50	10	1
115 mm	Dip 1	0.2	50	10	1
	Dip 2	0.1	100	10	1
	Dip 3	0.05	100	10	1
120 mm	Dip 1	0.5	100	10	1
	Dip 2	0.12	100	10	1
	Dip 3	0.11	400	10	1
125 mm	Dip 1	0.6	100	10	1
	Dip 2	0.05	500	10	1
	Dip 3	0.01	500	10	1
130 mm	Dip 1	0.3	100	10	1
	Dip 2	0.2	100	10	1
	Dip 3	0.1	100	1	1

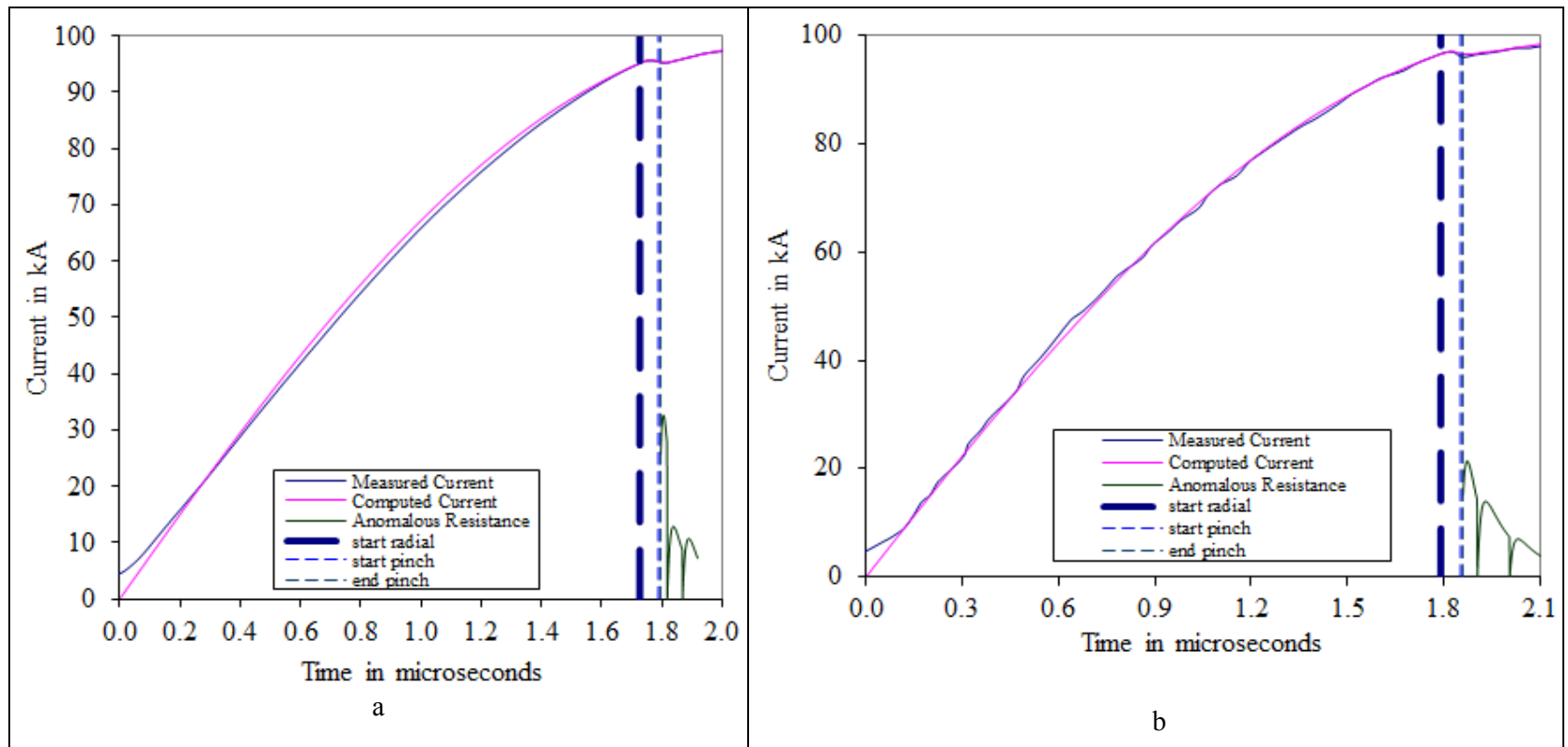


Figure 5.1 The measured current waveforms (anode of 110 mm-figure (a) and anode 115 mm-figure (b)) of INDIA-Mohanty's Plasma Focus machine at 25 kV, 0.3 Torr nitrogen gas compared with Lee Model 6-phase code computed current.

Note: The measured current waveform was obtained from the current derivative in Figure 3, page 013302-3 of reference [131].

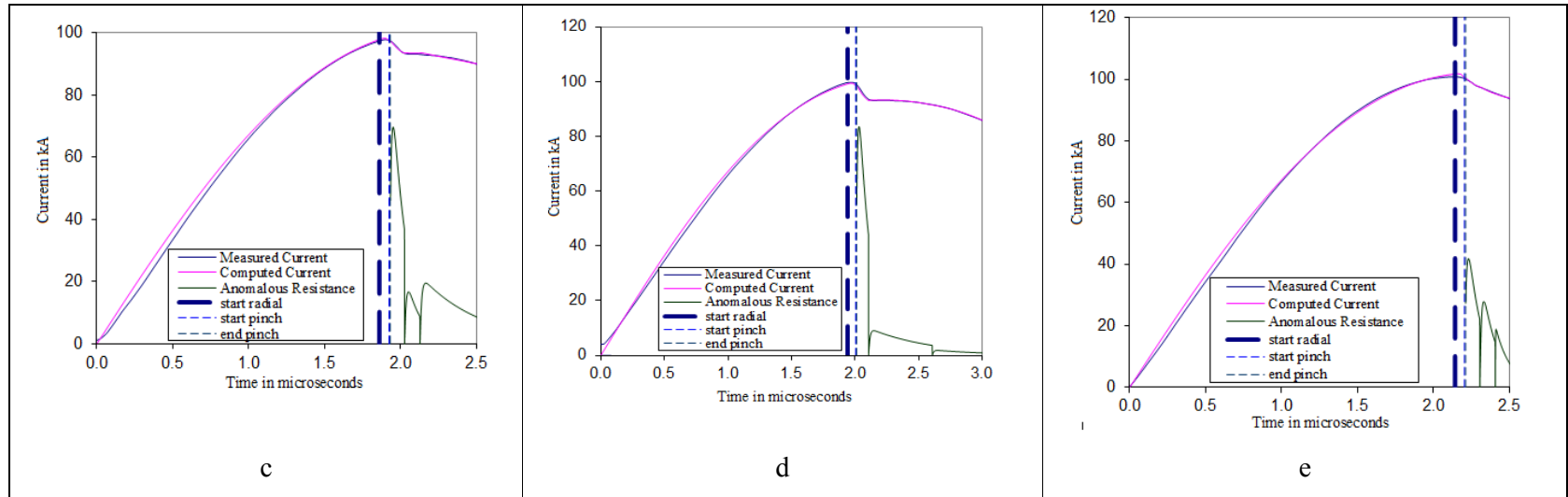


Figure 5.2 The measured current wave forms (anode of 120 mm-figure(c), anode 125 mm-figure (d) and anode 130 mm-figure (e)) of INDIA-Mohanty's Plasma Focus machine at 25 kV, 0.3 Torr nitrogen gas compared with Lee Model 6-phase code computed current.

Note: The measured current waveform was obtained from the current derivative in Figure 3, page 013302-3 of reference [131]

Table 5.3 Computed properties of INDIA-Mohanty's Plasma Focus machine using 5 different anode lengths.

Note: f_c , f_{mr} , f_{cr} for anode lengths of 110 and 115mm are the same; f_c , f_{mr} , f_{cr} for anode lengths of 120,125 and 130 mm are the same

Anode length	110mm	115mm		120mm	125mm	130mm	
Peak current (kA)	96	97	increase	98	99	102	increase
Pinch start current (kA)	67	68	increase	78	79	81	increase
Pinch minimum temperature (10^6 K)	9.06	9.3	increase	8.32	8.57	8.77	increase
Pinch maximum temperature (10^6 K)	9.08	9.33	increase	8.38	8.69	8.82	increase
Peak axial speed (cm/ μ s)	11.32	11.30	decrease	11.3	11.1	10.2	decrease
Peak radial shock speed (cm/ μ s)	34.6	35	increase	33.3	33.7	34.2	increase
Peak radial piston speed (cm/ μ s)	24.8	25.1	increase	23.8	24.1	24.5	increase
Final pinch radius r_{min} (cm)	0.11	0.11	constant	0.11	0.11	0.11	constant
Pinch length z_{max} (cm)	1.4	1.4	constant	1.4	1.5	1.4	constant
Pinch duration (ns)	7.4	7.3	decrease	7.6	7.6	7.5	decrease
Peak induced voltage (kV)	20.8	21.2	increase	23.1	23.5	24.1	increase
Soft X-ray yield (mJ)	0.23	0.22	decrease	0.60	0.56	0.55	decrease
Energy Inflow into Plasma (%)	2.4	2.5	increase	3.2	3.3	3.8	increase
Speed Factor ((kA/cm)/Torr ^{0.5})	175	177	increase	179	182	186	increase
Current per cm anode radius (kA/cm)	96	97	increase	98	99	102	increase

Table 5.3 shows the trends of various properties (speed, temperatures, pinch dimensions, lifetimes) with increase in anode length (range of 11-13 cm) for this device operated at 25 kV, 0.3 Torr nitrogen. The trends for each set of anodes are shown in the fourth and last columns of the table.

We now compare the result obtained from Lee Model 6-phase code to the published calculated values. From the computed result, the average speed is as shown in Table 5.4.

Table 5.4 Computed current sheath speed of INDIA-Mohanty's Plasma Focus machine at 25 kV, 0.3 Torr nitrogen gas.

Anode Length(cm)	Time to the end of pinch (μ s)	Average sheath speed (Axial speed) (cm/ μ s)
11	1.796	6.12
11.5	1.859	6.186
12	1.933	6.207
12.5	2.012	6.213
13	2.212	5.877

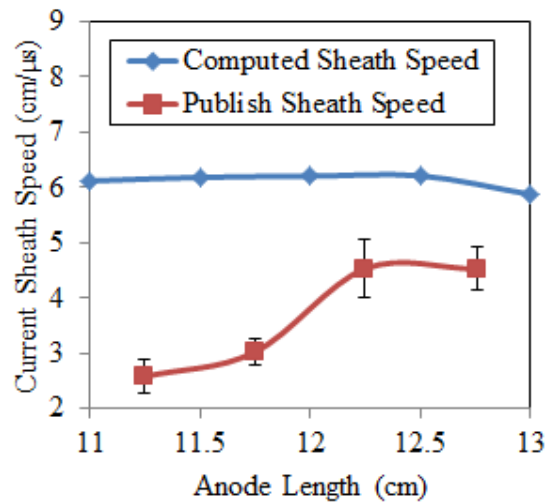


Figure 5.3 Published average speed compared to computed average speed for different anode lengths of INDIA-Mohanty's Plasma Focus machine at 25 kV, 0.3 Torr nitrogen gas.

Note: The published report (Figure 4, page 013302-3, reference [131]) took the average length of 2 anodes, for example 11 cm and 11.5 cm, as 11.25 cm

Figure 5.3 shows a vast different in the current sheath speed. The published results show that the average speed ranges from 2.6 to 4.55 cm/μs which is too low compared to the computed sheath speed from 5.9 cm/μs to 6.2 cm/μs. The difference in the calculation of the speed might be due computation error in the published report as shown in the Figure 5.4.

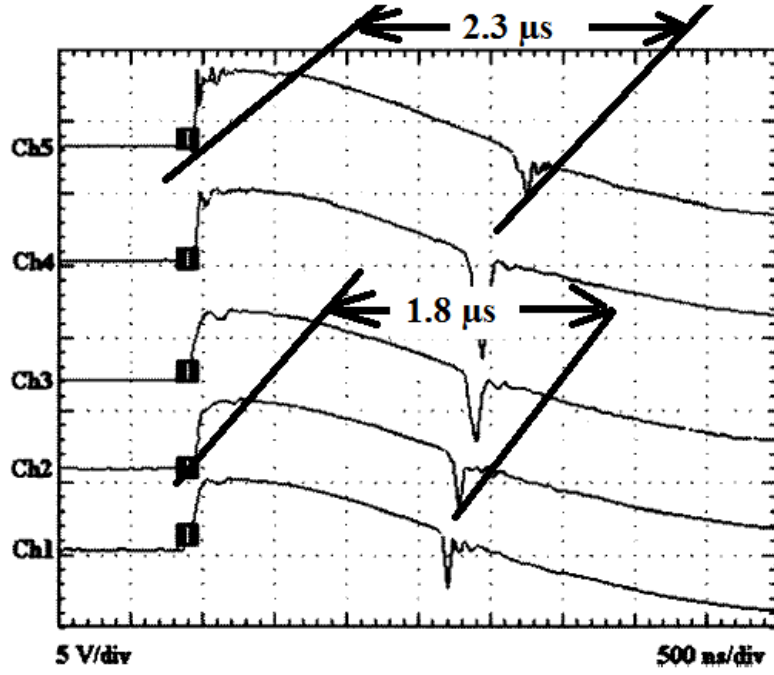


Figure 5.4 Ch5 shows the estimated time taken from the start of plasma to the pinch of a 13 cm anode and Ch 2 shows the estimated time taken in a 11.5 cm anode for INDIA-Mohanty's Plasma Focus machine [131].

From Figure 5.4, the estimated experimental average sheath speed (axial speed) should be $5.65 \text{ cm}/\mu\text{s}$ for 13 cm anode and $6.4 \text{ cm}/\mu\text{s}$ for 11.5 cm anode. These re-measured results agree with our computed values.

The next comparison between the numerical result and the published (measured) result is the soft X-ray yield.

When Lee Model 6-phase code was configured for 25 kV nitrogen with anode of 12.5 cm as shown in Figure 5.5, the soft X-ray yield was found to be 0.175 J at 1.7 Torr with the energy input into plasma at 3.6 %. The measured yield is published [131] as 4.5J into $4\pi\text{sr}$ at 0.7 Torr. Thus the measured result is far larger than our computed.

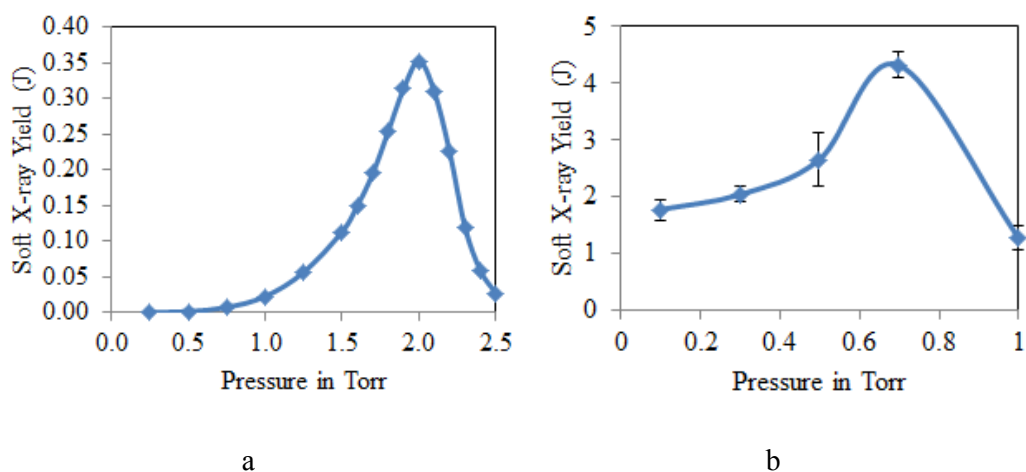


Figure 5.5 Computed soft X-ray yield versus pressure (a) and experimental soft X-ray yield versus pressure (b) for INDIA-Mohanty's Plasma Focus machine.

Note: The experimental soft X-ray yield versus pressure waveform was taken from Figure 7, page 013302-4, reference [131].

Figure 5.5 shows an unacceptably large difference between the computed and the experimental yield.

5.3 Sofia Plasma Focus machine (2.6 kJ - Type 1 machine - Air)

Sofia Plasma Focus machine is a Mather- type dense Plasma Focus machine belonging to Faculty of Physics at the University of Sofia, Bulgaria [132]. It is a 3 kJ Plasma Focus machine whose machine parameters are shown in Table 5.5.

Table 5.5 The machine, operational and fitting parameters for Sofia Plasma Focus machine.

Capacitance C_0 (μF) = 20	Fill gas (molecular weight) = 28
Static inductance L_0 (nH) = 80	Fill gas (atomic number) = 7
Circuit resistance r_0 (m Ω) = 6.5	Fill gas (molecule) = 2
Cathode radius 'b' (cm) = 3.1	The fitted model parameters are:
Anode radius 'a' (cm) = 1	Axial phase mass factor, $f_m = 0.094$
Anode length 'z ₀ ' (cm) = 14.5	Axial phase current factor, $f_c = 0.7$
Charging voltage V_0 (kV) = 16	Radial phase mass factor, $f_{mr} = 0.15$
Fill pressure P_0 (Torr) = 1.05	Radial phase current factor, $f_{cr} = 0.7$

Its static inductance is 80 nH. The Lee Model 5-phase code was used to fit a measured current waveform operating in 2.0 Torr air which we approximate as 2.0 Torr nitrogen. The current fitting is shown in Figure 5.6.

The maximum computed current is 104 kA and exhibits the radial start time at 2.798 μs for pinch duration of 0.094 μs . It has a speed factor of 202 (kA/cm)/Torr^{0.5} nitrogen and soft X-ray yield of 20.4 mJ. The detailed information is presented in Table 5.6.

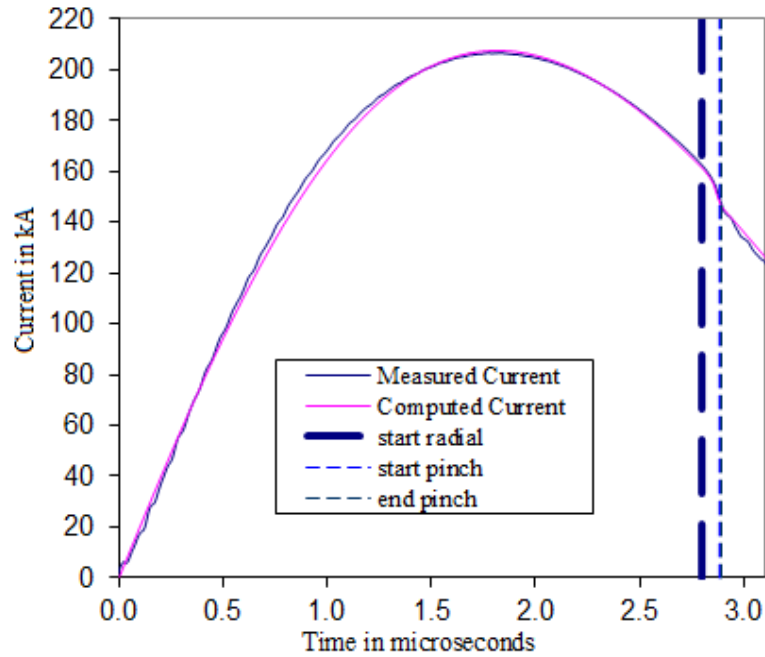


Figure 5.6 The measured current waveform for Sofia Plasma Focus machine at 16 kV, 1.05 Torr nitrogen gas compared with Lee Model 5-phase code computation.

Note: The measured current time derivative waveform was obtained from research collaboration with A. Blagoev [132]

Table 5.6 Information obtained from Lee Model 5-phase code configured for Sofia Plasma Focus machine at 16 kV, 1.05 Torr nitrogen gas.

Peak current (kA) = 207	Final pinch radius $r_{\min}$ (cm) = 0.11
Pinch start current (kA) = 104	Pinch length $z_{\max}$ (cm) = 1.4
Pinch minimum temperature (10^6 K) = 4.05	Pinch duration (ns) = 9.8
Pinch maximum temperature (10^6 K) = 4.18	Peak induced voltage (kV) = 17.6
Peak axial speed (cm/ μ s) = 6.8	Soft X-ray yield(J) = 0.02
Peak radial shock speed (cm/ μ s) = 25.1	Energy Inflow into Plasma (%) = 12.6
Peak radial piston speed (cm/ μ s) = 17.9	Speed Factor ((kA/cm)/Torr ^{0.5}) = 202
Current per cm anode radius (kA/cm) = 207	

Using the information obtained from the current waveform, the Lee model code was configured to operate the Sofia Plasma Focus machine from 1.0 Torr to 2.0 Torr nitrogen gas to obtain the soft x-ray versus pressure. The resulting curve is shown in Figure 5.7.

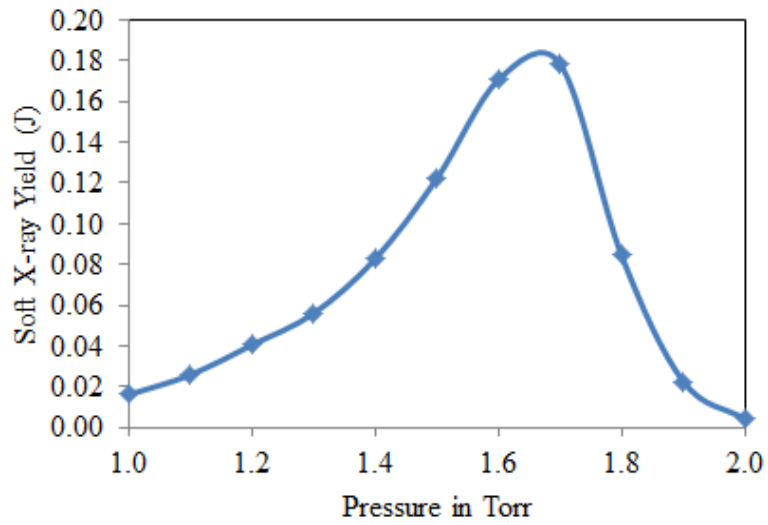


Figure 5.7 Variation of computed soft X-ray yield with respect to pressure operating at 16 kV in nitrogen for Sofia Plasma Focus machine.

From Figure 5.7, it is observed that the maximum soft X-ray yield occurs at 1.7 Torr and it produces 0.18 J soft X-ray with the energy input into plasma at 10.6 %.

5.4 Zimbabwe Plasma Focus machine (2.7 kJ - Type 2 machine - Air)

The Zimbabwe Plasma Focus machine has its plasma chamber made of a chromed steel tube that is 30 cm long and 16.3 cm in diameter. The central electrode is a copper tube 16 cm long and 1.9 cm in diameter and forms the anode of the focus tube. The outer electrode system which forms the cathode of the focus consists of six copper rods placed symmetrically around the anode at a radial distance of 3.05 cm. The diameter of each rod is 8 mm and the length is 15.0 cm [133]. Its static inductance is 100 nH and the Lee Model 6-phase code was used to fit its current waveform. Detailed parameters are presented in Table 5.7.

The computed and measured waveforms show a good fit as seen in Figure 5.8. The data we have of this device are of shots fired in air. As an approximation, nitrogen was used in the computation. As far as the dynamics is concerned the mass loading on the system computed in nitrogen is slightly less than the actual mass loading in air the difference being approximately 3 %. The speeds and trajectories would be affected by less than 2 % (speed being inversely proportional to square root of density).

The maximum computed current is 190 kA and exhibits the radial start time at 4.476 μs for pinch duration of 0.104 μs . It has a speed factor of 200 (kA/cm)/Torr^{0.5} nitrogen. The detailed information is presented in Table 5.8.

Table 5.7 The machine, operational and fitting parameters for Zimbabwe Plasma Focus machine.

Capacitance C_0 (μF) = 32.2	Fill gas (molecular weight) = 28			
Static inductance L_0 (nH) = 106.6	Fill gas (atomic number) = 7			
Circuit resistance r_0 ($\text{m}\Omega$) = 6	Fill gas (molecule) = 2			
Cathode radius ‘b’ (cm) = 3.05	The fitted model parameters are:			
Anode radius ‘a’ (cm) = 0.95	Axial phase mass factor, $f_m = 0.19$			
Anode length ‘z ₀ ’ (cm) = 16	Axial phase current factor, $f_c = 0.7$			
Charging voltage V_0 (kV) = 13	Radial phase mass factor, $f_{mr} = 0.15$			
Fill pressure P_0 (Torr) = 1	Radial phase current factor, $f_{cr} = 0.7$			
The anomalous resistance parameters are:				
	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time
Dip 1	0.01	150	1	1
Dip 2	0.05	50	1	1
Dip 3	0.02	50	1	1

Table 5.8 Information obtained from Lee Model 6-phase code configured for Zimbabwe Plasma Focus machine at 13 kV, 1.0 Torr nitrogen gas.
Peak current (kA) = 190 Pinch start current (kA) = 81 Pinch minimum temperature (10^6 K) = 2.89 Pinch maximum temperature (10^6 K) = 2.97 Peak axial speed (cm/μs) = 4.4 Peak radial shock speed (cm/μs) = 21.8 Peak radial piston speed (cm/μs) = 15.5 Final pinch radius $r_{\min}$ (cm) = 0.1 Pinch length $z_{\max}$ (cm) = 1.3 Pinch duration (ns) = 102 Peak induced voltage (kV) = 12.6 All-line yield (J) = 0.02 Energy Inflow into Plasma (%) = 10.2 Speed Factor ((kA/cm)/Torr^{0.5}) = 200 Current per cm anode radius (kA/cm) = 200

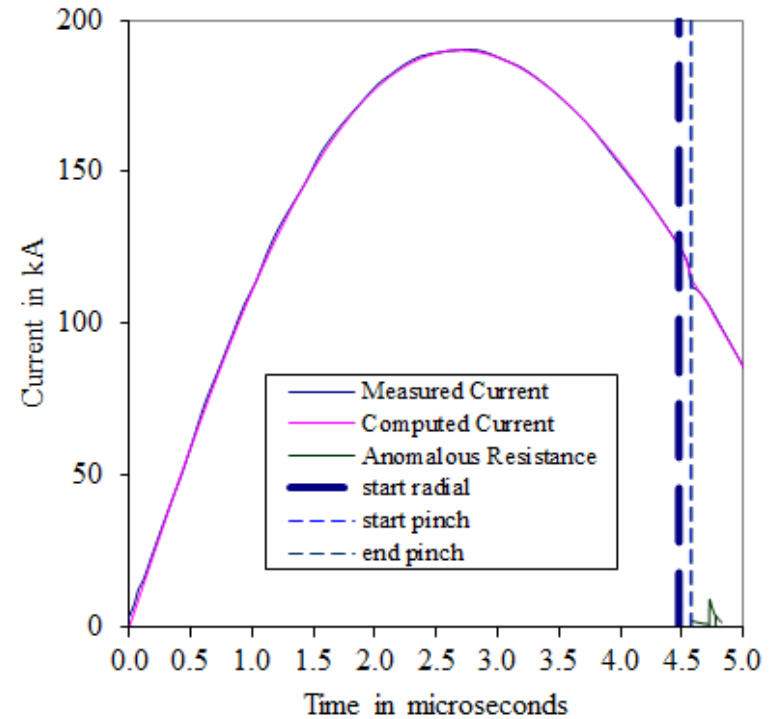


Figure 5.8 The measured current waveform for Zimbabwe Plasma Focus machine at 13 kV, 1.0 Torr nitrogen gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was extracted from Figure 2, page 4573 of reference [133].

Using the above current waveform Lee Model 6-phase code was configured to operate the Zimbabwe Plasma Focus machine from 0.5 and 1.0 Torr and the results are compared to the experimental data (using air as the medium).

Table 5.9 Information obtained from Lee Model 6-phase code configured for Zimbabwe Plasma Focus machine for nitrogen gas at 13 kV compared to the experimental values (air) obtained from Table 2, page 4574 of reference [133].

Pressure of nitrogen (Torr)	0.5	0.8	1
Axial transit time (Experimental) (μs)	3.45	3.5	3.6
Axial transit time (computed) (μs)	3.513	4.12	4.4
Current shedding Factor (Experimental)	0.69	0.68	0.68
Current shedding Factor (computed)	0.7	0.7	0.7

From Table 5.9, it is obvious that the trend shown by the experimental results are similar to the computed results.

5.5 Nano Focus machine (0.1 J - Hybrid machine - H gas)

The Nano Focus is a hybrid machine ($0.25 < \text{aspect ratio} < 5$) operated with 0.1-0.2 J in energy with an anode 0.8 mm in radius and a length of half mm [15] constructed at the Chilean Nuclear Energy Commission. It uses a pair of 200 mm diameter brass electrodes which acts as a capacitor to drive the discharge. We obtained the current waveform [24] and fitted it to the computed waveform using the Lee Model 5-phase code. The device uses an anode of 0.8 mm radius working in hydrogen at 6.5 kV. It was found from Lee Model 5-phase code that, in order to achieve a satisfactory fit, the effective anode length was 0.025cm instead of 0.04cm as stated in reference [24].

Table 5.10 The machine, operational and fitting parameters for the Nano Focus machine.

Capacitance C_0 (μF) = 0.0049	Fill gas (molecular weight) = 2
Static inductance L_0 (nH) = 4	Fill gas (atomic number) = 1
Circuit resistance r_0 ($\text{m}\Omega$) = 300	Fill gas (atom) = 1
Cathode radius 'b' (cm) = 0.15	The fitted model parameters are:
Anode radius 'a' (cm) = 0.08	Axial phase mass factor, $f_m = 0.029$
Anode length 'z ₀ ' (cm) = 0.025	Axial phase current factor, $f_c = 0.7$
Charging voltage V_0 (kV) = 6.5	Radial phase mass factor, $f_{mr} = 0.042$
Fill pressure P_0 (Torr) = 2.25	Radial phase current factor, $f_{cr} = 0.96$

This hybrid machine has static inductance L_0 of 4 nH (Obtained from Lee Model 5-phase code instead of 4.8 nH as stated in the reference [24]). The Lee Model 5-phase code was used for the fitting but found that the code needed to be modified at step 4000 where $D = D_{\text{real}}/10^9$ was used instead of $D_{\text{real}}/10^8$ in order to divide the radial phase into 10x smaller steps. The fitting obtained is shown in Figure 5.9 and the information obtained from Lee Model 5-phase codes are shown in Table 5.11.

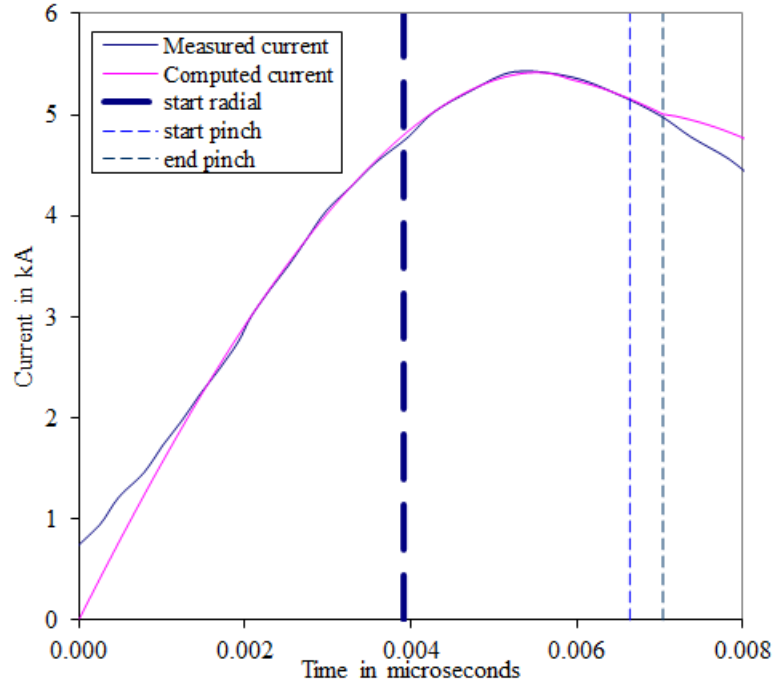


Figure 5.9 Measured current waveform of the Nano Focus machine at 6.5 kV, 2.25 Torr hydrogen gas compared with Lee Model 5-phase code computed current waveform.

Note: The measured current waveform was obtained from Figure 4, page 3 of reference [24].

Table 5.11 Information obtained from Lee Model 5-phase code configured for Nano Focus machine at 6.5 kV, 2.25Torr hydrogen gas.

Peak current (kA) = 5.4	Final pinch radius $r_{\min}$ (cm) = 0.01
Pinch start current (kA) = 4.9	Pinch length $z_{\max}$ (cm) = 0.1
Pinch minimum temperature (10^6 K) = 18.66	Pinch duration (ns) = 0.4
Pinch maximum temperature (10^6 K) = 19.8	Peak induced voltage (kV) = 1.8
Peak axial speed (cm/ μ s) = 11.6	All-line yield (J) = 0
Peak radial shock speed (cm/ μ s) = 60.7	Energy Inflow into Plasma (%) = 6.7
Peak radial piston speed (cm/ μ s) = 40.4	Speed Factor ((kA/cm)/Torr ^{0.5}) = 45
Current per cm anode radius (kA/cm) = 68	Peak ion number density ($10^{23}/\text{m}^3$) = 0.5

The computed peak current obtained is 5.4 kA as compared to 4.5 kA as obtained in the experiment. Thus this experiment shows that the Lee Model 5-phase code can be used to fit even the smallest hybrid type machine.

It should be noted that there is no other published information available where the computed results can be compared to the experimental results.

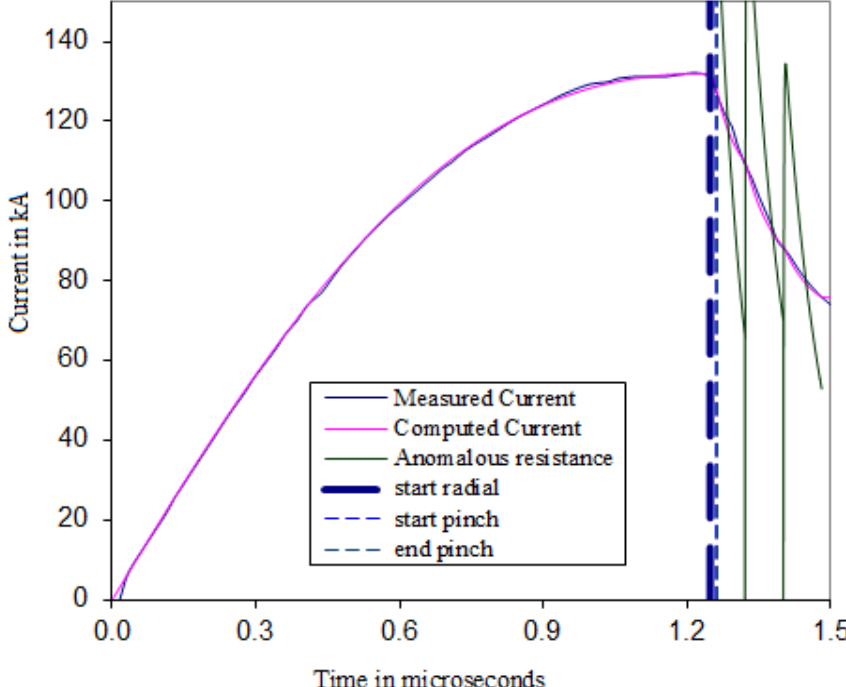
5.6 Quaid I Azam University Plasma Focus machine (1.8 kJ - Type 2 machine - H gas)

This Mather type machine with its static inductance of 100 nH from Pakistan has its machine parameters as shown in Table 5.12. We tried to fit with the 5-phase code; and it became obvious that the fit was incomplete. The Lee Model 6-phase code was used as shown in Figure 5.10.

The thesis [135] stated that the static inductance is 80 nH and the circuit resistance is 94 m Ω but the Lee Model 6-phase code fitted with 100 nH and 15 m Ω respectively. The detailed data computed from the Lee Model 6-phase code is presented in Table 5.13.

Table 5.12 The machine, operational and fitting parameters for Quaid I Azam University Plasma Focus machine.

Capacitance C_0 (μF) = 9	Fill pressure P_0 (Torr) = 0.38																
Static inductance L_0 (nH) = 100	Fill gas (molecular weight) = 2																
Circuit resistance r_0 (m Ω) = 15	Fill gas (atomic number) = 1																
Cathode radius ‘b’ (cm) = 3.5	Fill gas (molecule) = 2																
Anode radius ‘a’ (cm) = 0.9	The fitted model parameters are:																
Anode length ‘z ₀ ’ (cm) = 16	Axial phase mass factor, f_m = 0.16																
Taper starts at (cm) = 3	Axial phase current factor, f_c = 0.7																
Final tapered radius (cm) = 0.7	Radial phase mass factor, f_{mr} = 0.2																
Charging voltage V_0 (kV) = 20	Radial phase current factor, f_{cr} = 0.7																
The anomalous resistance parameters are:																	
	<table><tr><td>$R_0(\Omega)$</td><td>Characteristic of fall time τ_2 (ns)</td><td>Characteristic of rise time τ_1 (ns)</td><td>End fraction time</td></tr><tr><td>Dip 1</td><td>0.9</td><td>60</td><td>1</td></tr><tr><td>Dip 2</td><td>0.95</td><td>80</td><td>1</td></tr><tr><td>Dip 3</td><td>0.72</td><td>80</td><td>1</td></tr></table>	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time	Dip 1	0.9	60	1	Dip 2	0.95	80	1	Dip 3	0.72	80	1
$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time														
Dip 1	0.9	60	1														
Dip 2	0.95	80	1														
Dip 3	0.72	80	1														

Table 5.13 Information obtained from Lee Model 6-phase code configured for Quaid I Azam University Plasma Focus machine at 20 kV, 0.38 Torr hydrogen gas.	
Peak current (kA) = 132 Pinch start current (kA) = 90 Pinch minimum temperature (10^6 K) = 52.02 Pinch maximum temperature (10^6 K) = 52.77 Peak axial speed (cm/μs) = 20.9 Peak radial shock speed (cm/μs) = 105.8 Peak radial piston speed (cm/μs) = 76.4 Final pinch radius $r_{\min}$ (cm) = 0.12 Pinch length $z_{\max}$ (cm) = 1.2 Pinch duration (ns) = 2.3 Peak induced voltage (kV) = 59 All-line yield (J) = 0 Energy Inflow into Plasma (%) = 9.3 Speed Factor ((kA/cm)Torr^{0.5}) = 238	Figure 5.10 The measured current waveform of Quaid I Azam University Plasma Focus machine at 20 kV, 0.38 Torr hydrogen gas compared with Lee Model 6-phase code computed current. Note: The measured current waveform information was extracted from Figure 4.4(b) on page 85 of “Safdar Hussain PhD Thesis” [135].

It should be noted that no further suitable experimental values are given in the reference for comparison with the computed results.

5.7 EGYPT Plasma Focus machine (112.5 J - Type 2 machine - He gas)

The Egypt Plasma Focus machine is a Mather-type 112.5J Plasma Focus device. Its anode has a diameter of 18 mm and a length of 100 mm. whereas its cathode is formed from eight copper rods, each with a length of 130 mm and a diameter of 10 mm. The cylindrical insulator ring is 130 mm in diameter and 35 mm in thickness. The electrode system is enclosed in a vacuum chamber made of a stainless steel tank of 350 mm in length and 100 mm in diameter. The condenser bank of the plasma focus device consists of a condenser of 1 μF rated at 25 kV [136]. The parameters of this machine are shown in Table 5.14.

The static inductance is given as 6.5 μH ; but is found to be 5.5 μH by our fitting. The Lee Model 5-phase code was used to fit the measured current waveform as shown in Figure 5.11 since there is no pinching apparent from the computed results.

The detailed information is presented in Table 5.15.

Table 5.14 The machine, operational and fitting parameters for Egypt Plasma Focus machine.

Capacitance C_0 (μF) = 1	Fill gas (molecular weight) = 4
Static inductance L_0 (nH) = 5500	Fill gas (atomic number) = 2
Circuit resistance r_0 ($\text{m}\Omega$) = 100	Fill gas (atom) = 1
Cathode radius 'b' (cm) = 2.75	The fitted model parameters are:
Anode radius 'a' (cm) = 0.9	Axial phase mass factor, $f_m = 0.08$
Anode length 'z ₀ ' (cm) = 10	Axial phase current factor, $f_c = 0.7$
Charging voltage V_0 (kV) = 15	Radial phase mass factor, $f_{mr} = 0.12$
Fill pressure P_0 (Torr) = 0.8	Radial phase current factor, $f_{cr} = 0.7$

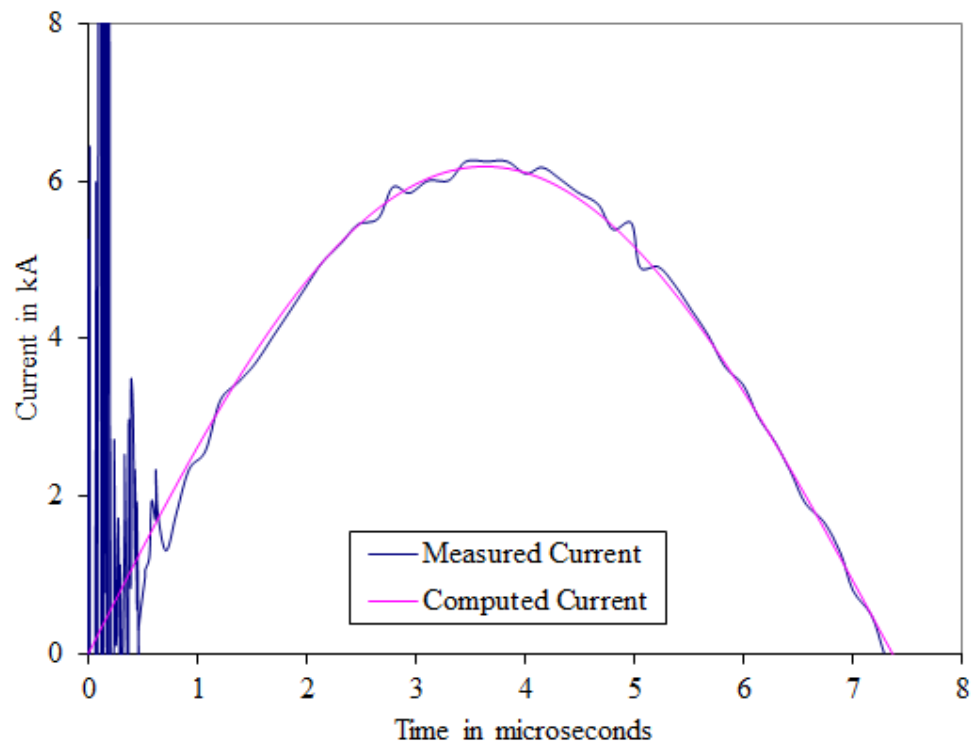


Figure 5.11 The measured current waveform of Egypt Plasma Focus machine at 15 kV, 0.8 Torr helium gas compared with Lee Model 5-phase code computed current waveform.

Note: The measured current waveform information was extracted from Figure 3, page 3 of reference [136].

Table 5.15 Information obtained from Lee Model 5-phase code configured for Egypt Plasma Focus machine at 15 kV, 0.8 Torr helium gas.

Peak current (kA) = 6	Speed Factor $((\text{kA}/\text{cm})/\text{Torr}^{0.5}) = 23$
Peak axial speed (cm/ μs) = 2	Current per cm anode radius (kA/cm) = 7

From Table 5.15, it is observed that the Egypt Plasma Focus machine has a speed factor of 23 (kA/cm)/Torr^{0.5}. This value is extremely low for the plasma to focus.

5.8 PFMA-1 Plasma Focus machine (71 kJ - Type 1 machine - He gas)

PFMA-1 (Plasma Focus for Medical Applications-1) [137] is a Plasma Focus device in Italy that is built for the production of radio-isotopes used for the diagnosis of cancer. This machine can be operated up to 30 kV, with 32 capacitors, 11.1 μ F each, charged in parallel for maximum total bank energy of 150 kJ. The cycle repetition rate for this device was designed for 1Hz. The machine parameters for PFMA-1 are presented in Table 5.16. It should be noted that the published paper [137] states that the inductance is 40 nH but fitting using the Lee Model 5-phase code found that the machine inductance is 30 nH. The experimental waveform shows the maximum current at about 1 MA whereas Lee Model 5-phase code computes the value at 1.17 MA.

The other information obtained from the Lee Model 5-phase code is shown in Table 5.17.

Table 5.16 The machine, operational and fitting parameters for PFMA-1 Plasma Focus machine working in helium gas.

Capacitance C_0 (μ F) = 355.2	Fill gas (molecular weight) = 4
Static inductance L_0 (nH) = 30	Fill gas (atomic number) = 2
Circuit resistance r_0 (m Ω) = 3	Fill gas (atom) = 1
Cathode radius 'b' (cm) = 8.5	The fitted model parameters are:
Anode radius 'a' (cm) = 4.8	Axial phase mass factor, $f_m = 0.144$
Anode length 'z ₀ ' (cm) = 16	Axial phase current factor, $f_c = 0.7$
Charging voltage V_0 (kV) = 20.5	Radial phase mass factor, $f_{mr} = 0.178$
Fill pressure P_0 (Torr) = 9.8	Radial phase current factor, $f_{cr} = 0.7$

Table 5.17 Information obtained from Lee Model 5-phase code configured for the PFMA-1 Plasma Focus machine at 20.5 kV, 9.8 Torr helium gas.	Figure 5.12 The measured current waveform of PFMA-1 Plasma Focus machine at 20.5 kV, 9.8 Torr helium gas compared with Lee Model 5-phase code computed current waveform. Note: The measured current waveform information was extracted from Figure 1, page 2 of reference [137].
Peak current (kA) = 1173 Pinch start current (kA) = 625 Pinch minimum temperature (10^6 K) = 2.8 Pinch maximum temperature (10^6 K) = 3.2 Peak axial speed (cm/μs) = 7.9 Peak radial shock speed (cm/μs) = 23.7 Peak radial piston speed (cm/μs) = 17.6 Final pinch radius $r_{\min}$ (cm) = 0.84 Pinch length $z_{\max}$ (cm) = 7.2 Pinch duration (ns) = 71.2 Peak induced voltage (kV) = 58 All-line yield (J) = 1.97 Energy Inflow into Plasma (%) = 16.4 Speed Factor ((kA/cm)/Torr^{0.5}) = 78 Current per cm anode radius (kA/cm) = 244	

5.9 Rico Plasma Focus machine (1 kJ - Type 1 machine showing Type 2 characteristics - O gas)

The machine parameters for the 1 kJ Rico Plasma Focus are presented in Table 5.18. Its static inductance is below 69 nH. The 5-phase model code was used to fit the measured current waveform and where necessary the Lee Model 6-phase code was used to complete fitting any observed extended current dip. A reasonable fit is shown in Figure 5.13.

The maximum computed current is 88 kA and exhibits the radial start time at 0.676 μ s for pinch duration of 0.08 μ s. It has a speed factor of 112 (kA/cm)/Torr^{0.5}Oxygen. The detailed information is presented in Table 5.19. Rico Plasma Focus device has both high axial speed and pinch temperature and this is not very suitable for X-ray generation. It is used as an ion beam source for material science application [138].

Table 5.18 The machine, operational and fitting parameters for Rico Plasma Focus machine.

Capacitance C_0 (μ F) = 3.86	Fill gas (molecular weight) = 16																
Static inductance L_0 (nH) = 69	Fill gas (atomic number) = 8																
Circuit resistance r_0 (m Ω) = 20	Fill gas (molecule) = 2																
Cathode radius ‘b’ (cm) = 4.9	The fitted model parameters are:																
Anode radius ‘a’ (cm) = 1.75	Axial phase mass factor, f_m = 0.0095																
Anode length ‘z ₀ ’ (cm) = 6.75	Axial phase current factor, f_c = 0.7																
Charging voltage V_0 (kV) = 14.9	Radial phase mass factor, f_{mr} = 0.014																
Fill pressure P_0 (Torr) = 0.2	Radial phase current factor, f_{cr} = 0.45																
The anomalous resistance parameters are:																	
	<table><tr><td>$R_0(\Omega)$</td><td>Characteristic of fall time τ_2 (ns)</td><td>Characteristic of rise time τ_1 (ns)</td><td>End fraction time</td></tr><tr><td>Dip 1</td><td>0.02</td><td>140</td><td>1</td></tr><tr><td>Dip 2</td><td>0.02</td><td>120</td><td>1</td></tr><tr><td>Dip 3</td><td>0.02</td><td>120</td><td>1</td></tr></table>	$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time	Dip 1	0.02	140	1	Dip 2	0.02	120	1	Dip 3	0.02	120	1
$R_0(\Omega)$	Characteristic of fall time τ_2 (ns)	Characteristic of rise time τ_1 (ns)	End fraction time														
Dip 1	0.02	140	1														
Dip 2	0.02	120	1														
Dip 3	0.02	120	1														

Table 5.19 Information obtained from Lee Model 6-phase code configured for Rico Plasma Focus machine at 14.9 kV, 0.2 Torr oxygen gas.

Peak current (kA) = 88
Pinch start current (kA) = 38
Pinch minimum temperature (10^6 K) = 7.14
Pinch maximum temperature (10^6 K) = 8.29
Peak axial speed (cm/ μ s) = 16.6
Peak radial shock speed (cm/ μ s) = 52.4
Peak radial piston speed (cm/ μ s) = 37.8
Final pinch radius $r_{\min}$ (cm) = 0.17
Pinch length $z_{\max}$ (cm) = 2.2
Pinch duration (ns) = 8.4
Peak induced voltage (kV) = 17.3
All-line yield (J) = 0.04 m
Energy Inflow into Plasma (%) = 6.6
Speed Factor ((kA/cm)/Torr ^{0.5}) = 112
Current per cm anode radius (kA/cm) = 50

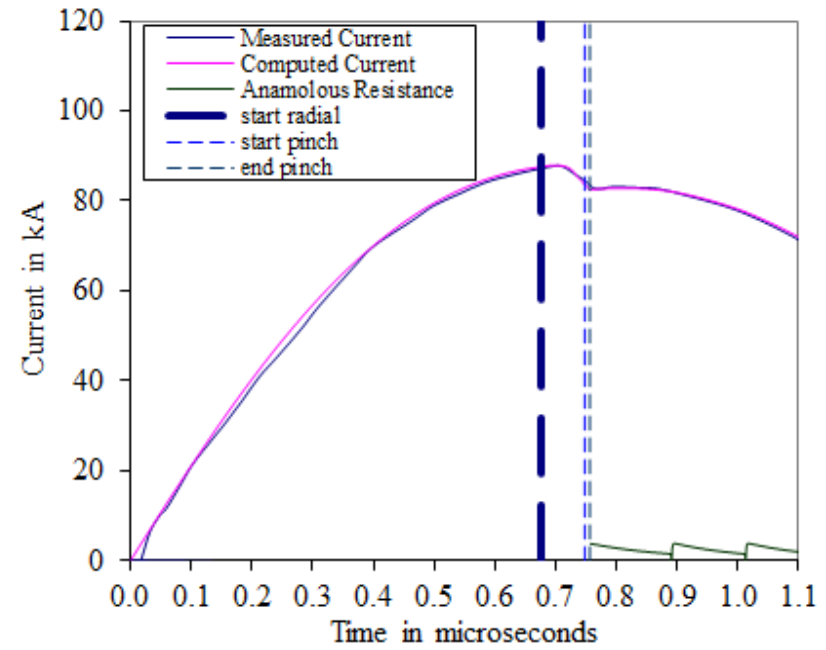


Figure 5.13 The measured current waveform of Rico Plasma Focus machine at 14.9 kV, 0.2 Torr oxygen gas compared with Lee Model 6-phase code computed current waveform.

Note: The measured current waveform information was extracted from Figure 5, page 227 of reference [138].

It should be noted that no further experimental values were given in the reference for comparison to be made.

5.10 PF3 Plasma Focus machine (373 kJ Filippov type machine - Ne gas)

The Lee Model 5 and 6-phase code has been shown to describe the Mather Plasma Focus very well. In other words the physical mechanisms assumed in the modeling: an axial phase described by a snow-plow model with fittable f_m and f_c ; and a radial phase of an accelerating and elongating plasma slug with fittable f_{mr} and f_{cr} ; appear to be adequate to represent the Mather Plasma Focus at least in the gross mass-consistent, energy-consistent and charge-consistent sense. We now pose the question: Does the Lee Model 5 and 6-phase code also adequately describe the Filippov Plasma Focus? This question had in effect already been asked by Siahpoush et al [60] who showed that the code needed to be adapted by adding in an inverse pinch (lift-off) phase in order to make a better attempt in modeling their Filippov Plasma Focus. To test their claim we configured our code for the Filippov Plasma Focus PF3 and attempted to fit the computed current trace to the measured current trace. The resulting study is instructive. Not only did we find that the Lee Model 5 and 6-phase code could not fit this Filippov Focus, we are able to deduce something important about the radial dynamics of this Filippov Focus.

Table 5.20 The machine, operational and fitting parameters for PF3 Plasma Focus machine.

Capacitance C_0 (μF) = 9200	Fill gas (molecular weight) = 20
Static inductance L_0 (nH) = 40	Fill gas (atomic number) = 10
Circuit resistance r_0 ($\text{m}\Omega$) = 1	Fill gas (atom) = 1
Cathode radius 'b' (cm) = 58	The fitted model parameters are:
Anode radius 'a' (cm) = 50	Axial phase mass factor, $f_m = 0.025$
Anode length 'z ₀ ' (cm) = 2	Axial phase current factor, $f_c = 0.7$
Charging voltage V_0 (kV) = 9	Radial phase mass factor, $f_{mr} = 0.03$
Fill pressure P_0 (Torr) = 1.5	Radial phase current factor, $f_{cr} = 0.5$

We have a current waveform of this Filippov's type machine at 9 kV at a pressure of 1.5 Torr neon gas [139]. The Lee Model 5-phase code was used to try to fit the computed to the measured current waveform as shown in Figure 5.14.

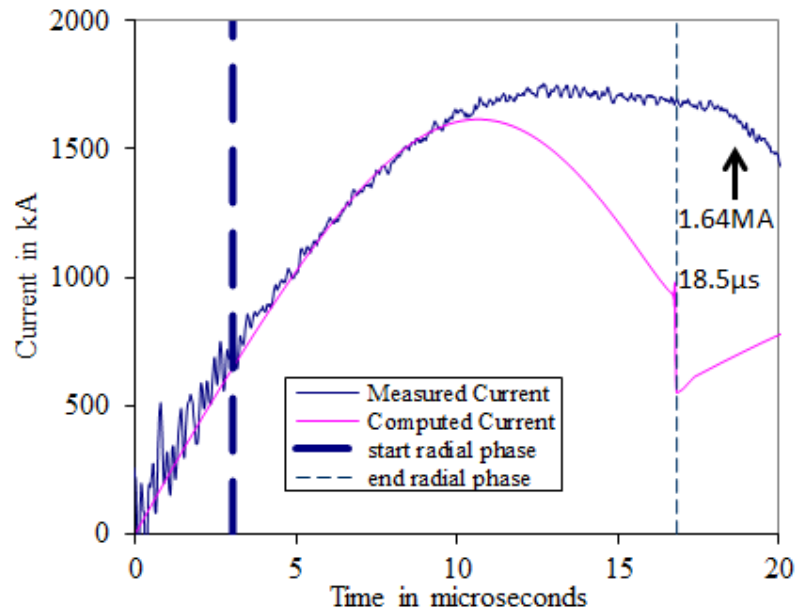


Figure 5.14 Measured current waveform of Filippov PF3 Plasma Focus at 9 kV, 1.5 Torr neon gas compared with Lee Model 5-phase code computed current waveform.

Note: The digital waveform for shot 4102 was obtained from Mitrofanov Konstantin and Krause Slava on 3/5/2011 who are the authors of reference [139].

From Figure 5.14 we notice that the ‘axial’ phase is fitted reasonably well. The radial phase is also very well fitted up to 9.6 μs (when the shock-piston slug is at 36 cm in its inward radial motion)- see attached Figure 5.15 (below); the timing and position are worked out from Figure 5.14 from the time of the start of the radial phase (as printed out in the active Sheet 1).

We discuss the results of the fitting. The Filippov PF3 can be fitted by the Lee Model 5-phase code by varying f_m and f_c for its very short ‘axial’ phase and by varying f_{mr} and f_{cr} in its radial phase from 50 cm to 36 cm. Beyond that the Lee Model code assumes a well-defined slug motion (of shock front and magnetic piston) which fails to describe the progression of events in the PF3. The measured current continues to rise to 1.7 MA at 12.7 μs then flattens out with a flat droop to 1.64 MA at 18.5 μs . The best computed fit sees the current peaking at 1.62 MA at a 10.6 μs then dropping with a relatively steep slope to a value of 0.95 MA at 16.7 μs

before a final abrupt dip caused likely by the radiative collapse mechanism [140] which is built into the Lee Model 5-phase code.

The divergence of computed to measured can be interpreted as follows: The PF3 can be modeled by a slug-like compression of the plasma up to 36 cm and then that slug-like behaviour is no longer a valid description. From the current trace of this shot the flattening of the current trace (instead of sharply dipping) indicates that the inward piston- driven plasma slug motion gives way, probably to some form of plasma diffusive mechanism.

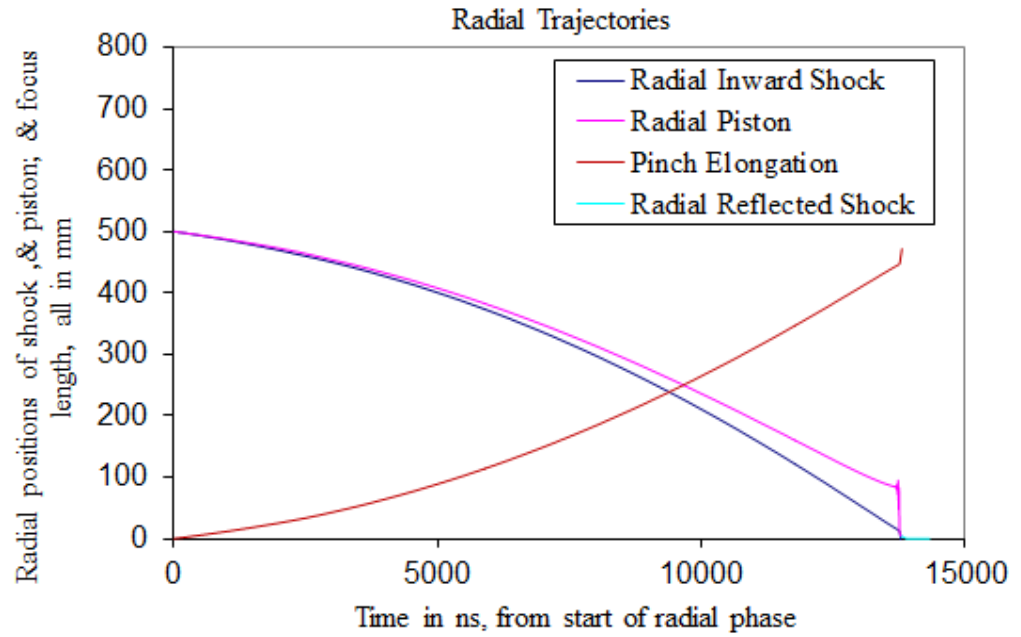


Figure 5.15 Radial trajectory of Filippov PF3 according to the Lee Model 5-phase code; taken from same numerical shot as Figure 5.14.

We note that based on the measured current traces of PF3 it is concluded that the PF3 Filippov PF agrees with the Lee Model 5-phase code in its lift-off phase but diverges from the Lee Model code dramatically in the crucial late radial and pinch phases.

We then proceed to also look at the DENA Filippov Plasma Focus machine [82].

5.11 Dena Plasma Focus machine (37 kJ Filippov type machine - Ar gas)

DENA is a Filippov Plasma Focus machine from Iran with the machine parameters as shown in Table 5.21.

Table 5.21 The machine, operational and fitting parameters for Dena Plasma Focus machine.

Capacitance C_0 (μF) = 288	Fill gas (molecular weight) = 40
Static inductance L_0 (nH) = 55	Fill gas (atomic number) = 18
Circuit resistance r_0 (m Ω) = 1	Fill gas (atom) = 1
Cathode radius 'b' (cm) = 35	The fitted model parameters are:
Anode radius 'a' (cm) = 24	Axial phase mass factor, $f_m = 0.001$
Anode length 'z ₀ ' (cm) = 12	Axial phase current factor, $f_c = 0.6$
Charging voltage V_0 (kV) = 16	Radial phase mass factor, $f_{mr} = 0.05$
Fill pressure P_0 (Torr) = 0.5	Radial phase current factor, $f_{cr} = 0.7$

This Filippov's type machine was charged to 16 kV at a pressure of 0.5 Torr argon gas. The Lee Model 5-phase code was used fit the measured current waveform as shown in Figure 5.16.

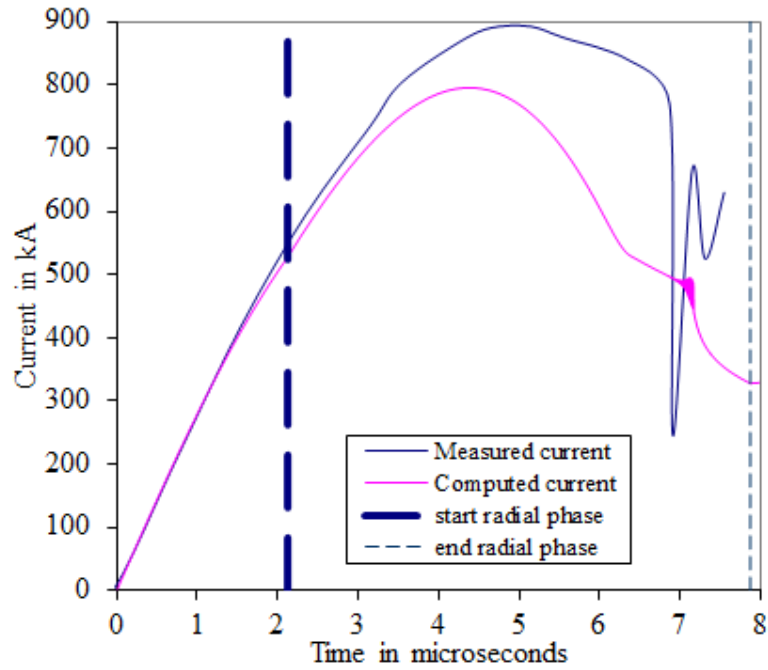


Figure 5.16 Measured current waveform of Filippov Dena Plasma Focus at 16 kV, 0.5 Torr argon gas compared with Lee Model 5-phase code computed current waveform.

Note: The measured current waveform was obtained from Figure 2, page 2 of the reference [141].

We find that as in the case of PF3, DENA [82, 141] can be fitted for the ‘axial’ phase and for the first part of the radial phase until 4 μ s at a modeled plasma slug radial position of 15 cm (less than half way in the radial compression). Then the modelled current trace drops significantly more than the measured current trace. The measured current trace has the same (as observed in PF3) flattening out characteristic which diverges from the modeled more sharply dropping current trace.

This divergence of the DENA computed from measured current trace follows the same pattern as the divergence of the PF3 computed from measured current trace.

From the fitting of both the PF3 and DENA Filippov Plasma Focus machines our conclusion is that the ‘axial’ part (including lift-off) of Filippov machines can be

adequately fitted by the Lee Model 5-phase code. The first part of the radial phase can also be fitted; but there comes a point about 0.7a for PF3 and 0.6a for DENA beyond which the slug-like radial collapse of the Lee Model 5-phase code no longer describes the Filippov machine. The conclusion is that to model the Filippov machine it is not so much that the axial (including lift-off) phase that needs to be modified, but rather more importantly the later radial phase just cannot be modeled by the Lee Model 5-phase code. It is the radial phase that needs to be re-modelled to include an obviously missing mechanism.

Since the machine was Filippov type, the axial phase is almost none existent since the anode is covered with an insulator right up to the anode phase. The five phase codes has been modified as explained by Siahpoush et al in the paper “Adaptation of Sing Lee’s model to the Filippov type plasma focus geometry” [60].

According to their paper [60] the following are the major differences between the Lee Model 5-phase code and Siahpoush ML (modified Lee) model:

(1) In Lee Model 5-phase code, the Current Sheath (CS) motion consists of two phases (i.e. axial and radial motions). But since the ML model should be applicable for Filippov type devices, in this model three CS motions have been considered (i.e. outward, inward and axial motions).

(2) In Lee Model 5-phase code both phases occur sequentially. But in the ML model all three CS motions have been assumed to occur simultaneously.

(3) In the ML model, the slug equations have been used to describe all three CS motions. However, these equations have been developed by Lee to describe the radial motion in his model.

The Siahpoush ML model is used to fit the Dena Plasma Focus current waveform as shown in Figure 5.17. It should be noted that this current fitting is different from that given in the article [60]. We took their comparative waveforms

and made small adjustment to their measured waveforms. For the experimental (measured) current waveform we apply a time-shift of $0.3 \mu\text{s}$ relative to the computed. We also multiplied the measured amplitude by factor of 0.99. With these small adjustments we significantly improved the fit. The result is shown in Figure 5.17.

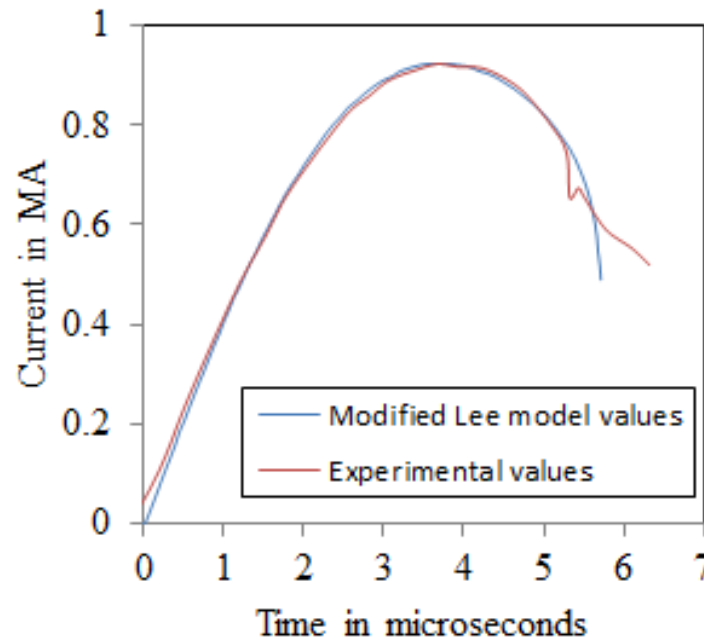


Figure 5.17 Siahpoush's Modified Lee [ML] code used to fit Dena Plasma Focus machine [60]; with our additional small adjustments: time shifted $0.3 \mu\text{s}$ and the amplitude reduced by 1 %. These small adjustments significantly improve the fit.

Note: The waveforms were obtained from Figure 5, page 1072 of reference number [60].

5.12 SPEED 2 Plasma Focus machine (67.5 kJ Mather type machine - D gas)

SPEED 2 Plasma Focus device [25] is based on Marx technology and was designed in the University of Duesseldorf (Germany). It consists of 40 Marx modules connected in parallel. Each module has 6 capacitors (50 kV, 0.625 μ F, 20 nH) and 3 spark gaps, so the pulse power generator SPEED 2 is a medium energy and large current device (4.1 μ F equivalent Marx generator capacity, 300 kV, 4 MA in short circuit, 187 kJ, 400 ns rise time, $di/dt \sim 10^{13}$ A/s). The SPEED 2 arrived at the CCHEN (Chile) in May 2001 from Duesseldorf University, Germany, and it is in operation since January 2002, being the most powerful and energetic device for dense transient plasma in South America. Table 5.22 shows the machine parameters for the SPEED 2 device.

We make a simplistic approximation of the Marx-bank to its simple capacitor-bank equivalent and use the Lee Model 5-phase code to fit the computed to the measured current waveform. Considering the simplistic approximation of the power source the fit as shown in Figure 5.18 may be deemed to be reasonably good. The extra oscillatory features of the measured current waveform are obviously due to the behavior of the Marx-bank, which are not factored into the computation.

Table 5.22 The machine, operational and fitting parameters for SPEED2 Plasma Focus machine.
Capacitance C_0 (μF) = 4.16
External or static inductance L_0 (nH) = 20
Circuit resistance r_0 ($\text{m}\Omega$) = 7
Cathode radius 'b' (cm) = 11
Anode radius 'a' (cm) = 5.4
Anode length 'z ₀ ' (cm) = 2.5
Charging voltage V_0 (kV) = 180
Fill pressure P_0 (Torr) = 1.875
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.5$
Axial phase current factor, $f_c = 0.7$
Radial phase mass factor, $f_{mr} = 0.5$
Radial phase current factor, $f_{cr} = 0.7$

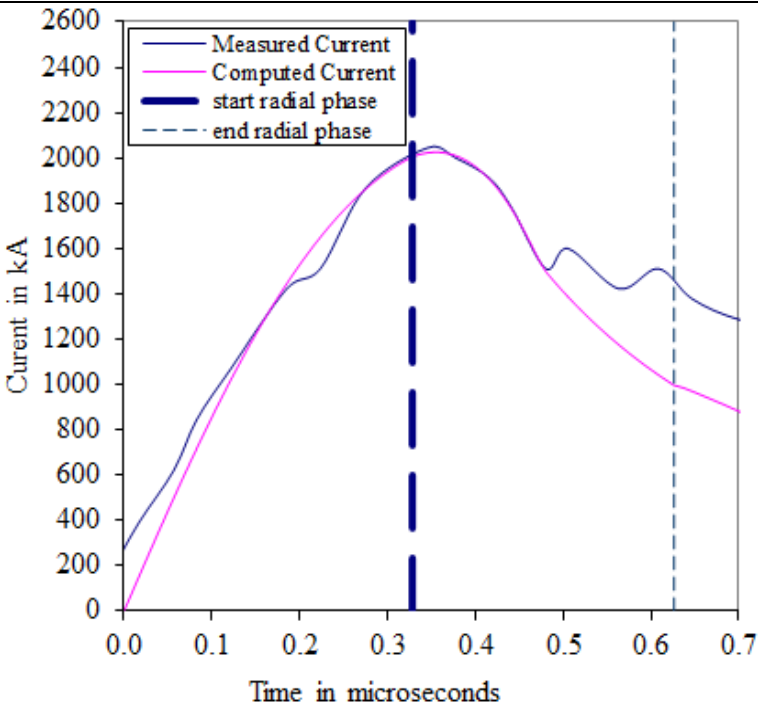


Figure 5.18 The measured current waveform at 180 kV, 1.875Torr deuterium gas compared with Lee Model 5-phase code computed current waveform.

Note: The measured current waveform information was extracted from Figure 5, page 1818 of reference [25]. This fit is reasonably good considering that we are approximating the Marx-generator of the machine to a simple equivalent capacitor bank.

5.13 University Malaya Plasma Focus machines UMDPF1 and UMDPF2 (12 kJ - Type 1 machines - D gas)

This survey of plasma focus machines concludes with the 12 kJ Plasma Focus machines which operated at University Malaya, Kuala Lumpur in the 1970's and 80's. That research was started in parallel with the development of the 40 kV, 60 μ F (48 kJ) capacitor bank which was completed and published in 1973 with a measured current of 1.91 MA [96], one of the biggest capacitor current in Asia at that time. The University Malaya Plasma Focus machine (UMDPF1) was constructed and used for studies of axial and radial phase dynamics in the period 1971-1973 [97-100] culminating with the measurement of D-D neutrons by time-of-flight method, reported in 1975 [101]. PhD theses using this plasma focus were completed in 1978 [102], 1987 [103] and 1989 [104]. Many features of our knowledge of the plasma focus (incorporated into the Lee Model code) such as mass shedding and current shedding, axial and radial phases, dimensions and lifetimes of the plasma focus was developed from these formative research efforts.

The UMDPF1 was designed for a convenient maximum operating voltage of 20 kV using the 60 μ F bank. A typical operating point for which we have a current trace [103] is the following (at 14 kV, 7.6 Torr (10 mbar) deuterium) which is shown in the Table 5.23. The current fitting is shown in Figure 5.19.

The maximum computed current at 14 kV was 285 kA and exhibits the radial start time at 2.26 μ s for pinch duration of 0.133 μ s. It has a speed factor of 83 (kA/cm)/Torr^{0.5} and neutron yield of 1.84×10^8 n/shot. The detailed information is presented in Table 5.24.

Using the information obtained from the current waveform, the UMDPF1 Plasma Focus machine was configured to work from 1.0 Torr to 25.0 Torr deuterium at a voltage of 14 kV to get the neutron yield versus pressure curve.

The waveform is presented in Figure 5.20.

Table 5.23 The machine, operational and fitting parameters for UMDPF1 Plasma Focus machine.

Capacitance C_0 (μF) = 62.2
Static inductance L_0 (nH) = 60
Circuit resistance r_0 (m Ω) = 3.5
Cathode radius 'b' (cm) = 4.25
Anode radius 'a' (cm) = 1.25
Anode length 'z ₀ ' (cm) = 15
Charging voltage V_0 (kV) = 14
Fill pressure P_0 (Torr) = 7.6
Fill gas (molecular weight) = 4
Fill gas (atomic number) = 1
Fill gas (molecule) = 2
The fitted model parameters are:
Axial phase mass factor, $f_m = 0.04$
Axial phase current factor, $f_c = 0.7$
Radial phase mass factor, $f_{mr} = 0.3$
Radial phase current factor, $f_{cr} = 0.7$

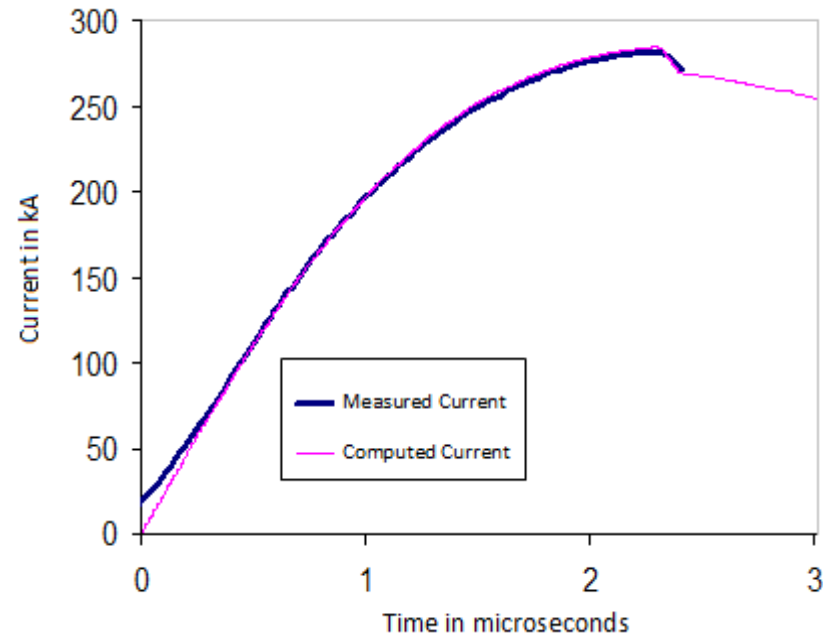


Figure 5.19 The measured current waveform compared with Lee Model 5-phase code computed.

Note: The measured current waveform information was taken from Figure 3, page 161 of reference [101].

Table 5.24 Information obtained from Lee Model 5-phase code configured for UMDPF1 Plasma Focus machine at 14 kV, 7.6 Torr deuterium gas.

Peak current (kA) = 285	Final pinch radius $r_{\min}$ (cm) = 0.18
Pinch start current (kA) = 191	Pinch length $z_{\max}$ (cm) = 2
Pinch minimum temperature (10^6 K) = 2.45	Pinch duration (ns) = 17.1
Pinch maximum temperature (10^6 K) = 2.51	Peak induced voltage (kV) = 22.3
Peak axial speed (cm/ μ s) = 10.6	Neutron yield ($\times 10^8$ n/shot) = 1.84
Peak radial shock speed (cm/ μ s) = 21.4	Energy Inflow into Plasma (%) = 12.2
Peak radial piston speed (cm/ μ s) = 14.6	Speed Factor ((kA/cm)/Torr ^{0.5}) = 83
Current per cm anode radius (kA/cm) = 228	

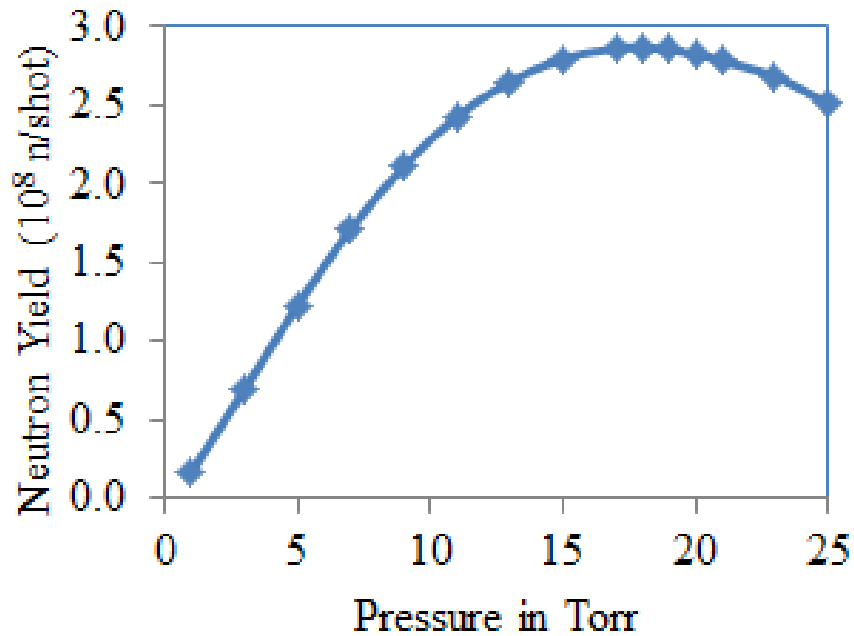


Figure 5.20 Variation of computed neutron yield with respect to pressure when capacitor in UMDPF1 Plasma Focus machine is charged to 14 kV in deuterium gas.

The maximum computed neutron yield obtained was 2.87×10^8 n/shot at 18.0 Torr whereas the energy input into plasma at this instant is 15.6 %. The measured yield for this machine at 14 kV, 7.6 Torr was recorded as 10^8 n/shot [101].

This compares with our computed yield of 1.8×10^8 n/shot at this operating condition.

In another series to optimize the neutron yield, the UMDPF1 delivered an optimum yield of 1.5×10^9 n /shot with 20 kV, 62 μ F, 12 kJ at 10.0 Torr [106]. This compared with our computed yield of 0.7×10^9 n/shot at 20 kV 10.0 Torr.

In their effort to optimize further, the tube dimensions, bank parameters and operational parameters of the machine were varied resulting in an optimum yield of 6.3×10^9 n/shot at 16.0 Torr deuterium at 33 kV, 22 μ F, 12 kJ using the dense plasma focus called UMDPF2 with configuration of b/a of 3.25 cm/1.25 cm, and anode length of 8.5 cm. A peak current of 760 kA was measured for that shot [106]. Using the new configuration and assuming the model parameters fitted for UMDPF1 we compute a neutron yield of 2.5×10^9 n/shot at 33 kV 16.0 Torr.

In summary our computed values of neutron yield differ from the measured neutron yield values of the Universiti Malaya Dense Plasma Focus machines UMDPF1 and UMDPF2 by a factor less than 3

CHAPTER 6

Discussion

6.1 Introduction

Plasma focus machines are available in various sizes and stored energy E_0 from the small Nano focus machine up to the large mega-Joule machine such as the PF1000.

Two different sizes of the machine are shown below. Figure 6.1 shows the UNU ICTP PFF 3 kJ device [14, 55] mounted on a 1 m by 1 m by 0.5 m trolley. The single capacitor is seen mounted on the trolley. In contrast, Figure 6.2 shows the 300 times larger (in terms of E_0) PF1000, the 1 MJ device at the ICDMP in Warsaw Poland [4]. Only the chamber and the cables connecting the plasma focus to the capacitors are shown. The capacitor bank with its 288 capacitors, switches and chargers are located in a separate hall.



Figure 6.1 3 kJ UNU ICTP PFF.



Figure 6.2 1 MJ PF1000 plasma focus.

Taken from “The Plasma Focus-Scaling Properties to Scaling laws” [92]

To understand plasma focus machines better, we will first discuss the results of the 44 machines we have analyzed in Chapters 3, 4 and 5.

6.2 Investigation

Mather type machine were normally classified either as Type 1 or Type 2 depending on their static inductance. A preliminary rather arbitrary classification [75] is to classify machines as Type 1 if their static inductance is below 100 nH and those above this value are classified as Type 2. It is suggested that Lee Model 5-phase code is adequate to fit Type 1 machines but Type 2 machines require the Lee Model 6-phase code. In discussing the demarcation of Type 1 and Type 2 machines by a value of $L_0=100$ nH, it was realized [75] that a more definite criterion should depend on energy distributions which may depend on relative sizes of various inductances rather than just L_0 .

During the course of our investigation, we found that some of the Type 1 machines have a much longer and deeper extended dip than what was normally anticipated. This occurs when the focus occurs at a too-early time before the natural current peak such that not enough of the capacitor bank energy has been converted to the magnetic field driving the plasma. In other words, if the plasma focus device is operated with the radial start time significantly coming before the quarter time period ($T/4=\pi\sqrt{LC}/2$) of the discharge current.

From the 44 machine investigated, we found examples of current waveforms from low L_0 machines operated with too-early current dip. These include the AASC [107, 108] ($L_0 = 15.8$ nH), NX1 [5, 6, 7] ($L_0 = 29$ nH), PF143 [125, 126 127] ($L_0 = 42$ nH), India's smallest seal type machine [109] ($L_0 = 46$ nH).

From considering the other machines, we tentatively find that when 'low-inductance' plasma focus machines (say 50 nH and below) are operating in energy optimised mode which means the axial phase ends close to the time of peak current (time-matched), then these machines almost invariably can be fitted with the 5-phase model and in this sense exhibit Type-1 characteristics. We suggest adding to the criterion of Type 1 characteristics not just 'low-inductance machines' but 'low inductance machines operating under time-matched conditions'.

The practical strategy in fitting is to use the 5-phase code and see if a good fit can be obtained of the computed current dip to the measured current dip. If there is a left-over part of the measured dip which cannot be fitted by the 5-phase, then we need to use the part of the code that extends into the anomalous resistive phases.

We also noticed that as the gas in the plasma focus gets contaminated (due to evaporation of electrode materials and the accumulation of other gases), the mass factor increases. This will result in slower axial speed as was observed in the India's smallest seal type Plasma Focus. The lower speed in this machine was due to 200 shots being fired without changing the gas in between the shots thus resulting in deteriorating neutron yield after each shot.

Lee Model 5-phase or 6-phase code is one of the best tools in understanding the dynamics of plasma and it can be used from the smallest Mather type plasma focus machine to the largest machines (PF1000). Lee Model 5-phase code has also been successfully used to fit the measured current waveform of the smallest hybrid machine to date, which is the Chile nanofocus machine. One of the technical steps we had to take to do this fitting was to reduce the step-size in computing the radial phase.

Finally, even though Lee Model 5-phase or 6-phase code is versatile for analyzing the plasma focus, it still has some limitations. Firstly, the codes were not designed to fit capacitors design using Marx modules connected in parallel such as in Speed 2 Plasma Focus machine [25] and secondly, the codes were not designed to fit Filippov type machine such as PF3 [139] and Dena [82, 141] plasma focus machine without modification as explained in the paper "Adaptation of Sing Lee's model to the Filippov type plasma focus geometry" [82].

6.3 Machines working in deuterium gas medium

We now continue our discussion by comparing the computed neutron yield of all the machines we have analysed in Tables 6.1 to 6.3. We also compare computed neutron yields with the measured yields for those machines which we have the data.

When optimum computed neutron yield was plotted as a function of the energy input into the plasma focus machine as shown in Figure 6.3, $Y_n = 5 \times 10^6 E_0^{1.9061}$ was obtained. This formula can also be rounded up as $Y_n \sim E_0^{1.91}$ which is close to the generally accepted neutron yield versus energy scaling law of $Y_n \sim E_0^2$ [67, 69, 73, 76, 142].

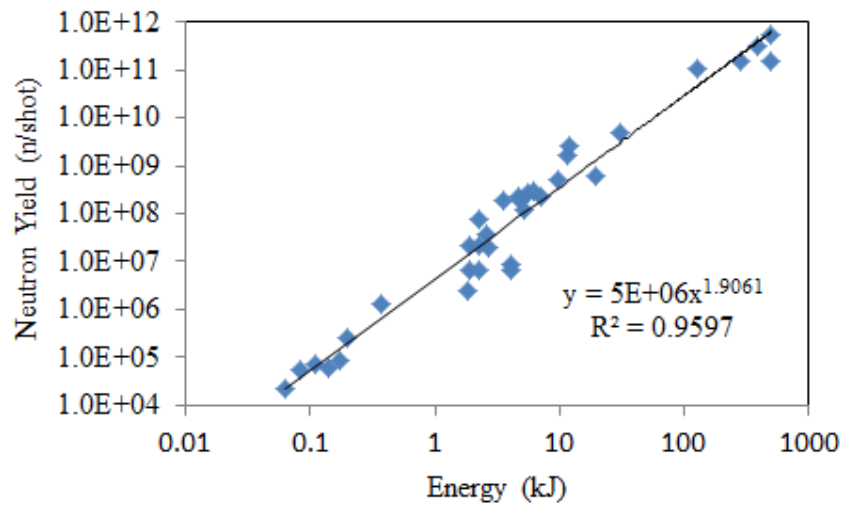


Figure 6.3 Computed neutron yield versus input energy in the plasma focus machine

Table 6.1 Parameters corresponding to optimum computed neutron yield for the machines named below (1/3).

Machine Name	PF50	PF50	AASC	Nano focus	FMPF-1	IS PF	PF400
Energy (J)	63	84	110	141	173	200	372
Optimum neutron yield ($\times 10^4$ n/shot) [computed]	2.16	5.23	7	6.2	8.8	25	125
Measured neutron yield ($\times 10^4$ n/shot)	1.2 ± 0.5	3.6 ± 1.6	26	NA	1 ± 0.27	7.8	106 ± 13
Pinch current (kA)	39	45	57	50	48	62	84
Peak current (kA)	54	63	87	54	69	82	125
Pinch duration (ns)	1.6	1.6	6.7	5.4	3.5	3.9	4.4

Note: The Nanofocus parameters are for the Argentina machine whereas “IS PF” is India’s smallest PF

Machine Name	KSU-PF2	Imperial	Paco	NX2T	Monte	DPF2.2	PF12
Energy (kJ)	1.8	1.9	1.9	2.2	2.2	2.3	2.6
Optimum neutron yield ($\times 10^6$ n/shot) [computed]	2.56	6.8	21	74.5	6.8	21	37.8
Measured neutron yield ($\times 10^6$ n/shot)	NA	20 ± 5	200	100	NA	31 ± 15	NA
Pinch current (kA)	94	115	148	188	124	133	142
Peak current (kA)	141	172	229	362	190	199	212
Pinch duration (ns)	5.2	7.3	14.9	23.1	7.1	9.3	13.3

Table 6.2 Parameters corresponding to optimum computed neutron yield for the machines named below (2/3).

Machine Name	UNU	BORA	Hanyang	Hanyang	GN1	FNII	Assam
Energy (kJ)	2.7	3.5	4.1	4.1	4.7	4.9	5.1
Optimum neutron yield ($\times 10^8$ n/shot) [computed]	0.2	1.83	0.09	0.067	2.2	1.89	1.22
Measured neutron yield ($\times 10^8$ n/shot)	0.5-1.0	NA	1.24	0.667	2.3	1.66 ± 0.14	3.3
Pinch current (kA)	110	197	108	107	252	217	198
Peak current (kA)	166	303	157	157	361	328	259
Pinch duration (ns)	7.9	28.1	12.6	9.1	15.7	29	21.7

Machine Name	GN1	UMDPF1	NX3	NX3	BARC	UMDPF2	PF143
Energy (kJ)	5.7	6.1	7.2	9.8	11.5	12	20.2
Optimum neutron yield ($\times 10^8$ n/shot) [computed]	2.6	2.87	2.37	5.22	16.3	25	6.1
Measured neutron yield ($\times 10^8$ n/shot)	2.3	≈ 1	NA	NA	15.8	63	NA
Pinch current (kA)	237	206	243	283	295	316	293
Peak current (kA)	390	314	422	493	474	467	697
Pinch duration (ns)	19.8	24.4	29.3	30.3	53.6	17.7	26.9

Table 6.3 Parameters corresponding to optimum computed neutron yield for the machines named below (3/3).

Machine Name	DPF78[136]	Tamu	Poseidon[16]	Poseidon[16]	PF1000	Poseidon[16]
Energy (kJ)	31	126	281	382	485	500
Optimum neutron yield ($\times 10^{11}$ n/shot) [computed]	0.05	1.09	1.6	3.11	1.47	5.43
Measured neutron yield ($\times 10^{11}$ n/shot)	0.08	2	0.9	1.4	2(Max)	2(Avg) 2.5(Max)
Pinch current (kA)	445	753	972	1125	826	1249
Peak current (kA)	891	1676	3468	4070	1980	4727
Pinch duration (ns)	29.6	124.8	116.2	120	385.9	132.5

When measured (experimental) neutron yield data was added to the optimum computed neutron yield data and plotted as a function of the energy input into the plasma focus machine as shown in Figure 6.4, $Y_n = 5 \times 10^6 E_0^{1.9479}$ was obtained. This formula can also be rounded up as $Y_n \sim E_0^{1.95}$ which is close to the generally accepted neutron yield versus energy scaling law of $Y_n \sim E_0^2$.

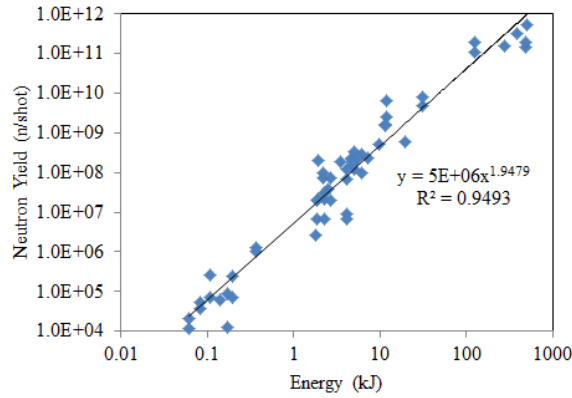


Figure 6.4 Neutron yield versus input energy (All data-computed and measured).

When optimum computed neutron yield was plotted as a function of the peak current into the plasma focus machine as shown in Figure 6.5, $Y_n = 0.016 I_{\text{peak}}^{3.8589}$ was obtained. This formula can also be rounded up as $Y_n \sim I_{\text{peak}}^{3.86}$ which is close to $Y_n \sim I_{\text{peak}}^{3.9}$ obtained by S Lee and SH Saw in their paper “Neutron Scaling Laws from Numerical Experiments” [73].

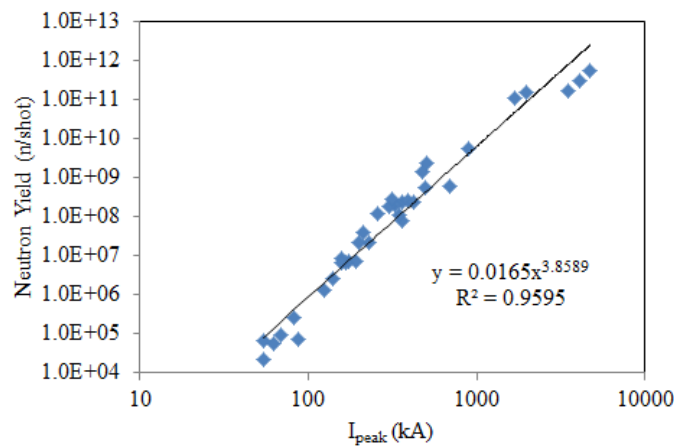


Figure 6.5 Computed neutron yield versus peak current.

When optimum computed neutron yield was plotted as a function of the pinch current into the plasma focus machine as shown in Figure 6.6, $Y_n = 0.0003 \cdot I_{\text{pinch}}^{4.9923}$ was obtained. This formula can also be rounded up as $Y_n \sim I_{\text{pinch}}^{5.0}$ which is close to $Y_n \sim I_{\text{pinch}}^{4.7}$ which was obtained by S Lee and S H Saw in their paper “Neutron Scaling Laws from Numerical Experiments” [73].

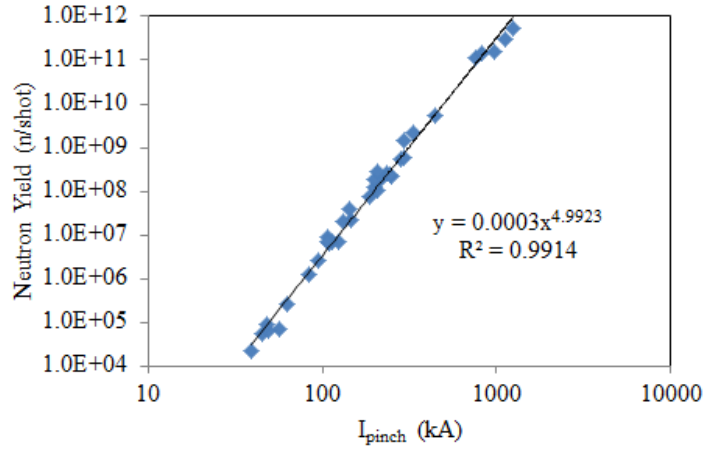


Figure 6.6 Computed neutron yield versus pinch current.

When computed peak current was plotted as a function of the energy storage E_0 of the plasma focus machines as shown in Figure 6.7, $I_{\text{peak}} = 155.42 \cdot E_0^{0.4794}$ was obtained. This formula can also be rounded up as $I_{\text{peak}} \sim E_0^{0.48}$.

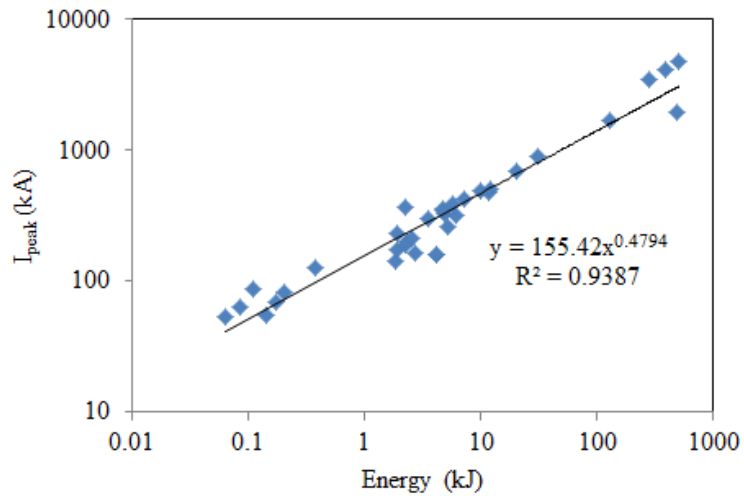


Figure 6.7 Computed peak current versus energy input into plasma.

When computed pinch current was plotted as a function of the energy storage E_0 of the plasma focus machines as shown in Figure 6.8, $I_{\text{pinch}} = 105.98 E_0^{0.3793}$ was obtained. This formula can also be rounded up as $I_{\text{pinch}} \sim E_0^{0.38}$.

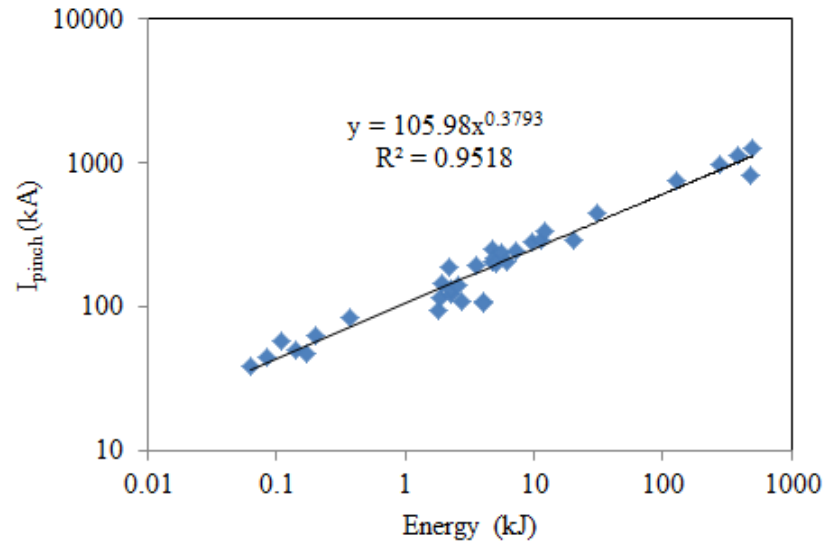


Figure 6.8 Computed pinch current versus energy input into plasma.

When optimum computed neutron yield was compared to the pinch duration (Pd) of the plasma as shown in Figure 6.9. The formula $Y_n = 3636.4 \text{ Pd}^{3.507}$ was obtained. This formula can also be rounded up as $Y_n \sim \text{Pd}^{3.5}$.

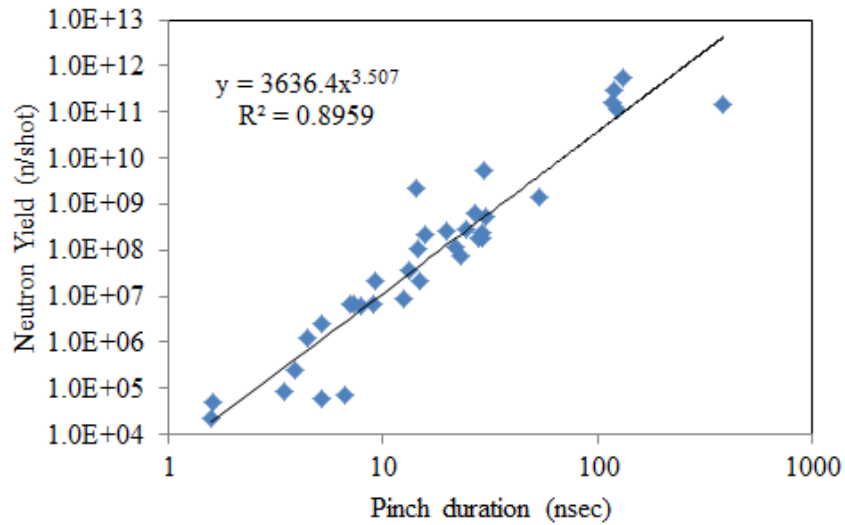


Figure 6.9 Computed neutron yield versus pinch duration.

6.4 Machines working in neon gas medium

We now continue our discussion by comparing the plasma focus machine operated in neon, whose current waveforms we have obtained. Lee Model 5/6-phase code was used to simulate that particular focus machine (done in Chapters 3, 4 and 5) for a range of pressure at a given voltage. The optimum neon soft X-ray yield was found. We tabulate these computed optimum neon SXR yields and other parameters in Table 6.4.

Table 6.4 Parameters corresponding to computed neon SXR for the machines named below.

Machine Name	NX1	NX2	KSU	NX2	Amir	UNU
Energy (kJ)	1.6	1.7	1.8	1.9	2.6	2.9
Peak Current (kA)	278	369	145	389	170	180
Pinch current (kA)	182	145	61	202	110	107
Plasma Column (cm)	0.11	0.16	0.05	0.16	0.08	0.06
Pinch Duration (ns)	16.7	30.2	7.5	29.6	13.9	8.9
EINP (%)	20.8	27.5	5.9	35.2	11	10.4
EINP(J)	324	466	106	671	286	305
Soft X- ray yield (J) -computed	23.6	26.2	0.8	44.7	7.4	4
Soft X- ray yield (J) -measured	19	19.1	NA	18	NA	5.4±1

We discuss the data in Table 6.4. When optimum computed neon soft X ray yield was plotted against pinch current in the plasma focus machine as shown in Figure 6.10 $Y_{\text{SXR}} = 8 \times 10^{-7} I_{\text{pinch}}^{3.3704}$ was obtained. This formula can also be rounded up as $Y_{\text{SXR}} \sim I_{\text{pinch}}^{3.4}$ which is closed to the results obtained from Saw and Lee in their paper “Scaling laws for plasma focus machines from numerical experiments” [67] in which they obtained $Y_{\text{SXR}} \sim I_{\text{pinch}}^{3.6}$.

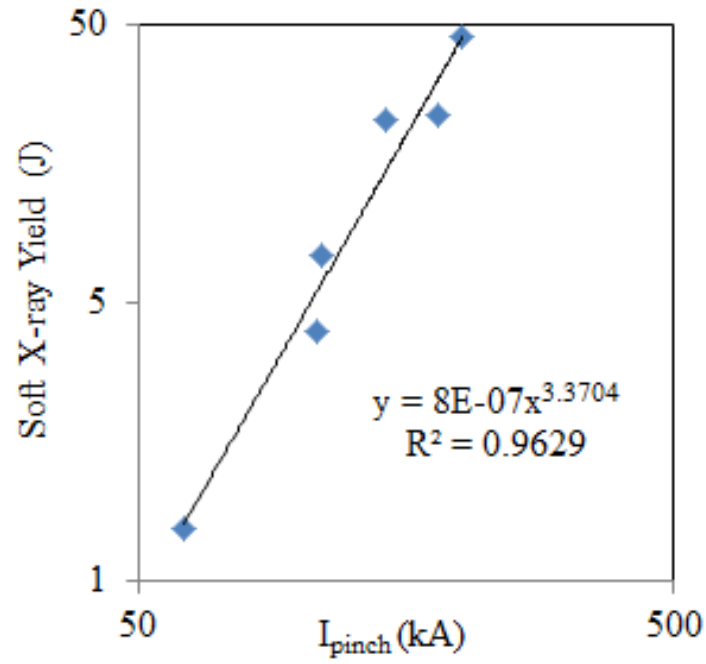


Figure 6.10 Computed neon soft X-ray yield versus pinch current.

When optimum computed neon soft X ray yield was compared to the peak current as shown in Figure 6.11. The formula $Y_{\text{SXR}} = 2 \times 10^{-7} I_{\text{peak}}^{3.1949}$ was obtained. This formula can also be rounded up as $Y_{\text{SXR}} \sim I_{\text{peak}}^{3.2}$ which is the results obtained by Saw and Lee in their paper “Scaling laws for plasma focus machines from numerical experiments” [67] in which they recorded their observed scaling law of $Y_{\text{SXR}} \sim I_{\text{peak}}^{3.2}$.

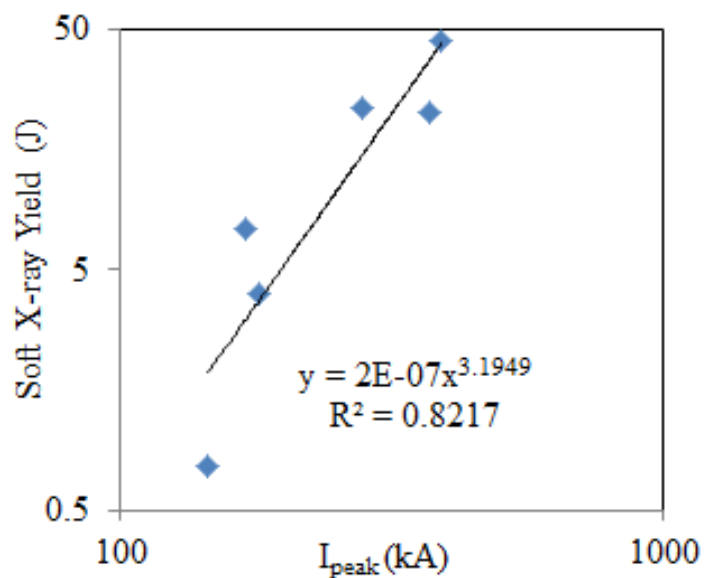


Figure 6.11 Computed neon soft X-ray yield versus peak current.

When optimum computed neon soft X-ray yield was plotted against the actual input of energy into plasma radial phase (EINP) as shown in Figure 6.12. The formula $Y_{\text{SXR}} = 2 \times 10^{-5} \text{EINP}^{2.2616}$ was obtained. This formula can also be rounded up as $Y_{\text{SXR}} \sim \text{EINP}^{2.26}$.

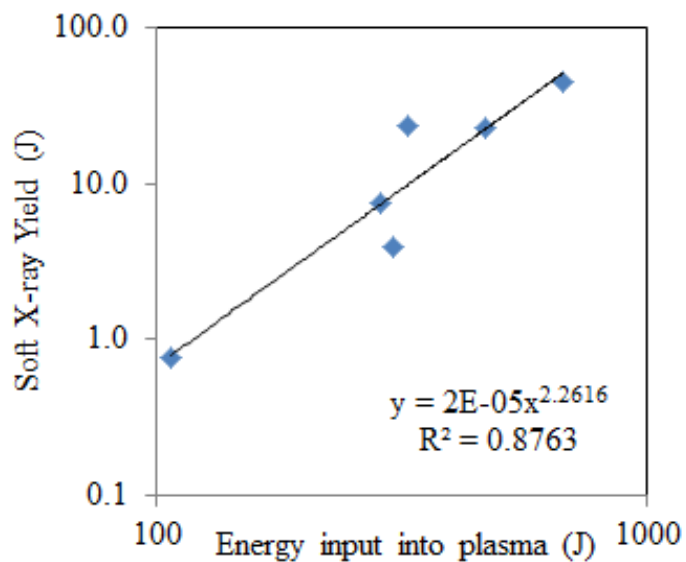


Figure 6.12 Computed neon soft X ray versus energy input into plasma.

When computed actual input of energy into plasma (EINP) was compared to the pinch duration (Pd) as shown in Figure 6.13. The formula $EINP = 25.327 Pd^{0.9175}$ was obtained. This formula can also be rounded up as $EINP \sim Pd^{0.92}$.

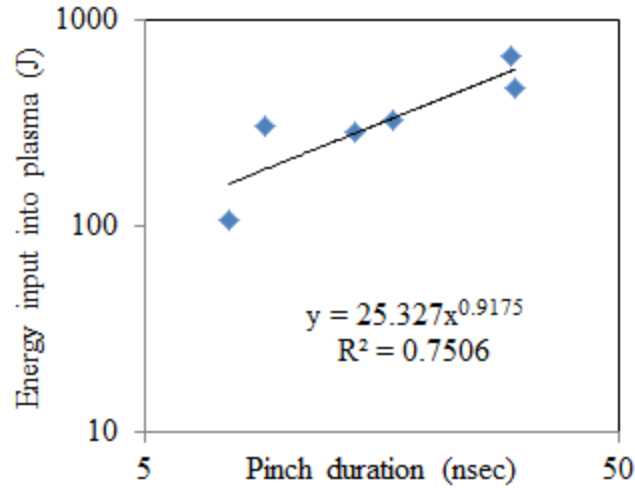


Figure 6.13 Computed energy input into plasma versus pinch duration

6.5 Machines working in argon gas

We now continue our discussion by comparing the plasma focus machine whose current waveforms we have obtained working in the argon gas medium. Lee Model 5 phase or 6-phase code was used to simulate that particular focus machine for a range of pressure at the given voltage the capacitor was charged to and the optimum all-line yield and other important parameters we want to compare are noted in Table 6.5. It should be noted that the computed argon all-line yield is different from the measured argon characteristic line yield (wavelength of 3-4 Å).

Table 6.5 Parameters corresponding to computed argon all-line yield for the machines named below.

Machine Name	PPRC	PPRC	PPRC	Tehran	UNU	AECS PF-1	AECS PF-2
Energy (kJ)	1	1.7	1.8	2	2.7	2.8	2.8
Peak current (kA)	96	127	131	152	171	58	108
Pinch current (kA)	66	81	76	77	101	36	68
Plasma column (cm)	0.02	0.02	0.02	0.02	0.02	0.03	0.02
Pinch duration (ns)	12.9	11.7	12.3	12.5	10.9	15.5	11.5
Pinch Voltage (kV)	69.7	112.8	93.5	90.3	134.4	27.4	87.3
EINP (%)	9.5	9.6	9.0	10.0	12.5	1.2	4.5
EINP (J)	96	164	162	199	342	34	127
All-line yield (J)	25.3	36.1	32.6	34.0	58.0	8.3	28.5

When optimum computed all-line yield was compared to the peak current in the plasma focus machine as shown in Figure 6.14. The formula $Y_{\text{all-line}} = 0.0121 I_{\text{peak}}^{1.6275}$

was obtained. This formula can also be rounded up as $Y_{\text{all-line}} \sim I_{\text{peak}}^{1.63}$

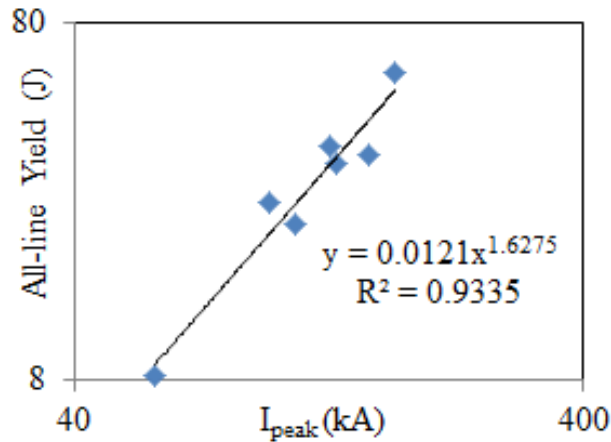


Figure 6.14 Computed all-line yield versus peak current.

When optimum computed all-line yield in argon was compared to the pinch current in the plasma focus machine as shown in Figure 6.15. The formula $Y_{\text{all-line}} = 0.0109 I_{\text{pinch}}^{1.8477}$

was obtained. This formula can also be rounded up as $Y_{\text{all-line}} \sim I_{\text{pinch}}^{1.85}$

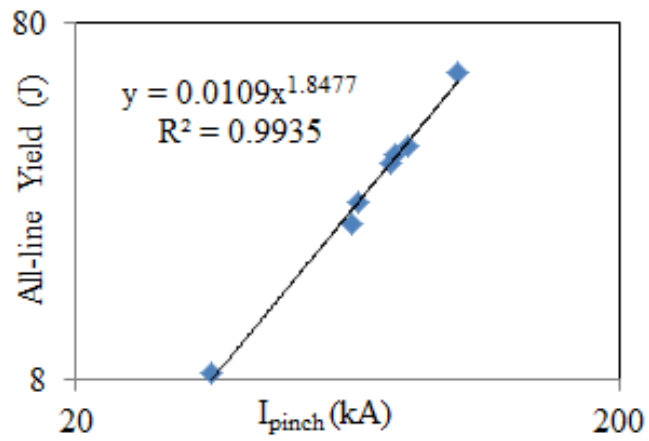


Figure 6.15 Computed all-line yield versus pinch current.

When optimum computed all-line yield was compared to the energy input into plasma as shown in Figure 6.16. The formula $Y_{\text{all-line}} = 0.4759 \text{ EINP}^{0.8258}$ was obtained. This formula can also be rounded up as $Y_{\text{all-line}} \sim \text{EINP}^{0.83}$.

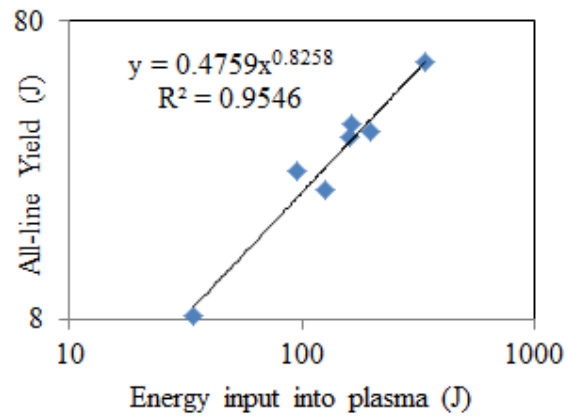


Figure 6.16 Computed all-line yield versus energy input into plasma.

CHAPTER 7

Conclusion

7.1 Introduction

The Lee Model code which was originally developed from 2 phase (axial and radial) model [53-56] in 1984 is a powerful and convenient numerical experiment facility as it can be configured to any conventional Mather-type plasma focus by inputting the relevant machine parameters which includes the inductance, capacitance, electrode radii and length and the operating parameters such as charging voltage and fill gas pressure as well as model parameters f_m , f_{mr} , f_c and f_{cr} . This model has helped in the analysis of Mather and Filippov type plasma focus machines (with some modification as explained in the paper “*Adaptation of Sing Lee’s model to the Filippov type plasma focus geometry*” [82]).

This thesis has analysed 44 Mather-type plasma focus machines operating in deuterium, neon and argon; obtaining good fit for all of them so that realistic plasma focus properties are computed and presented. Even the Speed 2 Plasma Focus machine [25] which uses Marx bank could be approximately modelled and analysed. This thesis also attempted to fit several Filippov-type plasma focus machines and found that the computed current traces could be fitted to the measured only down to about half the current dip. Thereafter the computed dip of the Filippov-type plasma focus machine could not be fitted.

7.2 Observation obtained from the comparative study

For optimised low-inductance plasma focus devices (typically those in the region of or below 50 nH) the Lee Model 5-phase code is found to be sufficient to fit the measured current traces. These Type-1 machines when not properly time-matched are sometimes found to require the Lee Model 6-phase code in order to complete the fitting of extended parts of the measured current dip. Type-2, high inductance plasma focus devices (typically those above 100 nH) invariably require

the Lee Model 6-phase code even when the measured current traces are taken from optimised shots. The range of plasma focus devices between 60-100 nH is rather more difficult to classify. The actual classification as to whether a machine is Type-1 or Type-2 may require a more subtle distinction based on energy distribution which depends more on relative inductances (for example static inductance, axial phase inductance, pinch inductance); as has been discussed by Lee et al [75]. This thesis uses the 5-phase code to start the analysis of a machine, and when necessary proceeds to use the 6-phase code to complete the fitting of any extended current dip which is found to be insufficiently fitted using the 5-phase code.

Care must be taken when considering the measured current waveforms of plasma focus devices which are operated with contamination due for example to evaporation of electrode materials leading to release of adsorbed gases. The unaccounted-for gases will present additional loading to the actual discharge. This may lead to unusually high values of mass fractions f_m and f_{mr} being needed to fit the measured current waveforms. Advantage could be taken of this to identify shots in which the discharge may be contaminated; such as observed in the smallest sealed Plasma Focus machine developed in India [109].

The code has also proven to be useful for testing whether machines are optimised or not. This has been the cases of the India's-Assam Plasma focus machine [124] with electrodes which were designed to work in an energy range different from its capacitor bank. This was also the case of the NX3 during its process of optimization [10]. In a sense most machines during optimization will find the code useful to assist in guiding the experimental process towards optimised operation.

7.3 Comparison of computed yields and measured yields

In analyzing the various machines we compared the computed neutron yields to the measured (where available) and in almost all cases for low inductance machines found that the computed optimised neutron yield agrees with the measured

maximum yield to a factor better than 3. Moreover in every case the computed neutron yield versus pressure curve has features of agreement with the measured neutron yield versus pressure curve. The exceptions are the FMPF-1 [12] and PACO [113]. The AASC [107] is a plasma focus with a transformer-coupled capacitor bank, and will not be considered for purpose of comparing computed with measured neutron yield since the transformer coupling mechanism is not modeled in detail. In the case of PACO, extensive studies [103] had concluded that PACO is exceptionally erratic in neutron yield. Moreover the results of Castillo et al [115] suggests that the average yield is more correctly stated to be less than 1×10^8 n/shot (rather than 2×10^8 , 3×10^8 or 4×10^8 n/shot which has been variously claimed for the machine). This has been carefully examined in section 3.14 of this thesis. Thus the difference between our computed PACO neutron yield is likely to be considerably less than we have stated based on the claimed value of 2×10^8 n/shot. In the case of FMPF-1, we believe that the low value of the measured neutron yield (1/9 of the computed optimum) indicates that the FMPF-1 had not been optimally adjusted in some of its machine characteristics (for example, backwall insulator configuration, electrode materials and possibly detailed geometry).

For high inductance machines we have less data and where we have data to compare, the agreement is not as good as the typical low inductance machine. For example for Hanyang plasma focus machine [122] a difference of 10-14 times was found; the computed yield being far lower than the measured. The other high inductance machine we have data on, the UNU/ICTP PFF [55] has a computed optimum neutron yield which is also lower, being 1/4 of the measured value. Looking at all data that we have, we are led to believe that high inductance machines, classified as Type 2 by Lee et al [75] may share this characteristic of having enhanced neutron yield (i.e. measured value much larger than computed) due to the extended dip (ED) attributed to anomalous resistivity [75]. This idea could be checked when more measured neutron yield data becomes available from high inductance machines.

We also compared computed neon SXR yields with the measured data we have access to and found the agreement to be typically good in both optimised yields as well as yield versus pressure curves.

7.4 Scaling and Related scaling laws obtained from this study

Finally, combining the computed data of the machines analysed, the following scaling laws were obtained:

Neutron yield Y_n in deuterium scaling laws as functions of stored energy E_0 , peak circuit current I_{peak} and pinch current I_{pinch} :

$$Y_n \sim E_0^{1.95}$$

$$Y_n \sim I_{\text{peak}}^{3.86}$$

$$Y_n \sim I_{\text{pinch}}^{5.0}$$

Related scaling laws: (Pd is the pinch duration)

$$I_{\text{peak}} \sim E_0^{1.95}$$

$$I_{\text{pinch}} \sim E_0^{0.38}$$

$$Y_n \sim Pd^{3.5}$$

Neon SXR Y_{SXR} yield scaling laws

$$Y_{\text{SXR}} \sim I_{\text{pinch}}^{3.4}$$

$$Y_{\text{SXR}} \sim I_{\text{peak}}^{3.2}$$

$$Y_{\text{SXR}} \sim \text{EINP}^{2.26}$$

Where EINP is the energy input by the magnetic piston (current sheet) into the radial phase plasma:

$$\text{EINP} \sim Pd^{0.92}$$

Argon all-line yield $Y_{\text{all-line}}$ scaling laws:

$$Y_{\text{all-line}} \sim I_{\text{peak}}^{1.63}$$

$$Y_{\text{all-line}} \sim I_{\text{pinch}}^{1.85}$$

$$Y_{\text{all-line}} \sim \text{EINP}^{0.83}$$

7.5 Observation of Slow Focus Mode in high pressure operation

One final note is made here. During our numerical experiments on the various machines we typically run through a range of pressures. For each machine when the pressure is high enough the outward moving reflected shock is unable to reach the incoming magnetic piston and the plasma focus pinching fails. There is always a pressure when the reflected shock just manages to reach the incoming piston. At that operating pressure the pinch radius is typically around two times that of the pinch radius at the pressure when the focus is time-matched. This is the Slow Focus Mode (SFM) which has been described by Saw et al [143, 144]. This confirms that in all the machines we have experimented with, the SFM produces a much larger pinch cross-section. The SFM also produces [143, 144] less intense ion beams so that target interaction is more with the Fast Plasma Stream FPS than with the fast Ion Beams [145, 146]. These features may be desirable for deposition of advanced materials and nano-materials.

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